

14. Which one of the following factors affects the electrical resistivity of a conductor ?

(A) Temperature

(B) Length

(C) Shape

(D) Thickness

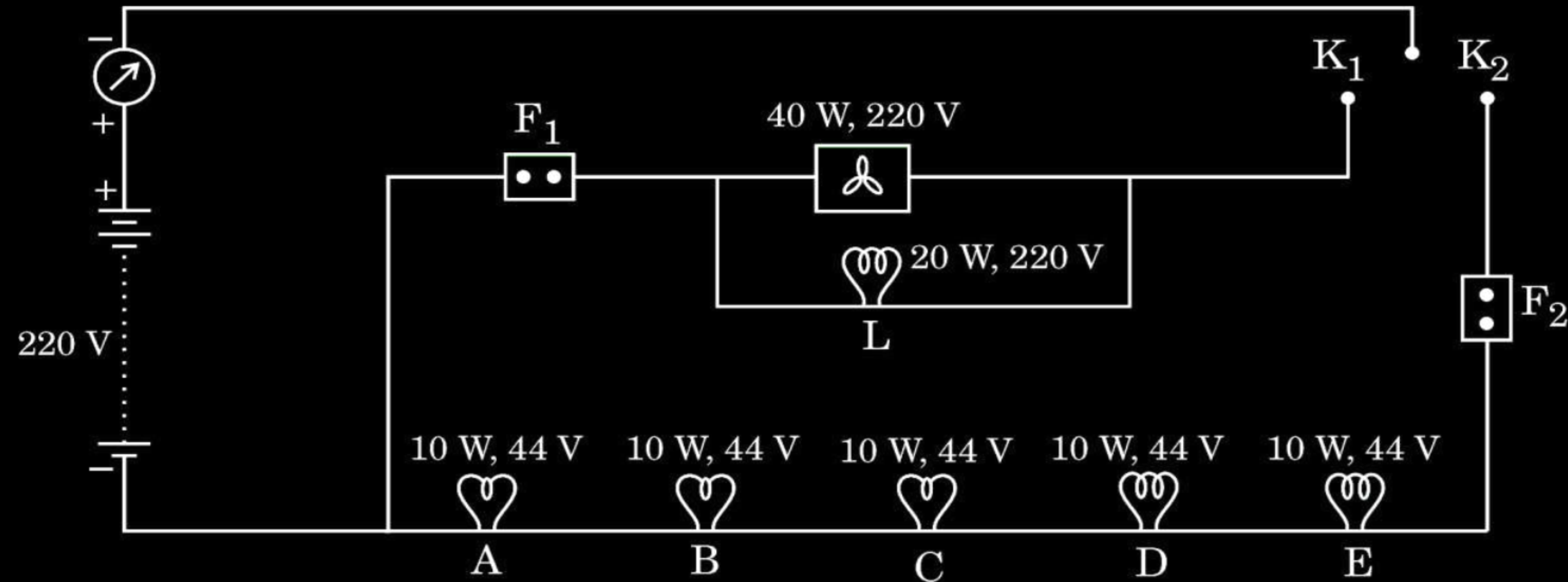
CBSE 2025 C

- 31.** (a) Write mathematical expression for Joule's law of heating.
- (b) Calculate the amount of heat generated while transferring 72000 coulomb of charge in two hours through a potential difference of 220 V.
- CBSE 2025 C

 1+2

38. In a domestic circuit, six LED bulbs and a ceiling fan are arranged as per the circuit shown in the figure. The source voltage is 220 volts and the power rating of each device is marked in the circuit diagram. Two fuses F_1 and F_2 are also provided in the circuit. Based on the circuit diagram, answer the following questions :

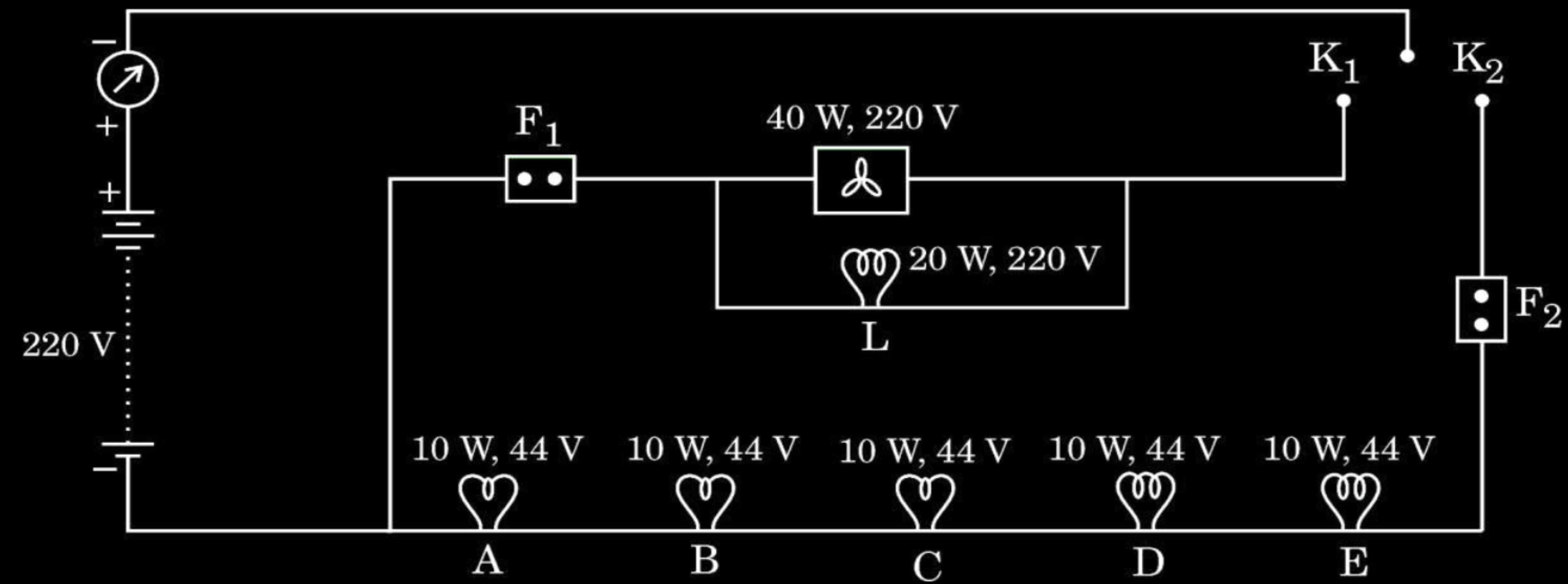
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- (a) State what happens when 1
- (i) only key K_1 is closed,
- (ii) and key K_2 is also closed.
- (b) Find the current drawn by the bulb L. 1
- (c) (i) Calculate the resistance of the bulb A. Find the total resistance offered by the circuit to the flow of current when only key K_2 is closed. 2

OR

- (c) (ii) What would happen to the glow of the bulbs in the circuit when K_1 and K_2 both are closed and the fuse F_1 melts 2



- 32.** Show the arrangement of three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ which are connected in such a way that the total/equivalent resistance of the combination is (a) $1\ \Omega$, and (b) $4\ \Omega$. Justify your answer in each case.

3

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15. Four wires a, b, c and d are made of the same materials. If ' l ' denotes length and ' r ' denotes radius, then the wire which offers least resistance to the flow of electric current has length and radius respectively as :

(A) $2l$ and $4r$

(B) l and $2r$

(C) $4l$ and $\frac{r}{4}$

(D) $\frac{l}{4}$ and $\frac{r}{2}$

- 25.** State Ohm's law. Give reason why the shape of voltage versus current ($V - I$) graph for a conductor should be a straight line passing through the origin.

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- 36.** (a) (i) If R_s and R_p respectively are the total resistances of the series and parallel combinations of the three resistors of resistances R_1 , R_2 and R_3 , then write the relationships among (I) R_s , R_1 , R_2 and R_3 and (II) R_p , R_1 , R_2 and R_3 .
- (ii) An electric oven operates on the mains voltage of 220 V. This oven consumes 2.2 kWh of electrical energy in one hour. Calculate :
- (I) power rating of the oven,
 - (II) current drawn by the oven, and
 - (III) resistance of the oven when in operation.

- (b) (i) Define the term electric power. Express it in terms of potential difference (V) and resistance (R).
- (ii) The resistance of a metal wire of length 6 m is $120\ \Omega$. If the area of cross-section of the wire is $4 \times 10^{-7}\ \text{m}^2$, calculate the electrical resistivity of the material of the wire. How would the resistivity be affected if this wire is gradually stretched so that its length is doubled ? Justify your answer.

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11. An electric bulb is connected to a power supply of 220 V. If the current drawn by the bulb from the supply is 500 mA, the power of the bulb is :

(A) 11 W

(B) 110 W

(C) 220 W

(D) 1100 W

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(B) 110 W

(C) 220 W

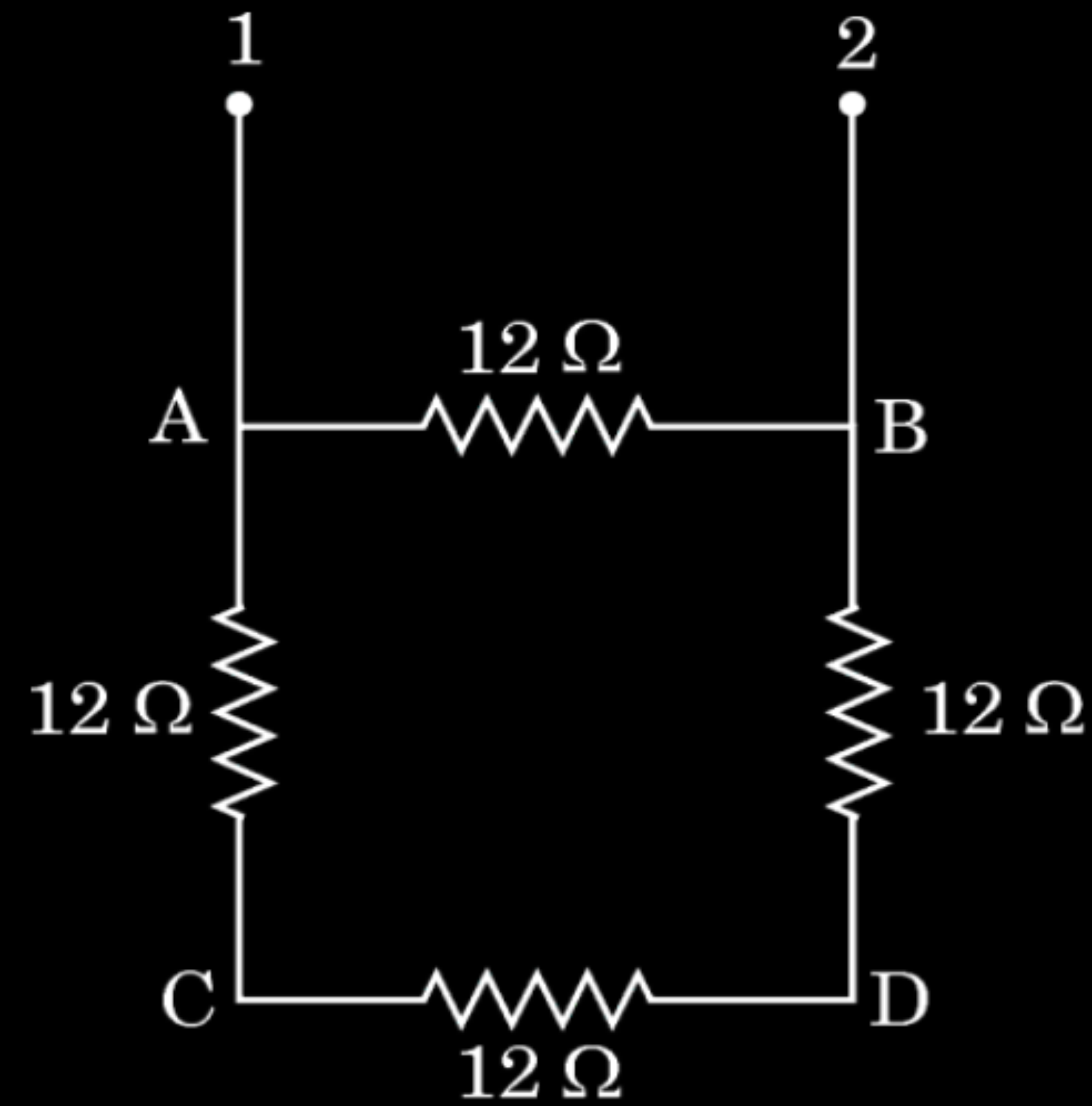
(D) 1100 W

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(B)/ 110 W

12. Four identical resistors of $12\ \Omega$ each are connected in series to form a square ABCD as shown in the figure. The resistance of the network between the two points 1 and 2 is :

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(A) $48\ \Omega$

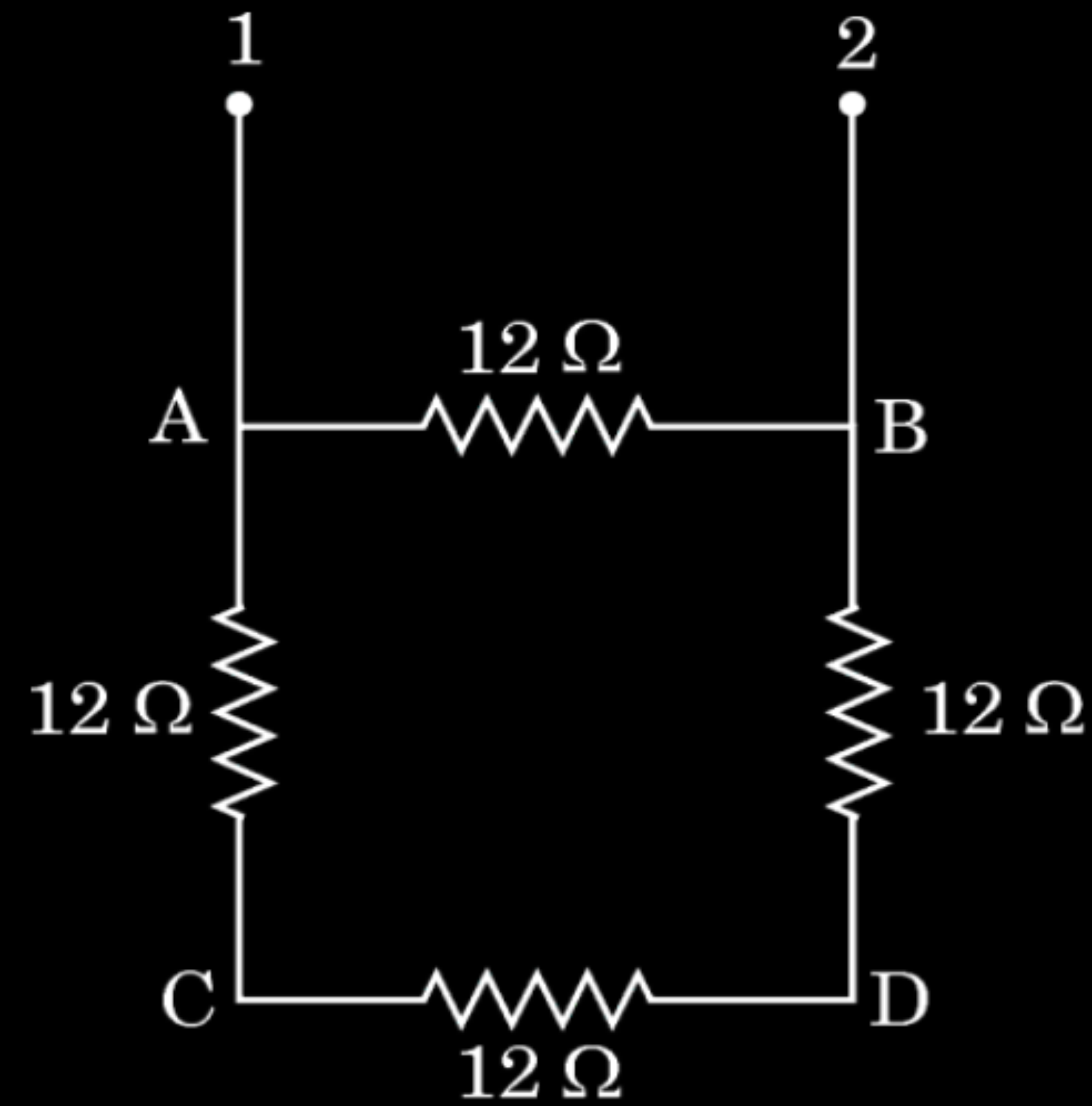
(B) $36\ \Omega$

(C) $9\ \Omega$

(D) $6\ \Omega$

12. Four identical resistors of $12\ \Omega$ each are connected in series to form a square ABCD as shown in the figure. The resistance of the network between the two points 1 and 2 is :

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(A) $48\ \Omega$

(B) $36\ \Omega$

(C) $9\ \Omega$

(D) $6\ \Omega$

(C) / $9\ \Omega$

19. *Assertion (A) :* The bending of a wire in the form of a coil of several turns does not affect its electrical resistance.

Reason (R) : The electrical resistance of a wire is directly proportional to its electrical resistivity.

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Reason (R) : The electrical resistance of a wire is directly proportional to its electrical resistivity.

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32. Three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ are connected in (i) series, and (ii) parallel. Draw the arrangements of the resistors and find the equivalent resistance of each arrangement.

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3

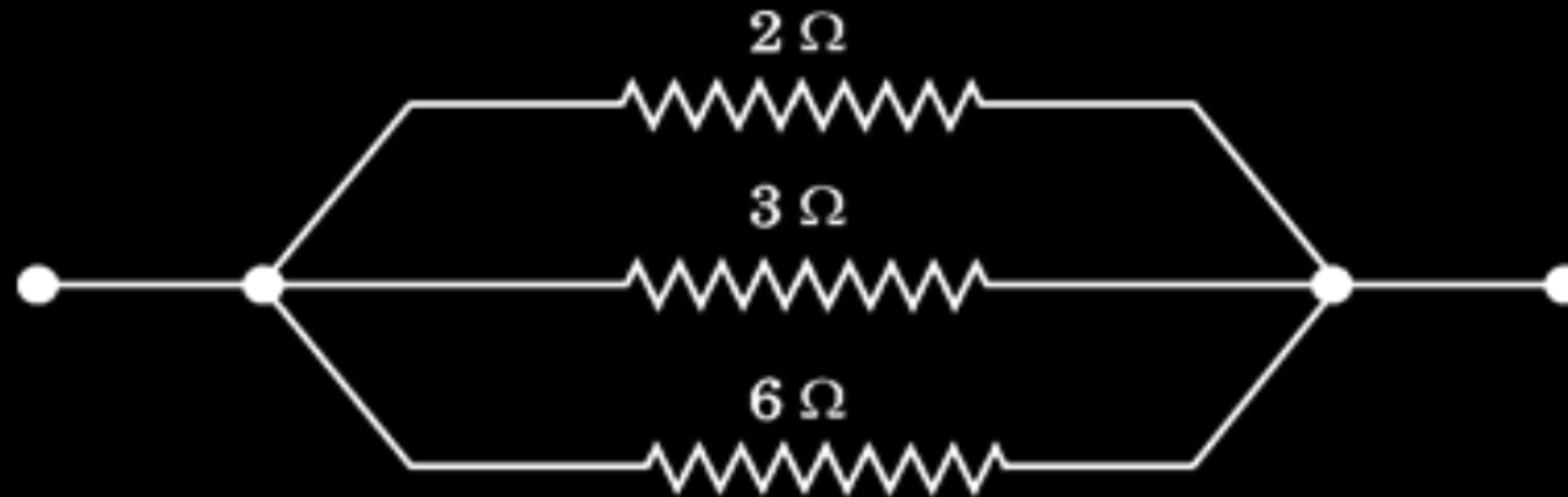
(i)



In series, $R_s = R_1 + R_2 + R_3$

$$= (2 + 3 + 6)\ \Omega = 11\ \Omega$$

(ii)



In parallel, $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$$

$$= \frac{3+2+1}{6}$$

$$R_p = 1.0\ \Omega$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

14. The resistivity of a wire made of an alloy is generally :
- (A) Lower than that of its constituent metals.
 - (B) Higher than that of its constituent metals.
 - (C) Decreases with increase in its area of cross-section.
 - (D) Increases with increase in its length.

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(B) / Higher than that of its constituent metals.

18. *Assertion (A) :* Nichrome is an alloy which is commonly used in electrical heating devices such as electric irons, toasters, etc.

Reason (R) : The resistivity of nichrome is high and its resistance decreases with increase in temperature.

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18. *Assertion (A) :* Nichrome is an alloy which is commonly used in electrical heating devices such as electric irons, toasters, etc.

Reason (R) : The resistivity of nichrome is high and its resistance decreases with increase in temperature.

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(C) /A is true, but R is false.

- 32.** Resistance of a wire of length 1 m is $35\ \Omega$ at 20°C . If the diameter of the wire is 0.2 mm, determine the resistivity of the material of the wire at that temperature. How will the resistivity of the wire change if the length and diameter of the wire both are doubled ? Justify your answer.

(Given that $\pi = \frac{22}{7}$)

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3

$$R = 35 \, \Omega$$

$$\text{Diameter } (2r) = 0.2 \, \text{mm} = 2 \times 10^{-4} \, \text{m} \quad \therefore r = 10^{-4} \, \text{m}$$

$$\text{Length of the wire} = 1 \, \text{m}$$

$$R = \rho \frac{l}{A}$$

$$\text{Area of cross section, } A = \pi r^2$$

$$\text{Resistivity, } \rho = \frac{RA}{l} = \frac{35 \times \frac{22}{7} \times (10^{-4})^2}{1}$$

$$= 110 \times 10^{-8} \, \Omega\text{m}$$

Resistivity of the wire will not change

as it is a characteristic property of the material which does not depend on the dimensions of the wire.

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

- 25.** (a) A wire of resistance R is cut into three equal parts. If these three parts are then joined in parallel, calculate the total resistance of the combination so formed.

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2

(a)

$$\text{Resistance of each part} = \frac{R}{3}$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{R/3} + \frac{1}{R/3} + \frac{1}{R/3}$$

$$\frac{1}{R_p} = \frac{3}{R} + \frac{3}{R} + \frac{3}{R} = \frac{9}{R}$$

$$\Rightarrow R_p = \frac{R}{9}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b) Define electric power. When do we say that the power consumed in an electric circuit is 1 watt ?

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2

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2

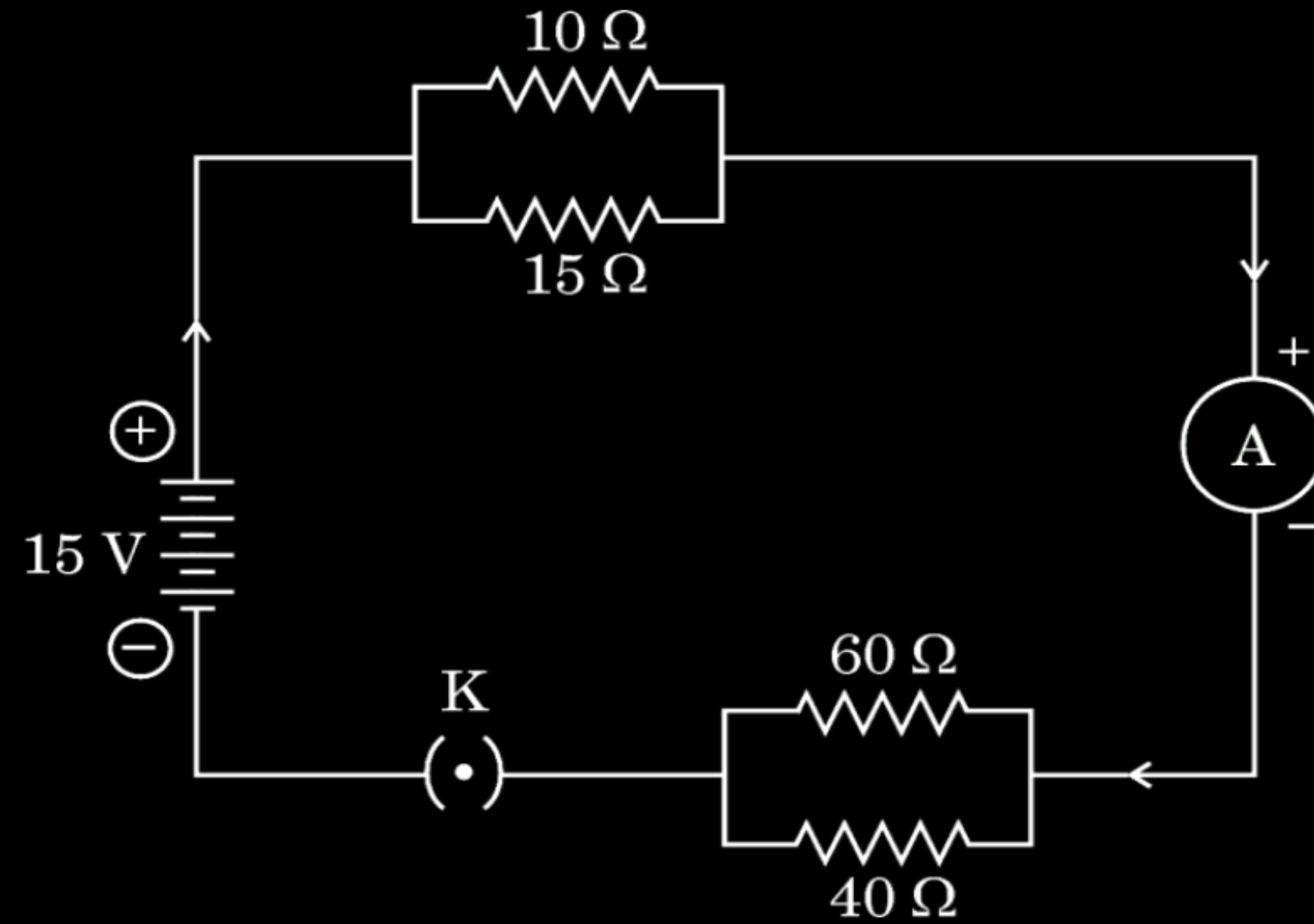
(b)

- Electric power is the rate at which electrical energy is consumed or dissipated in an electrical circuit. / Electric power is electrical energy consumed per unit time.
- When 1A current at 1V potential difference flows in a circuit. / Power consumed is 1W when 1J electrical energy is consumed in 1 s.

1

1

32. Consider the following electric circuit :



Calculate the values of the following :

- (a) The total resistance of the circuit
- (b) The total current drawn from the source
- (c) Potential difference across the parallel combination of 10 Ω and 15 Ω resistors

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(a)

$$\frac{1}{R_1} = \frac{1}{10\ \Omega} + \frac{1}{15\ \Omega} = \frac{1}{6\ \Omega} \Rightarrow R_1 = 6\ \Omega$$

$$\frac{1}{R_2} = \frac{1}{60\ \Omega} + \frac{1}{40\ \Omega} = \frac{100}{2400\ \Omega} \Rightarrow R_2 = 24\ \Omega$$

$\therefore R_1$ and R_2 are in series

$$\therefore R_{\text{total}} = R_1 + R_2 = (6 + 24) = 30\ \Omega$$

$$(b) \quad V = IR \quad \Rightarrow \quad I = \frac{V}{R} = \frac{15\ V}{30\ \Omega} = 0.5\ A$$

$$(c) \quad V = IR_1 = 0.5\ A \times 6\ \Omega = 3.0\ V$$

1

1

1

33. (a) Write the relationship between resistivity and resistance of a cylindrical conductor of length l and area of cross-section A . Hence derive the SI unit of resistivity.

(b) Why are alloys used in electrical heating devices ?

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(b) Why are alloys used in electrical heating devices ?

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3

(a)

- $R = \rho \frac{l}{A}$

$$\rho = \frac{RA}{l}$$

- SI unit of $\rho = \frac{\text{ohm} \times (\text{metre})^2}{\text{metre}}$
= ohm meter

1

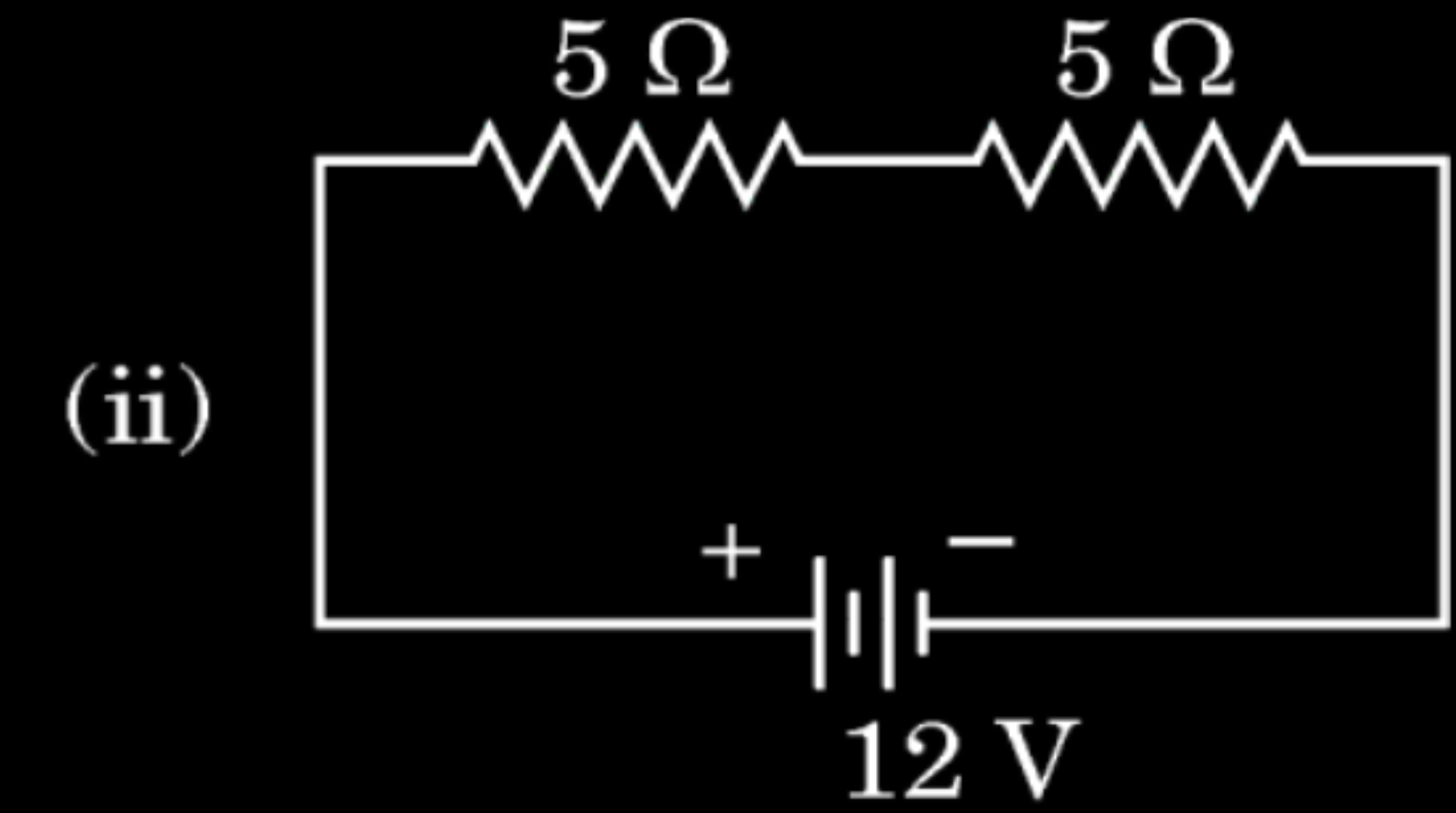
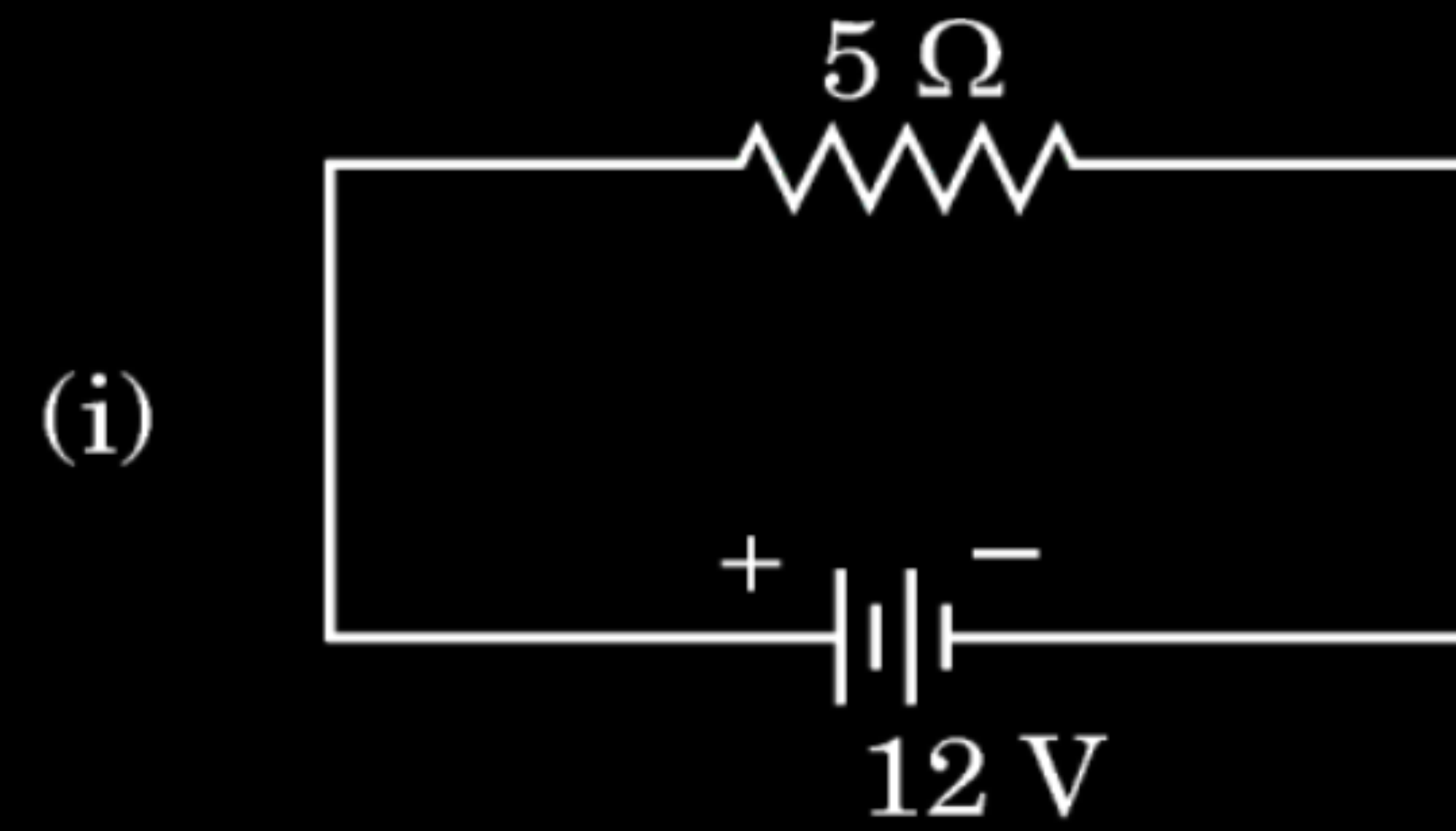
$\frac{1}{2}$

$\frac{1}{2}$

(b) Resistivity of alloy is higher than pure metals. / They do not oxidise (burn) readily at high temperatures.

1

22. (a) Consider the following circuits :



In which circuit will the power dissipated in the circuit be
(I) minimum (II) maximum ? Justify your answer.

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• (a) Power dissipated will be: • (I) minimum in circuit (ii) and • (II) maximum in circuit (iii) • • Justification : Power dissipated = $\frac{V^2}{R}$ • Since voltage is same in three cases, the power dissipation • will be inversely proportional to the resistance of the • circuit. • • (Award marks if explained through calculations)	1/2 1/2 1
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- (b) Two lamps, rated 100 W; 220 V and 60 W; 220 V are connected in parallel to electric main supply of 220 V. Find the current drawn by the two lamps from the supply.

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2

(b)	Voltage $V = 220V$	$\frac{1}{2}$
	$P = V \times I$	
	Power $P_1 = 100\text{ W}$ $P_2 = 60\text{ W}$	
	$\therefore I_1 = \frac{P_1}{V} = \frac{100\text{ W}}{220\text{ V}} = 0.45\text{ A}$	$\frac{1}{2}$
	$\therefore I_2 = \frac{P_2}{V} = \frac{60\text{ W}}{220\text{ V}} = 0.27\text{ A}$	$\frac{1}{2}$
	$\therefore$ Total current drawn $I = I_1 + I_2 = 0.45 + 0.27 = 0.72\text{ A}$	$\frac{1}{2}$

- 27.** (a) Define one ampere.
- (b) The resistance of a wire of 0.01 cm radius is $14\ \Omega$. If the resistivity of the material of the wire is $44 \times 10^{-8}\ \Omega\text{m}$, find the length of the wire. (Given $\pi = \frac{22}{7}$)

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27. (a) Define one ampere.
- (b) The resistance of a wire of 0.01 cm radius is 14 Ω . If the resistivity of the material of the wire is $44 \times 10^{-8} \Omega\text{m}$, find the length of the wire. (Given $\pi = \frac{22}{7}$)

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3

(a) When 1 coulomb of charge flows through any cross section of a conductor in 1 second, then the electric current flowing through it is 1 ampere. / $1\text{A} = \frac{1\text{C}}{1\text{s}}$

1

• (b) Radius of wire = 0.01 cm = 0.01×10^{-2} m

• $\rho = 44 \times 10^{-8} \Omega\text{m}$

• $R = 14 \Omega$

• $R = \rho \frac{l}{A}$

• $1 = \frac{R \times A}{\rho} = \frac{14 \times 22 \times (0.01 \times 10^{-2})^2}{7 \times 44 \times 10^{-8}}$

• $= \frac{10^{-8}}{10^{-8}}$
 $= 1.0 \text{ m}$

$\frac{1}{2}$

$\frac{1}{2}$

1

- 22.** (a) Consider two lamps A and B of rating 50 W; 220 V and 25 W; 220 V respectively. Find the ratio of the resistances of the two lamps (i.e. $R_A : R_B$).

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2

22. (a) Consider two lamps A and B of rating 50 W; 220 V and 25 W; 220 V respectively. Find the ratio of the resistances of the two lamps (i.e. $R_A : R_B$).

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2

(a) $P = \frac{V^2}{R}$	$\frac{1}{2}$
$R = \frac{V^2}{P}$ $R_A = \frac{(220)^2}{50}$	
$R_B = \frac{(220)^2}{25}$	$\frac{1}{2}$
$\frac{R_A}{R_B} = \frac{1}{2}$	1
/	
$P = \frac{V^2}{R}$ Since V is same means P is inversely proportional to R.	
Power of lamp A is double that of B hence Resistance of A is half that of B.	1
$\frac{R_A}{R_B} = \frac{1}{2}$	1

- (b) Heat produced per second due to a current in a resistor of $4\ \Omega$ is 400 joules. Calculate the potential difference across the resistor. 2

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(b)

$$H = \frac{V^2}{R} \times t$$

$$400 = \frac{V^2}{4} \times 1$$

$$V^2 = 1600$$

$$V = 40\text{V}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

27. (a) Define one volt potential difference between two points in an electric field.

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(b) Draw a schematic diagram of an electric circuit of a cell of 1.5 V, 5 Ω and 10 Ω resistor and a plug key, all connected in series. Calculate the current drawn from the cell when the key is closed.

3

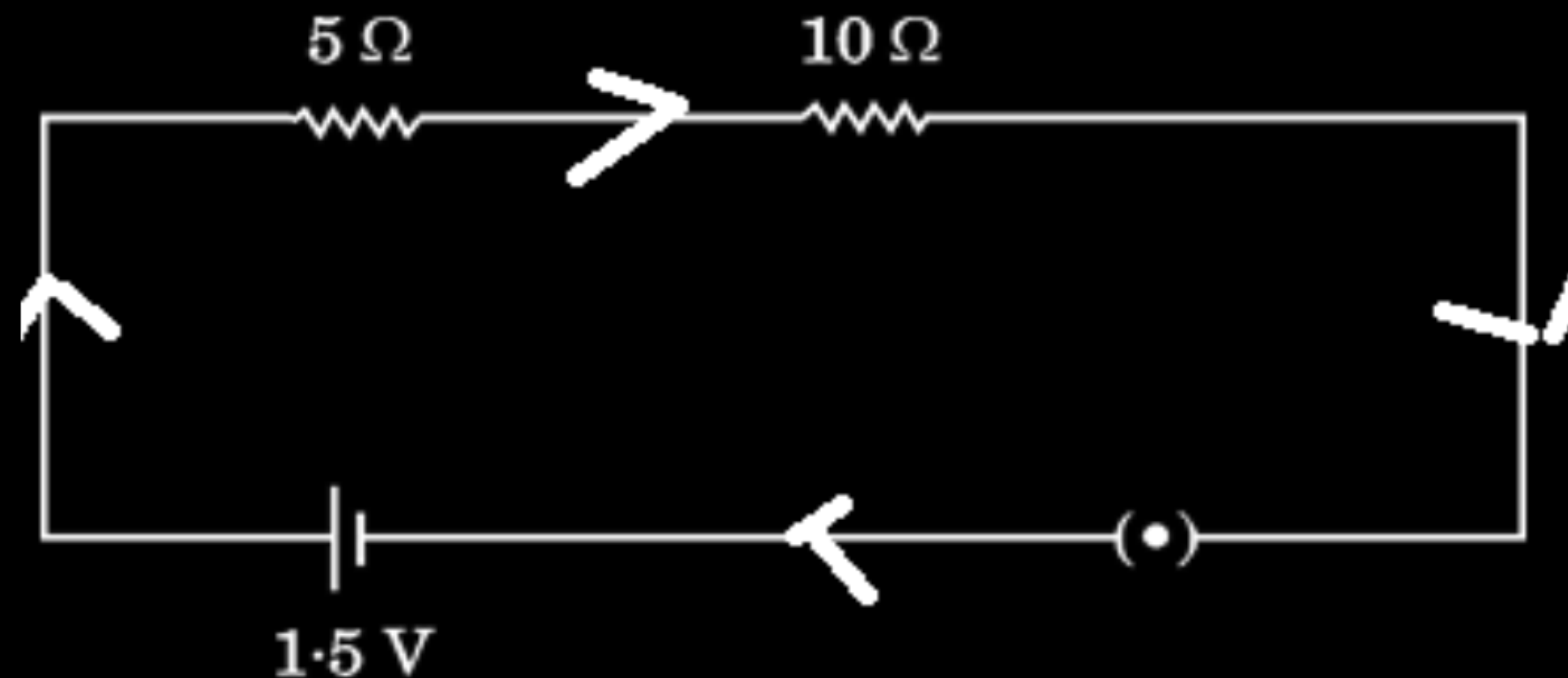
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3

(a) It is the amount of work done to bring one coulomb charge from one point to another in the field is 1 joule. / $1\text{ V} = \frac{1\text{ J}}{1\text{ C}}$

1

(b)



1

$$\text{Current} = \frac{\text{Potential difference}}{\text{Resistance}} = \frac{1.5\text{ V}}{(5\ \Omega + 10\ \Omega)}$$

$$= \frac{1.5\text{ V}}{15\ \Omega} = 0.1\text{ A}$$

1

12. A piece of wire of resistance 'R' is cut lengthwise into three identical parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R' , then the value of R/R' is :

1

(a) $1/9$

(b) $1/3$

(c) 3

(d) 9

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(a) $1/9$

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(c) 3

(d) 9

(d)/9

13. An electric bulb is rated 220 V; 11W. The resistance of its filament when it glows with a power supply of 220 V is :

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1

- (a) 4400 Ω
- (b) 440 Ω
- (c) 400 Ω
- (d) 20 Ω

13. An electric bulb is rated 220 V; 11W. The resistance of its filament when it glows with a power supply of 220 V is :

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1

- (a) 4400 Ω
- (b) 440 Ω
- (c) 400 Ω
- (d) 20 Ω

(a)/4400 Ω

14. The minimum number of identical bulbs of rating $4\text{V}; 6\text{W}$, that can work safely with desired brightness, when connected in series with a 240 V mains supply is :

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1

(a) 20

(b) 40

(c) 60

(d) 80

14. The minimum number of identical bulbs of rating 4V; 6W, that can work safely with desired brightness, when connected in series with a 240 V mains supply is :

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1

- (a) 20
- (b) 40
- (c) 60
- (d) 80

(c)/60

32. The electrical resistivity of three materials A, B and C at 20°C is given below :

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3

Material	Resistivity ($\Omega \text{ m}$)
A	10^{17}
B	44×10^{-6}
C	1.62×10^{-8}

- Classify these materials as conductor, alloy and insulator.
- Give one example of each of these materials and state one use of each material in the design of an electrical appliance say an electric stove or an electric iron.

(i)

A - Insulator

$\frac{1}{2}$

B - Alloy

$\frac{1}{2}$

C - Conductor

$\frac{1}{2}$

(ii)

A: Plastic - handle of an electric iron.

B: Nichrome – used as a heating element in an electric iron.

C: Copper - electric wires. /

A: Rubber– foot of the electric stove.

B: Nichrome – used as a heating element in an electric stove.

C: Copper- electric wires.

(any other example with its use in an electric appliance)

($\frac{1}{2}$ X3)

11. A wire of length ' l ' is gradually stretched so that its length increases to $3l$. If its original resistance is R , then its new resistance will be : **CBSE 2025**

(A) $3R$

(B) $6R$

(C) $9R$

(D) $27R$

11. A wire of length ' l ' is gradually stretched so that its length increases to $3l$. If its original resistance is R , then its new resistance will be : **CBSE 2025**

(A) $3R$

(B) $6R$

(C) $9R$

(D) $27R$

(C) / $9R$

- 26.** A voltage source sends a current of 2 A to a resistor of $40\ \Omega$ connected across it for 5 minutes. Calculate the electrical energy supplied by the source.

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- 26.** A voltage source sends a current of 2 A to a resistor of 40 Ω connected across it for 5 minutes. Calculate the electrical energy supplied by the source.

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Given $I = 2 \text{ A}$, $R = 40 \text{ } \Omega$, $t = 5 \text{ minutes} = 300 \text{ s}$

$$\text{Electrical energy} = I^2 R t$$

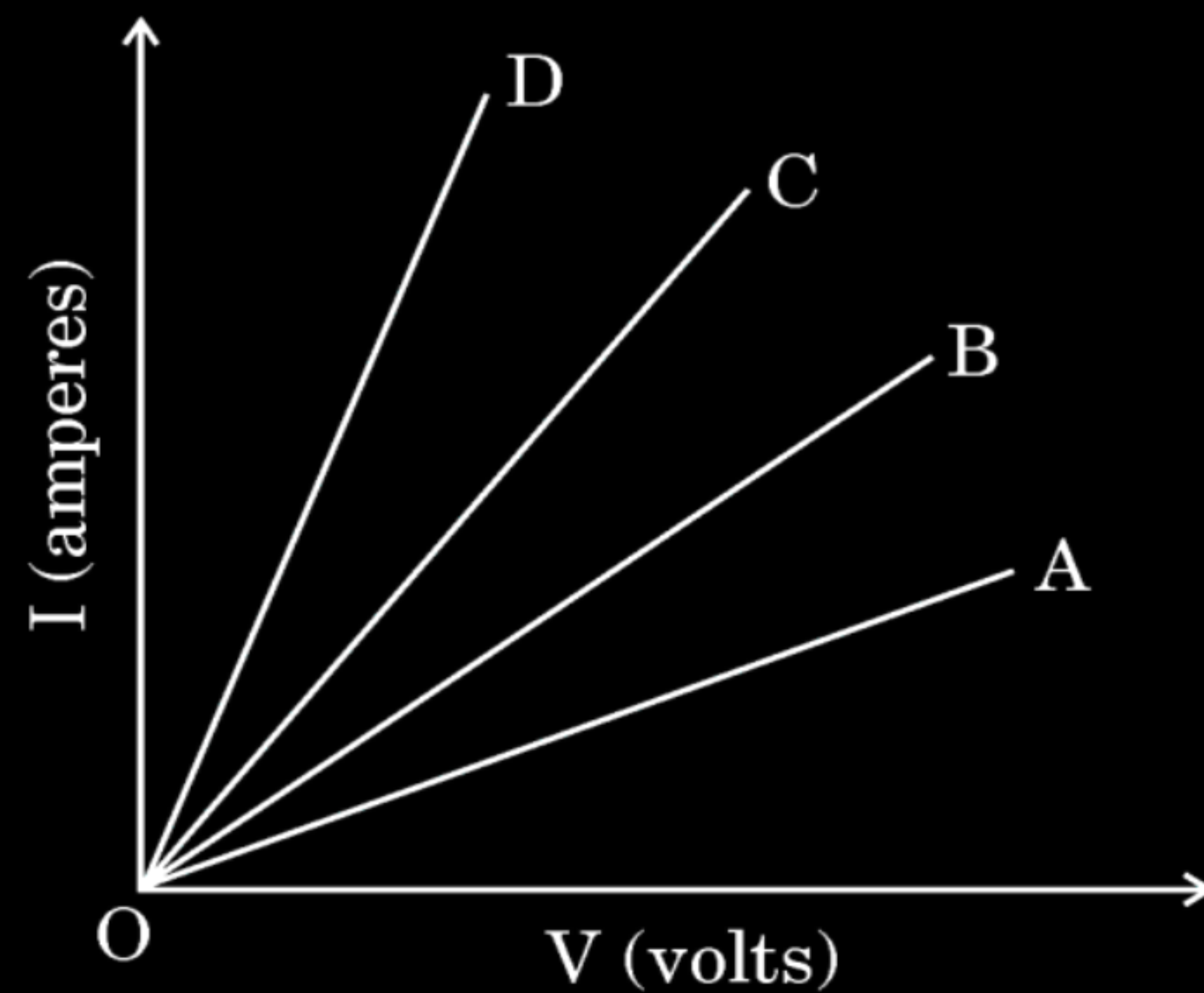
$$= (2 \text{ A})^2 \times 40 \text{ } \Omega \times 300 \text{ s}$$

$$= 48000 \text{ J}$$

$\frac{1}{2}$

1

$\frac{1}{2}$



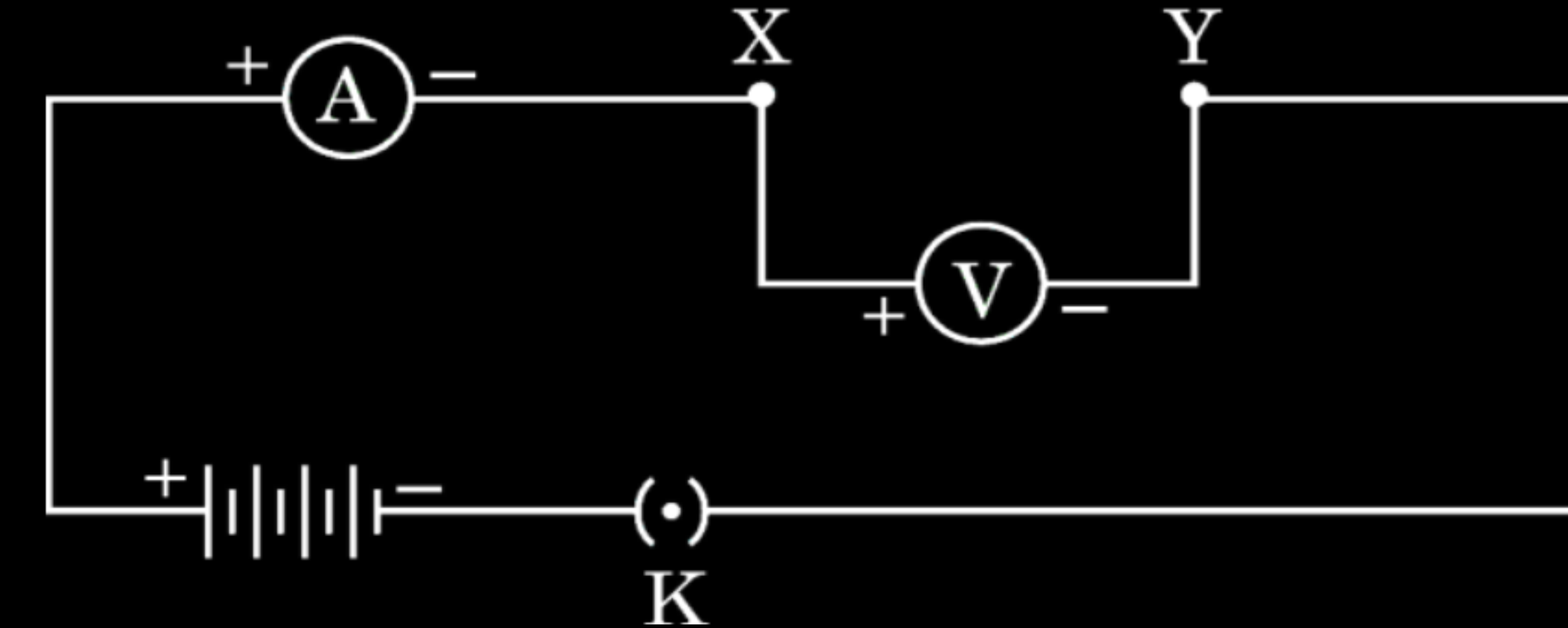
39. As shown in the diagram, an electric circuit consisting of an ammeter, a voltmeter, 4 cells of 1.5 V each, a plug key with a gap XY was set up. Voltmeter and ammeter readings were recorded in the observation table for four arrangements as given below :

Arrangement No. 1 – only resistor R_1 in gap XY

Arrangement No. 2 – only resistor R_2 in gap XY

Arrangement No. 3 – Resistors R_1 and R_2 in series in gap XY

Arrangement No. 4 – Resistors R_1 and R_2 in parallel in gap XY



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Based on the observations, four V – I graphs A, B, C and D as shown in figure were drawn. Study these graphs.

- (a) Which one of the graphs represents the series combination of R_1 and R_2 ? 1
- (b) Which one of these graphs represents the parallel combination of R_1 and R_2 ? 1
- (c) (i) Show an arrangement of three resistors, each of resistance $10\ \Omega$, so that the combination has a resistance of $15\ \Omega$. Give justification for your answer. 2

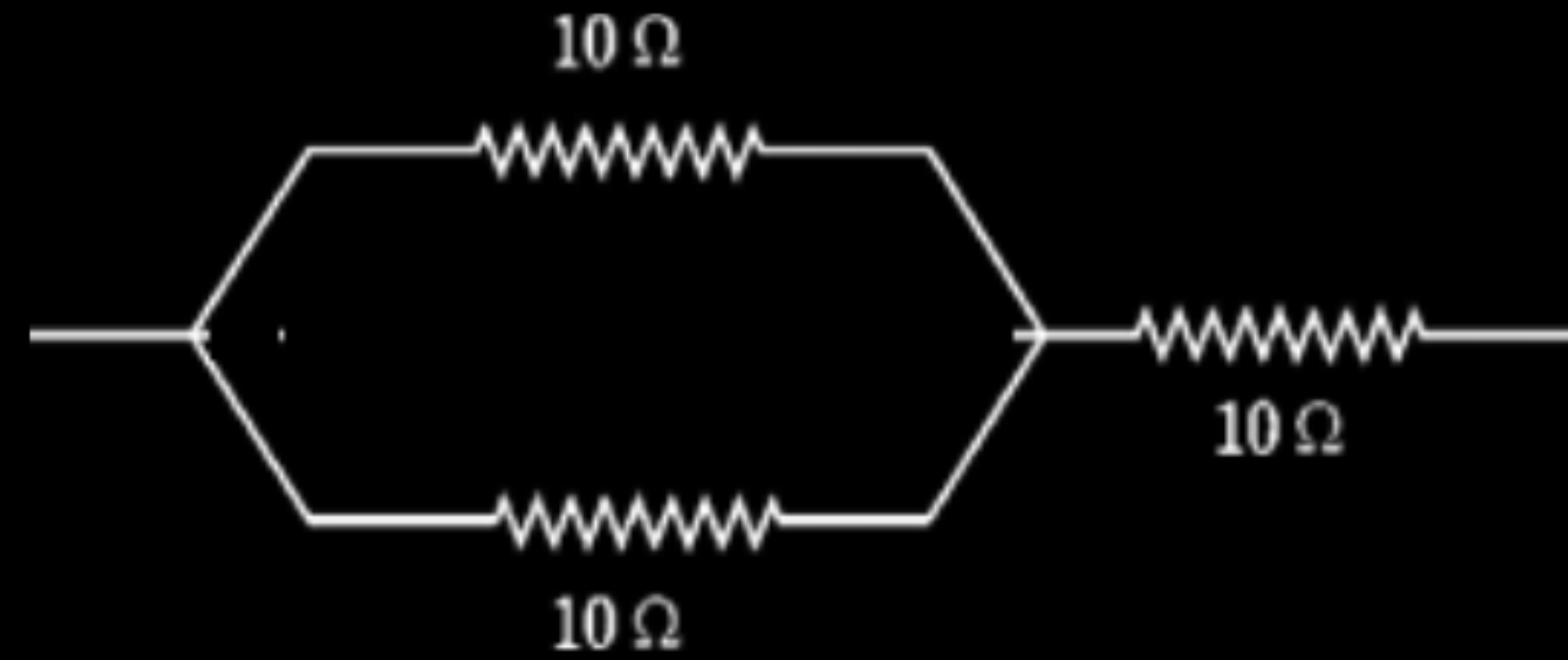
OR

- (c) (ii) A battery of 6 V is connected with a series combination of five resistors of $0.1\ \Omega$, $0.2\ \Omega$, $0.3\ \Omega$, $0.4\ \Omega$ and $0.5\ \Omega$. How much current would flow through the $0.3\ \Omega$ resistor ? Justify your answer. 2

(a) Graph A

(b) Graph D

(c) (i)



$$R = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

$$R = \left(\frac{10 \times 10}{10 + 10} + 10 \right) \Omega = 5\ \Omega + 10\ \Omega = 15\ \Omega$$

OR

(c)

(ii) • $I = \frac{V}{R} = \frac{6\text{ V}}{(0.1 + 0.2 + 0.3 + 0.4 + 0.5)\ \Omega} = \frac{6\text{ V}}{1.5\ \Omega} = 4.0\text{ A}$

• same current flows when resistors are connected in series.

26. An electric kettle is rated 230 V; 1000 W. Calculate the resistance of its heating element when in operation.

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2

26. An electric kettle is rated 230 V; 1000 W. Calculate the resistance of its heating element when in operation.

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2

$$P = 1000 \text{ W}, V = 230 \text{ V}; \text{ Formula } = P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P}$$

$$= \frac{(230)^2}{1000}$$

$$= 52.9 \, \Omega$$

$\frac{1}{2}$

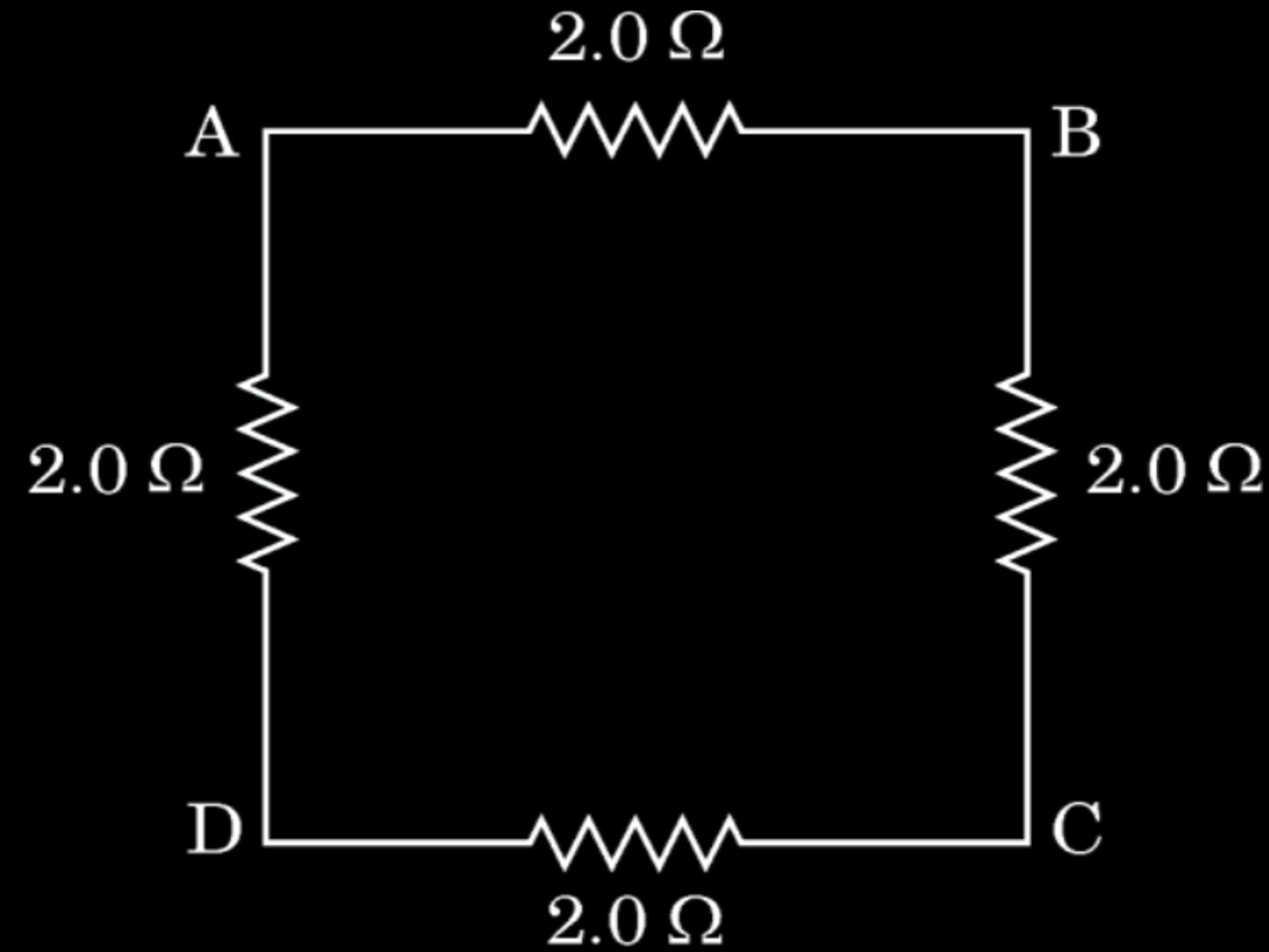
1

$\frac{1}{2}$

- 26.** Four resistors, each of resistance $2.0\ \Omega$, are joined end to end to form a square ABCD as shown. Using appropriate formula, determine the equivalent resistance of the combination between its two ends A and B.

2

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Between A and B, 3 resistors in series = AD + DC + CB in one branch

Equivalent resistance, $R_s = 2 \Omega + 2 \Omega + 2 \Omega = 6 \Omega$

Now one resistor in arm AD is in parallel combination with the other three.

Two branches:

$$R_1 = 2 \Omega$$

$$R_s = 6 \Omega$$

$$\therefore \frac{1}{R_p} = \frac{1}{6} + \frac{1}{2} = \frac{1+3}{6} = \frac{4}{6}$$

$$R_p = \frac{6}{4} = 1.5 \Omega$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

32. Define the term “potential difference” between two points in an electric circuit carrying current. Name and define its S.I. unit. Also express it in terms of S.I. unit of work and charge.

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32. Define the term “potential difference” between two points in an electric circuit carrying current. Name and define its S.I. unit. Also express it in terms of S.I. unit of work and charge.

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3

- The work done to move a unit charge from one point to other in a conductor. / ($V=W/Q$)
- volt (V)
- In a current carrying conductor, when one joule of work is done to move a charge of 1 coulomb from one point to another.
- 1 volt = 1 joule/1 coulomb or $1V=1 \text{ J C}^{-1}$

1

$\frac{1}{2}$

1

$\frac{1}{2}$

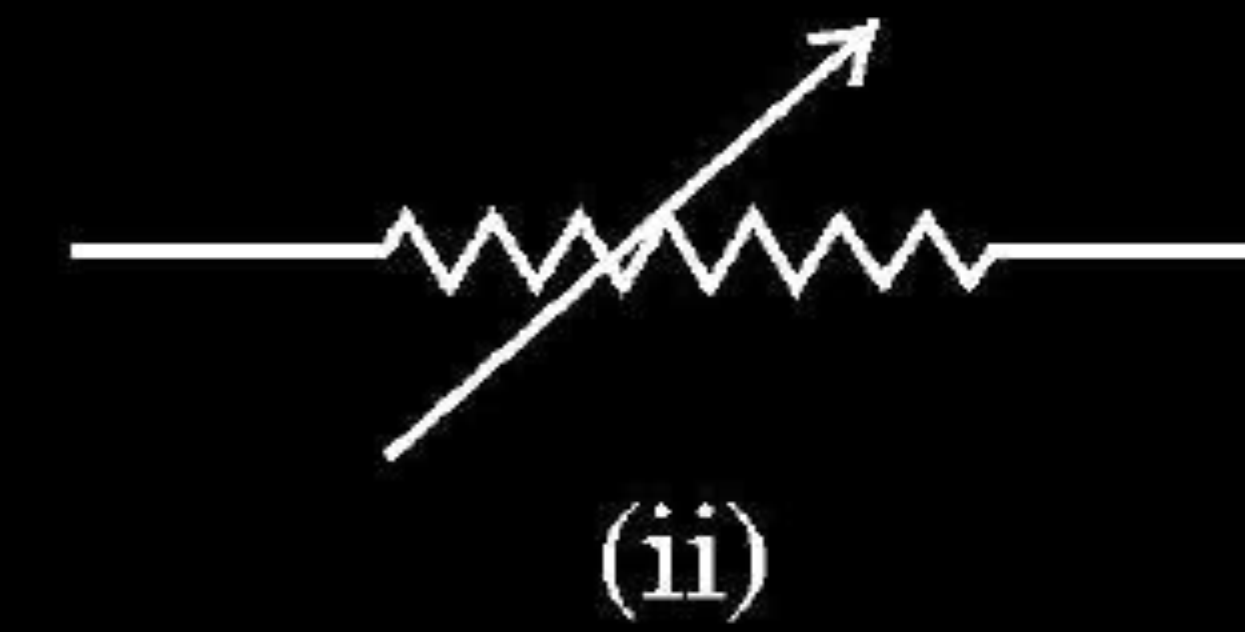
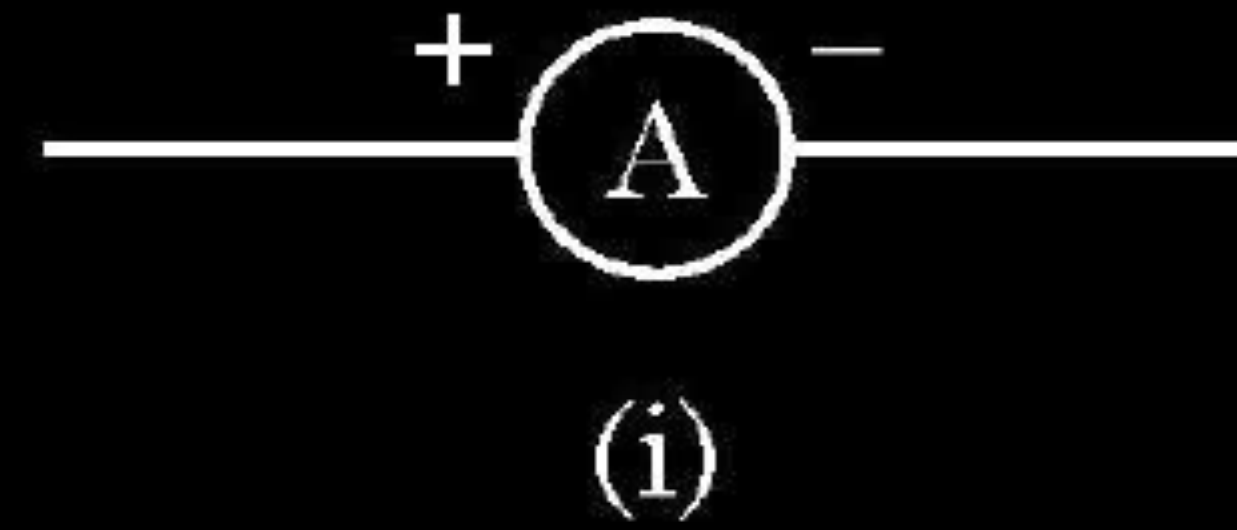
33. (a) Explain the statement “Potential difference between two points is 1 volt”.

(b) What do the symbols given below represent in an electric circuit ?

Write one function of each.

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3



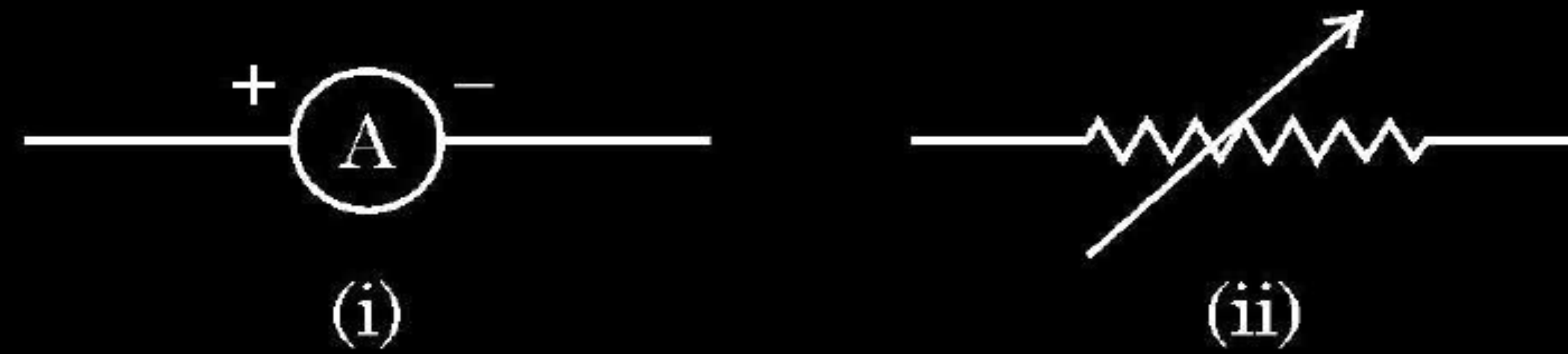
33. (a) Explain the statement “Potential difference between two points is 1 volt”.

(b) What do the symbols given below represent in an electric circuit ?

Write one function of each.

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3



(a) It means 1 joule of work is done to move a charge of 1 coulomb from one point of the conductor to the other.

(b) (i)

Ammeter.

Ammeter is used to measure electric current

(ii)

Rheostat or variable resistance

Rheostat is used in a circuit to vary the resistance of the circuit

1

$\frac{1}{2}$

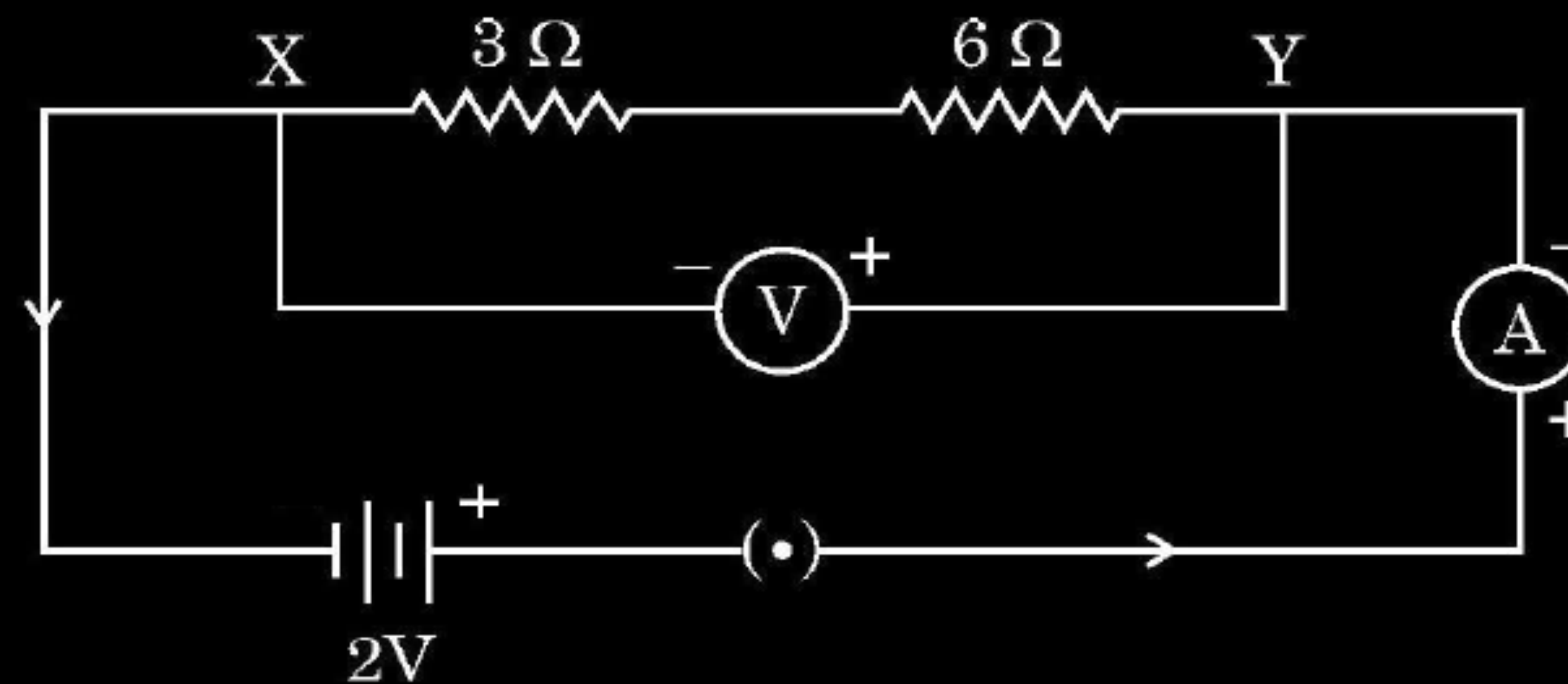
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39. Study the circuit shown in which two resistors X and Y of resistances $3\ \Omega$ and $6\ \Omega$ respectively are joined in series with a battery of 2 V .

4



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- (I) Draw a circuit diagram showing the above two resistors X and Y joined in parallel with same battery and same ammeter and voltmeter.
- (II) In which combination of resistors will the (i) potential difference across X and Y and (ii) current through X and Y, be the same ?
- (III) (a) Find the current drawn from the battery by the series combination of the two resistors (X and Y).

1

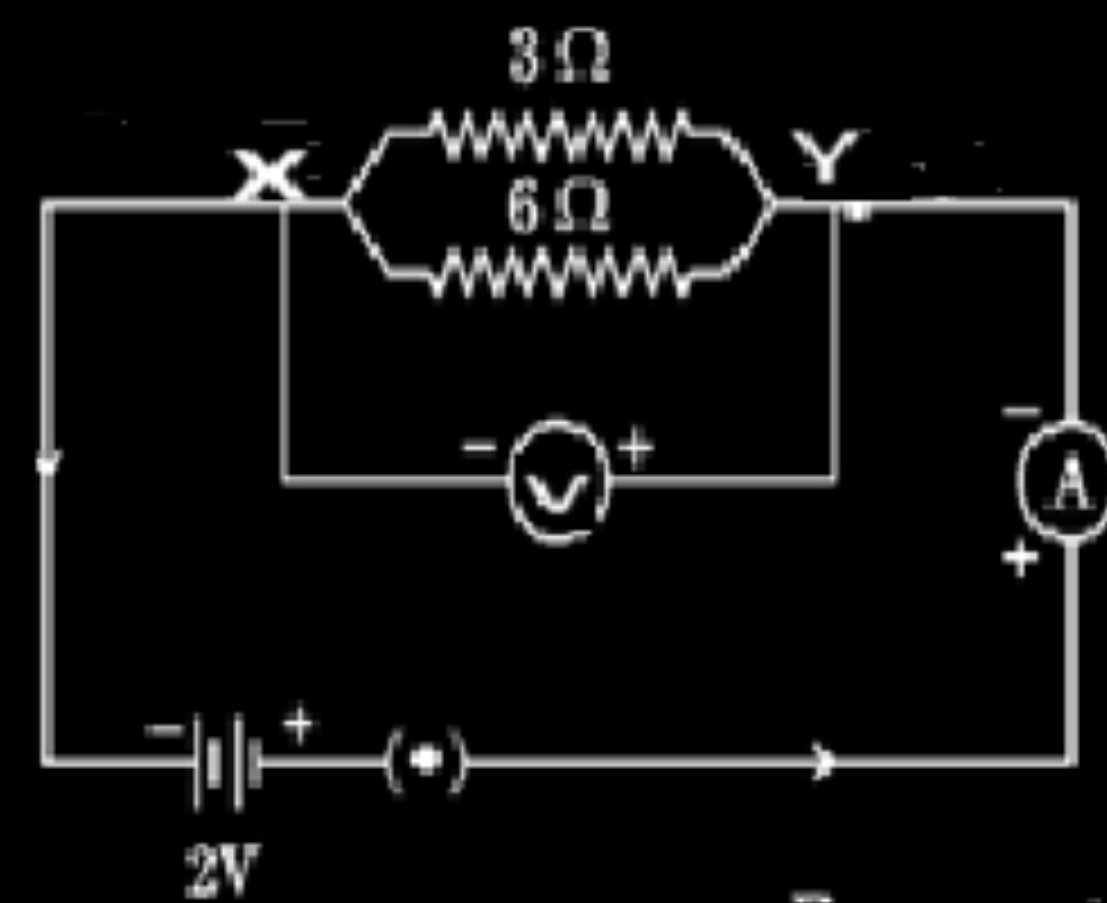
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2

OR

- (III) (b) Determine the equivalent resistance of the parallel combination of the two resistors (X and Y).

2



(or Any other way)

(II)

- (i) in parallel combination.
- (ii) in series combination.

(III) (a) Resistance $R = 3 \Omega + 6 \Omega = 9 \Omega$

$$V = 2V$$

$$I = \frac{V}{R} = \frac{2V}{9\Omega} = 0.22A$$

OR

$$(b) \quad \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R}$$

$$= \frac{1}{3\Omega} + \frac{1}{6\Omega}$$

$$= \frac{6+3}{18\Omega}$$

$$\therefore R = 2\text{ohm}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

1

1

$\frac{1}{2}$

$\frac{1}{2}$

1

2. In case of wires of same resistivity, the resistance of the wire will be maximum if the diameter and length of the wire respectively are

(A) D and L

(B) $\frac{D}{2}$ and $\frac{L}{4}$

(C) $\frac{D}{3}$ and $\frac{L}{6}$

(D) $\frac{D}{4}$ and $\frac{L}{12}$

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2. In case of wires of same resistivity, the resistance of the wire will be maximum if the diameter and length of the wire respectively are

(A) D and L

(B) $\frac{D}{2}$ and $\frac{L}{4}$

(C) $\frac{D}{3}$ and $\frac{L}{6}$

(D) $\frac{D}{4}$ and $\frac{L}{12}$

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(C)/ $D/3$ and $L/6$

15. The resistance of a wire is $40\ \Omega$. If 2 A of current flows through this wire for 50 seconds, the heat produced in the wire is

(A) 4000 J

(B) 8000 J

(C) 16000 J

(D) 80000 J

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15. The resistance of a wire is $40\ \Omega$. If 2 A of current flows through this wire for 50 seconds, the heat produced in the wire is

(A) 4000 J

(B) 8000 J

(C) 16000 J

(D) 80000 J

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(B)/ 8000 J

36. (a) Define Electric power. Express electric power of an electric appliance in terms of its resistance (R) and the voltage / potential difference (V) at which it operates.
- (b) The rated voltage of an air-conditioner is 220 V. This air-conditioner consumes 22 units of electric energy in 10 hours, calculate :
- (i) Power rating of the air-conditioner,
 - (ii) Current drawn by the air-conditioner, and
 - (iii) Resistance of the air-conditioner

5

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(a)	• Electric Power: It is the rate at which electric energy is dissipated or consumed.	1
	• Expression: $P = V^2/R$	1
(b)	(i) $\text{Power} = \frac{\text{Electrical energy consumed}}{\text{Time}}$	$\frac{1}{2}$
	$= \frac{22 \text{ kWh}}{10 \text{ h}} = 2.2 \text{ kW or } 2200 \text{ W}$	$\frac{1}{2}$
	(ii) $\text{Current} = \frac{\text{Power}}{\text{Voltage}}$	$\frac{1}{2}$
	$= \frac{2200 \text{ W}}{220 \text{ V}} = 10 \text{ A}$	$\frac{1}{2}$
	(iii) $\text{Resistance} = \frac{(\text{Voltage})^2}{\text{Power}}$	$\frac{1}{2}$
	$= \frac{(220)^2}{2200} = 22 \text{ } \Omega$	$\frac{1}{2}$

22. (A) Determine the maximum and minimum resistance which can be obtained by joining five resistors of $\frac{1}{5} \Omega$ each.

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22. (A) Determine the maximum and minimum resistance which can be obtained by joining five resistors of $\frac{1}{5} \Omega$ each.

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(A) The maximum resistance is obtained when resistors are connected in series combination.

$$\text{Thus } R_s = 5 \times \frac{1}{5} \Omega = 1 \Omega$$

The minimum resistance is obtained when resistors are connected in parallel combination.

$$\frac{1}{R_p} = \frac{1}{\frac{1}{5}\Omega} + \frac{1}{\frac{1}{5}\Omega} + \frac{1}{\frac{1}{5}\Omega} + \frac{1}{\frac{1}{5}\Omega} + \frac{1}{\frac{1}{5}\Omega}$$

$$\frac{1}{R_p} = (5 + 5 + 5 + 5 + 5)$$

$$\therefore R_p = \frac{1}{25} \Omega$$

1

1

(B) Calculate potential difference across a $4\ \Omega$ resistor that produces 100 J of heat every second.

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(B) Calculate potential difference across a $4\ \Omega$ resistor that produces 100 J of heat every second.

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(B) Heat, $H = 100\text{ J}$

$$R = 4\ \Omega$$

$$t = 1\text{ s}$$

$$H = \frac{V^2 t}{R}$$

$$\Rightarrow V = \sqrt{\frac{HR}{t}} = \sqrt{\frac{400}{1}} = 20\text{ V}$$

1

$\frac{1}{2} + \frac{1}{2}$

28. The resistance of a wire of 0.01 cm radius is 7 ohms. If the resistivity of the material of the wire is 44×10^{-6} ohm meter, calculate the length of the wire.

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28. The resistance of a wire of 0.01 cm radius is 7 ohms. If the resistivity of the material of the wire is 44×10^{-6} ohm meter, calculate the length of the wire.

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$$r=0.01 \text{ cm}=0.01 \times 10^{-2} \text{ m}, R=7 \text{ ohm}$$

$$\rho=44 \times 10^{-6} \Omega \text{m}$$

$$A=\pi r^2$$

$$R = \rho \frac{l}{A} \quad \Rightarrow l = \frac{R \times A}{\rho}$$

$$l = \frac{7 \Omega \times (0.01 \times 10^{-2})^2 \times \frac{22}{7} \text{m}^2}{44 \times 10^{-6} \Omega \text{m}}$$

$$l = 0.5 \times 10^{-2} \text{m}$$

1/2

1/2

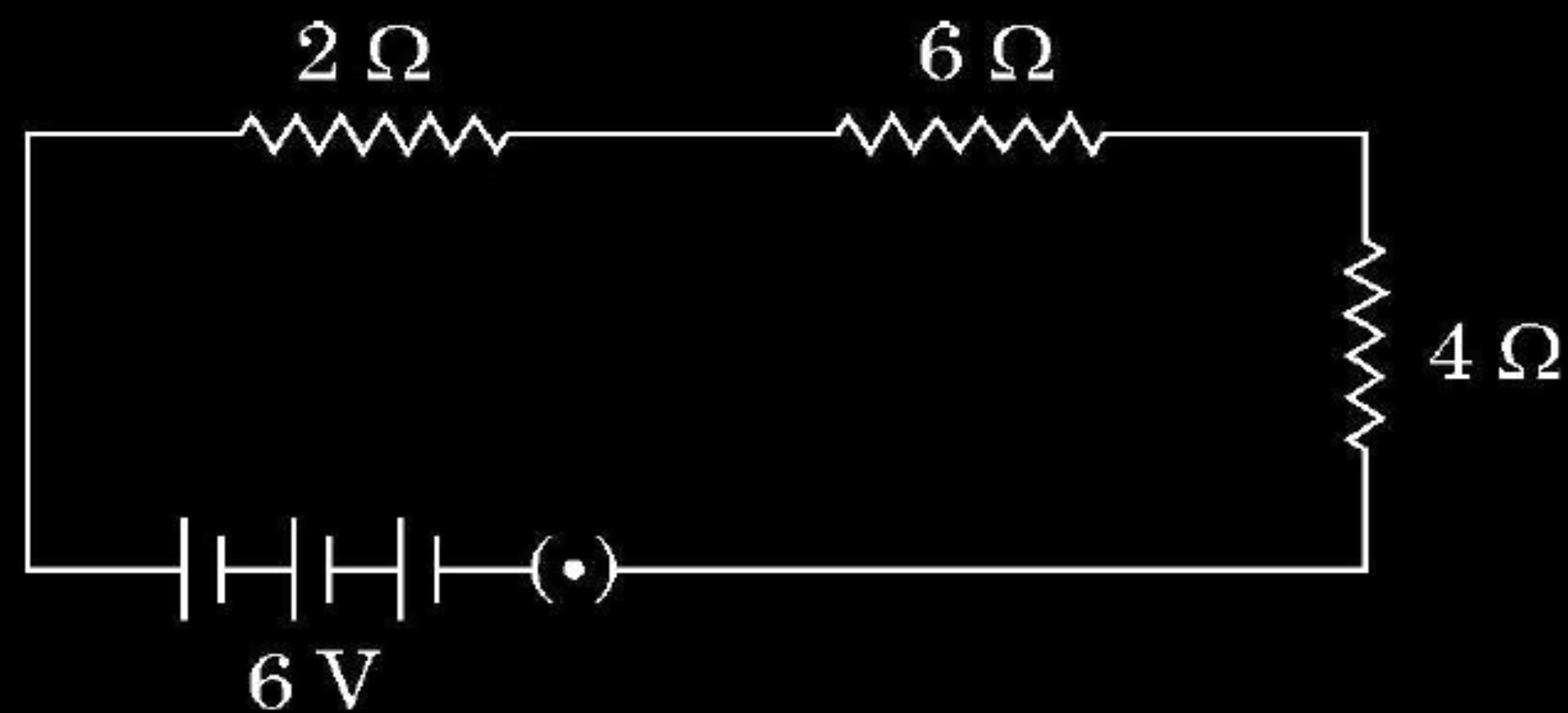
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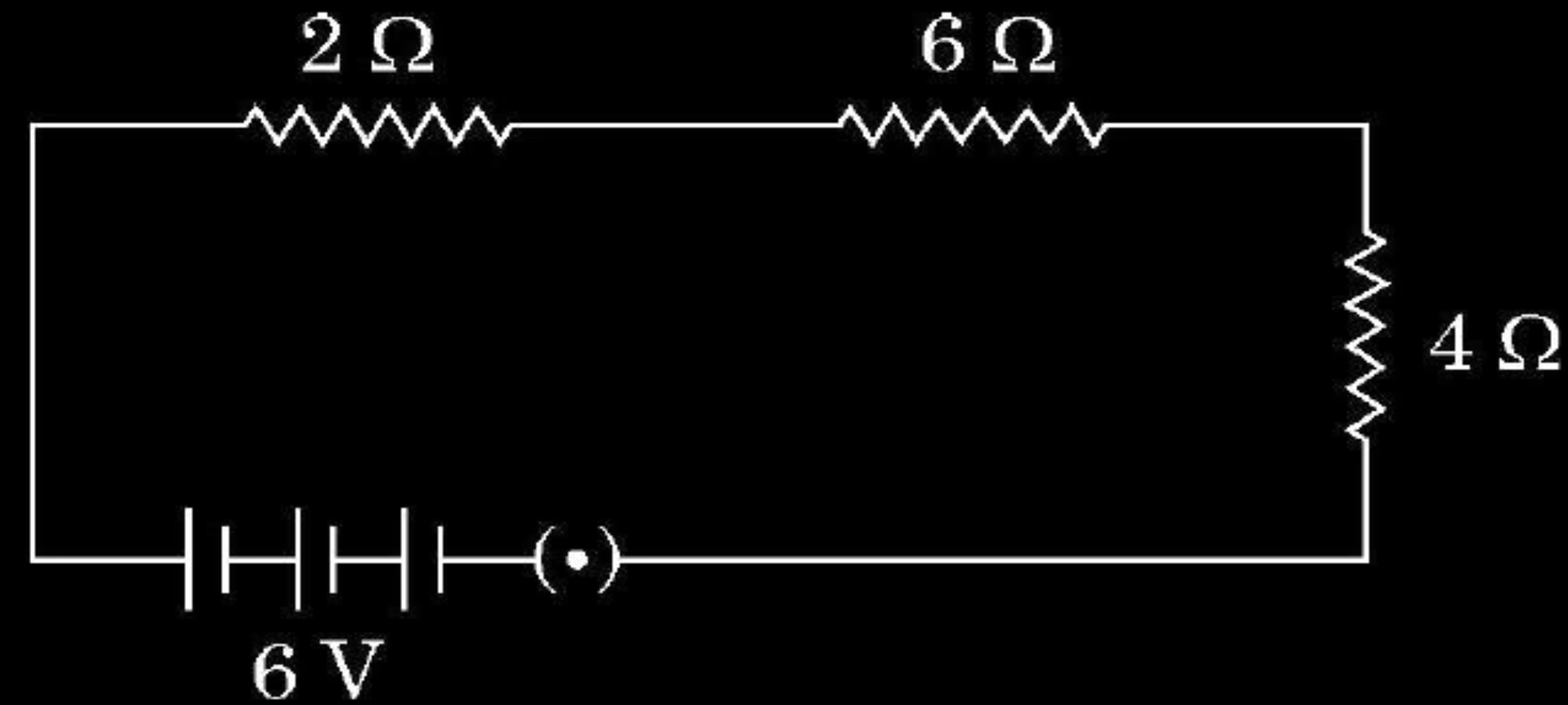
22. (A) Use Ohm's law to determine the potential difference across the $6\ \Omega$ resistor in the following circuit when key is closed :

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22. (A) Use Ohm's law to determine the potential difference across the $6\ \Omega$ resistor in the following circuit when key is closed :

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(A) Total resistance in the circuit

$$R = R_1 + R_2 + R_3 = 12\ \Omega$$

Voltage of the battery = 6 V

$$\therefore I = \frac{V}{R} = \frac{6}{12} = 0.5\ \text{A}$$

$$\therefore \text{Potential difference across } 6\ \Omega \text{ resistor} = 0.5\ \text{A} \times 6\ \Omega = 3.0\ \text{V}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

(B) Prove that if the current through a resistor is increased by 100%, then the increase in power dissipated through the resistor will be 300%.

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- (B) Prove that if the current through a resistor is increased by 100%, then the increase in power dissipated through the resistor will be 300%.

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$$(B) P_1 = I^2 R$$

$$P_2 = (2I)^2 R = 4I^2 R \text{ [100\% increase in current means current becomes } 2I]$$

$$\therefore \text{Increase in power dissipated} = P_2 - P_1 = 4I^2 R - I^2 R = 3I^2 R \\ = 3P_1$$

$$\text{Percentage increase in power dissipated} = \frac{3P_1}{P_1} \times 100 = 300\%$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

25. (A) State two applications of Joule's heating in domestic electric circuit.
- (B) (a) Establish the relationship between the commercial unit of electric energy and the SI unit of electric energy.

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25. (A) State two applications of Joule's heating in domestic electric circuit.

(B) (a) Establish the relationship between the commercial unit of electric energy and the SI unit of electric energy.

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(A) Electric bulb / electric iron / Electric fuse / Electric heater / electric Oven

(Any two)

(B) (a) $1 \text{ kWh} = 1000 \text{ watt} \times 3600 \text{ second}$
 $= 3.6 \times 10^6 \text{ watt second}$
 $= 3.6 \times 10^6 \text{ Joule (J)}$

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

- (b) Determine the total resistance of the parallel combination of three resistances of $2\ \Omega$, $4\ \Omega$ and $6\ \Omega$.

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(b) Determine the total resistance of the parallel combination of three resistances of $2\ \Omega$, $4\ \Omega$ and $6\ \Omega$.

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(b)

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{2\ \Omega} + \frac{1}{4\ \Omega} + \frac{1}{6\ \Omega}$$

$$\therefore R_p = \frac{12}{11}\ \Omega = 1.09\ \Omega$$

$\frac{1}{2}$

$\frac{1}{2}$

18. *Assertion (A) :* The commercial unit of electric energy is kilowatt hour (kWh).

Reason (R) : $1 \text{ kWh} = 10^6 \text{ joule (J)}$

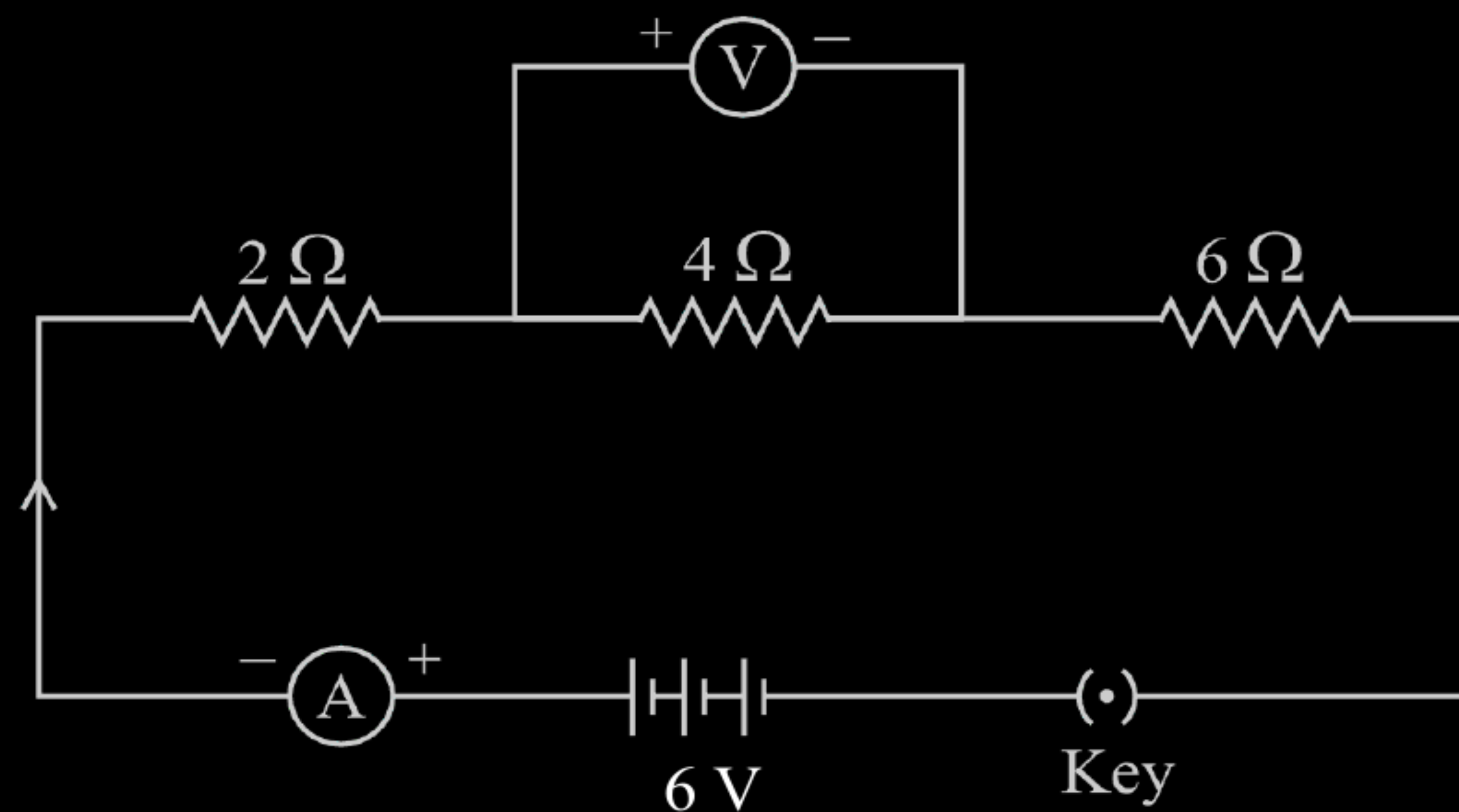
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18. *Assertion (A) :* The commercial unit of electric energy is kilowatt hour (kWh).

Reason (R) : $1 \text{ kWh} = 10^6 \text{ joule (J)}$

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29. In the given circuit when key is closed, determine the following :



- (i) Total resistance offered by the three resistors
- (ii) Reading of the ammeter
- (iii) Reading of the voltmeter

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(i) $R = 2\ \Omega + 4\ \Omega + 6\ \Omega = 12\ \Omega$	1
(ii) $I = \frac{V}{R} = \frac{6V}{12\Omega} = 0.5\ A$	1
(iii) $V = IR = 0.5A \times 4\Omega = 2V$	1

34. (a) (i) Write the relation between the electrical resistivity ' ρ ' and the resistance ' R ' of a cylindrical conductor of length ' l ' and area of cross-section ' A '. Use this relationship to determine the SI unit of ' ρ '.

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- (ii) Find the electrical resistivity of the material of a wire of length 120 cm and diameter 0.4 mm. The resistance of this wire is $3\ \Omega$.

(a) (i)

$$\bullet \quad R = \rho \frac{l}{A}$$
$$\rho = \frac{R \times A}{l}$$

1

$$\bullet \quad \text{SI unit of } \rho = \frac{\text{SI unit of } R \times \text{SI unit of } A}{\text{SI unit of } l} = \frac{\Omega \times \text{m}^2}{\text{m}}$$

$$= \Omega \text{ m}$$

1

(ii) $R = 3\Omega$; $l = 120 \text{ cm} = 1.2 \text{ m}$;

$d = 0.4 \text{ mm}$ or $r = 0.2 \text{ mm}$

$\frac{1}{2}$

$$\rho = \frac{R \times A}{l}$$

$$A = 3.14 \times (0.2 \text{ mm})^2 = 3.14 \times 4 \times 10^{-8} \text{ m}^2$$

$\frac{1}{2}$

$$= 3\Omega \times \frac{3.14 \times 4 \times 10^{-8} \text{ m}^2}{1.2 \text{ m}}$$

1

$$= 3.14 \times 10^{-7} \Omega \text{ m}$$

1

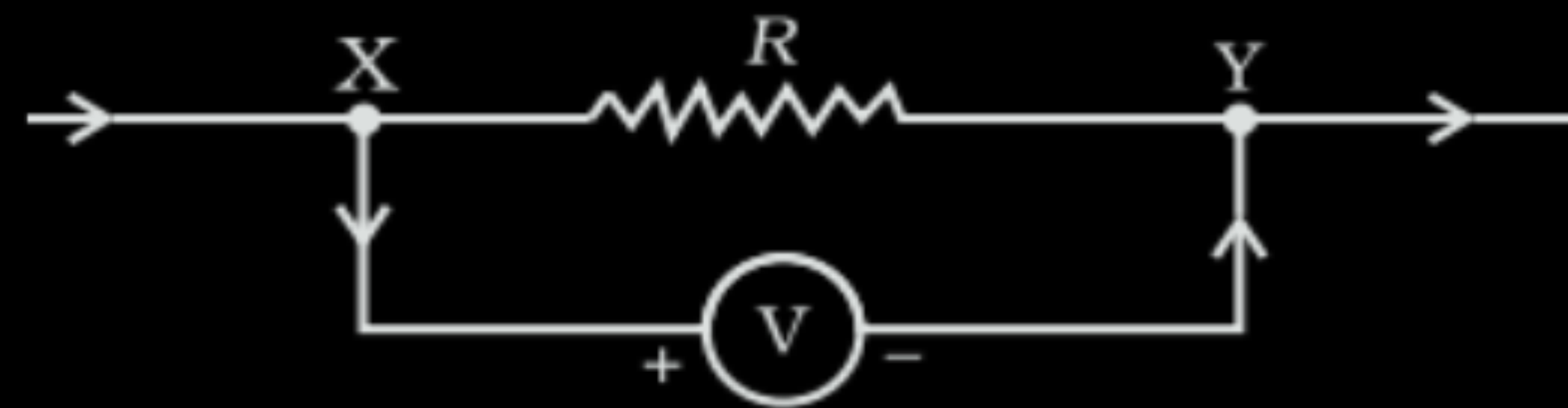
- (b) (i) What is meant by heating effect of electric current ? Derive an expression for the amount of heat produced in a conductor of resistance R through which a current I flows for a time t .

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- (ii) Calculate the amount of heat produced in a conductor of resistance $10\ \Omega$ through which a current of 2 A flows for 100 seconds.

(i)

- When electric current is passed through a resistor electric energy is dissipated and appears as heat energy.
- When charge Q flows through a conductor of resistance R having potential difference of V , the workdone W is given by-



$$\begin{aligned} W &= V \times Q = V \times I \times t \\ &= IR \times I \times t = I^2 R t \end{aligned}$$

($\because V = I \times R$ – Ohm's Law)

When circuit is purely resistive electric energy is converted into heat energy.

$$H = I^2 R t$$

(ii) Here $I = 2A$; $R = 10\Omega$; $t = 100s$

$$H = I^2 R t$$

$$= (2A)^2 \times 10\Omega \times 100s$$

36. (a) (i) State Ohm's law. Draw $V - I$ graph for an ohmic conductor.
- (ii) Electrical resistance of a metal wire of length 1 km is $200\ \Omega$. If the area of cross-section of the wire is $40 \times 10^{-8}\ \text{m}^2$, determine the electrical resistivity of the material of the wire.

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36. (a) (i) State Ohm's law. Draw V – I graph for an ohmic conductor.
- (ii) Electrical resistance of a metal wire of length 1 km is $200\ \Omega$. If the area of cross-section of the wire is $40 \times 10^{-8}\ \text{m}^2$, determine the electrical resistivity of the material of the wire.

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5

(a)

(i)

- Ohm's law: The potential difference, V , across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, provided its temperature remains the same. /

$$V \propto I$$

(ii) Here $l = 1\text{ km} = 1000\ \text{m}$

$R = 200\ \Omega$, $A = 40 \times 10^{-8}\ \text{m}^2$

$$\rho = \frac{R \times A}{l}$$

$$= 200\ \Omega \times \frac{40 \times 10^{-8}\ \text{m}^2}{1000\ \text{m}}$$

$$= 8 \times 10^{-8}\ \Omega\ \text{m}$$

1

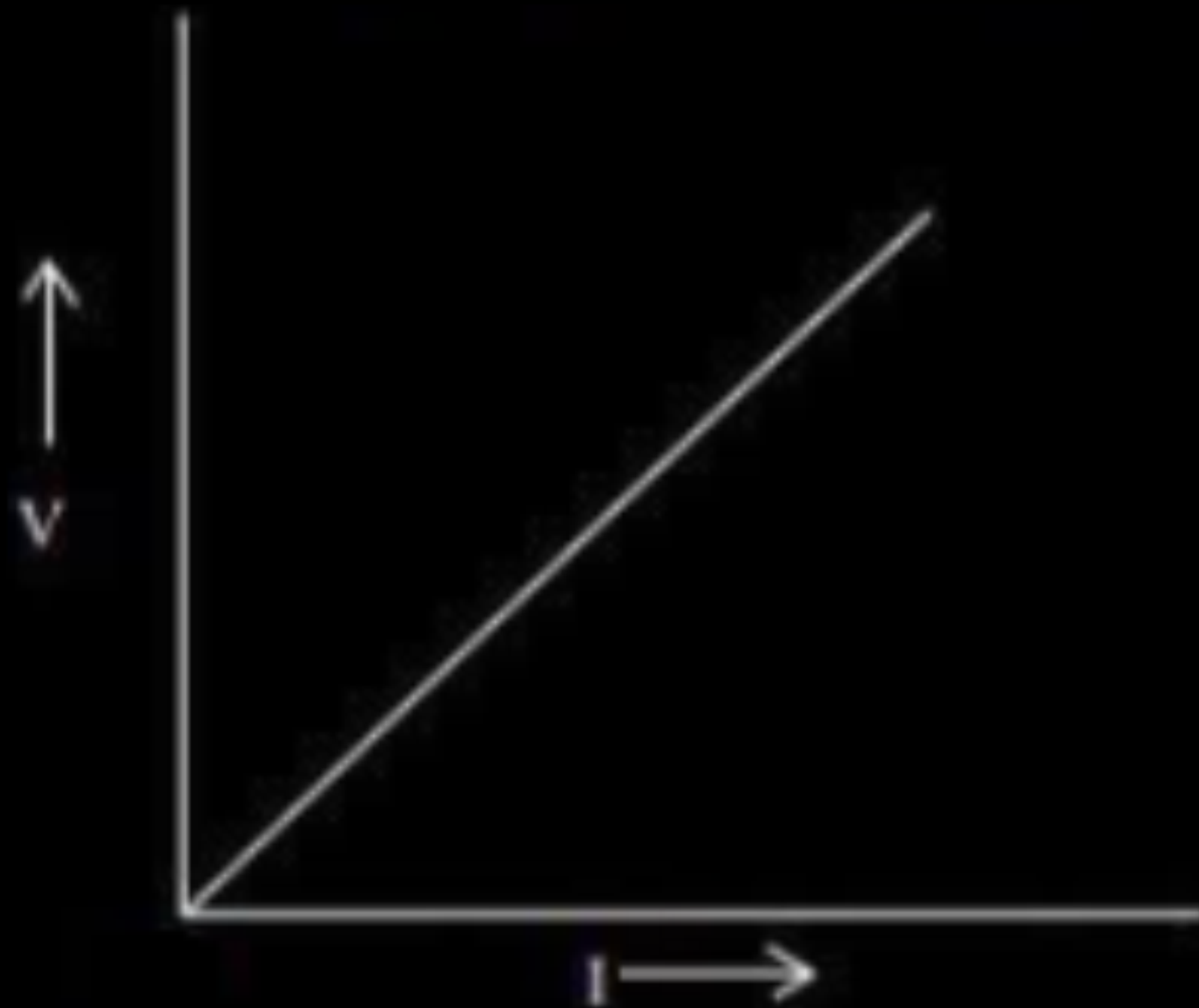
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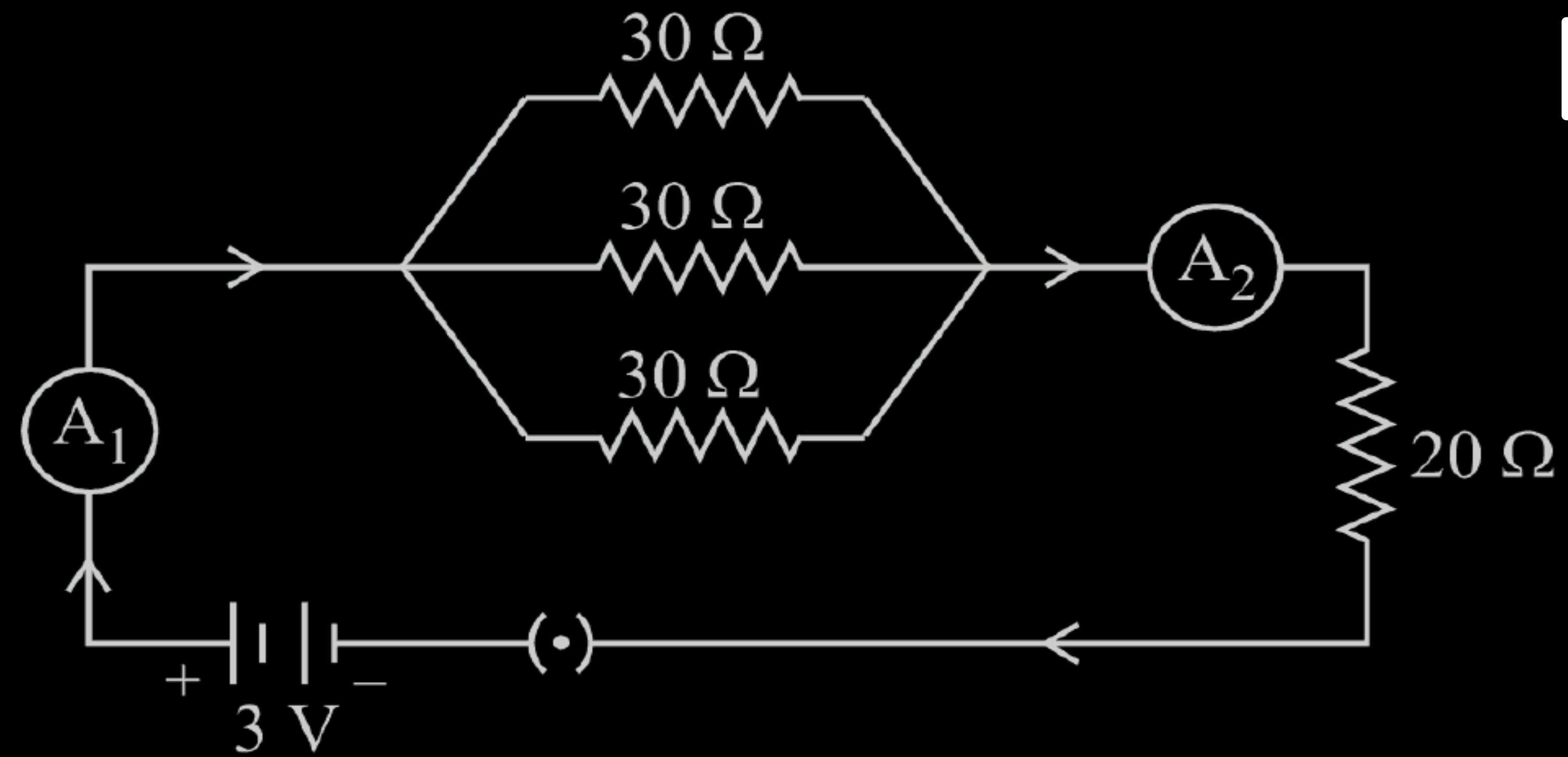
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1



- (b) (i) You have three resistors of resistance R_1 , R_2 and R_3 . If the equivalent resistance of series and parallel combination of the three resistors are R_s and R_p respectively, write the relationship to determine R_s and R_p in terms of R_1 , R_2 and R_3 .
- (ii) On the basis of the given circuit answer the questions given below :



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- (I) Write the relationship between the readings of the two ammeters A_1 and A_2 . Give reason for your answer.
- (II) What is the equivalent resistance of the parallel combination of three resistors ?
- (III) What is the value of current flowing in the circuit ?

(b)

(i)

$$R_s = R_1 + R_2 + R_3$$

1

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

1

(I)

- Both have same reading.
- As both are connected in series and in series current remains same.

$\frac{1}{2}$

$\frac{1}{2}$

$$(II) \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{30} + \frac{1}{30} + \frac{1}{30}$$

$$= 10 \Omega$$

1

(III)

$$I = \frac{V}{R} = \frac{3V}{30\Omega} = \frac{1}{10} = 0.1A$$

1

14. You have four resistance wires of same material. Select the wire which will offer the least resistance to the flow of current in an electric circuit :

- (A) Length $2l$ and diameter $2d$
- (B) Length l and diameter d
- (C) Length $l/2$ and diameter $d/2$
- (D) Length $l/4$ and diameter $d/4$

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14. You have four resistance wires of same material. Select the wire which will offer the least resistance to the flow of current in an electric circuit :

CBSE 2024 C

- (A) Length $2l$ and diameter $2d$
- (B) Length l and diameter d
- (C) Length $l/2$ and diameter $d/2$
- (D) Length $l/4$ and diameter $d/4$

16. The resistance of a resistor is increased four times its initial value. If other parameters (current and time) remain unchanged, the percentage change in the amount of heat produced in the resistor would be :

(A) 100%

(B) 200%

CBSE 2024 C

(C) 300%

(D) 400%

16. The resistance of a resistor is increased four times its initial value. If other parameters (current and time) remain unchanged, the percentage change in the amount of heat produced in the resistor would be :

(A) 100%

(B) 200%

CBSE 2024 C

(C) 300%

(D) 400%

(C)

- 36.** (a) List two factors on which the resistance of a conductor in the shape of a straight cylinder depends. Write the relation between resistance (R) and the resistivity (ρ) of the material of the conductor. Use this relation to obtain the SI unit of resistivity.
- (b) Resistance of a metal wire of length 1 m is $40\ \Omega$. If the area of cross-section of the wire is $6.5 \times 10^{-8}\ \text{m}^2$, find the resistivity of the material of the wire.

(a) • Two factors (any two)

(i) Length (l)

(ii) area of cross-section (A)

(iii) Material of conductor

$$\bullet R = \rho \frac{l}{A} \quad \text{or} \quad \rho = R \frac{l}{A}$$

$$\bullet \text{ SI unit of } \rho = \text{ SI unit of (R) } \times \frac{\text{SI unit of (A)}}{\text{SI unit of (l)}}$$

$$= \text{ohm} \times \frac{\text{metre}^2}{\text{metre}} = \text{ohm} \cdot \text{metre}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$1$$

$$1$$

$$(b) \rho = R \frac{l}{A}$$

$$= 40 \, \Omega \times \frac{6.5 \times 10^{-8} \, \text{m}^2}{1 \, \text{m}}$$

$$1$$

$$= 26 \times 10^{-7} \, \Omega \text{m}$$

$$1$$

3. Two resistors of $1\ \Omega$ and $3\ \Omega$ are connected in parallel in a circuit. If a current of 1 ampere is flowing through the circuit, the current flowing through the $3\ \Omega$ resistor is :

CBSE 2024

- (A) 1.0 ampere
- (B) 0.75 ampere
- (C) 0.33 ampere
- (D) 0.25 ampere

3. Two resistors of $1\ \Omega$ and $3\ \Omega$ are connected in parallel in a circuit. If a current of 1 ampere is flowing through the circuit, the current flowing through the $3\ \Omega$ resistor is : CBSE 2024

- (A) 1.0 ampere
- (B) 0.75 ampere
- (C) 0.33 ampere
- (D) 0.25 ampere

(D)/ 0.25 ampere

4. Two wires of same metal have the same length but their cross-sectional areas are in the ratio of 4 : 1. They are joined in series. The resistance of the thicker wire is $20\ \Omega$. The total resistance of the combination is :

CBSE 2024

- (A) $25\ \Omega$
- (B) $100\ \Omega$
- (C) $16\ \Omega$
- (D) $340\ \Omega$

4. Two wires of same metal have the same length but their cross-sectional areas are in the ratio of 4 : 1. They are joined in series. The resistance of the thicker wire is $20\ \Omega$. The total resistance of the combination is :

CBSE 2024

- (A) $25\ \Omega$
- (B) $100\ \Omega$
- (C) $16\ \Omega$
- (D) $340\ \Omega$

(B) / $100\ \Omega$

- 34.** (a) (i) Write the relation between resistivity (ρ) and resistance (R) for a uniform metallic conductor of length (l) and area of cross-section (A). Use this relation to obtain the SI unit of resistivity.
- (ii) The resistance of a metallic wire of 100 m is $20\ \Omega$. If the radius of the wire is 5 mm, find the resistivity of the metal of the wire.

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$$(a) (i) \quad R = \rho \frac{l}{A} \Rightarrow \rho = \frac{RA}{l}$$

$$\text{SI unit of resistivity } (\rho) = \frac{\text{SI unit of } R \times \text{SI unit of } A}{\text{SI unit of } l}$$

$$= \frac{\text{ohm} \times (\text{metre})^2}{\text{metre}}$$

$$= \text{ohm metre or } \Omega \text{ m}$$

$$(ii) \quad l = 100 \text{ m}; R = 20 \Omega; r = 5 \text{ mm} = \frac{5}{1000} \text{ m}$$

$$\therefore A = \pi r^2 = 3.14 \times \left(\frac{5}{1000} \right)^2 \text{ m}^2$$

$$\therefore \rho = R \frac{A}{l}$$

$$= 20 \Omega \times \frac{3.14 \times \left(\frac{5}{1000} \right)^2 \text{ m}^2}{100 \text{ m}}$$

$$= 157 \times 10^{-7} \Omega \text{ m}$$

1

1/2

1/2

1

1

1

- (b) (i) Obtain an expression for the amount of heat produced in a conductor of resistance R when a steady current I flows through it for a time t .
- (ii) An electrical heater of resistance $10\ \Omega$ draws a steady current of 5 A from the service mains for 2 hours. Find the power of this heater and the amount of energy consumed by the heater in kilowatt hour.

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(i) When Q amount of charge flows through a potential difference of V, Then the amount of work done, $W = V Q$ -----(1)	$\frac{1}{2}$
$P = W/t$ $= VQ/t$ [From (1)] $= V I$ [I = Q/t] -----(2)	$\frac{1}{2}$
Work done=heat produced, $H = P \times t$ $= VIt$ [From (2)] $= I^2Rt$ [From Ohm's law; $V = IR$]	$\frac{1}{2}$
(ii) Here, $R = 10 \Omega$, $I = 5 A$, $t = 2 h$, $P = ?$; $E = ?$ $P = I^2 R$ $= (5A)^2 \times 10 \Omega = 25 \times 10 W$ $= 250 W$	$\frac{1}{2}$
Energy / Work done = $P \times t$ $= 250 W \times 2 h = 500 Wh$ $= 0.5 kWh$	1
	$\frac{1}{2}$
	1

17. *Assertion (A) :* Electrons move from lower potential to higher potential in a conductor.

Reason (R) : A dry cell maintains electric potential difference across the ends of a conductor.

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17. *Assertion (A) :* Electrons move from lower potential to higher potential in a conductor.

Reason (R) : A dry cell maintains electric potential difference across the ends of a conductor.

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(B) / Both Assertion (A) and Reason (R) are the true , but Reason (R) is not a correct explanation of Assertion (A).

26. The filament of an electric lamp draws a current of 0.5 A , which lights for 2 hours. Calculate the charge that flows through the circuit.

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2

- 26.** The filament of an electric lamp draws a current of 0.5 A, which lights for 2 hours. Calculate the charge that flows through the circuit. CBSE 2024

2

$$I = 0.5 \text{ A}, t = 2 \text{ hours} = 2 \times 3600 \text{ s}$$

$$I = \frac{Q}{t}$$

$$\therefore Q = I \times t = 0.5 \text{ A} \times 2 \times 3600 \text{ s} = 3600 \text{ C}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

32. A 2000 W heater has a resistance of about $25\ \Omega$, whereas a 100 W bulb has a resistance of $500\ \Omega$. When 220 V is applied on these, then which of the two

- (a) can carry large currents ?
- (b) may be used with an electrical circuit having 1.0 A rating ?
- (c) will be fitted with a 15 A electric board and not with a 5 A electric board ?

Justify your answer in each case.

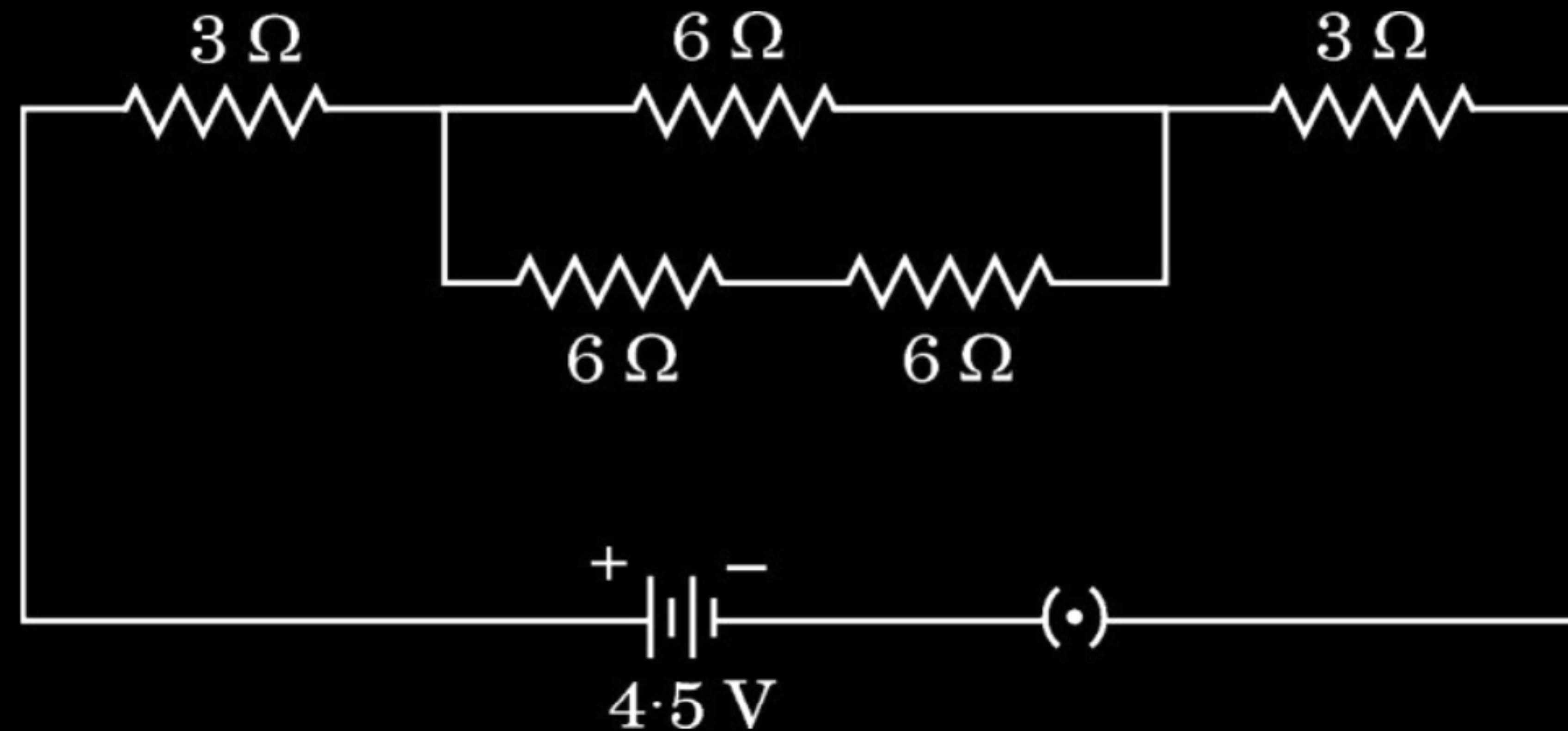
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3

(a) 2000 W heater , For heater, $I_1 = \frac{P}{V} = 9.09 \text{ A}$; For Bulb, $I_2 = \frac{P}{V} = 0.45 \text{ A}$ $I_1 > I_2$	$\frac{1}{2}$ $\frac{1}{2}$
(b) 100 W bulb $I_2 = \frac{P}{V} = 0.45 \text{ A}$ As it draws only 0.45 A which is less than 1 A.	$\frac{1}{2}$ $\frac{1}{2}$
(c) 2000 W heater $I_1 = \frac{P}{V} = 9.09 \text{ A}$ As the current drawn is 9.09 A which is higher than 5.0 A.	$\frac{1}{2}$ $\frac{1}{2}$

- 36.** (a) (i) Which type of circuits — series or parallel, should be used when you have to operate different electrical gadgets in your house ? List two reasons for your answer.
- (ii) Find the current flowing through the following electric circuit :

5



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(a)	(i) Parallel Circuit • Each electrical appliance has its own switch due to which it can be turned ON and OFF separately. • If one electrical appliance stops working, others remain unaffected. • Each appliance has equal potential difference and draws current as per its requirement. • The total resistance in parallel circuit decreases. (any two)	1
		$\frac{1}{2} + \frac{1}{2}$
(ii)	Combined resistance of the series, $R_1 = 6\ \Omega + 6\ \Omega = 12\ \Omega$	$\frac{1}{2}$
	Combined resistance of parallel grouping of $6\ \Omega$ and $R_1 = 12\ \Omega$, resistors is R_2, where	
	$\frac{1}{R_2} = \frac{1}{6} + \frac{1}{12} = \frac{9}{20}$	
	$R_2 = 4.0\ \Omega$	$\frac{1}{2}$
	Total resistance of circuit = $R = 3 + 4 + 3 = 10\ \Omega$	1
	Current flowing = $I = \frac{V}{R}$	$\frac{1}{2}$
	$= \frac{4.5\ \text{V}}{10\ \Omega}$ $= 0.45\ \text{A}$	$\frac{1}{2}$

- (b) The values of current I flowing in a given resistor for the corresponding values of potential difference V across the resistor are given in the following table :

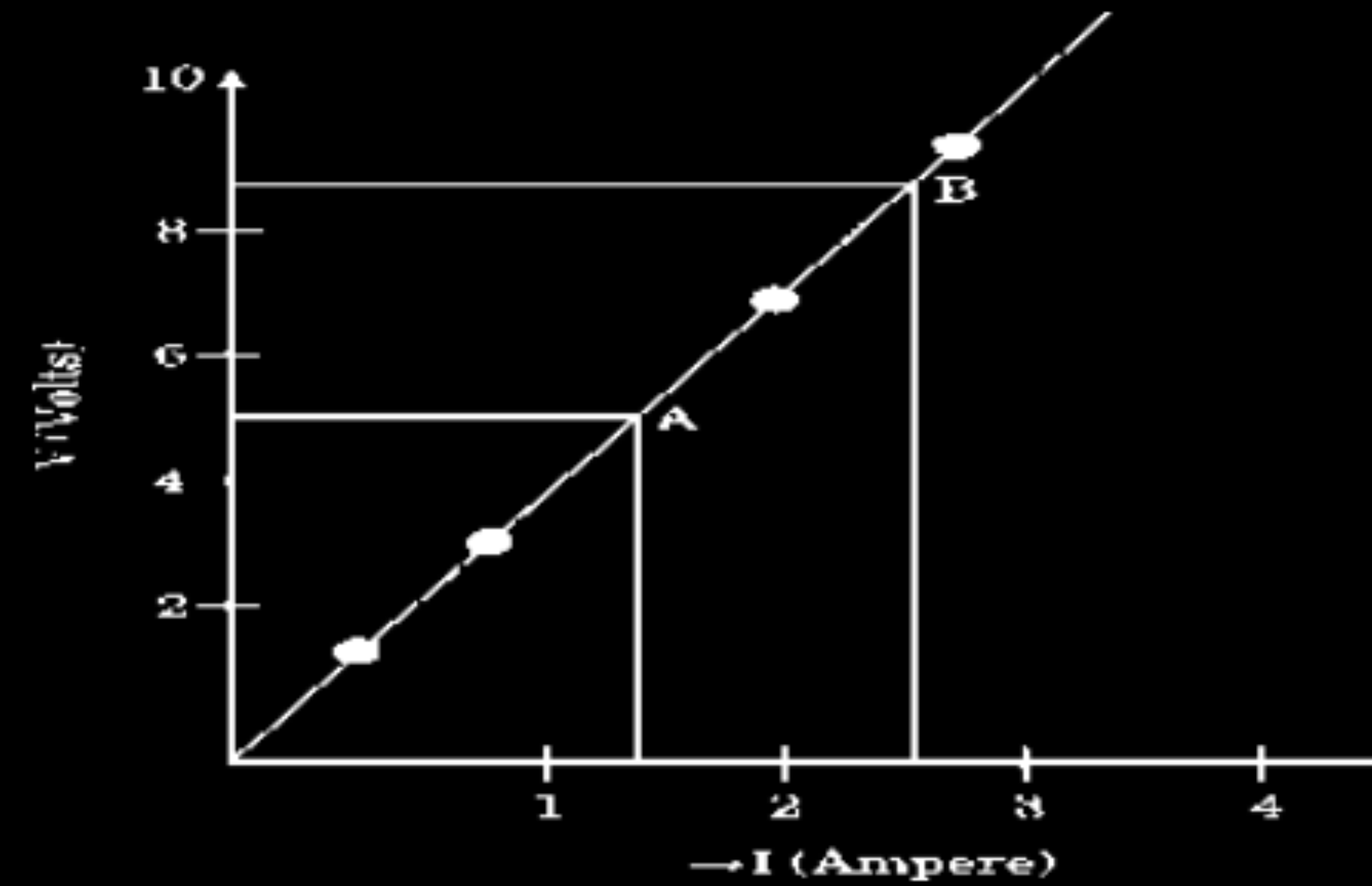
I (Amperes)	0.5	1.0	2.0	3.0	3.5
V (Volts)	1.5	3.0	6.2	9.3	10.8

- (i) Plot a graph between V and I .
- (ii) Calculate the resistance of the resistor with the help of the graph.
- (iii) What does the graph represent ?
- (iv) Why should this graph pass through the origin ?

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(b)

(i)



(ii) Resistance of resistor = $\frac{V_2 - V_1}{I_2 - I_1} = \frac{8.3 - 5.2}{2.5 - 1.5} = 3.1 \Omega$

(iii) The given resistor obeys Ohm's law./ Resistance remains constant.

(iv) Because when the value of $V = 0$, the current $I = 0$.

1

1+1

1

1

- 26.** An electric source can supply a charge of 750 coulomb. If the current drawn by a device is 15 mA, find the time in which the electric source will be discharged completely.

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2

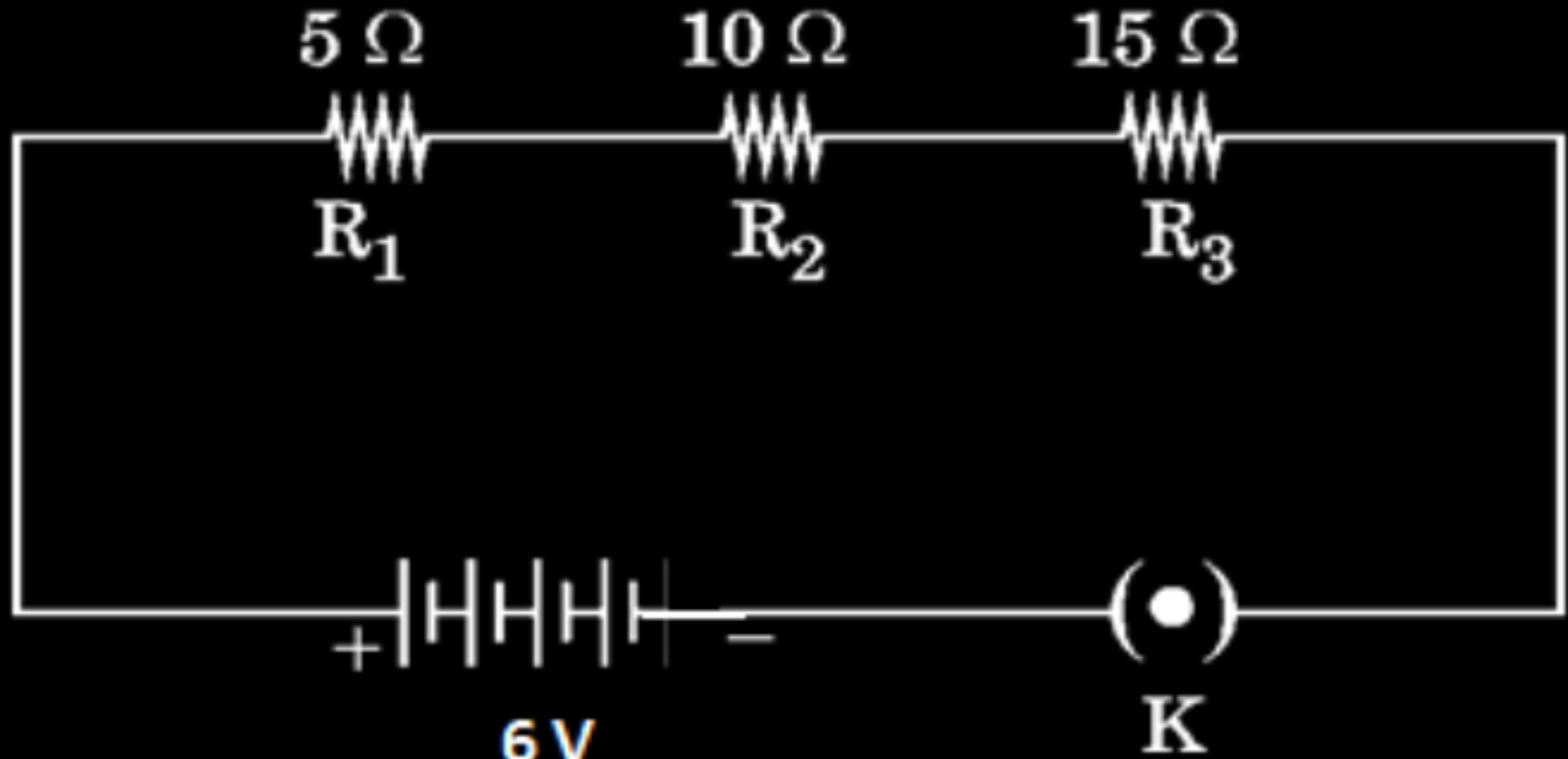
26. An electric source can supply a charge of 750 coulomb. If the current drawn by a device is 15 mA, find the time in which the electric source will be discharged completely.

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2

- | | |
|---|---|
| • $Q = I \times t \Rightarrow t = \frac{Q}{I}$ | 1 |
| • $\therefore t = \frac{750}{\frac{15}{1000}} = \frac{750 \times 1000}{15} = 50000 \text{ s}$ | 1 |

- 35.** (a) (i) The potential difference across the two ends of a circuit component is decreased to one-third of its initial value, while its resistance remains constant. What change will be observed in the current flowing through it ? Name and state the law which helps us to answer this question.
- (ii) Draw a schematic diagram of a circuit consisting of a battery of four 1.5 V cells, a $5\ \Omega$ resistor, a $10\ \Omega$ resistor and a $15\ \Omega$ resistor and a plug key, all connected in series. Now find (I) the electric current passing through the circuit, and (II) potential difference across the $10\ \Omega$ resistor when the plug key is closed.

(a)	(i) • Current becomes one-third of its initial value. • Ohm's Law The potential difference across the ends of a conductor is directly proportional to the current flowing through it, provided its temperature remains the same.	$\frac{1}{2}$ $\frac{1}{2}$
(ii)	 The diagram shows a rectangular circuit loop. At the top, three resistors are connected in series. The first resistor is labeled $5\ \Omega$ and R_1. The second is labeled $10\ \Omega$ and R_2. The third is labeled $15\ \Omega$ and R_3. At the bottom, there is a 6V battery (represented by four cells) and a key labeled K (represented by a circle with a dot). The battery is on the left and the key is on the right.	1
	Total Voltage = $V = 4 \times 1.5\text{ V} = 6\text{ V}$ Total resistance, $R(s) = R_1 + R_2 + R_3$ $= 5\ \Omega + 10\ \Omega + 15\ \Omega = 30\ \Omega$	
(I)	Current, $I = \frac{V}{R} = \frac{6\text{ V}}{30\ \Omega} = 0.2\text{ A}$	1
(II)	$V = IR = 0.2\text{ A} \times 10\ \Omega = 2\text{ V}$	1

- (b) (i) When is the potential difference between two points said to be 1 volt ?
- (ii) A copper wire has a diameter of 0.2 mm and resistivity of $1.6 \times 10^{-8} \Omega \text{ m}$. What will be the length of this wire to make its resistance 14Ω ? How much does the resistance change, if the diameter of the wire is doubled ?

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(b)

- (i) When 1 joule of work is done to move a charge of 1 coulomb from one point to the other.

$$d = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}; R = 14 \Omega$$

$$\rho = 1.6 \times 10^{-8} \Omega \text{ m}; A = \frac{\pi d^2}{4}$$

$$R = \frac{\rho l}{A} = \frac{4\rho l}{\pi d^2} \text{ or } l = \frac{\pi d^2 R}{4\rho}$$

$$l = \frac{22}{7} \times \frac{(2 \times 10^{-4})^2}{4 \times 1.6 \times 10^{-8}} \times 14$$
$$= \frac{22 \times 14}{7 \times 1.6} = 27.5 \text{ m}$$

When the diameter is doubled, $d' = 2d$

$$A' = 4A$$

$$\frac{R'}{R} = \frac{A}{A'} \text{ or } R' = \frac{RA}{A'} = \frac{RA}{4A}$$

$$R' = \frac{R}{4} = \frac{14 \Omega}{4} = 3.5 \Omega$$

$$\text{Change } (14.0 - 3.5) = 10.5 \Omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

1

$\frac{1}{2}$

1. An electric iron of resistance $20\ \Omega$ draws a current of $5\ \text{A}$. The heat developed in the iron in 30 seconds is :

(A) $15000\ \text{J}$

(B) $6000\ \text{J}$

(C) $1500\ \text{J}$

(D) $3000\ \text{J}$

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1. An electric iron of resistance $20\ \Omega$ draws a current of $5\ \text{A}$. The heat developed in the iron in 30 seconds is :

(A) $15000\ \text{J}$

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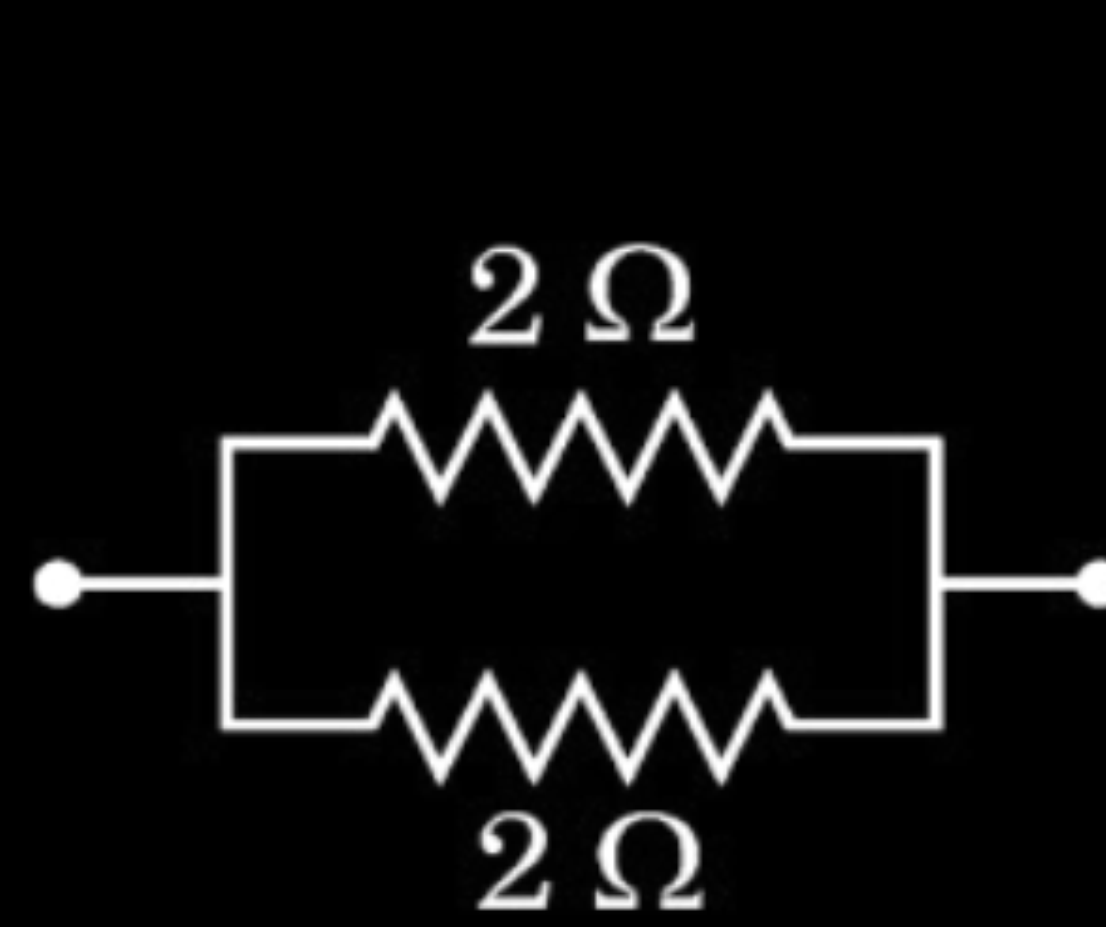
(C) $1500\ \text{J}$

(D) $3000\ \text{J}$

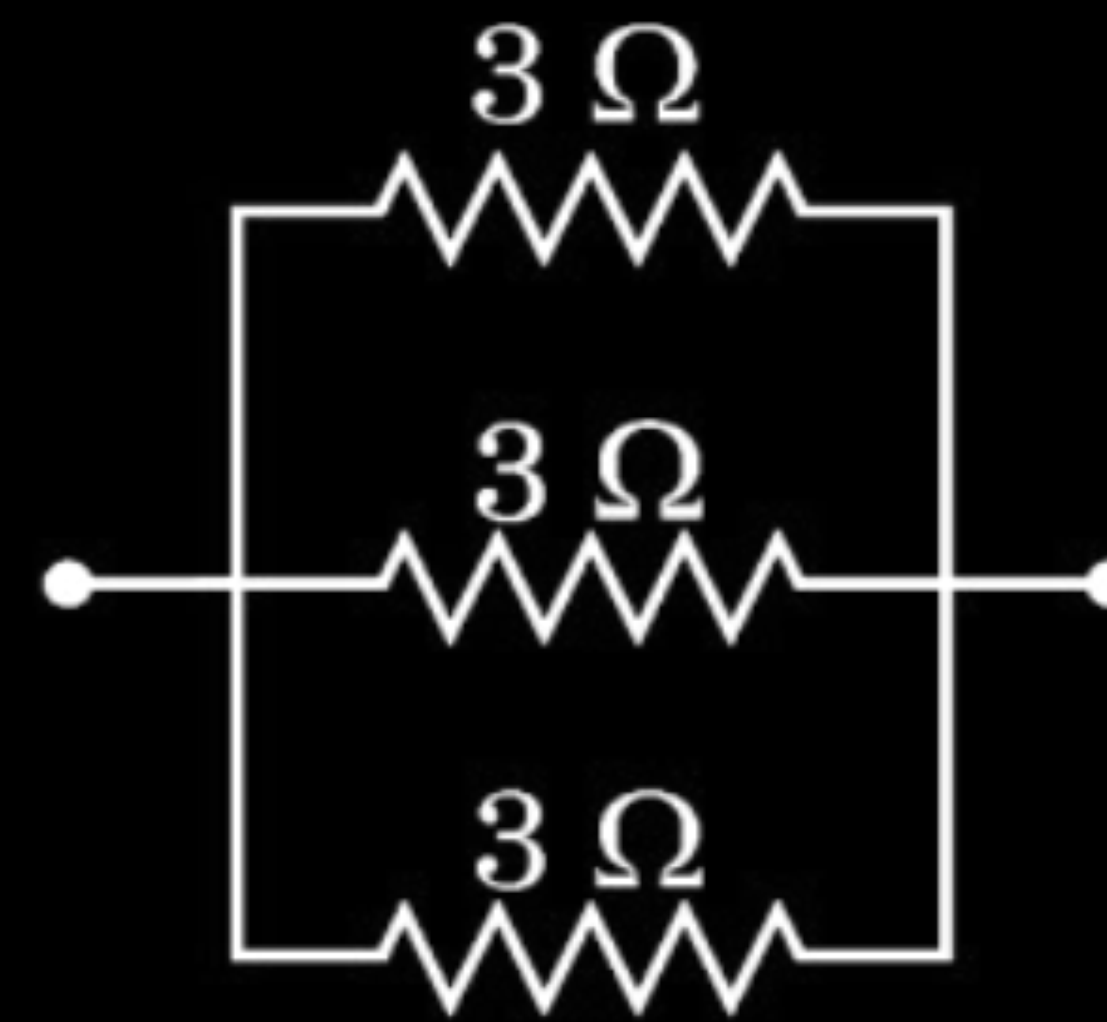
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(A)/15000 J

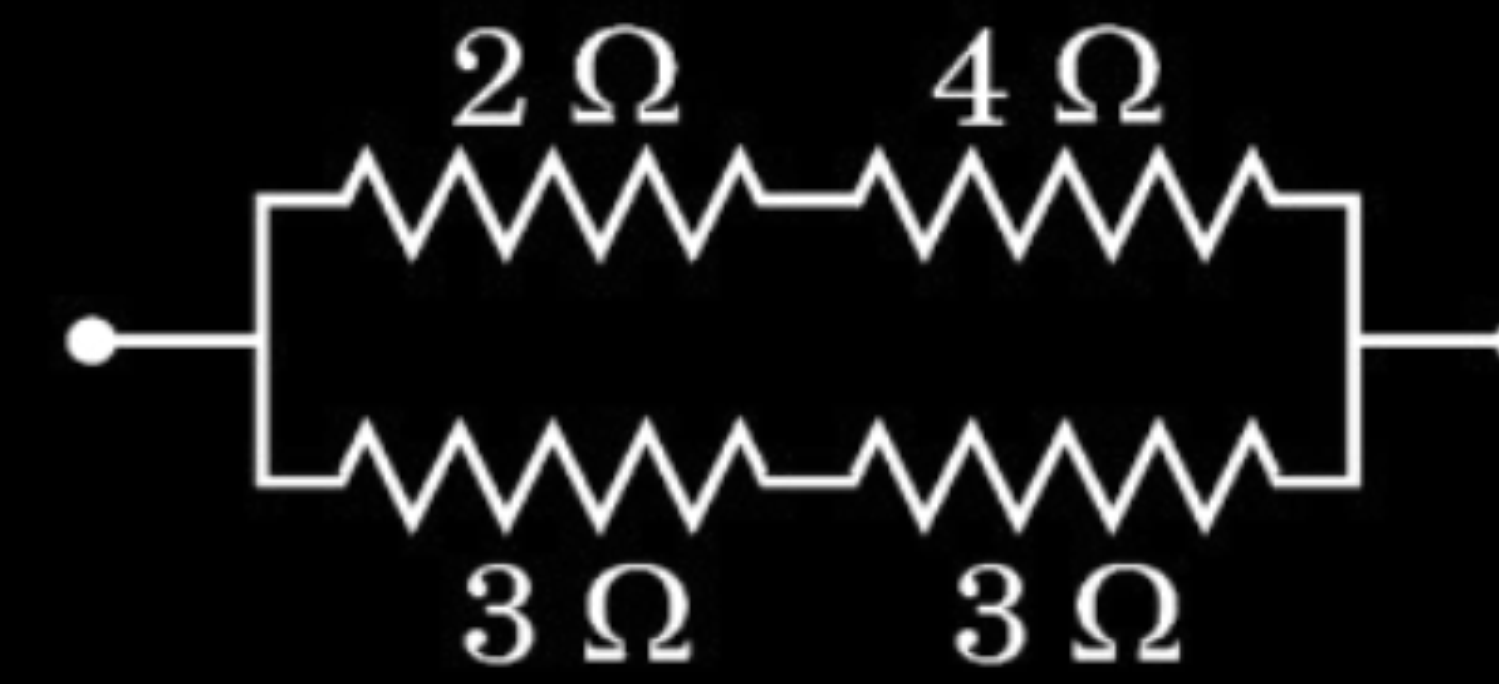
16. Consider the following combinations of resistors :



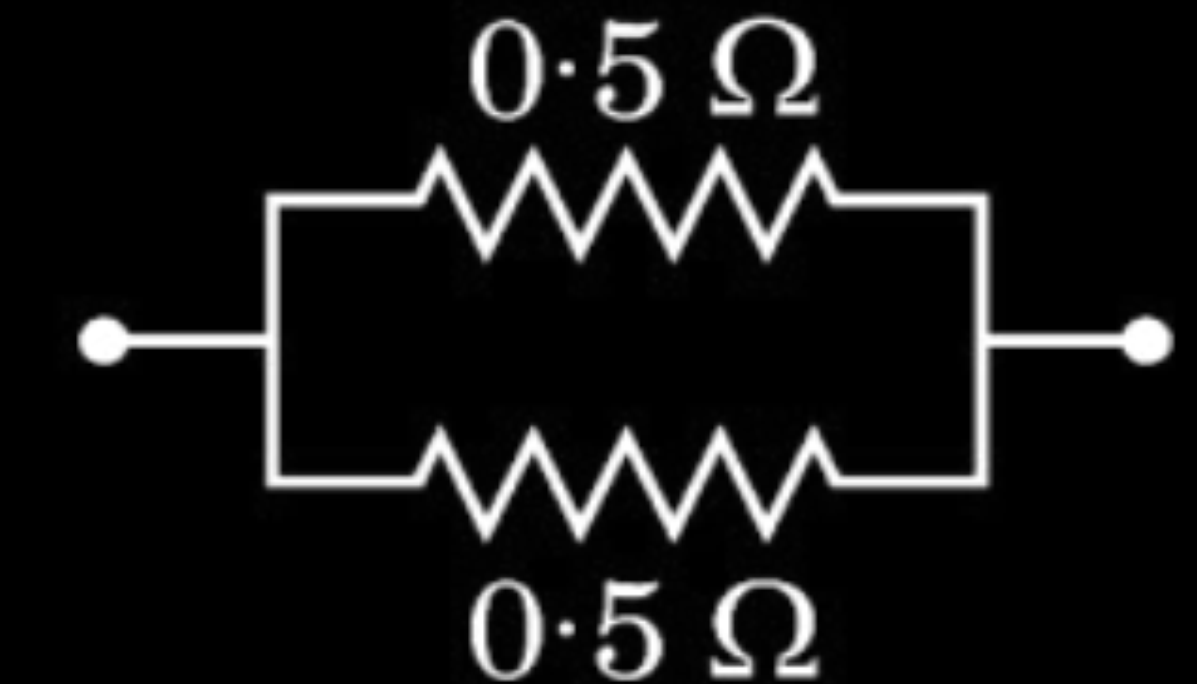
I



II



III



IV

The combinations having equivalent resistance $1\ \Omega$ is/are :

(A) I and IV

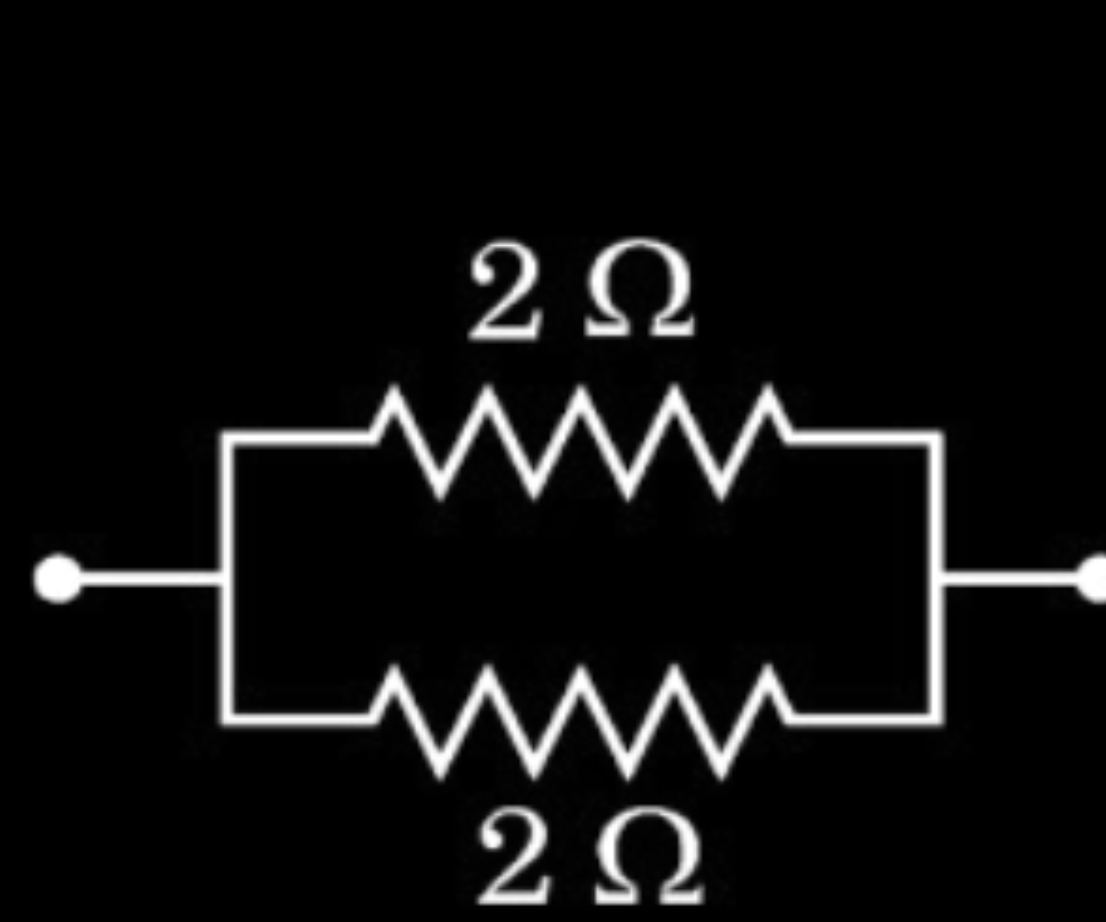
(B) Only IV

(C) I and II

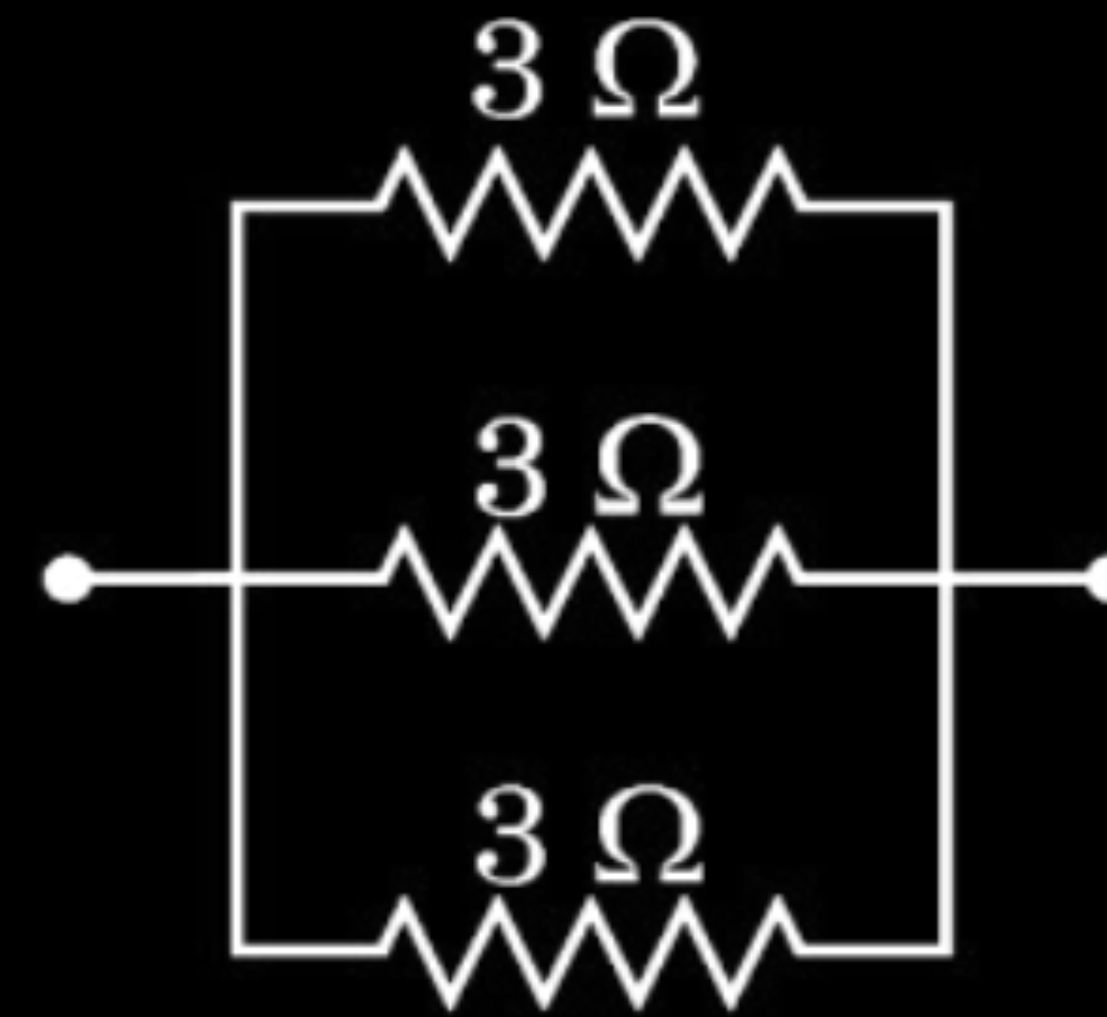
(D) I, II and III

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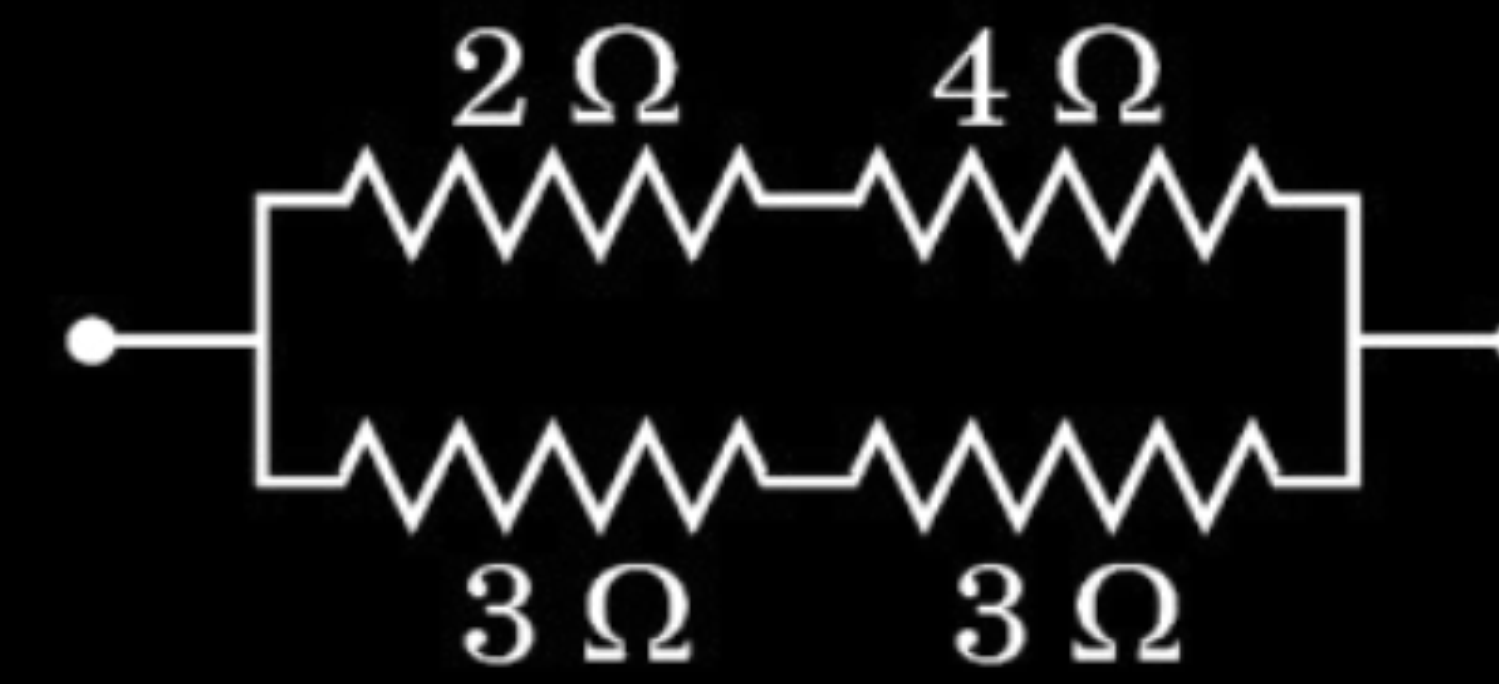
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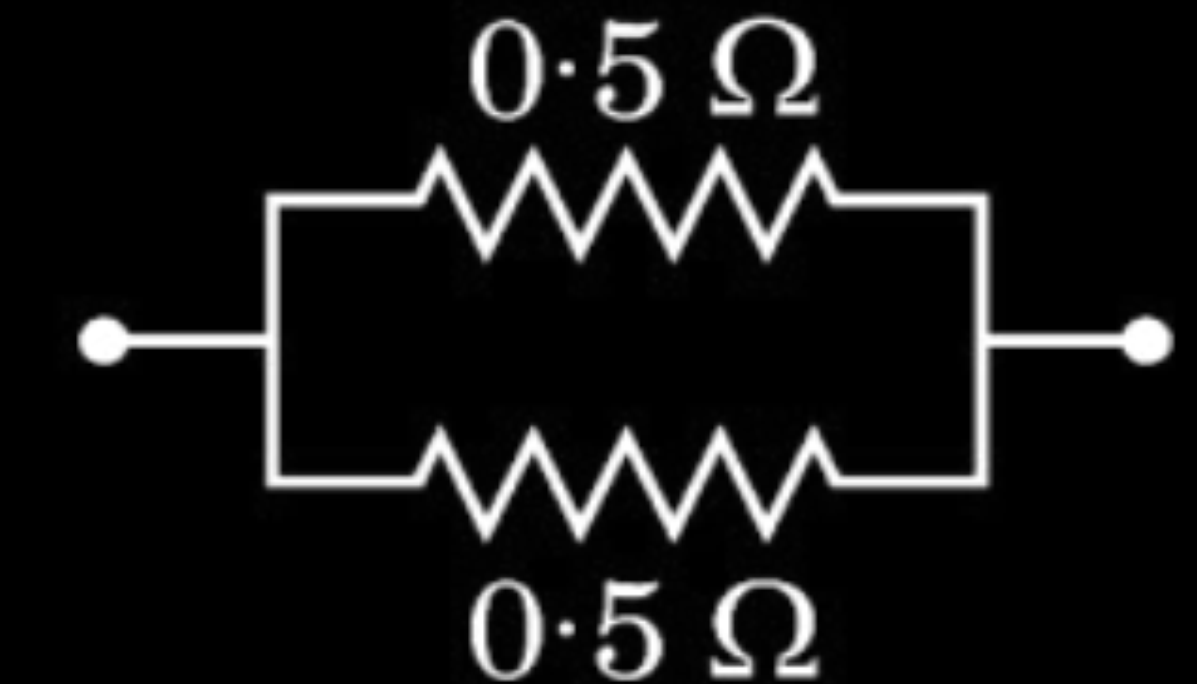
I



II



III



IV

The combinations having equivalent resistance $1\ \Omega$ is/are :

(A) I and IV

(B) Only IV

(C) I and II

(D) I, II and III

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(C)/I and II

26. Calculate the resistance of a copper wire of length 1000 m and area of cross-section 2 mm^2 . Resistivity of copper is $1.6 \times 10^{-8} \Omega \text{ m}$.

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2

26. Calculate the resistance of a copper wire of length 1000 m and area of cross-section 2 mm^2 . Resistivity of copper is $1.6 \times 10^{-8} \Omega \text{ m}$.

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2

$$R = \rho \frac{l}{A}$$

$$\rho = 1.6 \times 10^{-8} \Omega \text{ m}$$

$$A = 2 \times (10^{-3} \text{ m})^2$$

$$l = 1000 \text{ m}$$

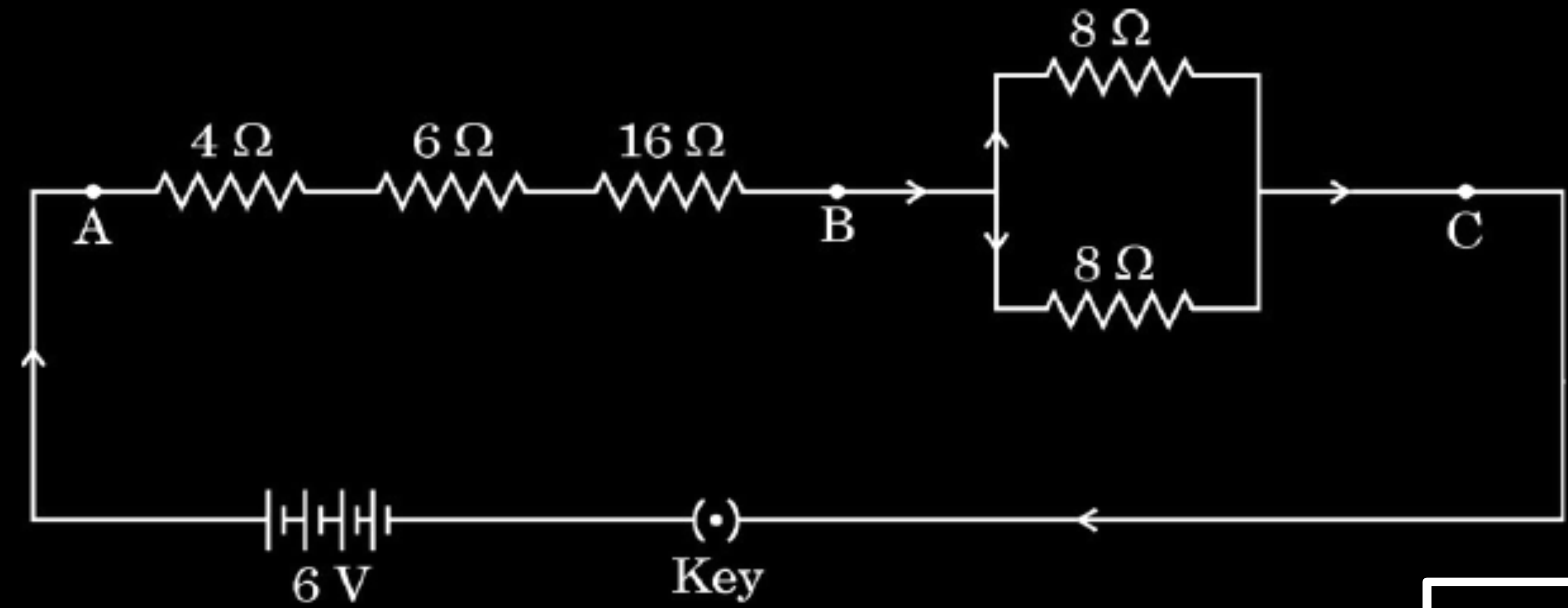
$$\begin{aligned} \therefore R &= (1.6 \times 10^{-8} \Omega \text{ m}) \times \frac{1000 \text{ m}}{2 \times (10^{-3} \text{ m})^2} \\ &= 8.0 \Omega \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

37. Study the following circuit :



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On the basis of this circuit, answer the following questions :

- (a) Find the value of total resistance between the points A and B. 1
- (b) Find the resistance between the points B and C. 1
- (c) (i) Calculate the current drawn from the battery, when the key is closed. 2

OR

- (c) (ii) In the above circuit, the $16\ \Omega$ resistor or the parallel combination of two resistors of $8\ \Omega$, which one of the two will have more potential difference across its two ends ? Justify your answer. 2

$$(a) \quad R_S = 4 \, \Omega + 6 \, \Omega + 16 \, \Omega = 26 \, \Omega$$

$$(b) \quad \frac{1}{R_P} = \frac{1}{8 \, \Omega} + \frac{1}{8 \, \Omega} = \frac{1}{4} \, \Omega$$

$$R_P = 4 \, \Omega$$

$$(c) (i) \quad \text{Total resistance} = 26 \, \Omega + 4 \, \Omega = 30 \, \Omega$$

$$\text{Potential difference} = V = 6V$$

$$\text{Current } I = \frac{V}{R}$$

$$\frac{6}{30} = \frac{1}{5} \, A \quad \text{or } 0.2 \, A.$$

OR

$$(c)(ii) \quad 16 \, \Omega$$

Justification: According to Ohm's law when same current flows, the potential difference across a higher resistance is always higher./

$$\text{Potential difference across } 16 \, \Omega = V = IR = 0.2 \times 16 = 3.2V$$

$$\text{Potential difference across } 8 \, \Omega = V = IR_{(total)} = 0.2 \times 4 = 0.8V$$

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

1

1

26. State Joule's law of heating. How is this effect useful in electric circuits where fuse is used as a safety device ?

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2

26. State Joule's law of heating. How is this effect useful in electric circuits where fuse is used as a safety device ?

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2

- | | |
|---|-------------------|
| <ul style="list-style-type: none">• Joule's Law – Heat produced in a resistor is directly proportional to:<ul style="list-style-type: none">-Square of current for a given resistance-Resistance for a given conductor and-Time for which the current flows through the resistor,• If any unduly high electric current flows through the circuit, the temperature of the fuse wire increases. This melts the fuse wire and breaks the circuit. | <p>1</p> <p>1</p> |
|---|-------------------|

- 26.** Two wires A and B of same material, having same lengths and diameters 0·2 mm and 0·3 mm respectively, are connected one by one in a circuit. Which one of these two wires will offer more resistance to the flow of current in the circuit ? Justify your answer.

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2

26. Two wires A and B of same material, having same lengths and diameters 0.2 mm and 0.3 mm respectively, are connected one by one in a circuit. Which one of these two wires will offer more resistance to the flow of current in the circuit ? Justify your answer.

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2

- Wire A will offer more resistance

Justification:

- $R \propto \frac{l}{A}$ /thinner the wire, more resistance to the flow of current.

1

1

13. In case of four wires of same material, the resistance will be minimum if the diameter and length of the wire respectively are

1

(A) $D/2$ and $L/4$

(B) $D/4$ and $4L$

(C) $2D$ and L

(D) $4D$ and $2L$

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13. In case of four wires of same material, the resistance will be minimum if the diameter and length of the wire respectively are

1

(A) $D/2$ and $L/4$

(B) $D/4$ and $4L$

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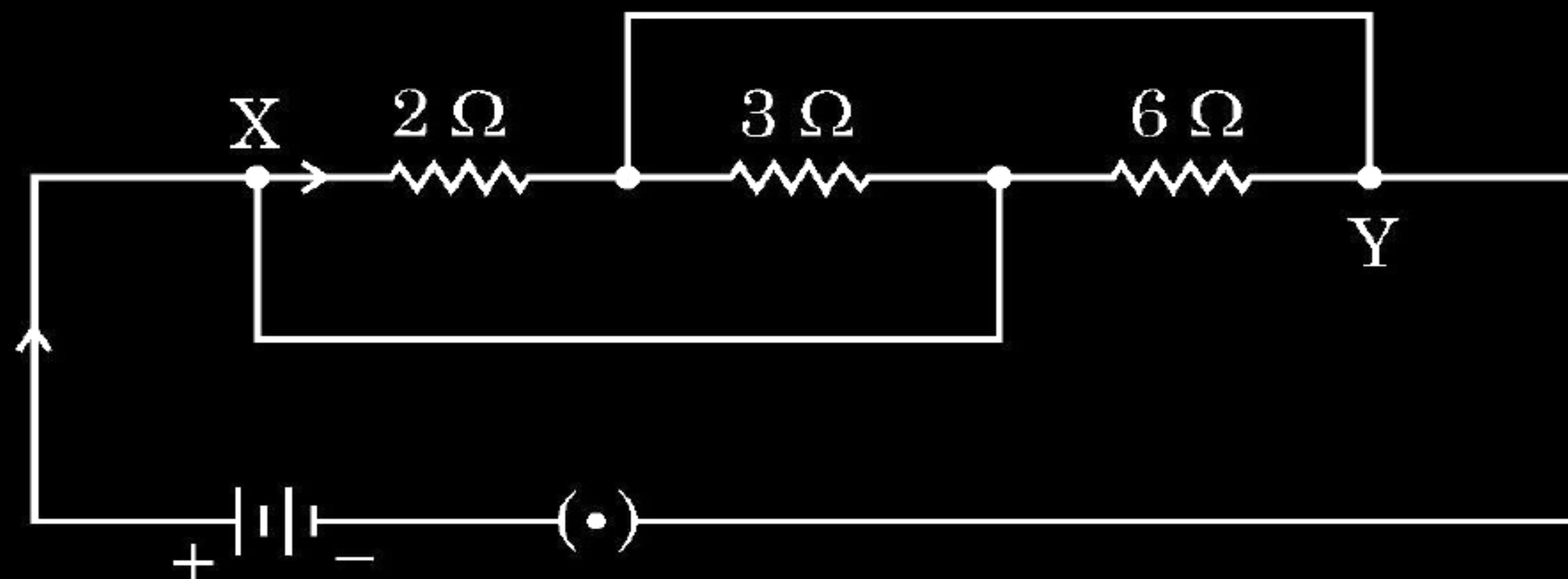
(D) $4D$ and $2L$

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(D) / $4D$ and $2L$

14. In the given circuit the total resistance between X and Y is :

1



(A) $12\ \Omega$

(B) $4\ \Omega$

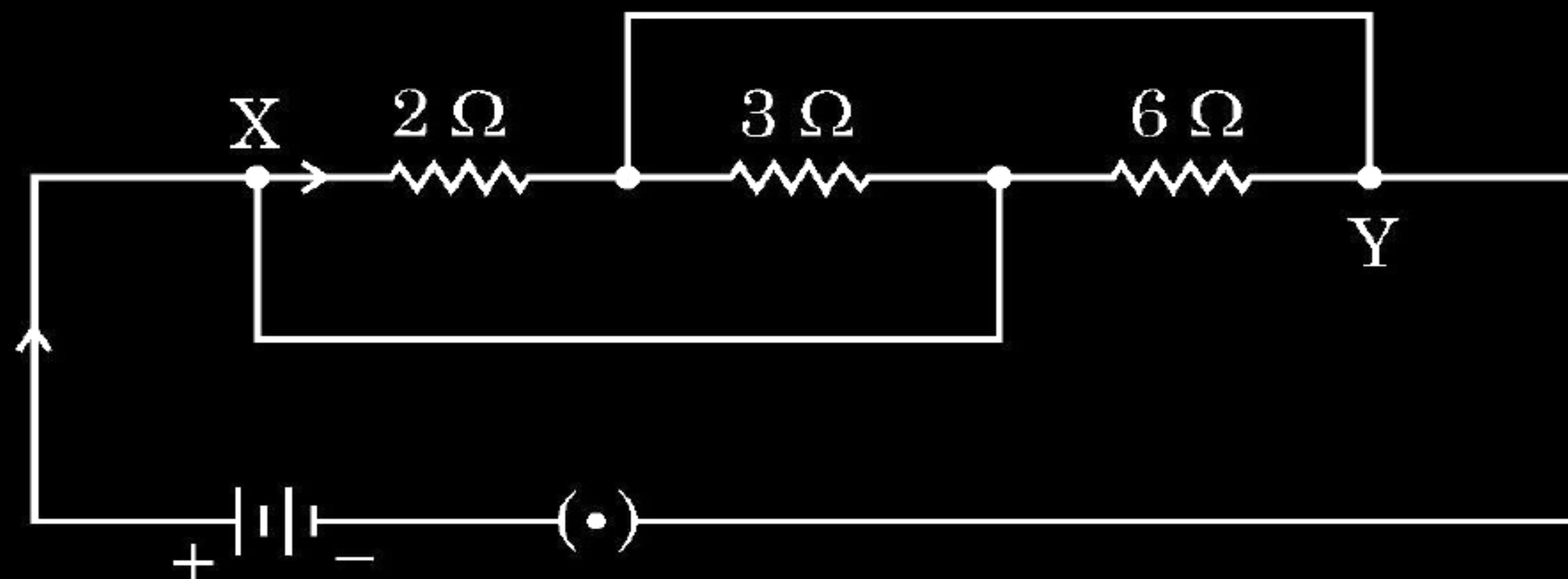
(C) $6\ \Omega$

(D) $1\ \Omega$

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14. In the given circuit the total resistance between X and Y is :

1



(A) $12\ \Omega$

(B) $4\ \Omega$

(C) $6\ \Omega$

(D) $1\ \Omega$

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(D) / $1\ \Omega$

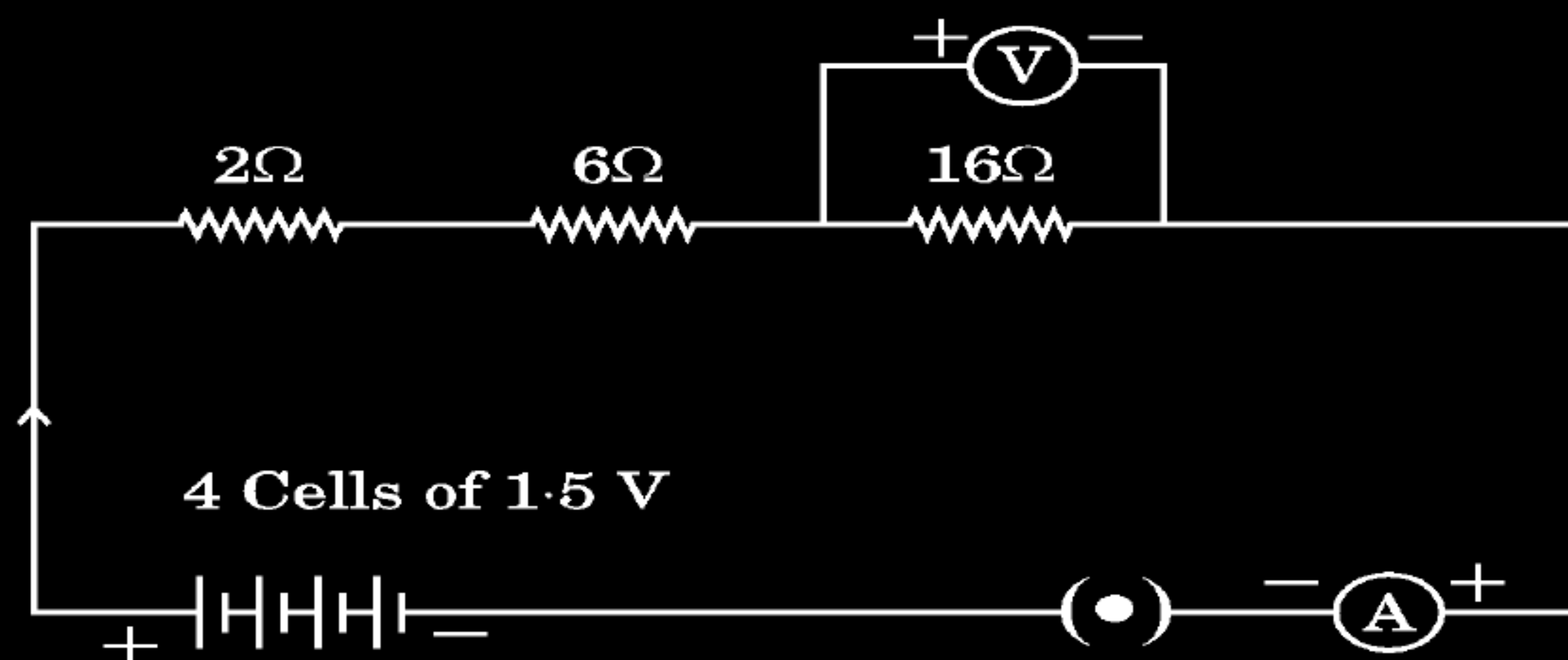
32. Draw a schematic diagram of a circuit consisting of a battery of four dry cells of 1.5 V each, a $2\ \Omega$ resistor, a $6\ \Omega$ resistor, $16\ \Omega$ resistor and a plug key all connected in series. Put an ammeter to measure the current in the circuit and a voltmeter across the $16\ \Omega$ resistor to measure potential difference across its two ends. Use Ohm's law to determine

3

(a) ammeter reading, and

(b) voltmeter reading when key is closed.

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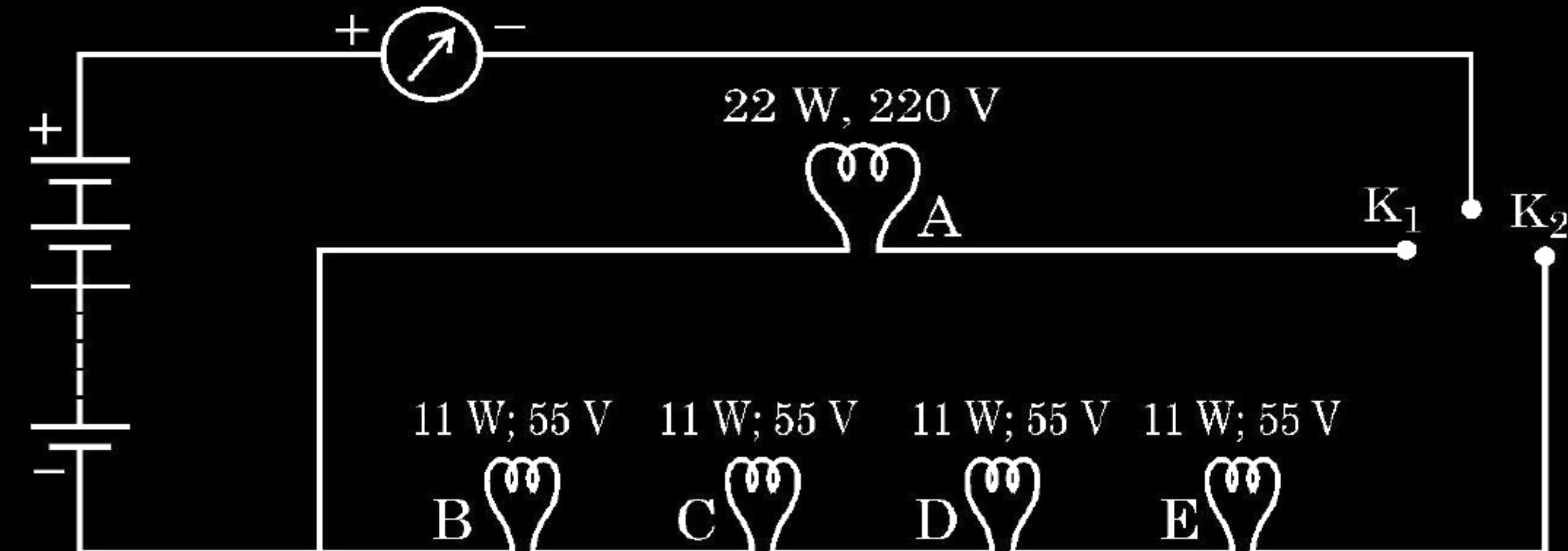
1

(a) Current (I) = $\frac{V}{R} = \frac{4 \times 1.5 \text{ V}}{2\Omega + 6\Omega + 16\Omega} = \frac{6 \text{ V}}{24\Omega} = \frac{1}{4} \text{ A}$

1

(b) Potential Difference across 16 Ω resistor = $\frac{1}{4} \text{ A} \times 16 \Omega = 4 \text{ V}$

1



39. In a domestic circuit five LED bulbs are arranged as shown. The source voltage is 220 V and the power rating of each bulb is marked in the circuit diagram. Based on the following circuit diagram, answer the following questions :

(a) State what happens when

(i) key K_1 is closed.

(ii) key K_2 is closed.

(b) Find the current drawn by the bulb B when it glows.

(c) Calculate

(i) the resistance of bulb B , and

(ii) total resistance of the combination of four bulbs B, C, D and E .

OR

(c) What would happen to the glow of all the bulbs in the circuit when keys K_1 and K_2 both are closed and the bulb C suddenly get fused ?

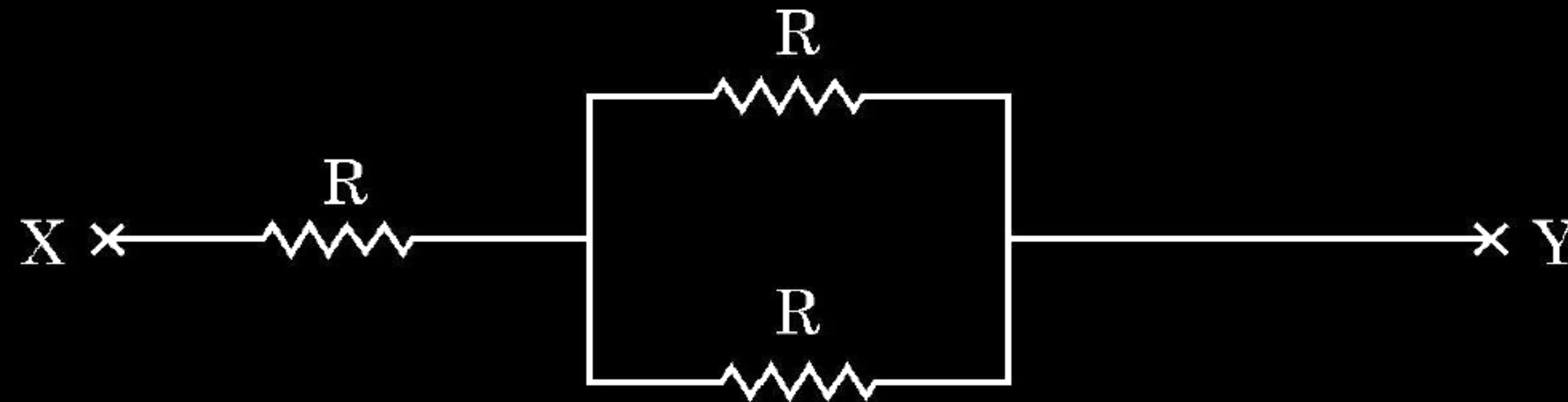
Give reason to justify your answer.

(a)		
(i)	Bulb A glows	$\frac{1}{2}$
(ii)	Bulbs B, C, D and E glow	$\frac{1}{2}$
(b)	$P = V \times I$	$\frac{1}{2}$
	$11 = 55 \times I$	
	$I = \frac{1}{5} = 0.2 \text{ amp}$	$\frac{1}{2}$
(c)		
(i)	Resistance of bulb B, $R = \frac{V}{I} = \frac{55 \text{ V}}{0.2 \text{ A}} = 275 \Omega$	1
	(alternative formula for calculation $R = \frac{V^2}{P}$)	
(ii)	Total resistance of the series combination of four bulbs $= 4 \times 275 = 1100 \Omega$ OR	1
(c)		
•	Bulb A will keep glowing with same brightness.	$\frac{1}{2}$
•	Other bulbs i.e., B, D and E will stop glowing.	$\frac{1}{2}$
	Reason :	
	As the bulbs B, D and E are connected in series with fused bulb C, so no current flows through them and thus they will not glow. The bulb A remains unaffected as it is connected in parallel combination.	1

33. (a) State Ohm's law. Write formula for the equivalent resistance R_p of the parallel combination of three resistors of values R_1 , R_2 and R_3 .

(b) Find the resistance of the following network of resistors :

3

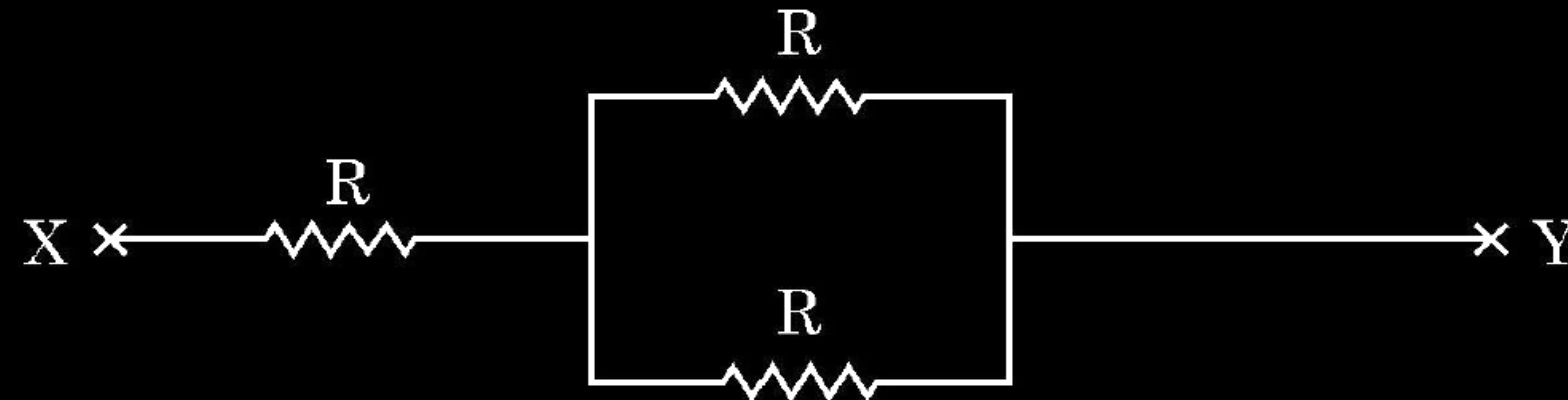


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33. (a) State Ohm's law. Write formula for the equivalent resistance R_p of the parallel combination of three resistors of values R_1 , R_2 and R_3 .

(b) Find the resistance of the following network of resistors :

3



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(a) Ohm's Law – The potential difference, V , across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, provided its temperature remains the same.

Formula :-
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(b)
$$R + \frac{R}{2} = \frac{3R}{2}$$

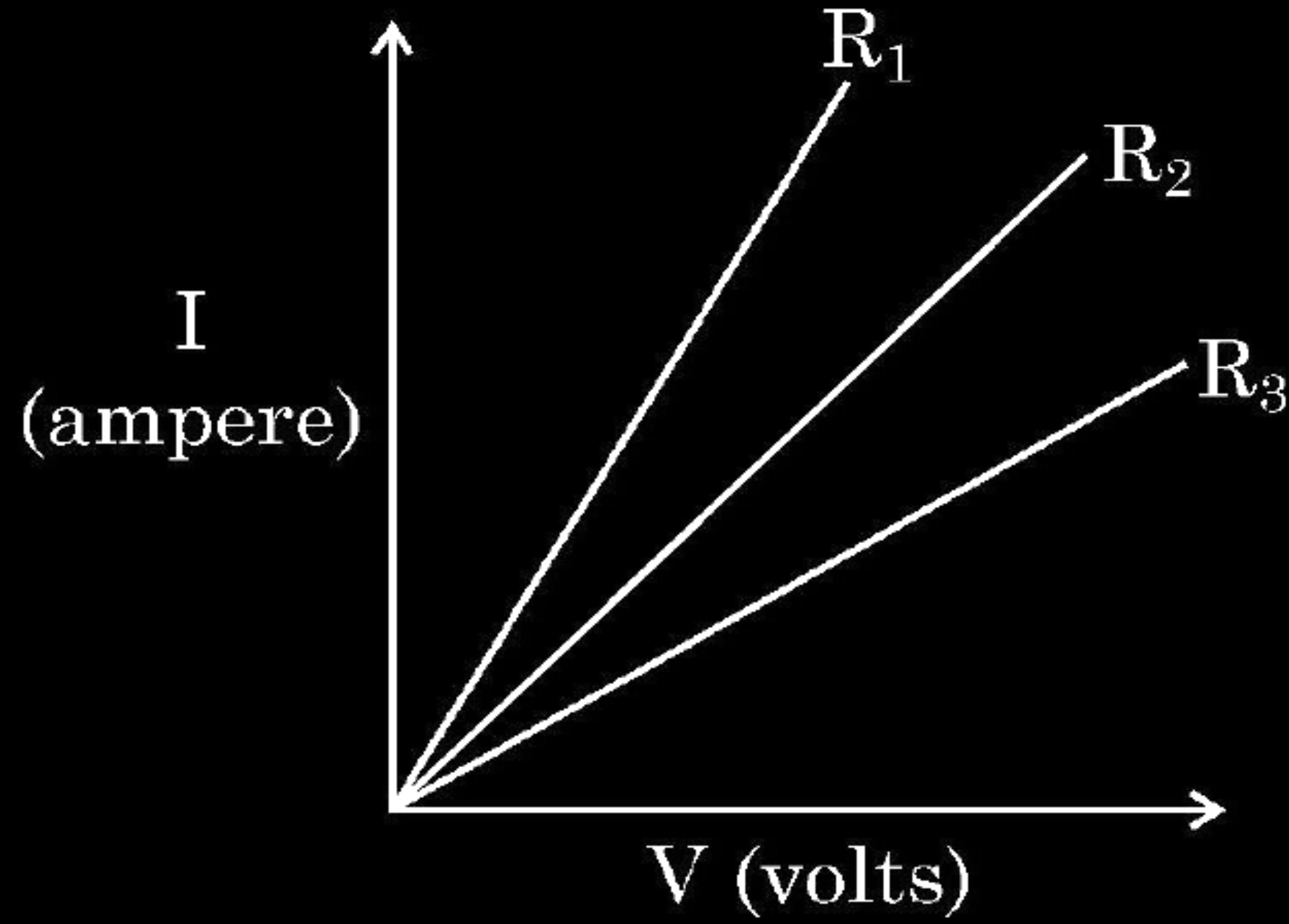
1

1

1

13. Study the I-V graph for three resistors of resistances R_1 , R_2 and R_3 and select the correct statement from the following :

1



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(A) $R_1 = R_2 = R_3$

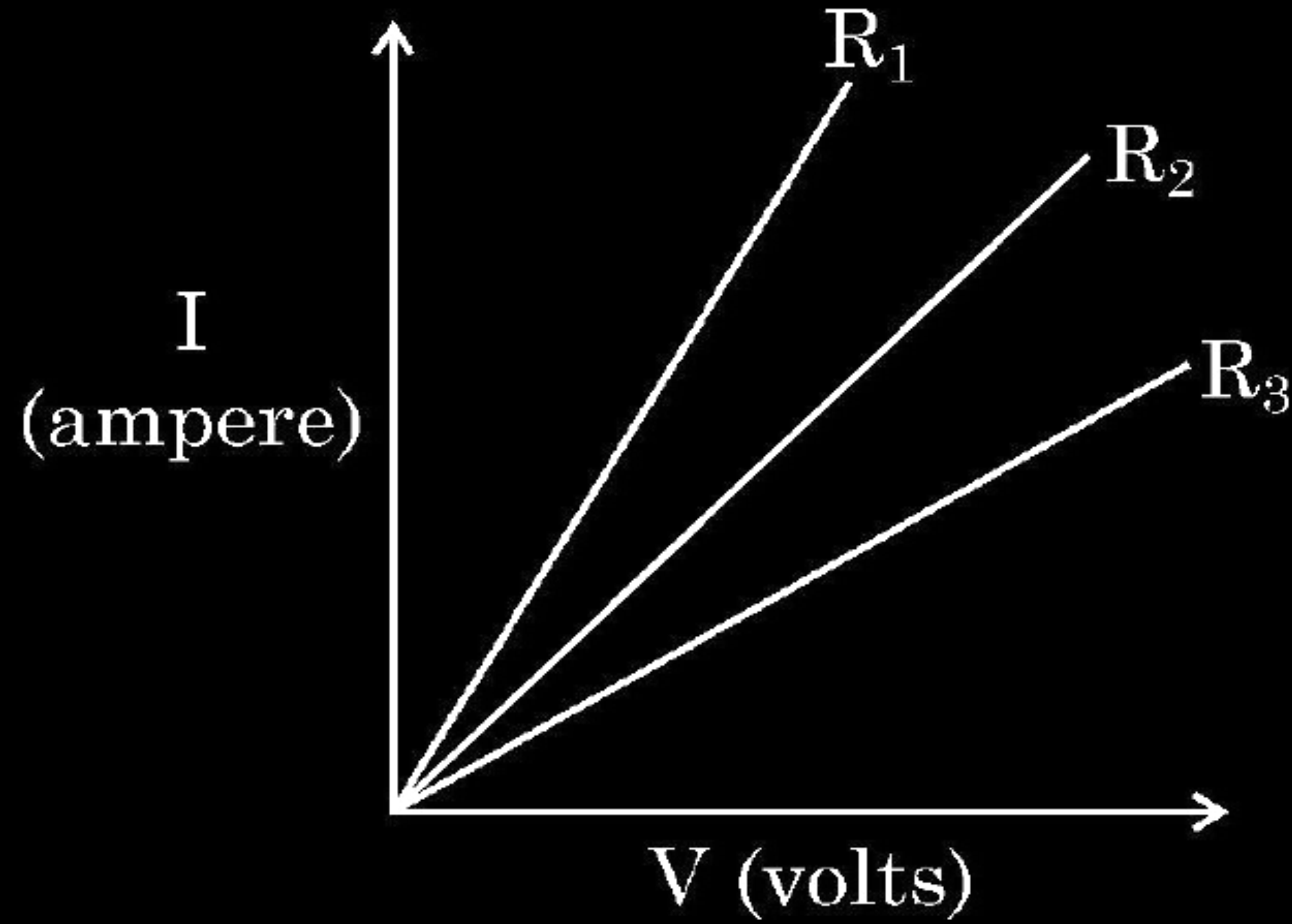
(B) $R_1 > R_2 > R_3$

(C) $R_3 > R_2 > R_1$

(D) $R_2 > R_3 > R_1$

13. Study the I-V graph for three resistors of resistances R_1 , R_2 and R_3 and select the correct statement from the following :

1



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- (A) $R_1 = R_2 = R_3$ (B) $R_1 > R_2 > R_3$
(C) $R_3 > R_2 > R_1$ (D) $R_2 > R_3 > R_1$

(C) $R_3 > R_2 > R_1$

14. The maximum resistance of a network of five identical resistors of $\frac{1}{5} \Omega$

each can be –

(A) 1Ω

(B) 0.5Ω

(C) 0.25Ω

(D) 0.1Ω

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14. The maximum resistance of a network of five identical resistors of $\frac{1}{5} \Omega$ each can be –

(A) 1Ω

(B) 0.5Ω

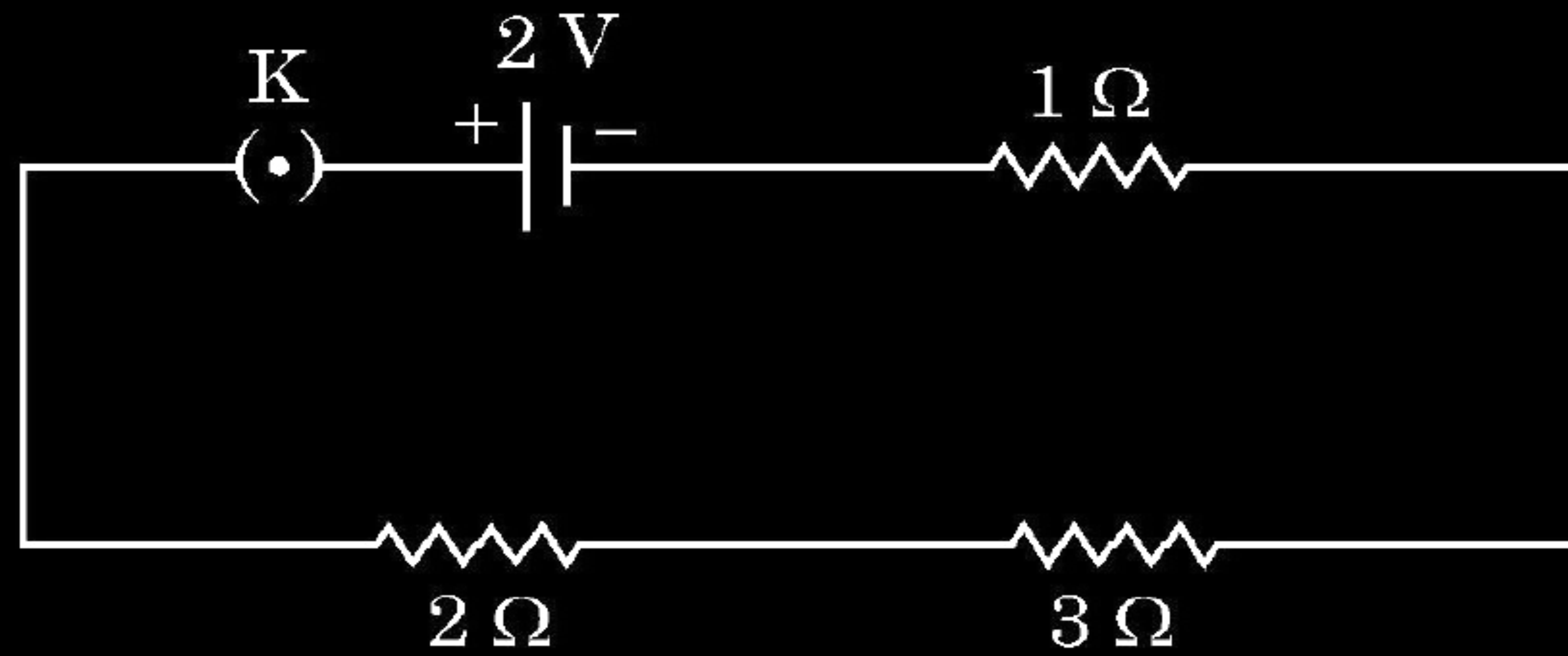
(C) 0.25Ω

(D) 0.1Ω

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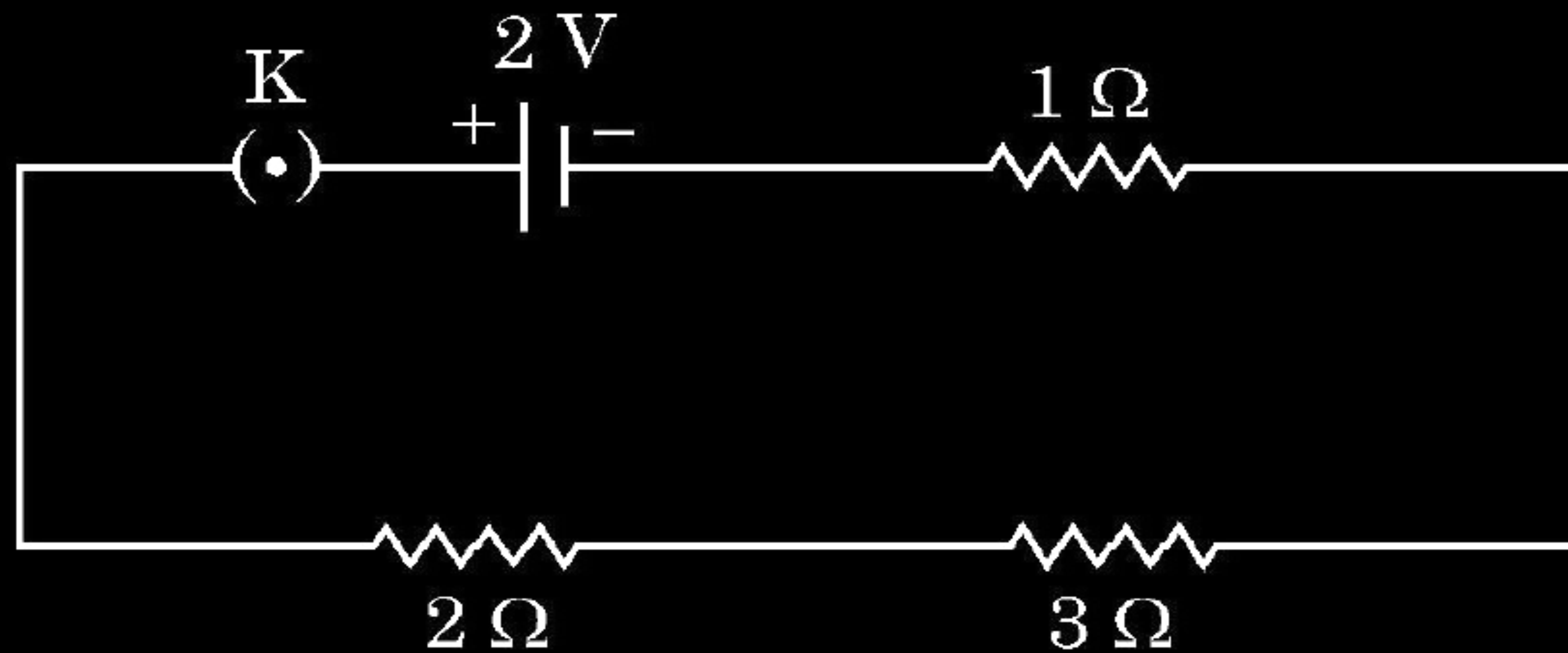
(A) / 1Ω

24. Use Ohm's law to determine the potential difference across the $3\ \Omega$ resistor in the circuit shown in the following diagram when key is closed : **2**



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24. Use Ohm's law to determine the potential difference across the $3\ \Omega$ resistor in the circuit shown in the following diagram when key is closed : **2**



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$$R_s = R_1 + R_2 + R_3$$
$$= 1 + 2 + 3 = 6\ \Omega$$

$$I = \frac{V}{R}$$
$$= \frac{2V}{6\Omega} = \frac{1}{3} A$$

$$V = IR$$
$$= \frac{1}{3} A \times 3(\Omega) = 1\ V$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

38. When electric current flows in a purely resistive circuit electrical energy gets fully converted into heat energy. The amount of heat produced (H) in the circuit is found to be directly proportional to (i) the square of current (I^2) (ii) the resistance (R) of the conductor and (iii) the time (t) for which current flows. In other words $H = I^2Rt$. Electrical devices such as an electric fuse, electric heater, electric iron etc. are all based on this effect called heating effect of electric current. 4
- (a) List two properties of heating elements. 1
- (b) List two properties of electric fuse. 1
- (c) Name the principle on which an electric fuse works. Explain how a fuse wire is capable of saving electrical appliances from getting damaged due to accidentally produced high currents. 2

OR

- (c) The power of an electric heater is 1100 W. If the potential difference between the two terminals of the heater is 220 V, find the current flowing in the circuit. What will happen to an electric fuse of rating 5 A connected in this circuit ?

• Higher resistivity than pure metals • Do not oxidise (burn) at high temperature.	$\frac{1}{2}$, $\frac{1}{2}$
(c) • Higher resistivity than pure metals • Low melting point.	$\frac{1}{2}$, $\frac{1}{2}$
(c) • Heating effect of electric current	1
• When high current flows in the circuit accidentally, the fuse wire melts and breaks the circuit .	1
OR	
(c) $P = 1100\text{ W}$; $V = 220\text{ V}$, $I = ?$	
$P = VI$	$\frac{1}{2}$
$I = \frac{P}{V} = \frac{1100\text{ W}}{220\text{ V}} = 5\text{A}$	$\frac{1}{2}$
• No effect on the fuse of 5A rating.	1

14. S.I. unit of electrical resistivity is

1

(A) ohm per metre³

(B) ohm per metre²

(C) ohm . metre

(D) ohm . metre³

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14. S.I. unit of electrical resistivity is

1

(A) ohm per metre³

(B) ohm per metre²

(C) ohm . metre

(D) ohm . metre³

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(C) /ohm.metre

15. The minimum resistance which can be made using five resistors each of resistance $10\ \Omega$ is :

1

(A) $\frac{1}{50}\ \Omega$

(B) $\frac{1}{5}\ \Omega$

(C) $2\ \Omega$

(D) $1\ \Omega$

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15. The minimum resistance which can be made using five resistors each of resistance $10\ \Omega$ is :

1

(A) $\frac{1}{50}\ \Omega$

(B) $\frac{1}{5}\ \Omega$

(C) $2\ \Omega$

(D) $1\ \Omega$

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(C) / $2\ \Omega$

- 34. (A) (i)** State whether the currents and potential difference in all the bulbs will be same or different when in a circuit three bulbs of :
- (a) same wattage are connected in series.
 - (b) same wattage are connected in parallel.
 - (c) different wattage are connected in series.
 - (d) different wattage are connected in parallel.
- (ii)** Two identical resistors of $24\ \Omega$ each are connected to a battery of 6 V. Calculate the ratio of the power consumed by the resulting combinations with (a) minimum resistance and (b) maximum resistance.

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- (A)(i)(a) same current and same potential difference.
 (b) same current and same potential difference
 (c) same current but different potential difference
 (d) different current but same potential difference.

$\frac{1}{2} \times 4$

- (ii) (a) Minimum resistance – When resistors are in parallel

$$\frac{1}{R} = \frac{1}{24} + \frac{1}{24}$$

$\frac{1}{2}$

$$\therefore R_P = 12 \Omega$$

$\frac{1}{2}$

$$\begin{aligned} \text{Power consumed } P_1 &= \frac{V^2}{R_P} \\ &= \frac{6V \times 6V}{12 \Omega} \\ &= 3W \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

- (b)

$$\therefore R_S = 24 \Omega + 24 \Omega = 48 \Omega$$

$$\begin{aligned} \text{Power consumed } P_2 &= \frac{V^2}{R_S} \\ &= \frac{6V \times 6V}{48 \Omega} \\ &= \frac{3}{4}W \end{aligned}$$

$\frac{1}{2}$

$$\therefore \frac{P_1}{P_2} = \frac{3}{\frac{3}{4}} = \frac{4}{1} \quad \text{from } P_1 \text{ and } P_2$$

$$\Rightarrow P_p : P_s = 4 : 1$$

$\frac{1}{2}$

(B) Draw a schematic diagram of a circuit consisting of a battery of six 2V cells, a $6\ \Omega$ resistor, a $12\ \Omega$ resistor and a $18\ \Omega$ resistor and a plug key all connected in series. Calculate the following (when key is closed) :

5

- (i) Electric current flowing in the circuit.
- (ii) Potential difference across $18\ \Omega$ resistor.
- (iii) Electric power consumed in $18\ \Omega$ resistor.

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(b)

$$\therefore R_s = 24 \Omega + 24 \Omega = 48 \Omega$$

$$\begin{aligned} \text{Power consumed } P_2 &= \frac{V^2}{R_s} \\ &= \frac{6 V \times 6 V}{48 \Omega} \\ &= \frac{3}{4} W \end{aligned}$$

 $\frac{1}{2}$

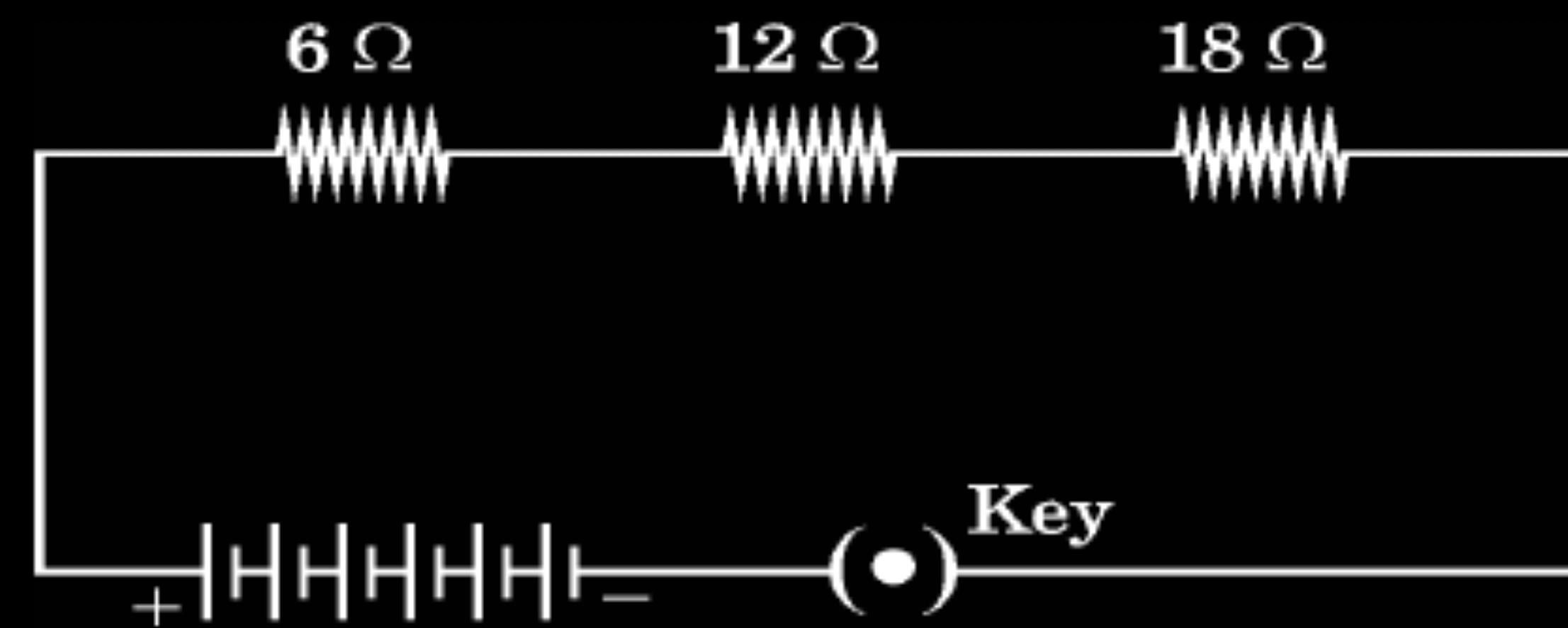
$$\therefore \frac{P_1}{P_2} = \frac{\frac{3}{4}}{\frac{3}{4}} = \frac{4}{1}$$

from P_1 and P_2

$$\Rightarrow P_p : P_s = 4 : 1$$

 $\frac{1}{2}$ **OR**

(B)



12 V (Six cells of 2 V each)

2

$$(i) \text{Current} = \frac{V}{R} = \frac{12}{(6+12+18) \Omega} = \frac{1}{3} A$$

1

$$(ii) \text{Potential difference across } 18 \Omega \text{ resistor} = I \times R = \frac{1}{3} A \times 18 \Omega = 6 V$$

1

$$(iii) \text{Power consumed in } 18 \Omega \text{ resistor} = V \times I = 6 V \times \frac{1}{3} A = 2 W$$

1

22. (A) Show how you would connect three resistors each of resistance $6\ \Omega$, so that the combination has a resistance of $9\ \Omega$. Also justify your answer.

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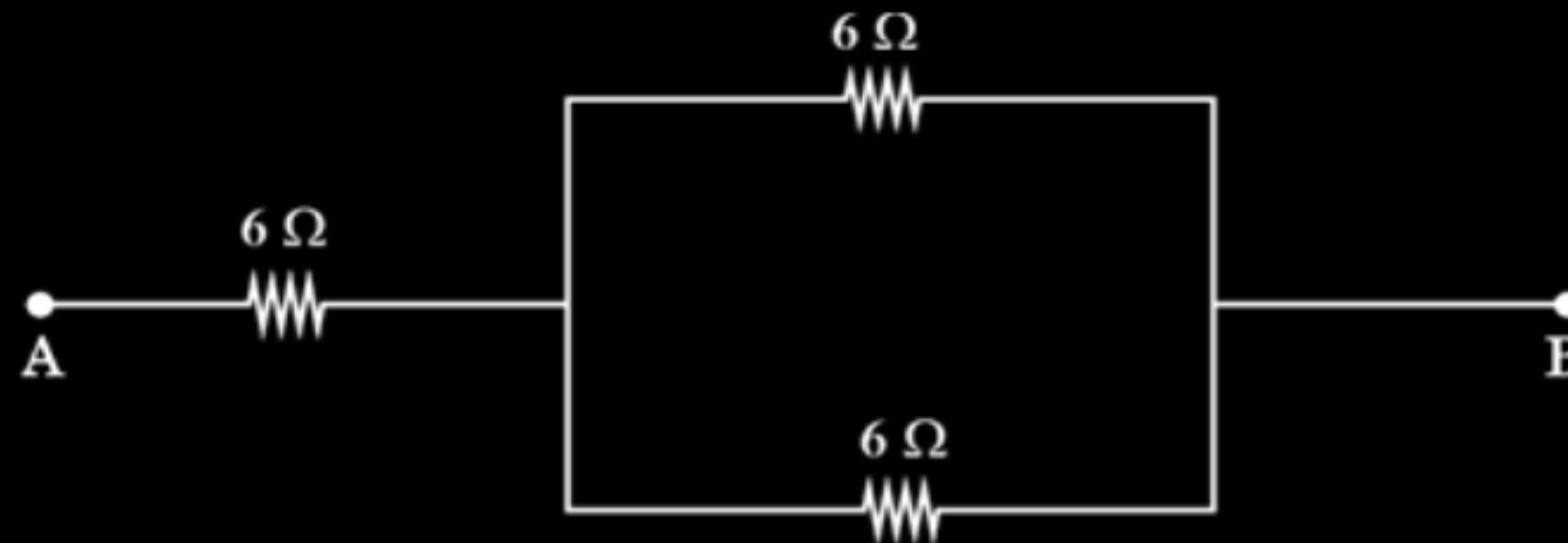
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22. (A) Show how you would connect three resistors each of resistance $6\ \Omega$, so that the combination has a resistance of $9\ \Omega$. Also justify your answer.

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2

(A) When two $6\ \Omega$ resistances are connected in parallel and the third resistance of $6\ \Omega$ is connected in series combinations to this, then equivalent resistance will be $9\ \Omega$ /



[Award marks for writing the statement or drawing the diagram]

$$\frac{1}{R_P} = \frac{1}{6\ \Omega} + \frac{1}{6\ \Omega}$$

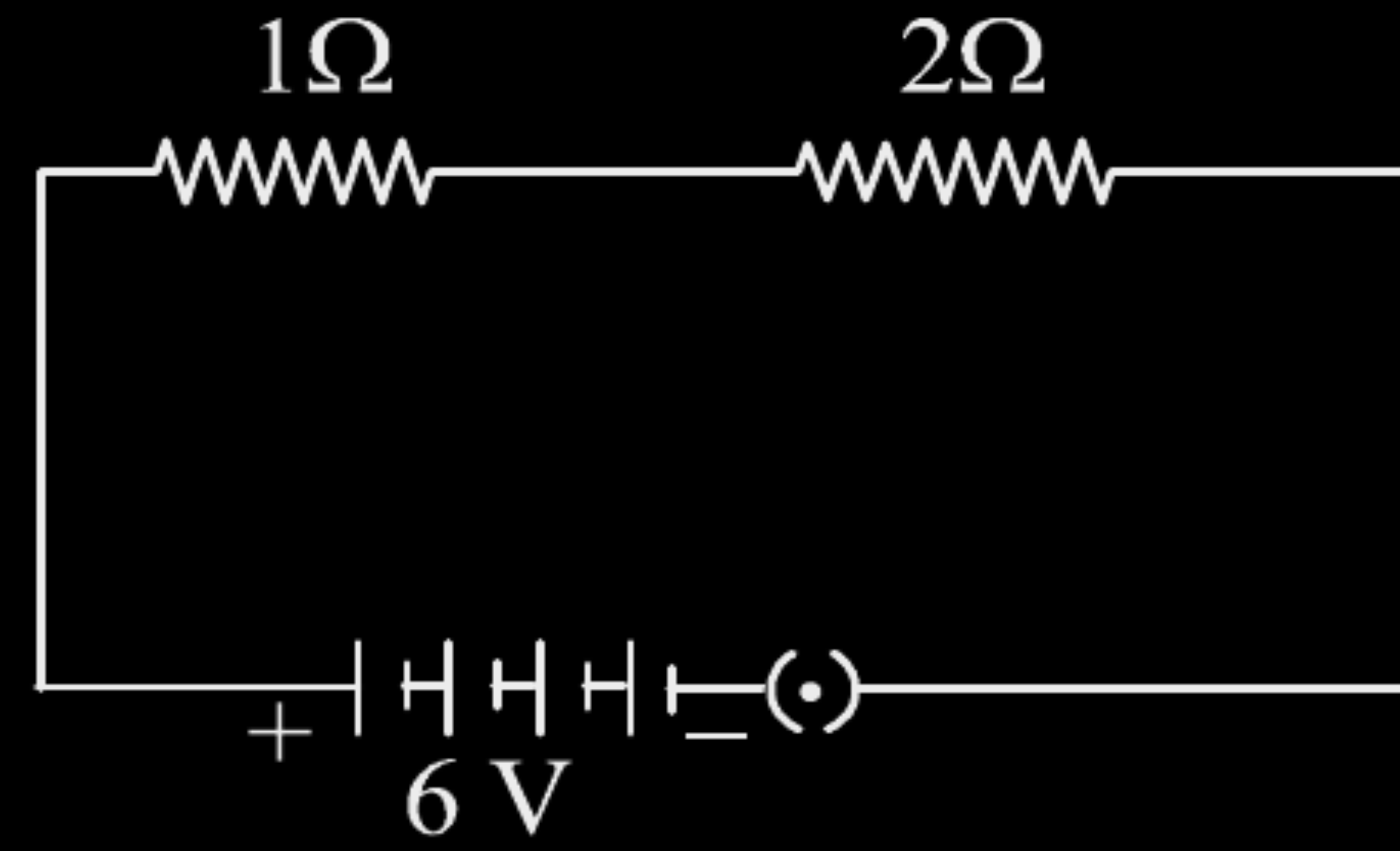
$$\therefore R_P = 3\ \Omega$$
$$R_S = 6 + 3 = 9\ \Omega$$

1

1

(B) In the given circuit calculate the power consumed in watts in the resistor of $2\ \Omega$:

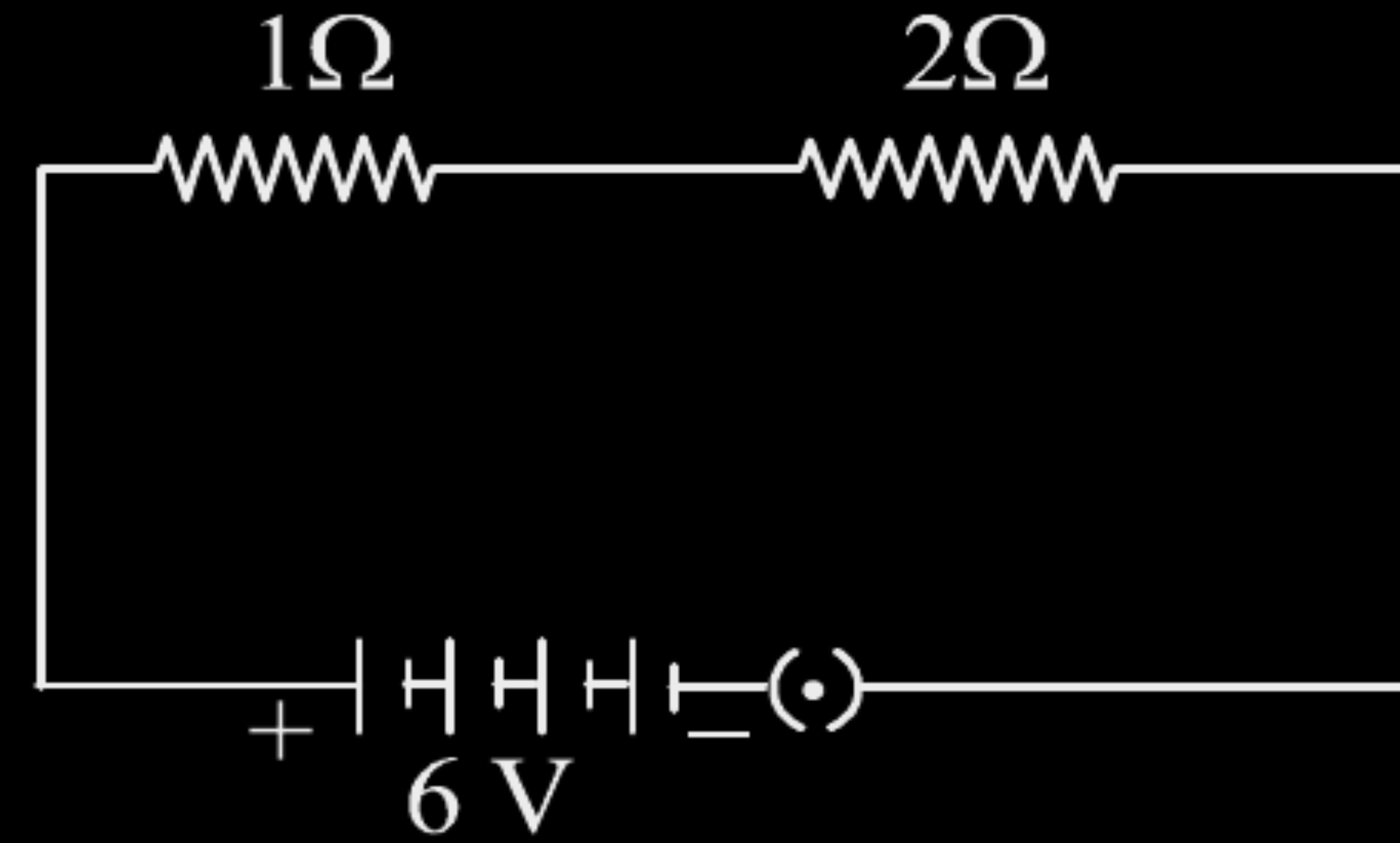
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(B) In the given circuit calculate the power consumed in watts in the resistor of $2\ \Omega$:

2



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(B) Equivalent resistance = $R_1 + R_2 = 1\ \Omega + 2\ \Omega = 3\ \Omega$

$\frac{1}{2}$

$$I = \frac{V}{R}$$

$$= \frac{6\text{ V}}{1\ \Omega + 2\ \Omega} = \frac{6\text{ V}}{3\ \Omega} = 2\text{ A}$$

$\frac{1}{2}$

Electric power, $P = I^2 R$

$\frac{1}{2}$

$$= (2\text{ A})^2 \times 2\ \Omega = 4 \times 2\text{ W} = 8\text{ W}$$

$\frac{1}{2}$

35. (A) (i) Define electric power. Express it in terms of potential difference (V) and resistance (R). **5**

(ii) An electric oven is designed to work on the mains voltage of 220 V. This oven consumes 11 units of electrical energy in 5 hours. Calculate :

- (a) power rating of the oven
- (b) current drawn by the oven
- (c) resistance of the oven when it is red hot

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(A) (i)

- Electric power : Rate at which electrical energy is dissipated or consumed / Rate of supplying energy to maintain the flow of current through a circuit.

1

- $P = \frac{V^2}{R}$

1

(ii) (a) (1 unit = 1kWh)

$$\text{Power, } P = \frac{\text{Electrical energy consumed}}{\text{Time}}$$

$\frac{1}{2}$

$$= \frac{11\text{kWh}}{5\text{h}} = 2.2\text{kW or } 2200 \text{ W}$$

$\frac{1}{2}$

(b) $I = \frac{P}{V}$

$\frac{1}{2}$

$$= \frac{2200}{220} = 10\text{A}$$

$\frac{1}{2}$

(c) $R = \frac{V^2}{P}$

$\frac{1}{2}$

$$= \frac{(220)^2}{2200} = 22 \Omega$$

$\frac{1}{2}$

(Alternate formula can be used)

- (B)** (i) Write the relation between resistance R and electrical resistivity ρ of the material of a conductor in the shape of cylinder of length l and area of cross-section A . Hence derive the SI unit of electrical resistivity.
- (ii) The resistance of a metal wire of length 3 m is $60\ \Omega$. If the area of cross-section of the wire is $4 \times 10^{-7}\ \text{m}^2$, calculate the electrical resistivity of the wire.
- (iii) State how would electrical resistivity be affected if the wire (of part 'ii') is stretched so that its length is doubled. Justify your answer.

(B)

(i) $R = \rho \frac{l}{A}$

1

$$\rho = \frac{R \times A}{l}$$
$$= \text{Ohm} \times \frac{(\text{metre})^2}{\text{metre}}$$

$$= \text{ohm meter} / \Omega\text{m}$$

1

(ii) Here $l = 3 \text{ m}$, $A = 4 \times 10^{-7} \text{ m}^2$, $R = 60 \Omega$

$$\rho = \frac{R \times A}{l}$$
$$= \frac{60 \times 4 \times 10^{-7}}{3}$$
$$= 80 \times 10^{-7} \Omega\text{m}$$

1

(iii)

- Resistivity will not change.
- because Resistivity does not depend on the dimension of the conductor / It only depends on the nature of the material.

1

1

11. The resistance of a wire does ***not*** depend on its :

- (a) Length
- (b) Area of cross-section
- (c) Shape
- (d) Material

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11. The resistance of a wire does ***not*** depend on its :

- (a) Length
- (b) Area of cross-section
- (c) Shape
- (d) Material

CBSE 2023 C

14. How much work is done in moving a charge of 3 C across two points having a potential difference of 12 V ?

- (a) 4 J
- (b) 36 J
- (c) 54 J
- (d) 216 J

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14. How much work is done in moving a charge of 3 C across two points having a potential difference of 12 V ?

- (a) 4 J
- (b) 36 J
- (c) 54 J
- (d) 216 J

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(b)

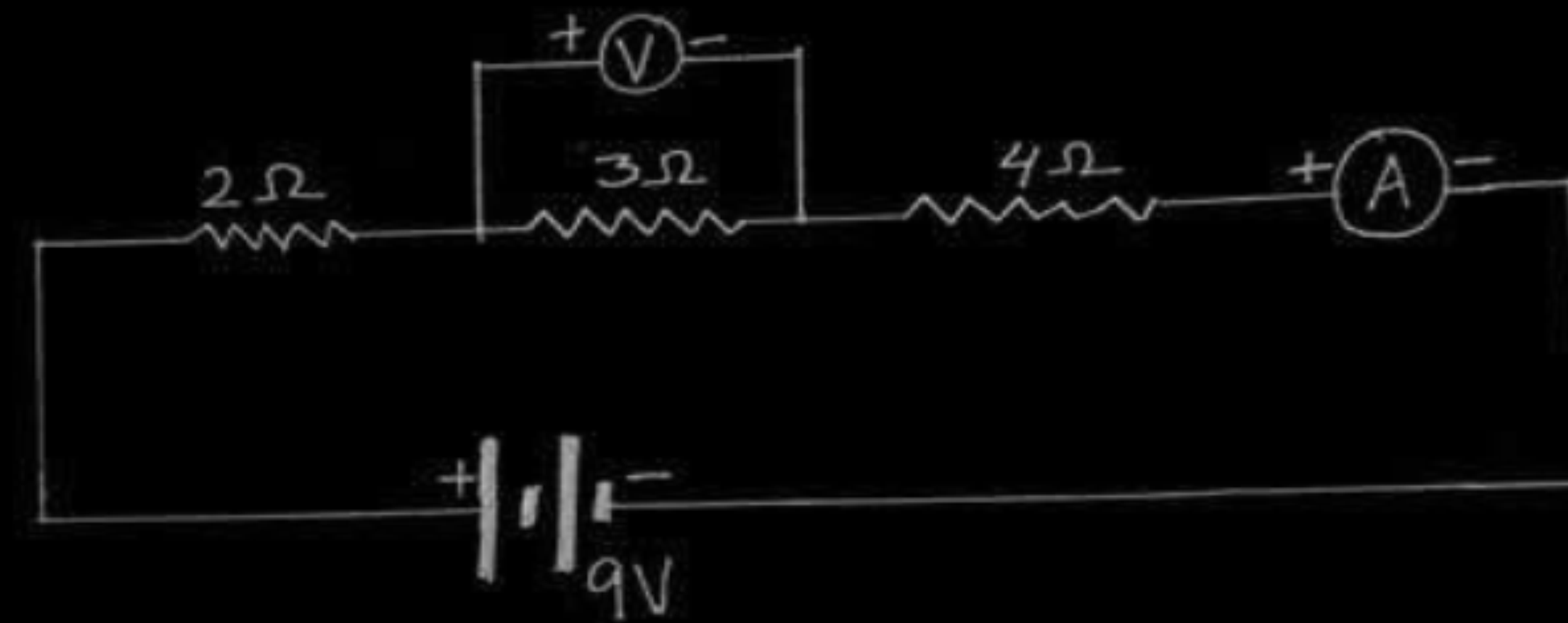
- 36.** (a) You have three resistors R_1 , R_2 and R_3 . Write the formula for the equivalent resistance of the combination, when these resistors are joined (i) in series and (ii) in parallel.
- (b) Three resistors of $2\ \Omega$, $3\ \Omega$ and $4\ \Omega$ are connected in series and the combination is connected to a battery of $9\ \text{V}$. An ammeter and a voltmeter are also connected in the circuit. Draw a circuit diagram to show the connection of the above said devices in a correct order. Determine (i) current shown by the ammeter and (ii) potential difference across the $3\ \Omega$ resistor.

(a) (i) $R_s = R_1 + R_2 + R_3$

(ii) $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$

(b)

•



(i) $R_s = 2\ \Omega + 3\ \Omega + 4\ \Omega = 9\ \Omega$

$$V = 9V$$

$$I = \frac{v}{R_s}$$

$$= \frac{9v}{9\Omega} = 1\ A$$

Ammeter reading = 1.0 A

(ii) Potential Difference across 3 Ω resistor

$$V = I R$$

$$= 1.0\ A \times 3.0\ \Omega$$

$$= 3\ V$$

9. A material used for making heating elements of electrical heating devices should have :

- (a) High resistivity and high melting point.
- (b) High resistivity and low melting point.
- (c) Low resistivity and high melting point.
- (d) Low resistivity and low melting point.

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9. A material used for making heating elements of electrical heating devices should have :

- (a) High resistivity and high melting point.
- (b) High resistivity and low melting point.
- (c) Low resistivity and high melting point.
- (d) Low resistivity and low melting point.

CBSE 2023 C

- 36.** (a) Calculate the resistance of an aluminium wire of length 1 m and area of cross-section 2 mm^2 . Resistivity of aluminium is $2.63 \times 10^{-8} \Omega\text{m}$.
- (b) From the values given below, plot a graph of I versus V. Show that the data is in conformity of the Ohm's law.

5

Current – I (ampere)	0.1	0.2	0.3	0.4
Potential Difference – V (volt)	1.2	2.4	3.6	4.8

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$$(a) \ l = 1 \text{ m}$$

$$A = 2\text{mm}^2 = 2 \times (10^{-3}\text{m})^2 = 2 \times 10^{-6} \text{ m}^2$$

$$\rho = 2.63 \times 10^{-8} \ \Omega\text{m}$$

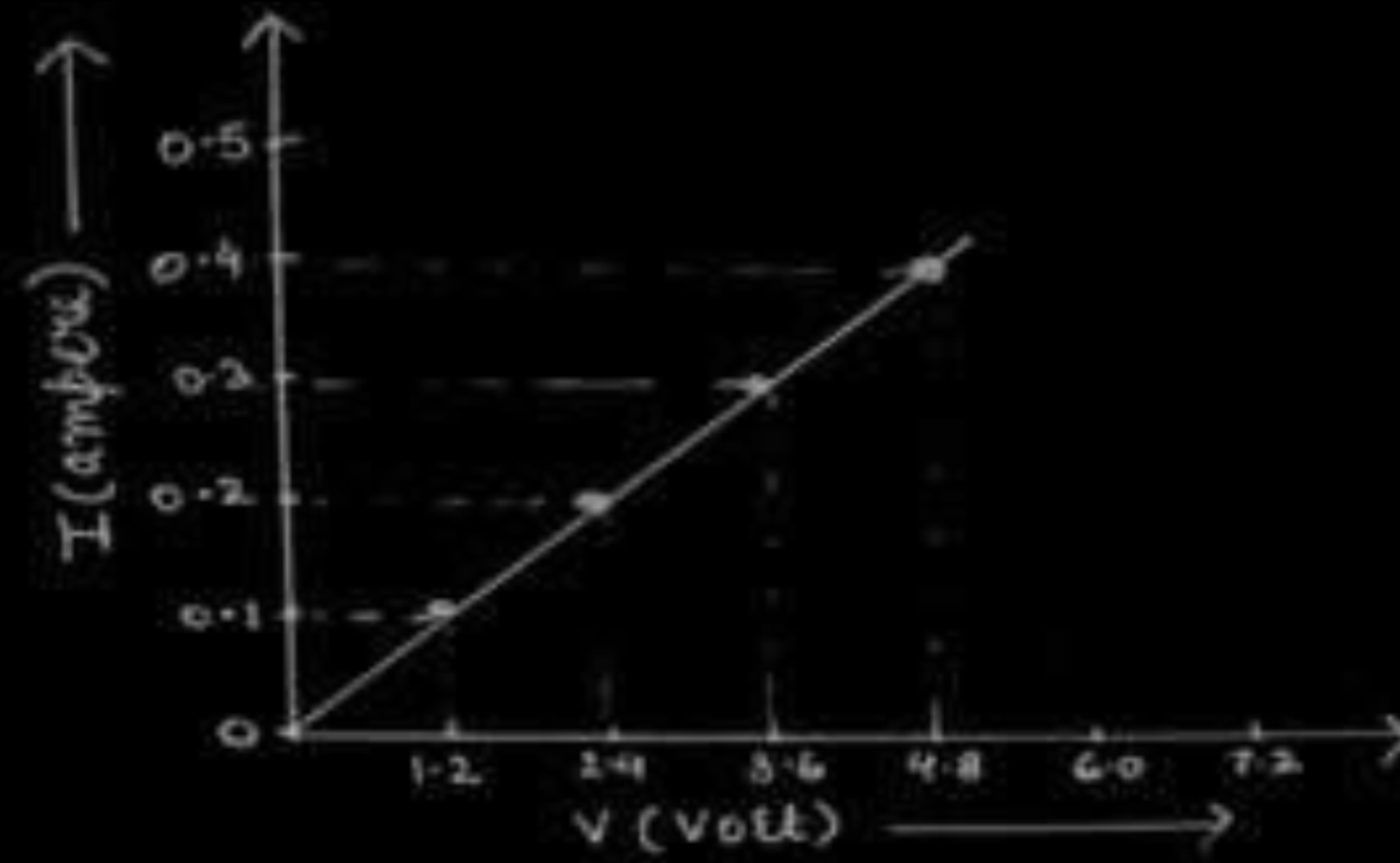
$$R = \rho \frac{l}{A}$$

$$R = (2.63 \times 10^{-8} \ \Omega\text{m}) \times \frac{1\text{m}}{2 \times 10^{-6} \text{ m}^2}$$

$$= \frac{2.63}{2} \times 10^{-2}$$

$$= 1.315 \times 10^{-2} \Omega$$

(



Since I – V graph is a straight line passing through the origin it shows that $I \propto V$ which is in accordance with the Ohm's law.

- 36.** (a) The commercial unit of electric energy is commonly known as 'unit'. Write its actual name and establish the relationship between this unit and the SI unit of energy.
- (b) The current rating of a circuit is 1·0 A. How many LED lamps of power rating 8 W; 220 V can safely be used simultaneously in this circuit ?

CBSE 2023 C

- 36.** (a) The commercial unit of electric energy is commonly known as 'unit'. Write its actual name and establish the relationship between this unit and the SI unit of energy.
- (b) The current rating of a circuit is 1.0 A. How many LED lamps of power rating 8 W; 220 V can safely be used simultaneously in this circuit ?

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5

(a)

- kilowatt hour or kWh
- $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ seconds}$
 $= 3.6 \times 10^6 \text{ watt seconds}$
 $1 \text{ kWh} = 3.6 \times 10^6 \text{ Joule}$

1

$\frac{1}{2}$

1

(b) $P = 8 \text{ W}$; $V = 220 \text{ V}$; Current rating = 1.0 A

$$\text{Current through each lamp} = \frac{8 \text{ W}}{220 \text{ V}}$$

1

$$\begin{aligned} \text{Number of lamp} &= \frac{\text{current rating}}{\text{current through each lamp}} \\ &= \frac{1.0 \text{ A}}{\frac{8 \text{ W}}{220 \text{ V}}} = \frac{220}{8} = 27.75 \end{aligned}$$

$\frac{1}{2}$

So, 27 lamps can safely be used in the circuit.

1

13. Potential difference can also be represented as :

(a) $\frac{\text{Work done}}{\text{Charge}}$

(b) $\text{Work done} \times \text{Charge}$

(c) $\text{Work done} \times \text{Charge} \times \text{Time}$

(d) $\frac{\text{Work done} \times \text{Charge}}{\text{Time}}$

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13. Potential difference can also be represented as :

(a) $\frac{\text{Work done}}{\text{Charge}}$

(b) $\text{Work done} \times \text{Charge}$

(c) $\text{Work done} \times \text{Charge} \times \text{Time}$

(d) $\frac{\text{Work done} \times \text{Charge}}{\text{Time}}$

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(a)

14. Two LED bulbs of 12 W and 6 W are connected in series. If the current flowing through 6 W bulbs is 0.005 A, the current that flows through 12 W bulb is :

(a) 0.005 A

(b) 0.01 A

(c) 0.0025 A

(d) 0.02 A

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14. Two LED bulbs of 12 W and 6 W are connected in series. If the current flowing through 6 W bulbs is 0.005 A, the current that flows through 12 W bulb is :

(a) 0.005 A

(b) 0.01 A

(c) 0.0025 A

(d) 0.02 A

CBSE 2023 C

- 32.** (a) State Ohm's law. List two factors on which the resistance of a cylindrical conductor at a given temperature depends. A wire was drawn into a wire of double its length by melting it. If the resistance of the old wire was R , what is the resistance of the new wire ?

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32. (a) State Ohm's law. List two factors on which the resistance of a cylindrical conductor at a given temperature depends. A wire was drawn into a wire of double its length by melting it. If the resistance of the old wire was R , what is the resistance of the new wire ? **CBSE 2023 C**

3

- (b) (i) Write the formula for finding the value of the equivalent resistance of three resistors of resistance R_1 , R_2 and R_3 , when the resistors are connected in (i) series, and (ii) parallel.
- (ii) List two advantages of connecting electrical devices in parallel with the source of electricity instead of connecting them in series.

- (b) (i) Write the formula for finding the value of the equivalent resistance of three resistors of resistance R_1 , R_2 and R_3 , when the resistors are connected in (i) series, and (ii) parallel.
- (ii) List two advantages of connecting electrical devices in parallel with the source of electricity instead of connecting them in series.

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3

(b) (i) Formula for Equivalent resistance in series : $R_s = R_1 + R_2 + R_3$ Equivalent resistance in parallel : $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$	1 1
(ii) (1) Each device can draw current as per need (2) Separate 'ON/OFF' switch for every device can be put (3) Total resistance of the combination decreases hence larger current can be drawn from the source. (4) If one device fails, other devices keep on working (any two)	2 x ½

13. SI unit of electrical resistivity is :

(a) ohm metre

(b) ohm per metre

(c) ohm / metre²

(d) ohm / metre³

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13. SI unit of electrical resistivity is :

- | | |
|------------------------------|------------------------------|
| (a) ohm metre | (b) ohm per metre |
| (c) ohm / metre ² | (d) ohm / metre ³ |

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—
(a)
—

14. You have a thin resistance wire having a thick plastic insulation. A decrease in the resistance of this wire would be produced if somehow we increase the :

- (a) thickness of the plastic insulation
- (b) temperature of the wire
- (c) length of the wire
- (d) diameter of the wire

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14. You have a thin resistance wire having a thick plastic insulation. A decrease in the resistance of this wire would be produced if somehow we increase the :

- (a) thickness of the plastic insulation
- (b) temperature of the wire
- (c) length of the wire
- (d) diameter of the wire

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(d)

- 36.** (a) Derive an expression for the amount of heat produced when a current flows through a resistor for a given time. Use this expression to state Joule's law of heating.
- (b) 200 joules of heat energy is produced each second in a resistor of $8\ \Omega$. Find :
- (i) The current flowing through the resistor, and
 - (ii) The potential difference across the ends of the resistor.

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(a)		
• Consider a current I flowing through a resistor of resistance R having a potential difference V across its ends.		
Let t be the time during which charge Q flows across		
∴ Work done in moving charge W = VQ		
∴ Power input to the circuit by the source $P = \frac{VQ}{t} = VI$		
∴ Energy supplied by the source to the circuit in time t = VIt	1/2	
$= IR \cdot It$	1/2	
This energy is dissipated in the form of heat if the circuit is purely resistive.		
$\therefore H = I^2Rt$	1/2	
This is the required expression for Joule's law of heating which implies that the heat produced H in a resistor is		
(i) directly proportional to square of current (I ²) for given R		1/2
(ii) directly proportional to resistance (R) for given I		1/2
(iii) directly proportional to time t for which the current flows through the resistor		1/2
(b) H = 200 J, t = 1 s , R = 8 Ω		
(i) Using the formula $H = I^2Rt$, we have $I^2 = \frac{H}{Rt}$		
$= \frac{200 \text{ J}}{8 \times 1 \Omega \text{ t}} =$		
$= 25$		
$\Rightarrow I = 5\text{A}$		1
(ii) $V = IR$		1/2
$= 5\text{A} \times 8 \Omega = 40\text{V}$		1/2

21. Let the resistance of an electrical device remain constant, while the potential difference across its two ends decreases to one fourth of its initial value. What change will occur in the current through it ? State the law which helps us in solving the above stated question.

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2

21. Let the resistance of an electrical device remain constant, while the potential difference across its two ends decreases to one fourth of its initial value. What change will occur in the current through it ? State the law which helps us in solving the above stated question.

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2

- Current becomes one fourth of its original value.
- According to the Ohm's law – potential difference is directly proportional to the current flowing through the conductor provided temperature remains constant. /

$$V \propto I \text{ (Temperature remaining constant for a given conductor)}$$

1

1

34. (a) (i) What is meant by resistance of a conductor ? Define its SI unit.
- (ii) List two factors on which the resistance of a rectangular conductor depends.
- (iii) How will the resistance of a wire be affected if its
- (1) length is doubled, and
 - (2) radius is also doubled ?

Give justification for your answer.

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(a) (i) The property of conductor to resist the flow of charges through it.	1
If Potential difference across the two ends of a conductor is 1V and the current through it is 1A, then resistance 'R' of the conductor is 1Ω.	1
Alternate answer $1\Omega = \frac{1 \text{ volt}}{1 \text{ ampere}} \quad / \quad \frac{1 \text{ V}}{1\text{A}}$	
(ii) • Length of the conductor • Area of cross-section of the conductor • Nature of the material • Temperature (any two)	1/2 , 1/2
(iii) (1) The resistance will become one half of its original value.	
$R = \rho \frac{l}{A} = \rho \frac{l}{\pi r^2}$	1/2
$R' = \frac{\rho \cdot 2L}{\pi (2r)^2}$	1/2
(2) $R' = \frac{\rho \cdot l}{\pi (2r)^2} \cdot \frac{2}{4} = \frac{R}{2}$	1/2
Resistance will reduce to one half.	1/2

(b) In an electric circuit three bulbs of 100 W each are connected in series to a source. In another circuit set of three bulbs of the same wattage are connected in parallel to the same source. 5

- (i) Will the bulb in the two circuits glow with the same brightness ? Justify your answer.
- (ii) Now, let one bulb in both the circuits get fused. Will the rest of the bulbs continue to glow in each circuit ? Give reason for your answer.

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- (b) In an electric circuit three bulbs of 100 W each are connected in series to a source. In another circuit set of three bulbs of the same wattage are connected in parallel to the same source. 5
- (i) Will the bulb in the two circuits glow with the same brightness ? Justify your answer.
- (ii) Now, let one bulb in both the circuits get fused. Will the rest of the bulbs continue to glow in each circuit ? Give reason for your answer.

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(b) (i) No

- In series combination overall resistance will increase hence decreasing the current . Potential difference also divides. Therefore power consumption is less by each bulb and glows with less brightness.
- In parallel combination each bulb will get the required potential difference hence the required current and will glow with its normal brightness.

(ii) None of the bulb glows in series combination as the circuit gets broken and current stops flowing.

In parallel combination the other two bulbs will glow with same brightness as the same voltage is available to them.

1

1

1

1

1

24. Three resistors of $6\ \Omega$, $4\ \Omega$ and $4\ \Omega$ are connected together so that the total resistance is $8\ \Omega$. Draw a diagram to show this arrangement and give reason to justify your answer.

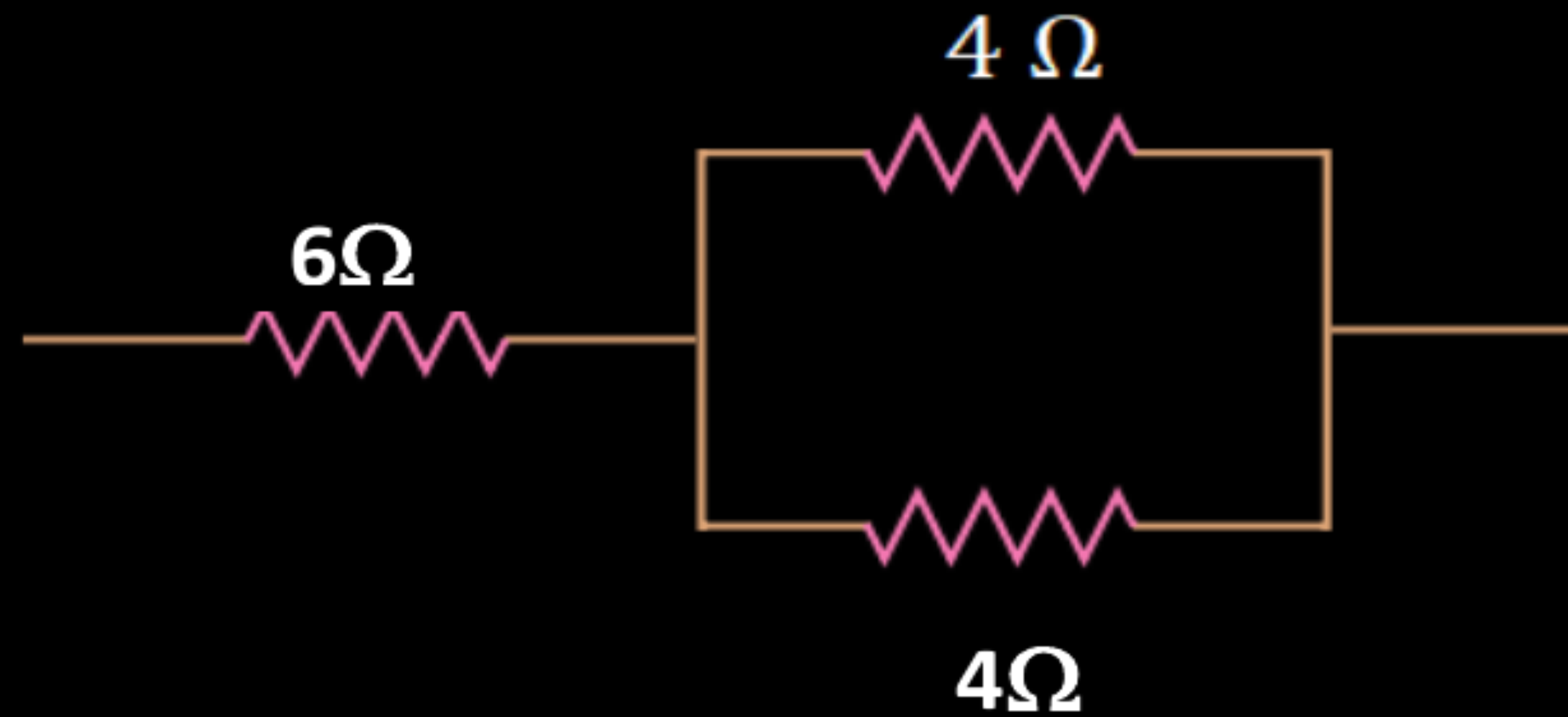
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2

24. Three resistors of $6\ \Omega$, $4\ \Omega$ and $4\ \Omega$ are connected together so that the total resistance is $8\ \Omega$. Draw a diagram to show this arrangement and give reason to justify your answer.

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2



$$\frac{1}{R_p} = \left(\frac{1}{4} + \frac{1}{4} \right) = \frac{2}{4} \Rightarrow R_p = 2\Omega$$

$$R = 6\Omega + 2\Omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

14. The expressions that relate (i) Q, I and t and (ii) Q, V and W respectively are (Here the symbols have their usual meanings) :

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(a) (i) $I = \frac{Q}{t}$ (ii) $W = \frac{V}{Q}$

(b) (i) $Q = I \times t$ (ii) $W = V \times Q$

(c) (i) $Q = \frac{I}{t}$ (ii) $V = \frac{W}{Q}$

(d) (i) $I = \frac{Q}{t}$ (ii) $Q = \frac{V}{W}$

14. The expressions that relate (i) Q, I and t and (ii) Q, V and W respectively are (Here the symbols have their usual meanings) :

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(a) (i) $I = \frac{Q}{t}$ (ii) $W = \frac{V}{Q}$

(b) (i) $Q = I \times t$ (ii) $W = V \times Q$

(c) (i) $Q = \frac{I}{t}$ (ii) $V = \frac{W}{Q}$

(d) (i) $I = \frac{Q}{t}$ (ii) $Q = \frac{V}{W}$

- 35.** (a) Define electric power and state its SI unit. The commercial unit of electrical energy is known as ‘unit’. Write the relation between this ‘unit’ and joule.
- (b) In a house, 2 bulbs of 50 W each are used for 6 hours daily and an electric geyser of 1 kW is used for 1 hour daily. Calculate the total energy consumed in a month of 30 days and its cost at the rate of ₹ 8.00 per kWh.

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(a) • The rate at which electrical energy is dissipated / consumed in a circuit.	1
• Watt or W	$\frac{1}{2}$
• 1 unit (1 kWh) = 3.6×10^6 joule (J)	$\frac{1}{2}$
(b) $E = P \times T$	$\frac{1}{2}$
Electrical energy, E_1 , consumed by 2 bulbs of 50 W each is	$\frac{1}{2}$
$E_1 = 50 \times 2 \times 6 = 600 \text{ Wh}$	
Electrical energy, E_2 , consumed by electric geyser of 1 kW is	
$E_2 = 1000 \times 1 = 1000 \text{ Wh}$	$\frac{1}{2}$
Total energy consumed in the month of 30 days	
$[E_1 + E_2] \times 30 = [600 + 1000] \times 30 \text{ Wh}$	
$= 1600 \times 30 = 48000 \text{ Wh}$	
$= 48 \text{ kWh}$	$\frac{1}{2}$
Cost of using these electrical devices at the rate of ₹ 8.00 per kwh is	
₹ $48 \times 8 = ₹ 384$	1

34. There are three resistors of $10\ \Omega$, $20\ \Omega$ and $30\ \Omega$ joined in parallel in a circuit. The potential difference across the electric circuit is 10 V .

- (a) Draw a circuit diagram for the above case.
- (b) Find the total resistance of the combination of resistors.
- (c) Calculate the electric current drawn from the same source.

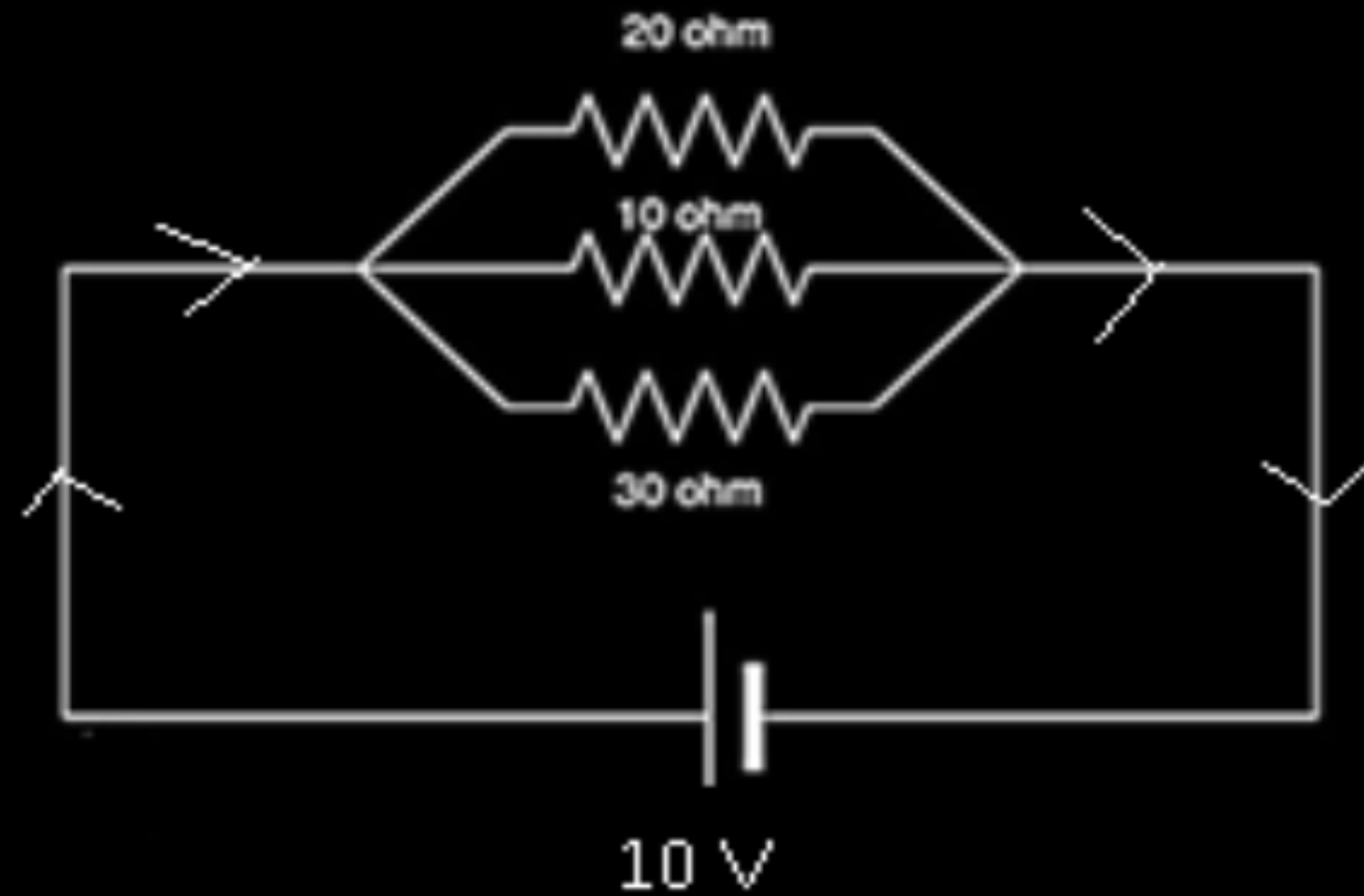
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- 34.** There are three resistors of $10\ \Omega$, $20\ \Omega$ and $30\ \Omega$ joined in parallel in a circuit. The potential difference across the electric circuit is 10 V .

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- (a) Draw a circuit diagram for the above case.
- (b) Find the total resistance of the combination of resistors.
- (c) Calculate the electric current drawn from the same source.

5



2

(b)
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$

$$= \frac{6+3+2}{60}$$

$\frac{1}{2}$

(c)
$$\therefore R = \frac{60}{11}\ \Omega \quad / 5.45\ \Omega$$

$$I = \frac{V}{R}$$

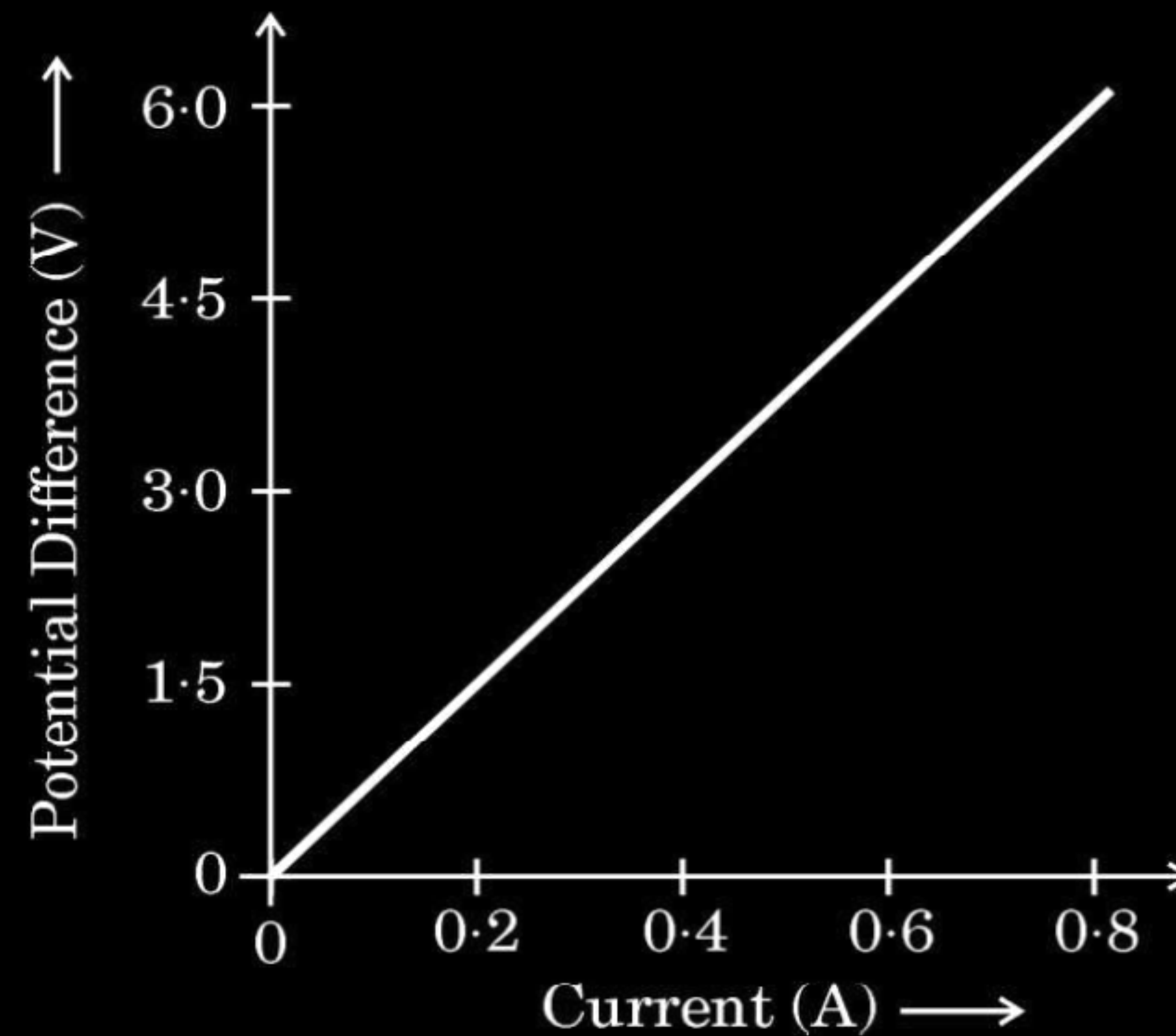
1

$\frac{1}{2}$

$$I = \frac{10\text{ V}}{\frac{60}{11}\ \Omega} = \frac{110}{60}\text{ A} = 1.83\text{ A}$$

1

36. (a) State Ohm's Law.
- (b) Name and define the physical quantity determined by the slope of $V - I$ curve given in the diagram. Use this graph to find the value of this physical quantity in SI units.



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- (c) Establish the relationship between 1 kWh and 1 joule.

(a) The potential difference V across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it provided its temperature remains the same. / $V \propto I$ or $V = IR$ (temperature remaining constant)	1
(b) Resistance Resistance is the property of a conductor to resist the flow of charges. $R = \text{slope} = \frac{y_2 - y_1}{x_2 - x_1} / \frac{V}{I} = \frac{6 - 0}{0.8 - 0}$ $= \frac{60}{8} = 7.5 \, \Omega$	1 1 $\frac{1}{2}$ $\frac{1}{2}$
(c) $1 \text{ kWh} = 1000 \frac{J}{\text{sec}} \times 3600 \text{ sec}$ $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$	1

14. Two LED bulbs of 12W and 6W are connected in series. If the current through 12W bulb is 0.06A the current through 6W bulb will be :

1

(a) 0.04A

(b) 0.06A

(c) 0.08A

(d) 0.12A

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14. Two LED bulbs of 12W and 6W are connected in series. If the current through 12W bulb is 0.06A the current through 6W bulb will be : 1
- (a) 0.04A (b) 0.06A (c) 0.08A (d) 0.12A

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(b)

16. The resistance of a resistor is reduced to half of its initial value. If other parameters of the electrical circuit remain unaltered, the amount of heat produced in the resistor will become :

1

- (a) four times
- (b) two times
- (c) half
- (d) one fourth

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16. The resistance of a resistor is reduced to half of its initial value. If other parameters of the electrical circuit remain unaltered, the amount of heat produced in the resistor will become :

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- (a) four times
- (b) two times
- (c) half
- (d) one fourth

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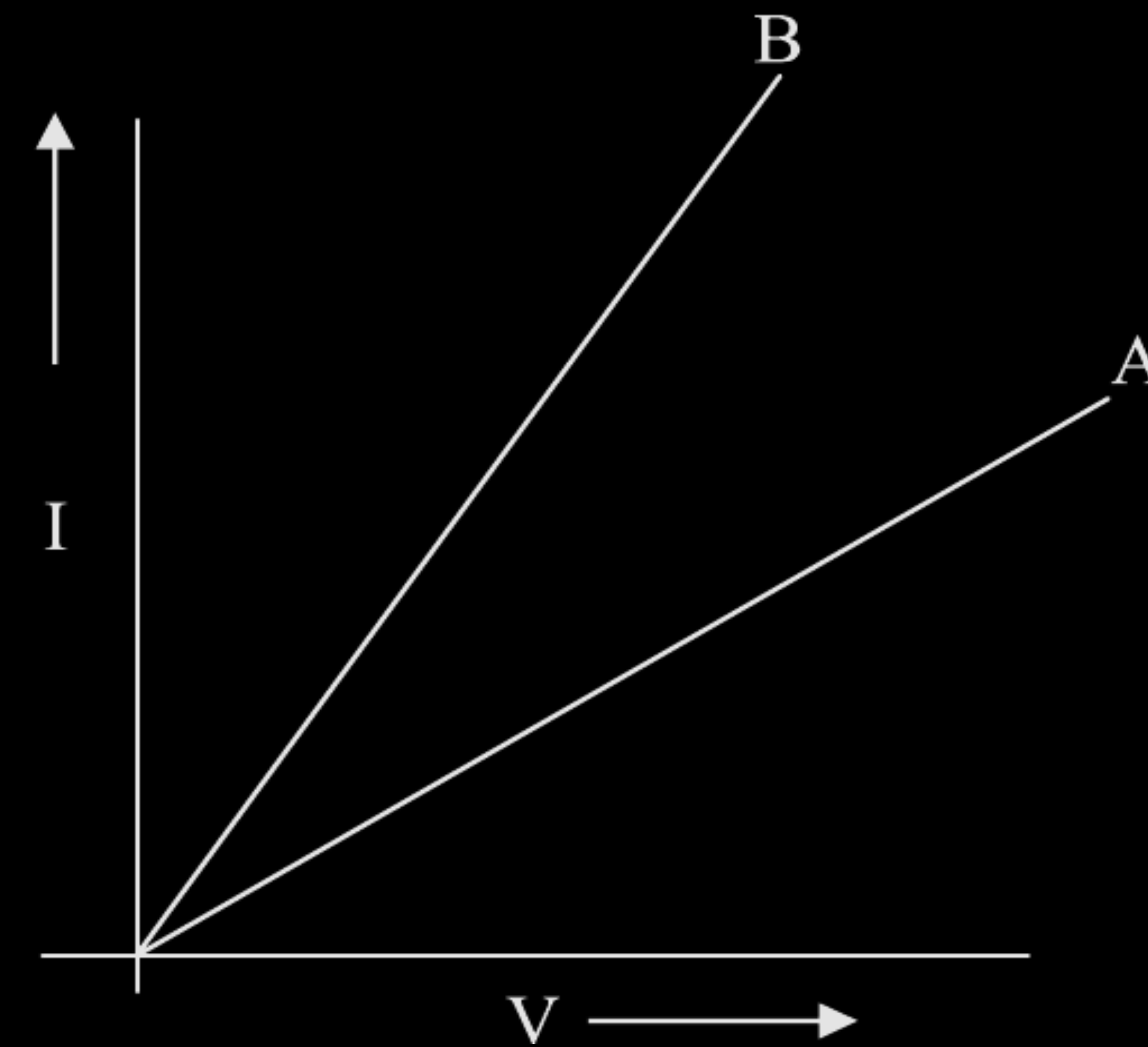
(b)

35. (i) How is electric current related to the potential difference across the terminals of a conductor ? **5**

Draw a labelled circuit diagram to verify this relationship.

(ii) Why should an ammeter have low resistance ?

(iii) Two V - I graphs A and B for series and parallel combinations of two resistors are as shown. Giving reason state which graph shows (a) series, (b) parallel combination of the resistors.



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16. Two LED bulbs of 10W and 5W are connected in series. If the current flowing through 5W bulb is 0.005A, the current flowing through 10W bulb is :

- (a) 0.02A
- (b) 0.01A
- (c) 0.005A
- (d) 0.0025A

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16. Two LED bulbs of 10W and 5W are connected in series. If the current flowing through 5W bulb is 0.005A, the current flowing through 10W bulb is :

1

- (a) 0.02A
- (b) 0.01A
- (c) 0.005A
- (d) 0.0025A

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8. For verifying Ohm's law, we design an electric circuit diagram in which we show the arrangement of different circuit components. We find that with respect to the resistor, the :

- (a) ammeter is connected in parallel and the voltmeter in series.
- (b) ammeter is connected in series and the voltmeter in parallel.
- (c) ammeter and voltmeter are both connected in series.
- (d) ammeter and voltmeter are both connected in parallel.

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8. For verifying Ohm's law, we design an electric circuit diagram in which we show the arrangement of different circuit components. We find that with respect to the resistor, the :

- (a) ammeter is connected in parallel and the voltmeter in series.
- (b) ammeter is connected in series and the voltmeter in parallel.
- (c) ammeter and voltmeter are both connected in series.
- (d) ammeter and voltmeter are both connected in parallel.

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16. In a resistive circuit if the current is increased to two times, the percentage change in the amount of heat dissipated in the circuit would be :

(a) 400%

(b) 300%

(c) 200%

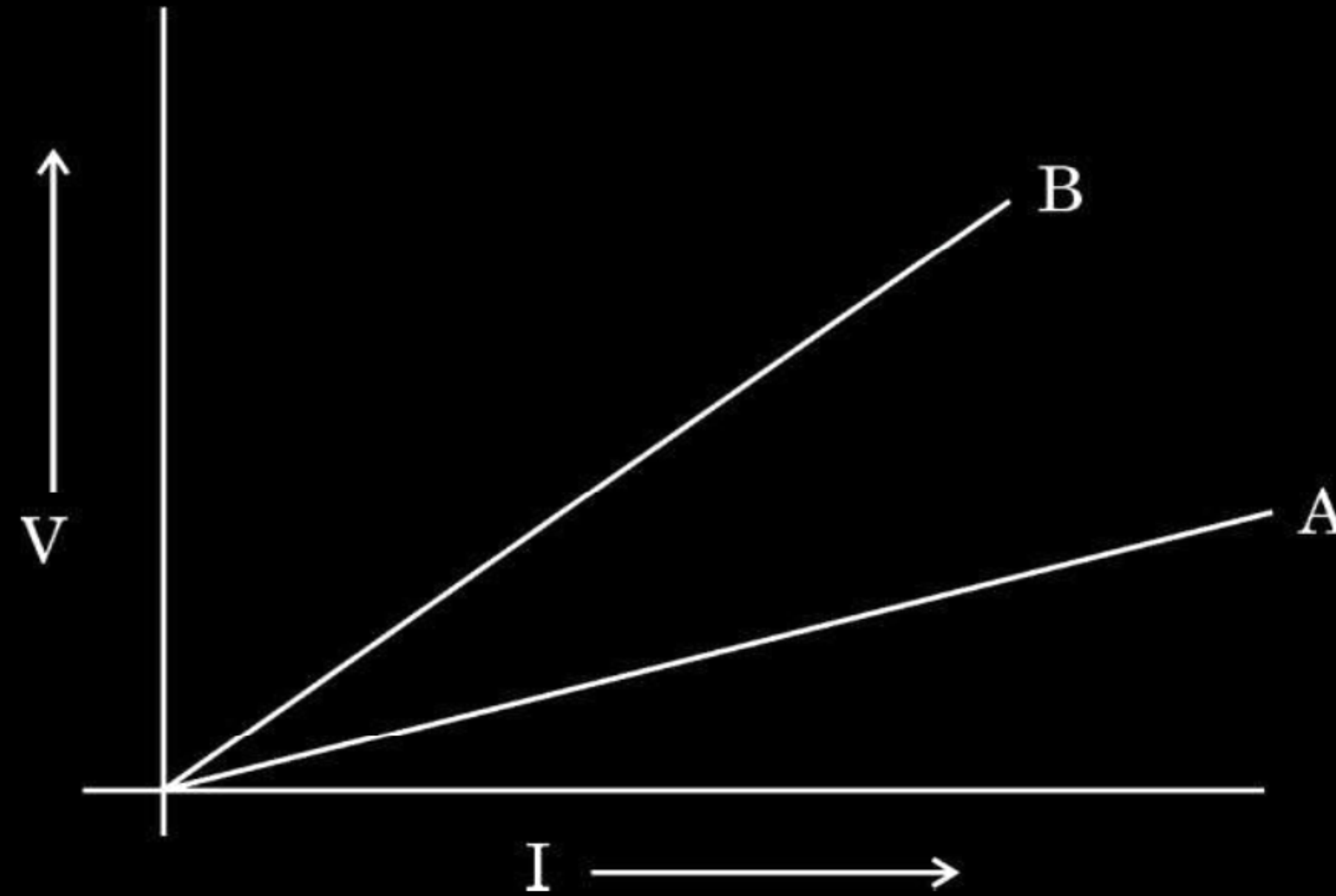
(d) 100%

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(b)

21. $V - I$ graph for two conducting wires A and B are as shown. If both wires are of the same length and same diameter, which of the two is made of a material of high resistivity ? Give reasons to justify your answer.

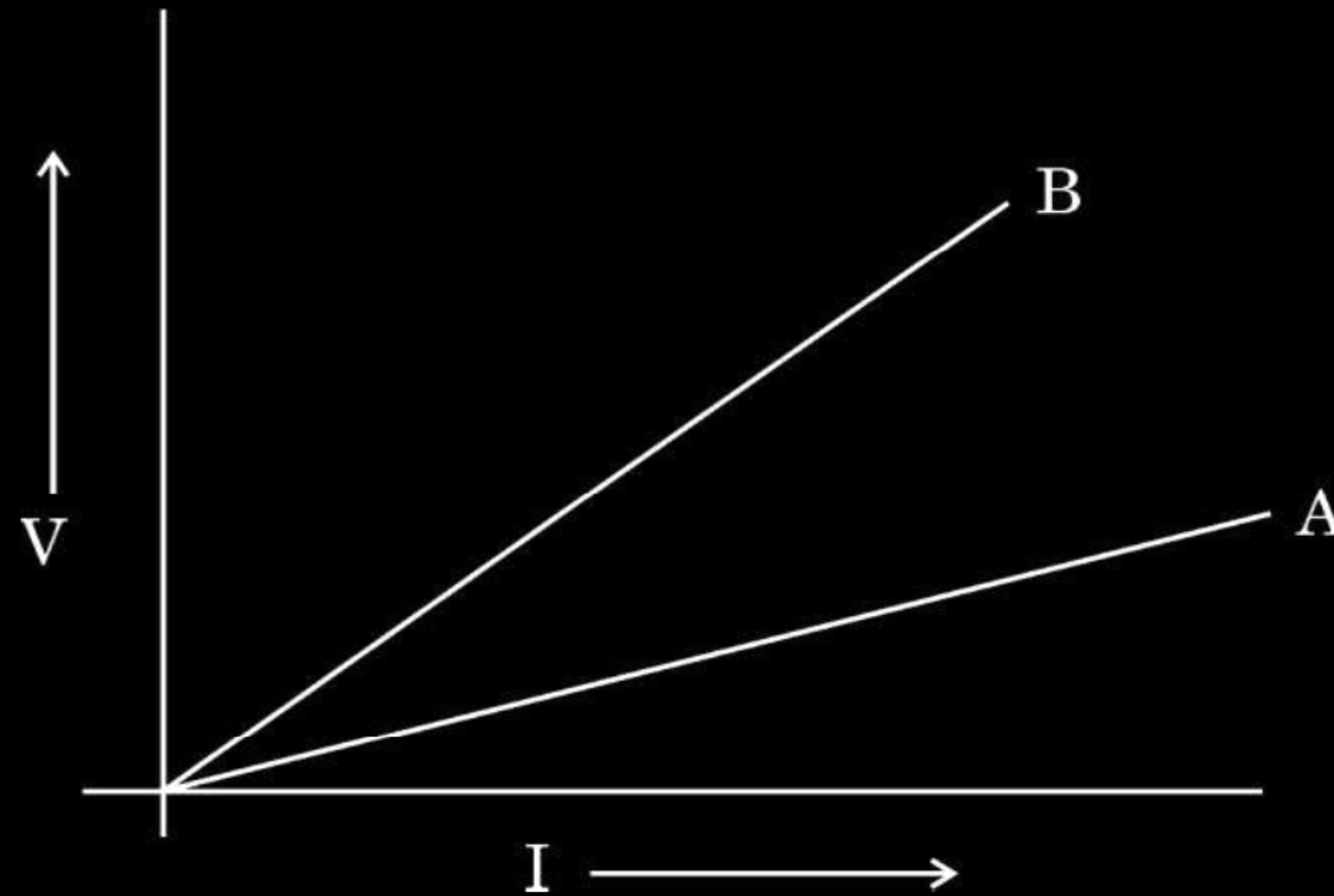
2



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21. $V - I$ graph for two conducting wires A and B are as shown. If both wires are of the same length and same diameter, which of the two is made of a material of high resistivity ? Give reasons to justify your answer.

2



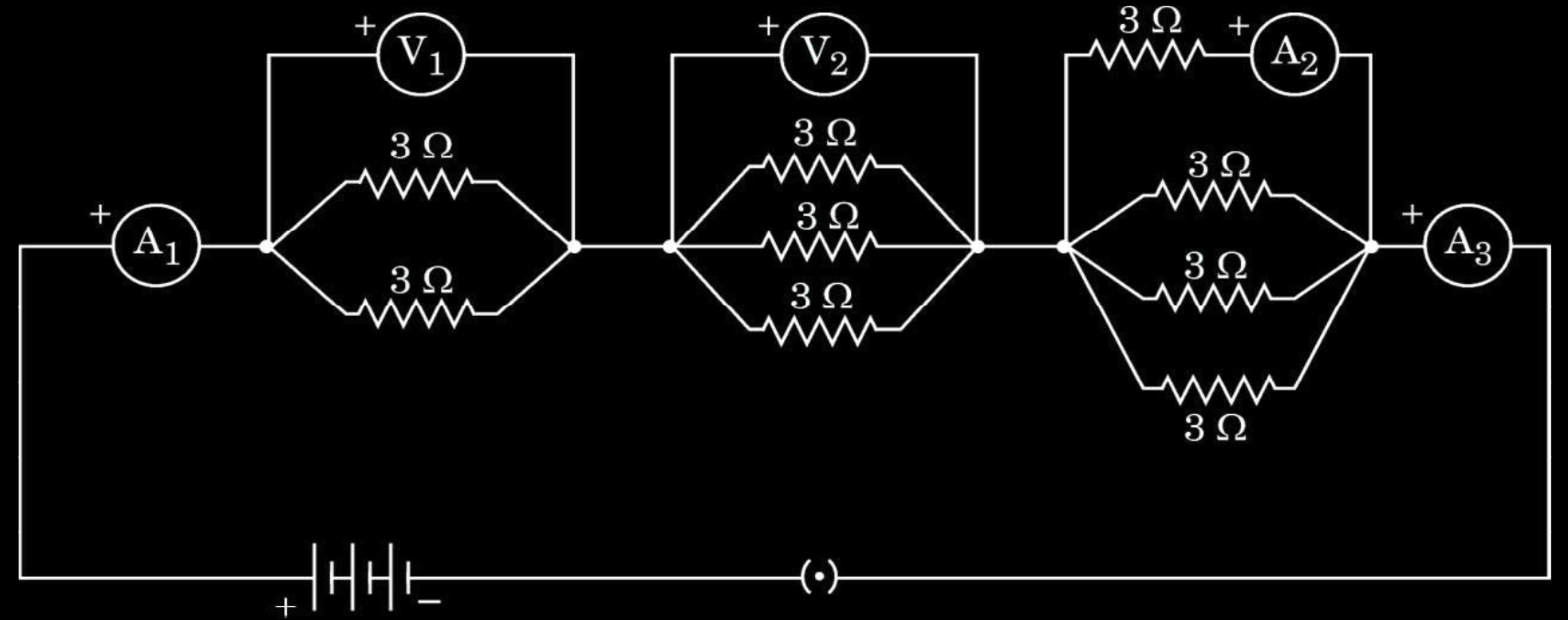
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- Wire B.
- For the conductors of the same dimensions greater the resistance, greater is the resistivity.

1

1

39. Consider the following electrical circuit diagram in which nine identical resistors of $3\ \Omega$ each are connected as shown. If the reading of the ammeter A_1 is 1 ampere, answer the following questions :



- (a) What is the relationship between the readings of A_1 and A_3 ? Give reasons for your answer. 1
- (b) What is the relationship between the readings of A_2 and A_3 ? 1
- (c) Determine the reading of the voltmeter V_1 . 2

OR

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- (c) Find the total resistance of the circuit. 2

(a) • Both have same reading / $A_1 = A_3$	$\frac{1}{2}$
• Both are connected in series	$\frac{1}{2}$
(b) Reading of $A_2 = \frac{1}{4}A$ as current is equally divided in the four identical resistors . / Reading of $A_2 = \frac{1}{4}$ times Reading of A_3 . / $A_2 = 0.25 A$ / $A_2 < A_3$	1
(c) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ / $R_p = \frac{R}{n}$	$\frac{1}{2}$
$\frac{1}{R_p} = \frac{1}{3\Omega} + \frac{1}{3\Omega}$	$\frac{1}{2}$
$R_p = \frac{3}{2}\Omega$	$\frac{1}{2}$
$V = IR$	$\frac{1}{2}$
$V_1 = 1A \times \frac{3}{2}\Omega = \frac{3}{2}V = 1.5V$	$\frac{1}{2}$
OR	
$\frac{1}{R_p} = \frac{1}{3\Omega} + \frac{1}{3\Omega}$	
$\therefore R_p = \frac{3}{2}\Omega$	$\frac{1}{2}$
$\frac{1}{R_p} = \frac{1}{3\Omega} + \frac{1}{3\Omega} + \frac{1}{3\Omega}$	
$\therefore R_{p2} = 1\Omega$	$\frac{1}{2}$
$\frac{1}{R_p} = \frac{1}{3\Omega} + \frac{1}{3\Omega} + \frac{1}{3\Omega} + \frac{1}{3\Omega}$	
$\therefore R_{p3} = \frac{3}{4}\Omega$	$\frac{1}{2}$
$\therefore R = R_{p1} + R_{p2} + R_{p3} = \left(\frac{3}{2} + 1 + \frac{3}{4}\right)\Omega = \frac{13}{4}\Omega / 3.25\Omega$	$\frac{1}{2}$

14. If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance R_s , and then connected in parallel so as to give an effective resistance R_p , then the ratio $\frac{R_s}{R_p}$ is

1

(a) 32

(b) 2

(c) 0.5

(d) 16

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14. If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance R_s , and then connected in parallel so as to give an effective resistance R_p , then the ratio $\frac{R_s}{R_p}$ is

1

(a) 32

(b) 2

(c) 0.5

(d) 16

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(d)

16. If the current I through a resistor is increased by 100% (at constant temperature), the increase in power dissipated will be 1

(a) 100%

(b) 200%

(c) 300%

(d) 400%

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16. If the current I through a resistor is increased by 100% (at constant temperature), the increase in power dissipated will be 1

(a) 100%

(b) 200%

(c) 300%

(d) 400%

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(c)

35. (a) An electric iron consumes energy at a rate of 880 W when heating is at the maximum rate and 330 W when the heating is at the minimum. If the source voltage is 220 V, calculate the current and resistance in each case.
- (b) What is heating effect of electric current ?
- (c) Find an expression for the amount of heat produced when a current passes through a resistor for some time.

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(a) When heating is at maximum rate.

Power, $P = 880 \text{ W}$

Voltage, $V = 220 \text{ V}$

$$\text{Current, } I = \frac{P}{V} = \frac{880}{220} = 4 \text{ A}$$

$$\text{Resistance, } R = \frac{V}{I} = \frac{220}{4} = 55 \, \Omega$$

When heating is at minimum rate

Power, $P = 330 \text{ W}$

Voltage, $V = 220 \text{ V}$

$$\text{Current, } I = \frac{P}{V} = \frac{330}{220} = \frac{3}{2} = 1.5 \text{ A}$$

$$\text{Resistance, } R = \frac{V}{I} = \frac{220}{1.5} = 146.6 \, \Omega$$

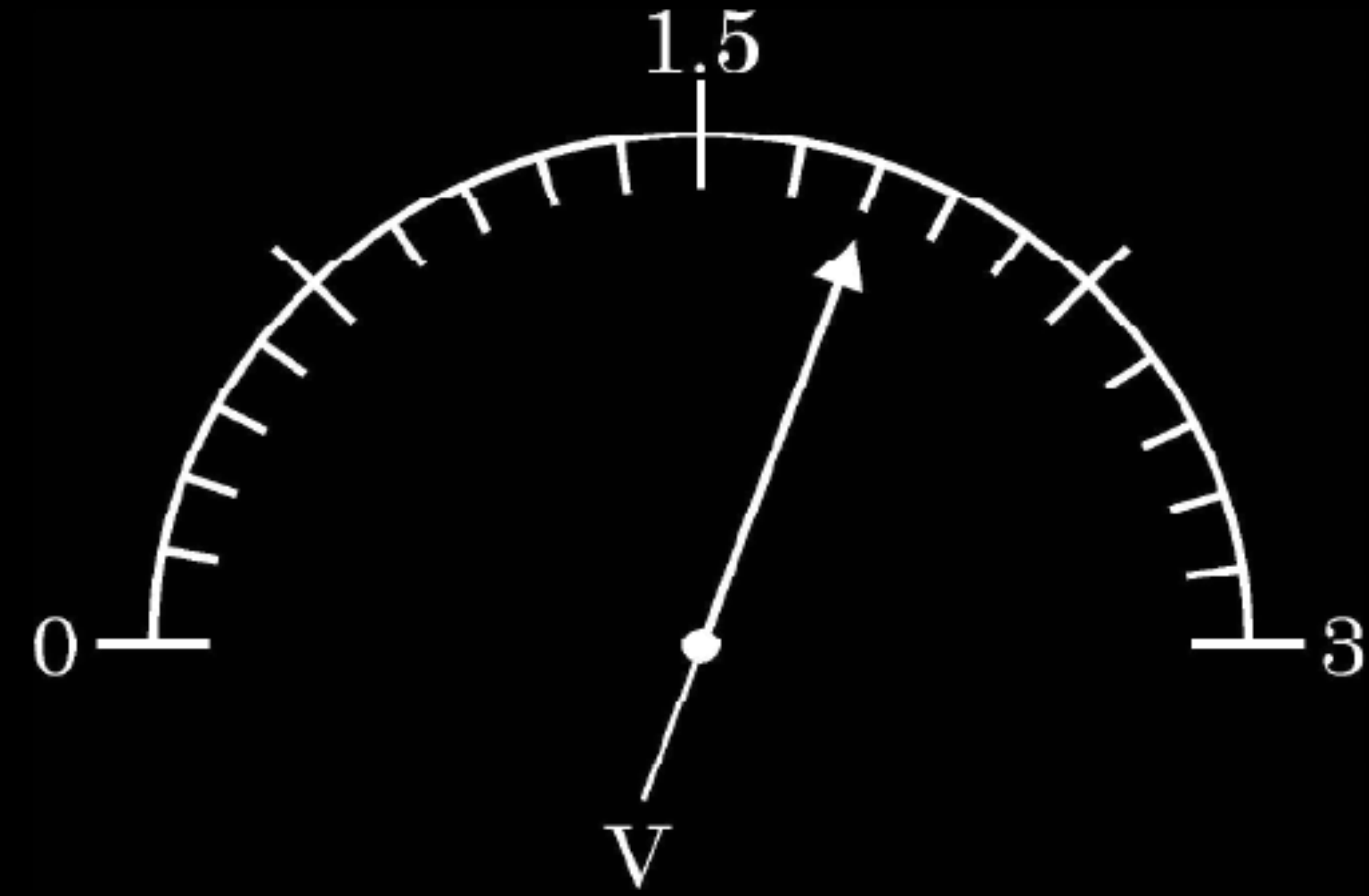
(b) When electric current is passed through a resistor, electrical energy is dissipated and appears as heat energy.

$$(c) \quad H = I^2 R t / H = V I t$$

15. In the following diagram, the position of the needle is shown on the scale of a voltmeter. The least count of the voltmeter and the reading shown by it respectively are :

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1

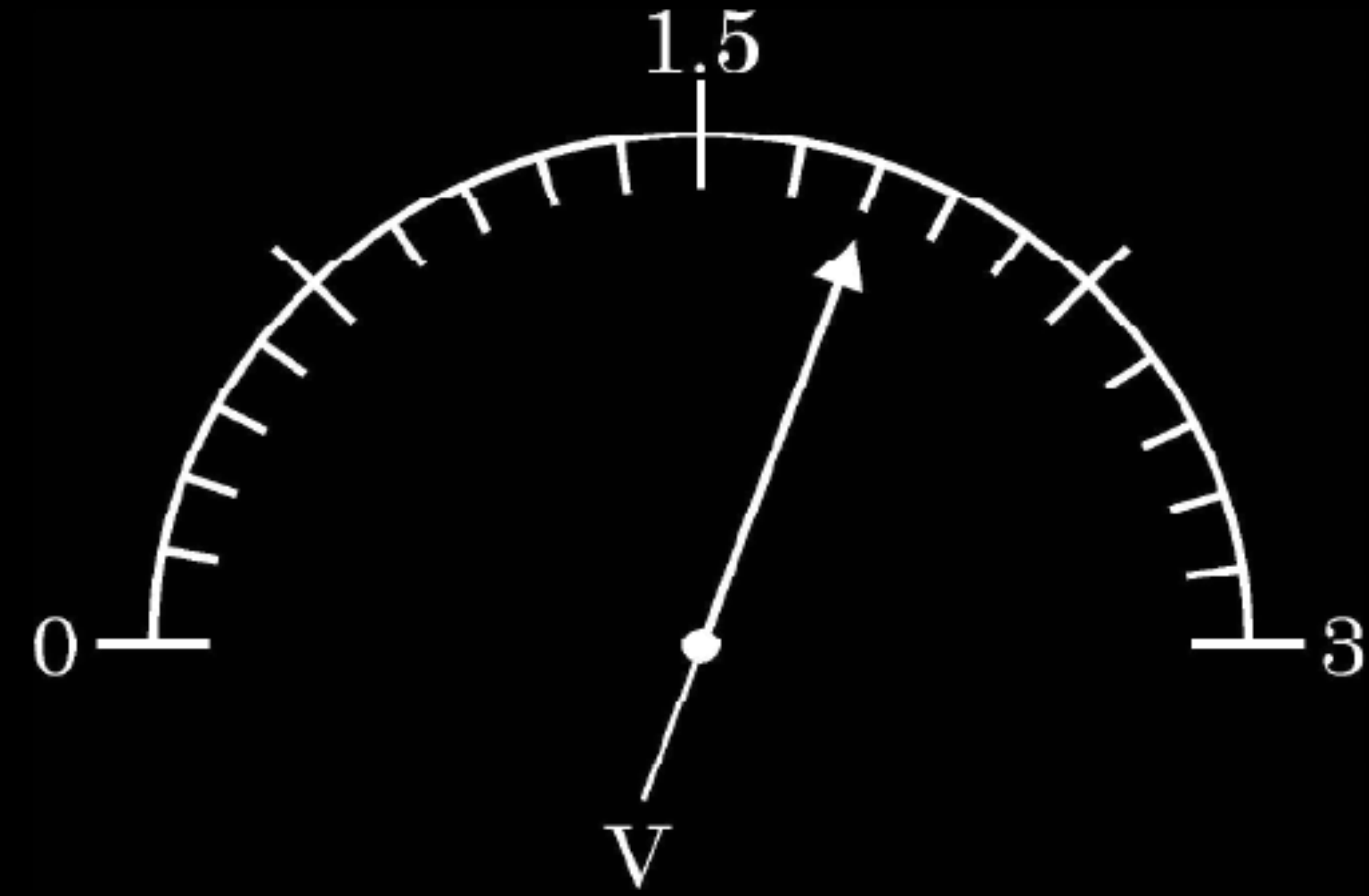


- (a) 0.15 V and 1.6 V (b) 0.05 V and 1.6 V
(c) 0.15 V and 1.8 V (d) 0.05 V and 1.8 V

15. In the following diagram, the position of the needle is shown on the scale of a voltmeter. The least count of the voltmeter and the reading shown by it respectively are :

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1



- (a) 0.15 V and 1.6 V
(c) 0.15 V and 1.8 V

- (b) 0.05 V and 1.6 V
(d) 0.05 V and 1.8 V

(c)

- 6.** (a) State Ohm's law. A student has drawn $V - I$ graph on the basis of his experimental observations. Why should this graph be a straight line passing through the origin ? 2

OR

- (b) When a 6.0 V battery is connected across a resistor there is a current of 30 mA in the circuit. Find the value of the resistance of the resistor in ohms. 2

CBSE 2022 C

6. (a) State Ohm's law. A student has drawn V – I graph on the basis of his experimental observations. Why should this graph be a straight line passing through the origin ? 2

OR

- (b) When a 6.0 V battery is connected across a resistor there is a current of 30 mA in the circuit. Find the value of the resistance of the resistor in ohms. CBSE 2022 C 2

(a) Current flowing through a given conductor is directly proportional to potential difference applied across the conductor provided that temperature remains constant. / $V \propto I$ (when T is constant)

• According to Ohm's law, there should be no current in conductor if potential difference is zero. That is why, this graph be straight line passing through the origin.

OR

(b) $V=6.0 \text{ V}$ $I=30 \text{ mA}$

$$R = \frac{V}{I}$$

$$R = \frac{6.0 \text{ V}}{30 \times 10^{-3} \text{ A}}$$

$$R = 200 \, \Omega$$

1

1

$\frac{1}{2}$

1

$\frac{1}{2}$

12. For a heater rated 4.4 kW; 220 V, calculate the following : 3

(a) The current

(b) The resistance of the heater

(c) The energy consumed in kWh in 5 hours

CBSE 2022 C

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(a) The current

(b) The resistance of the heater

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CBSE 2022 C

(a) $P = 4.4\text{kW}$ $V = 220\text{ V}$

$$P = VI$$

$$I = \frac{4.4 \times 1000\text{ W}}{220\text{ V}}$$

$$I = 20\text{ A}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

(b) $P = \frac{V^2}{R}$

$$R = \frac{V^2}{P}$$

$$R = \frac{220\text{ V} \times 220\text{ V}}{4.4 \times 10^3\text{ W}}$$

$$R = 11\ \Omega$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

(c) $E = P \times t$

$$E = 4.4\text{kW} \times 5\text{ hours}$$

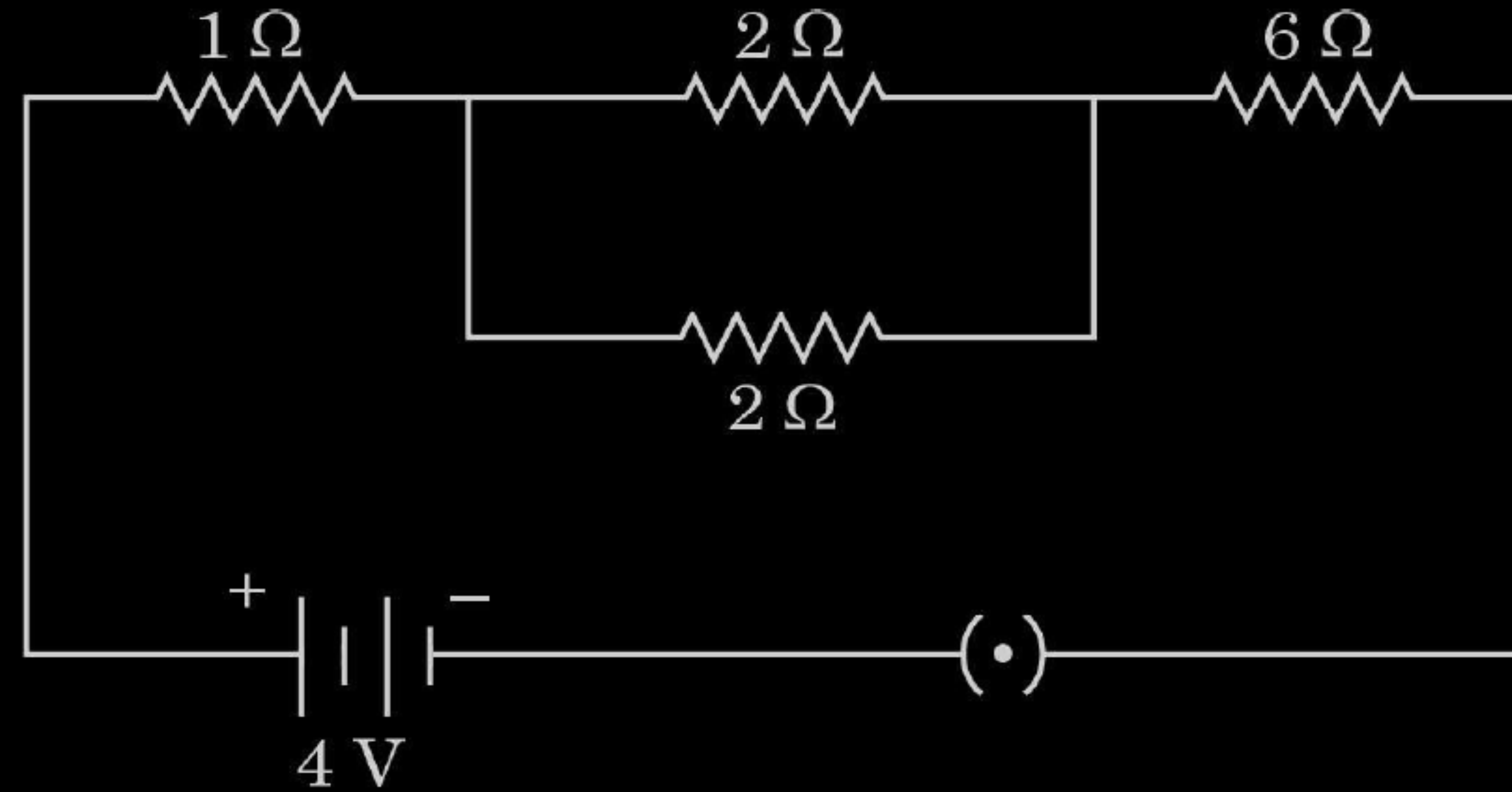
$$= 22.0\text{ kWh}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

10. (a) Find the current flowing through the following electric circuit :

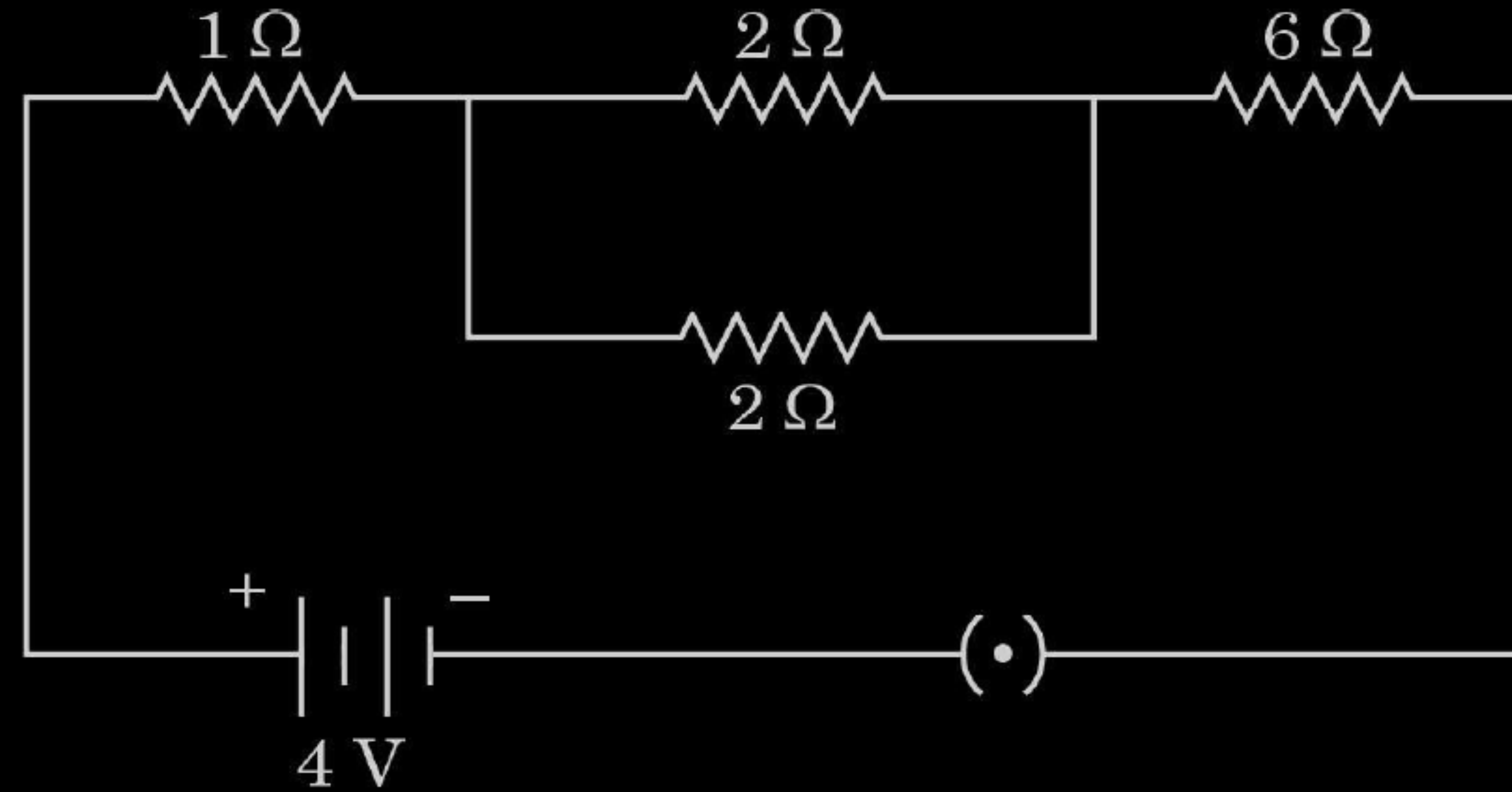
3



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10. (a) Find the current flowing through the following electric circuit :

3



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• Parallel combination,

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_p} = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} \Omega$$

$$R_p = 1 \Omega$$

• Total resistance,

$$R = 1 \Omega + 1 \Omega + 6 \Omega = 8 \Omega$$

• Current,

$$I = \frac{V}{R} = \frac{4V}{8 \Omega} = 0.5 A$$

$\frac{1}{2}$

$\frac{1}{2}$

1

1

- 13.** Define the term electric power. An electric device of resistance R when connected across an electric source of voltage V draws a current I . Derive an expression for the power in terms of resistance R and voltage V . What is the power of a device of resistance $400\ \Omega$ operating at 200 V ?

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13. Define the term electric power. An electric device of resistance R when connected across an electric source of voltage V draws a current I . Derive an expression for the power in terms of resistance R and voltage V . What is the power of a device of resistance $400\ \Omega$ operating at 200 V ?

3

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• Rate at which electrical energy is dissipated / consumed in an electric circuit.	1
• Power $P = VI$	$\frac{1}{2}$
$= \frac{V^2}{R} \quad (I = V/R)$	$\frac{1}{2}$
• $P = \frac{V^2}{R} = \frac{(200\text{ V})^2}{400\ \Omega} = 100\ \Omega$	1

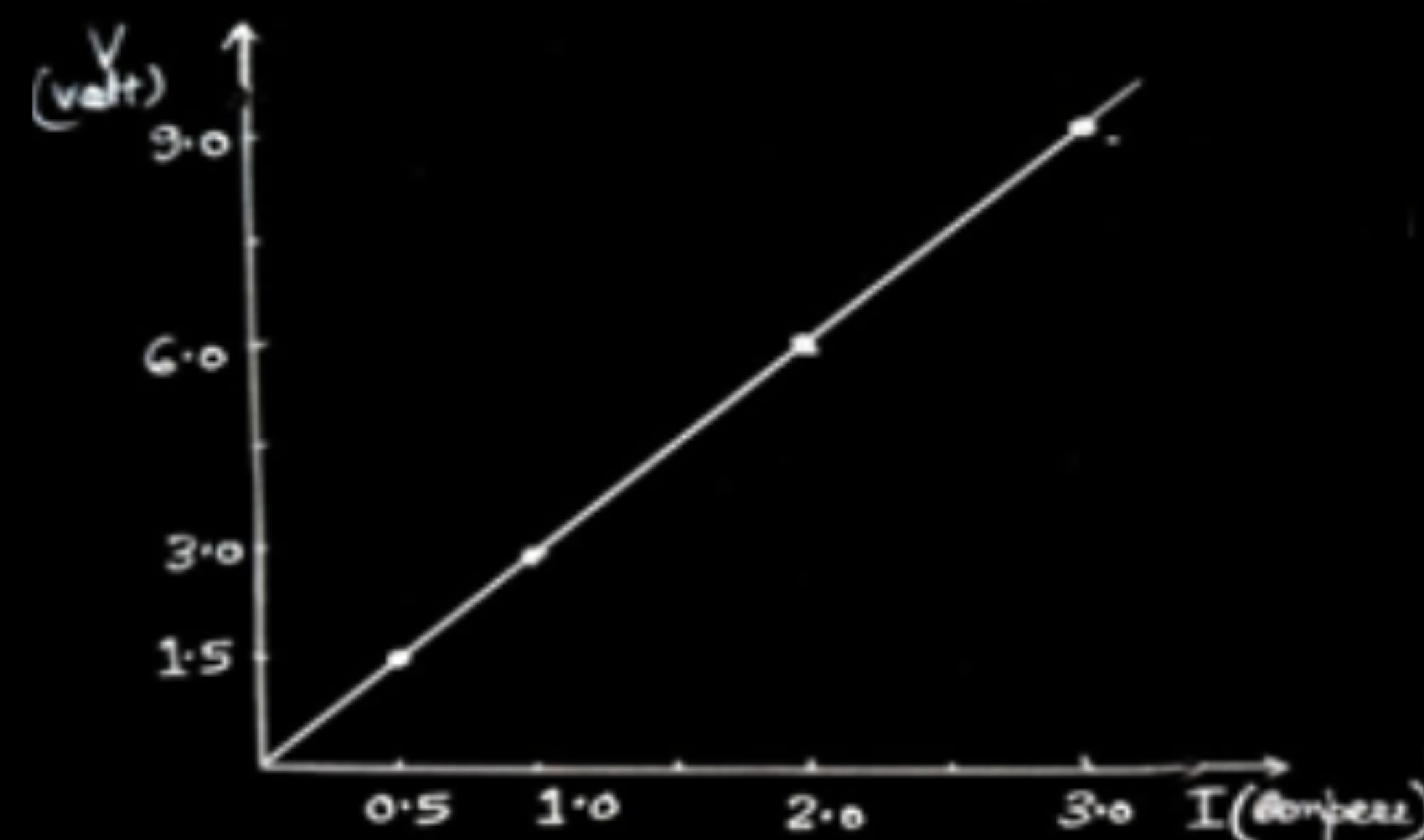
- 10.** (a) The values of current I flowing through a resistor for the corresponding values of potential difference V across it are given below :

V (volts)	1.5	3.0	6.0	9.0
I (amperes)	0.5	1.0	2.0	3.0

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- (i) Plot a graph between V and I .
- (ii) Why should this graph pass through the origin ?
- (iii) Name and state the law which is represented by the graph. 3

(a) (i)



(ii) According to Ohm's law, there should be no current in conductor if potential difference is zero. That is why, this graph be straight line passing through the origin.

(iii) Ohm's Law

The current flowing through a given conductor is directly proportional to the potential difference applied across the ends of conductor provided that temperature of conductor remains constant. / $V \propto I$

1

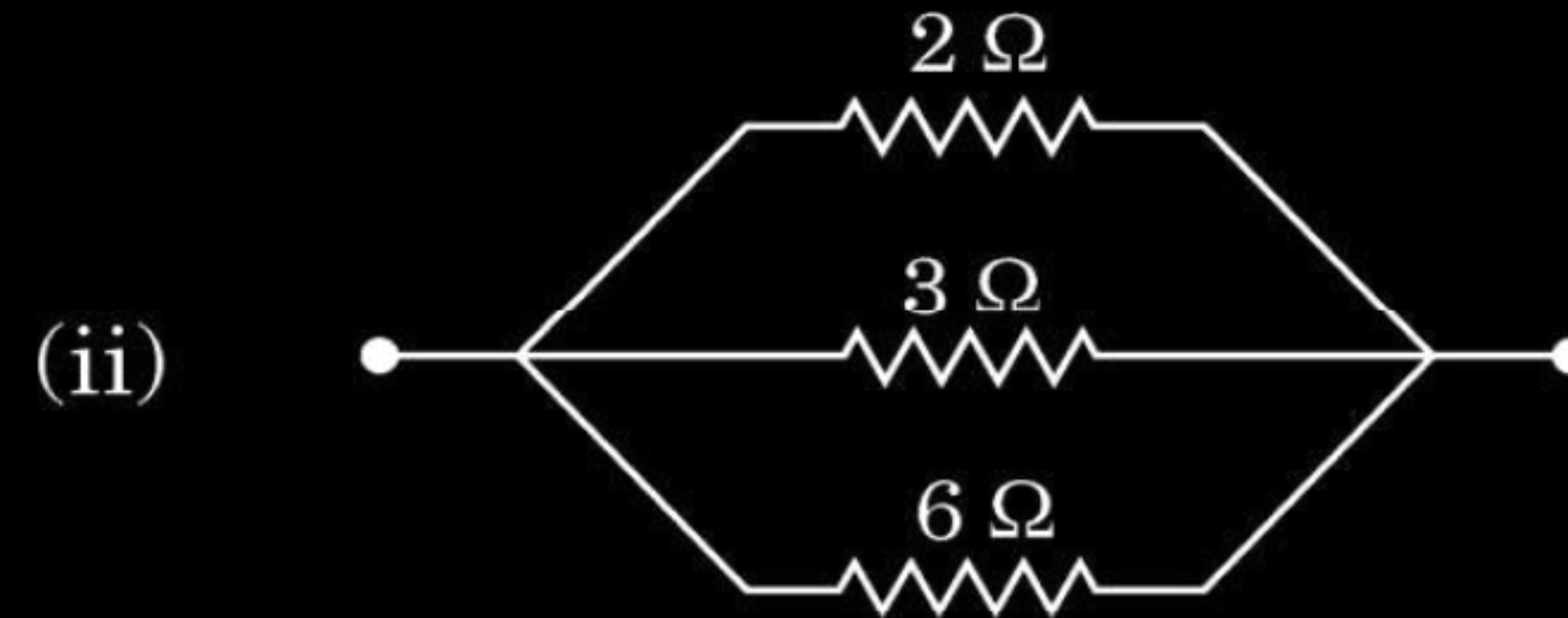
1

$\frac{1}{2}$

$\frac{1}{2}$

- (b) Find the equivalent resistance of the following combinations of resistors :

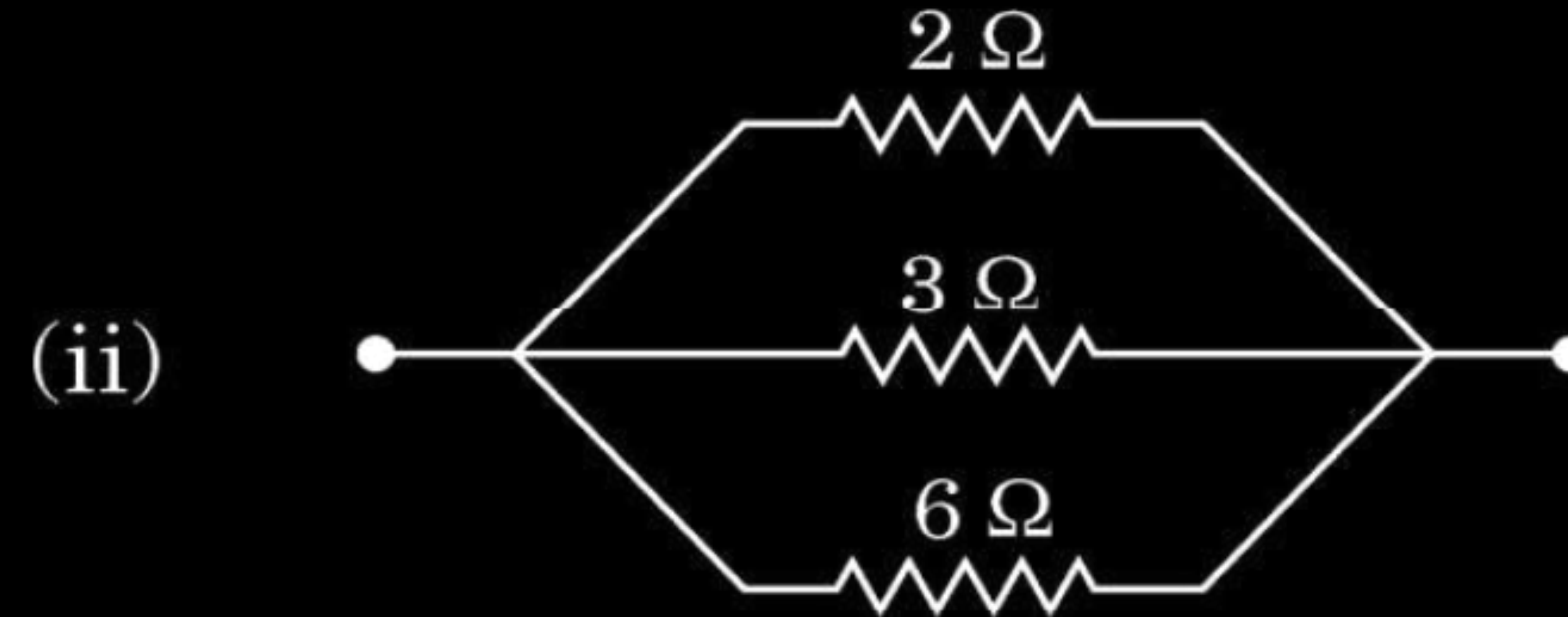
3



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(b) Find the equivalent resistance of the following combinations of resistors :

3



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$$\begin{aligned} \text{(b) (i) } R_s &= R_1 + R_2 + R_3 \\ &= 2 \, \Omega + 3 \, \Omega + 6 \, \Omega = 11 \, \Omega \end{aligned}$$

$$\begin{aligned} \text{(ii) } \frac{1}{R_p} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\ &= \frac{1}{2 \, \Omega} + \frac{1}{3 \, \Omega} + \frac{1}{6 \, \Omega} = 1 \, \Omega \end{aligned}$$

$\frac{1}{2}$
1

$\frac{1}{2}$

1

11. (a) (i) In the following figure, three cylindrical conductors A, B and C are shown along with their lengths and areas of cross-section.



If these three conductors are made of the same material and R_A , R_B and R_C be their respective resistances, then find (I) R_A / R_B , and (II) R_A / R_C .

- (ii) If the conductor A is made of copper and the conductor C is made of constantan (alloy of copper and nickel), then which one of the two will have more electrical resistance and why ?

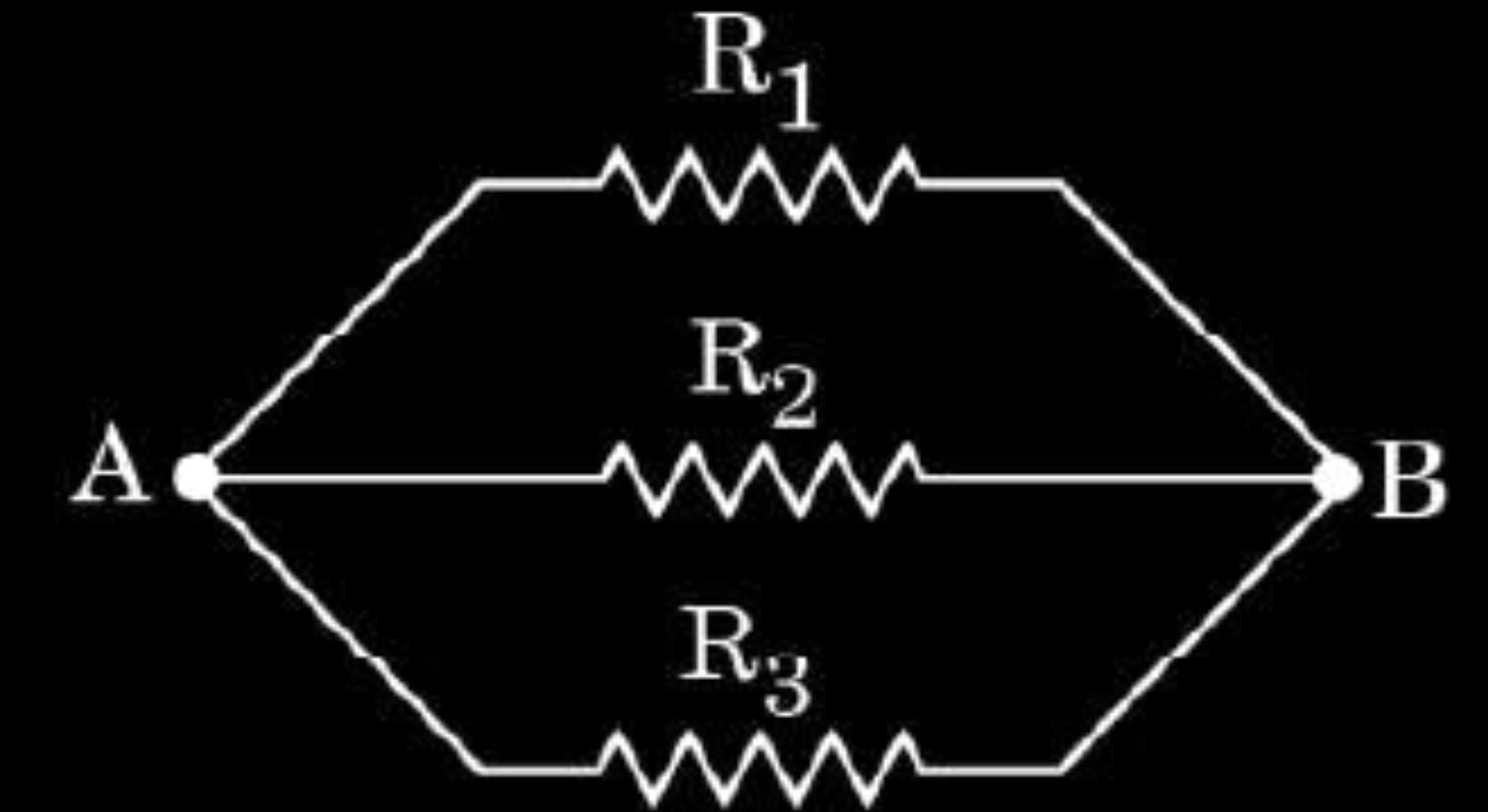
3

(a) (i) (I) $\frac{R_A}{R_B} = \frac{\rho \cdot \left(\frac{L}{A}\right)}{\rho \cdot \left(\frac{L}{4A}\right)}$ $= 4$ (II) $\frac{R_A}{R_C} = \frac{\rho \cdot \left(\frac{L}{A}\right)}{\rho \cdot \left(\frac{L}{2}\right) / \left(\frac{A}{2}\right)}$ $= 1$ (ii) • C • The resistivity of alloys is generally higher than that of the pure metals.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
---	---

- (b) (i) Write the formula for determining the equivalent resistance between A and B of the two combinations (I) and (II) of three resistors R_1 , R_2 and R_3 arranged as follows :



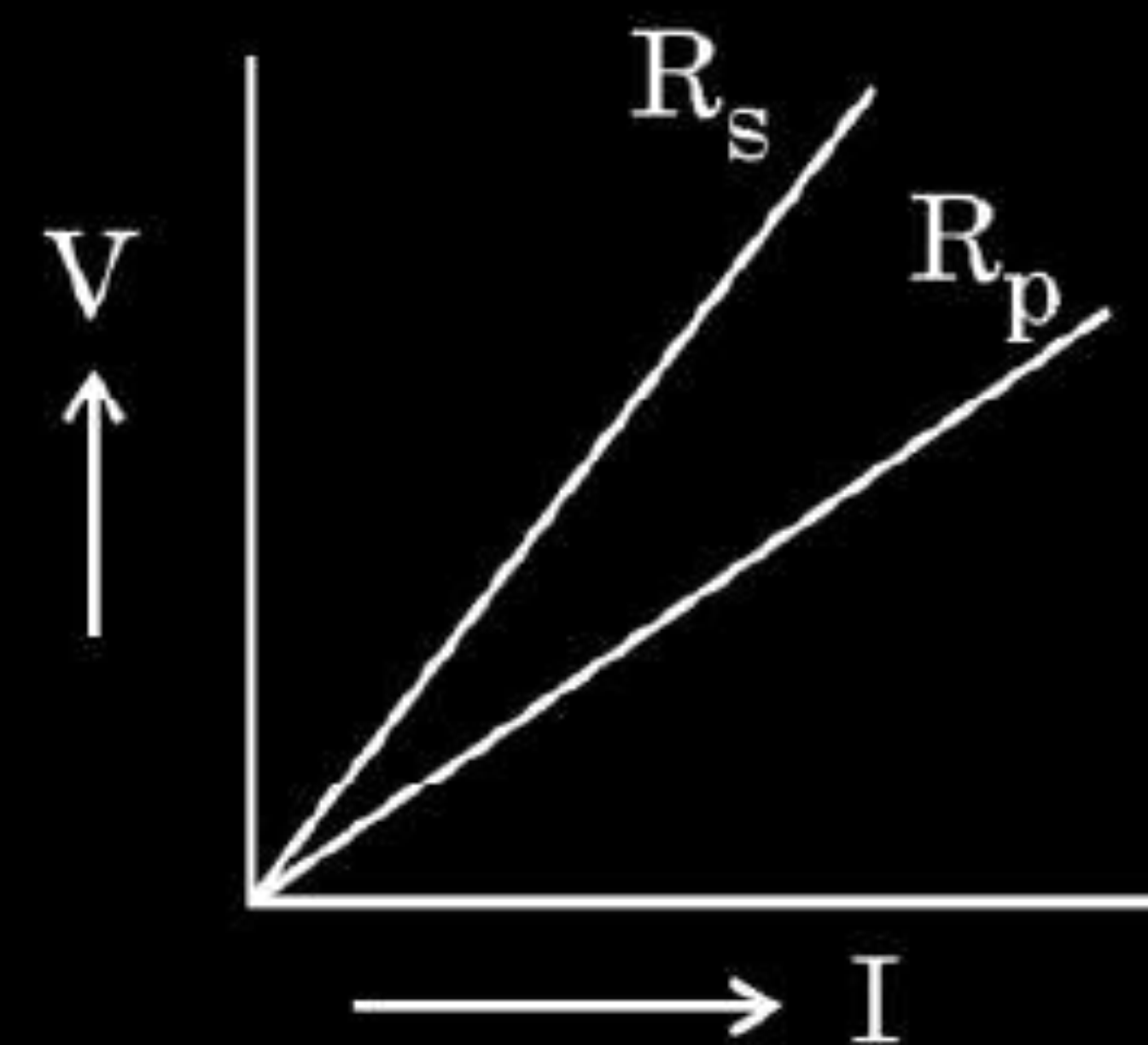
(I)



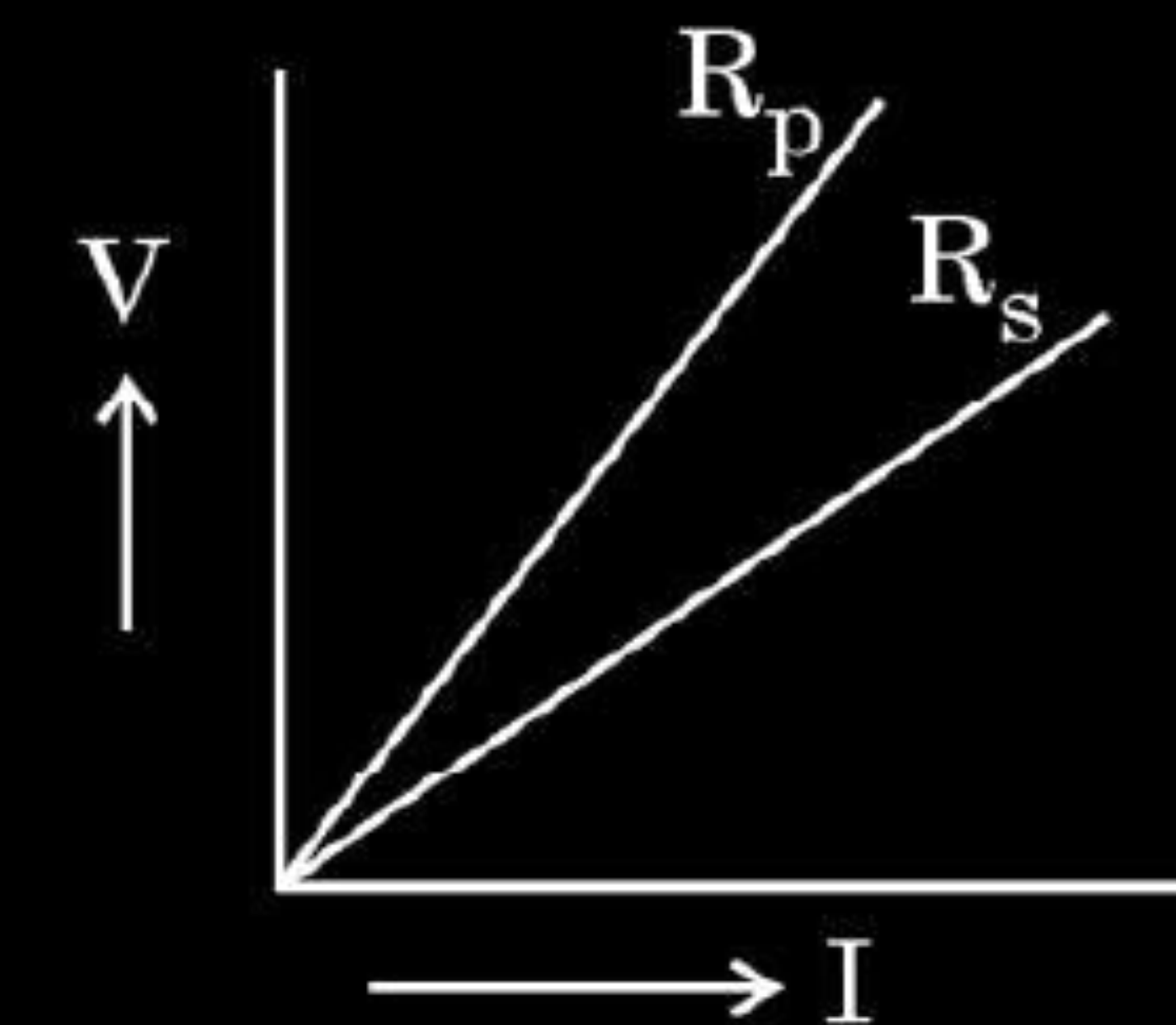
(II)

- (ii) If the equivalent resistance of the arrangements (I) and (II) are R_s and R_p respectively, then which one of the following $V - I$ graphs is correctly labelled ? Justify your answer.

3



(I)



(II)

(b) (i) (I) $R_s = R_1 + R_2 + R_3$	$\frac{1}{2}$
(II) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$	$\frac{1}{2}$
(ii) Graph (I)	1
Reason : The resistance for the series combination is more than that of parallel combination and it is directly proportional to slope of V-I graph. As in the graph (I), slope of R_s is greater than R_p , hence graph I is correct.	1

12. Give reason for the following :

- (a) Tungsten is used exclusively for the filaments of electric lamps.
- (b) Heating elements of electric heating devices are made of alloys rather than a pure metal.
- (c) Series arrangement is not used for domestic electric circuits.

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12. Give reason for the following :

- (a) Tungsten is used exclusively for the filaments of electric lamps.
- (b) Heating elements of electric heating devices are made of alloys rather than a pure metal.
- (c) Series arrangement is not used for domestic electric circuits.

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(a) Tungsten has a high melting point. So it does not melt when electric current flows through it.	1
(b) Alloys used for heating element have generally high melting point / high resistivity / Do not oxidise even at high temperature.	1
(c) If one appliance is switched off (or damaged) all other appliances stop working. / Different appliances require different current which is not available in series arrangement. (or any other)	1

- 15.** A student has designed an electric circuit containing a 6 V battery, a key, an ammeter, and a resistor of $30\ \Omega$ in series with a parallel combination of two resistors of $60\ \Omega$ each and a voltmeter across the $30\ \Omega$ resistor.
- (a) Find the equivalent resistance of the parallel combination of two $60\ \Omega$ resistors.
 - (b) Determine the total resistance of the circuit.
 - (c) (i) Calculate the current that will flow through the ammeter when the key is closed.

OR

- (ii) Will the potential difference across the $30\ \Omega$ resistor be the same as that across the parallel combination of two $60\ \Omega$ resistors ? Justify your answer.

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(a) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ $\frac{1}{R_p} = \frac{1}{60\ \Omega} + \frac{1}{60\ \Omega} = \frac{2}{60\ \Omega}$ $R_p = 30\ \Omega$	$\frac{1}{2}$
(b) $R_S = R_P + R_3$ $R_S = 30\ \Omega + 30\ \Omega = 60\ \Omega$	$\frac{1}{2}$
(c) (i) $I = \frac{V}{R}$ $= \frac{6V}{60\ \Omega} = 0.1A$	$\frac{1}{2}$
Deduct $\frac{1}{2}$ mark for no / wrong unit. OR	
(c) (ii) Yes Justification should include following three points. - Both combinations in series so same current flow. - Resistance of the parallel combination of two $60\ \Omega$ resistors is also $30\ \Omega$ $V = I \times R$ Current is same and the resistance of the parallel combination of two $60\ \Omega$ resistors is also $30\ \Omega$ (or any other correct explanation)	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

8. (a) Write the relationship between electrical resistance and electrical resistivity for a metallic conductor of cylindrical shape. Hence derive the SI unit of electrical resistivity.

3

(b) Find the resistivity of the material of a metallic conductor of length 2m and area of cross-section $1.4 \times 10^{-6} \text{m}^2$. The resistance of the conductor is 0.04 ohm.

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8. (a) Write the relationship between electrical resistance and electrical resistivity for a metallic conductor of cylindrical shape. Hence derive the SI unit of electrical resistivity.

3

(b) Find the resistivity of the material of a metallic conductor of length 2m and area of cross-section $1.4 \times 10^{-6} \text{m}^2$. The resistance of the conductor is 0.04 ohm.

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$$(a) \rho = R \frac{A}{l}$$

$\frac{1}{2}$

$$\begin{aligned} \text{SI unit of } \rho &= \text{ohm} \times \frac{\text{m}^2}{\text{m}} \\ &= \text{ohm} \times \text{metre} / \Omega \text{ m} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

(b)

$$\begin{aligned} \rho &= R \frac{A}{l} \\ &= \frac{0.04 \Omega \times 1.4 \times 10^{-6} \text{m}^2}{2 \text{m}} \\ &= 2.8 \times 10^{-8} \Omega \text{m} \end{aligned}$$

1

$\frac{1}{2}$

9. An electric motor rated 1100 W is connected to 220 V mains. Find :

3

- (i) The current drawn from the mains,
- (ii) Electric energy consumed if the motor is used for 5 hours daily for 6 days.
- (iii) Total cost of energy consumed if the rate of one unit is ₹ 5.

CBSE 2022

9. An electric motor rated 1100 W is connected to 220 V mains. Find :

3

- (i) The current drawn from the mains,
- (ii) Electric energy consumed if the motor is used for 5 hours daily for 6 days.
- (iii) Total cost of energy consumed if the rate of one unit is ₹ 5.

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Power (P) = 1100 W, V = 220 V

(i) Current drawn $= I = \frac{P}{V}$
 $= \frac{1100 \text{ W}}{220 \text{ V}} = 5 \text{ A}$

1/2

1/2

(ii) $E = P \times t$
 $= 1100 \text{ W} \times 5 \text{ h} \times 6 = 33000 \text{ Wh}$

1/2

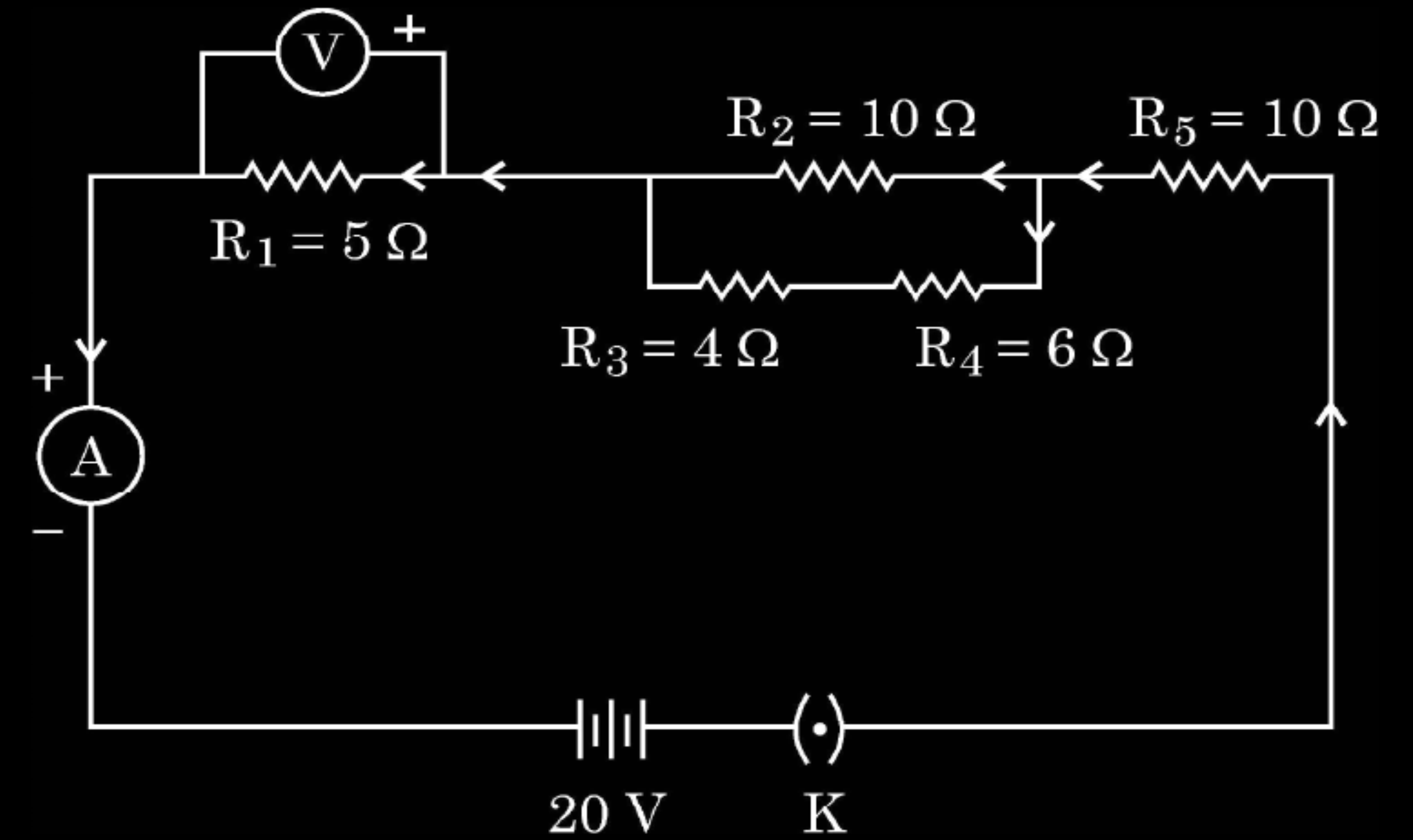
1/2

(iii) Cost of one commercial unit = ₹ 5
 Energy consumed = 33 kWh = 33 unit = $118.8 \times 10^6 \text{ J}$
 Cost of 33 unit = $33 \times 5 = ₹ 165$

1/2

1/2

Study the following circuit and find :



- (i) Effective resistance of the circuit
- (ii) Current drawn from the battery
- (iii) Potential difference across the $5\ \Omega$ resistor

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(i) Effective resistance of the circuit

$$R_s = R_3 + R_4 = 4\Omega + 6\Omega = 10\Omega$$

$$\frac{1}{R_p} = \frac{1}{R_s} + \frac{1}{R_2} = \frac{1}{10\Omega} + \frac{1}{10\Omega} = \frac{2}{10\Omega} = \frac{1}{5\Omega}$$

$$R_p = 5\Omega$$

$$\text{Total resistance of the circuit} = R_1 + R_p + R_5 = 5 + 5 + 10 = 20\Omega$$

(ii) Current drawn from the battery

$$V = 20V, R = 20\Omega$$

$$I = \frac{V}{R} = \frac{20V}{20\Omega}$$

$$I = 1 \text{ A}$$

(iii) Reading in voltmeter connected across 5Ω Resistance

$$V = I R$$

$$I = 1 \text{ A}$$

$$R = 5\Omega$$

$$V = 1 \text{ A} \times 5\Omega = 5 \text{ V}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

9. You are provided with a resistor, a key, an ammeter, a voltmeter, four cells of 1.5 V each and few connecting wires. Using these circuit components, draw a labelled circuit diagram to show the setup to study the Ohm's law.

State the relationship between potential difference (V) across the resistor and the current (I) flowing through it. Also draw V-I graph, taking V on the X-axis.

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Electric circuit

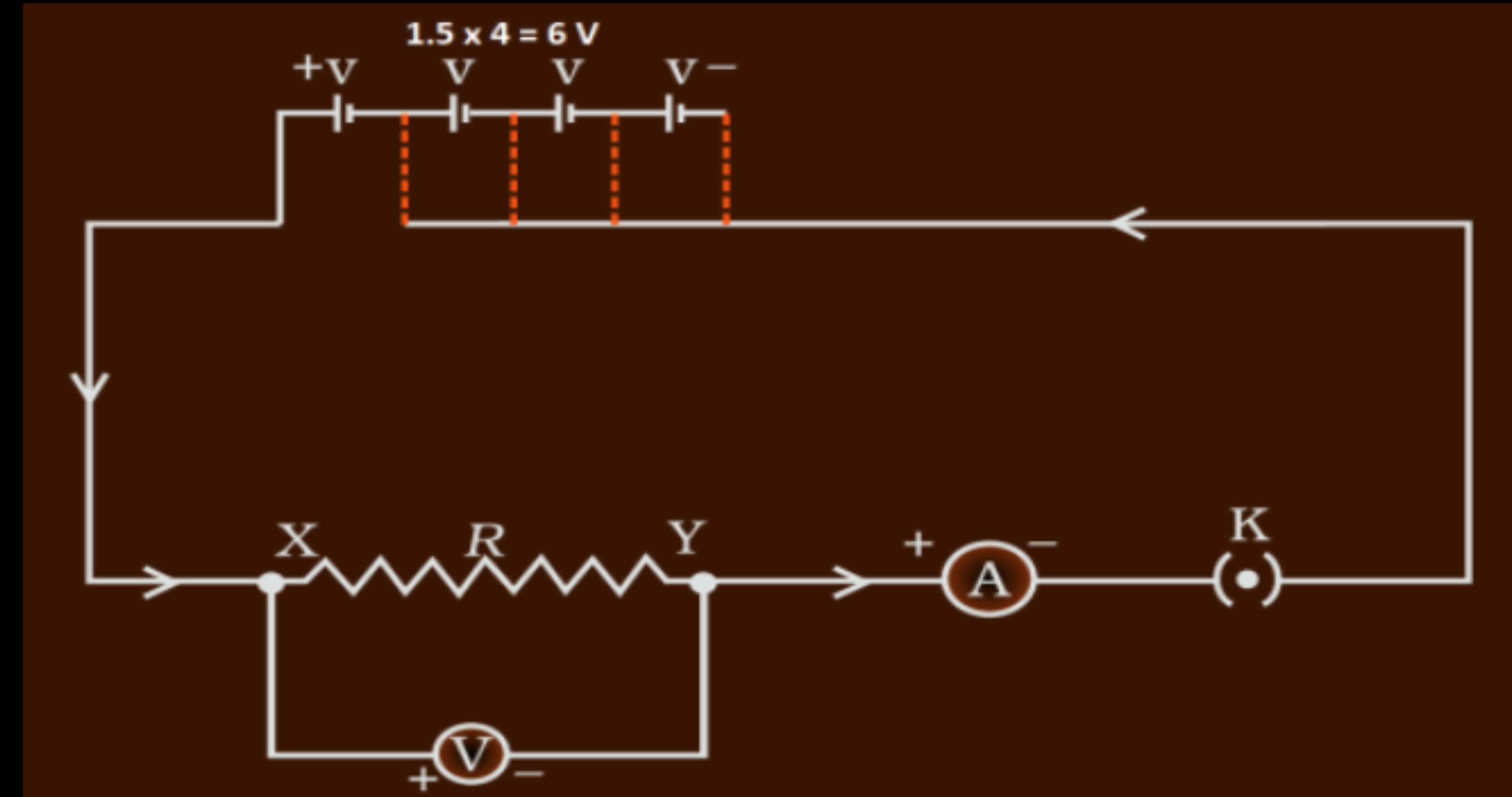
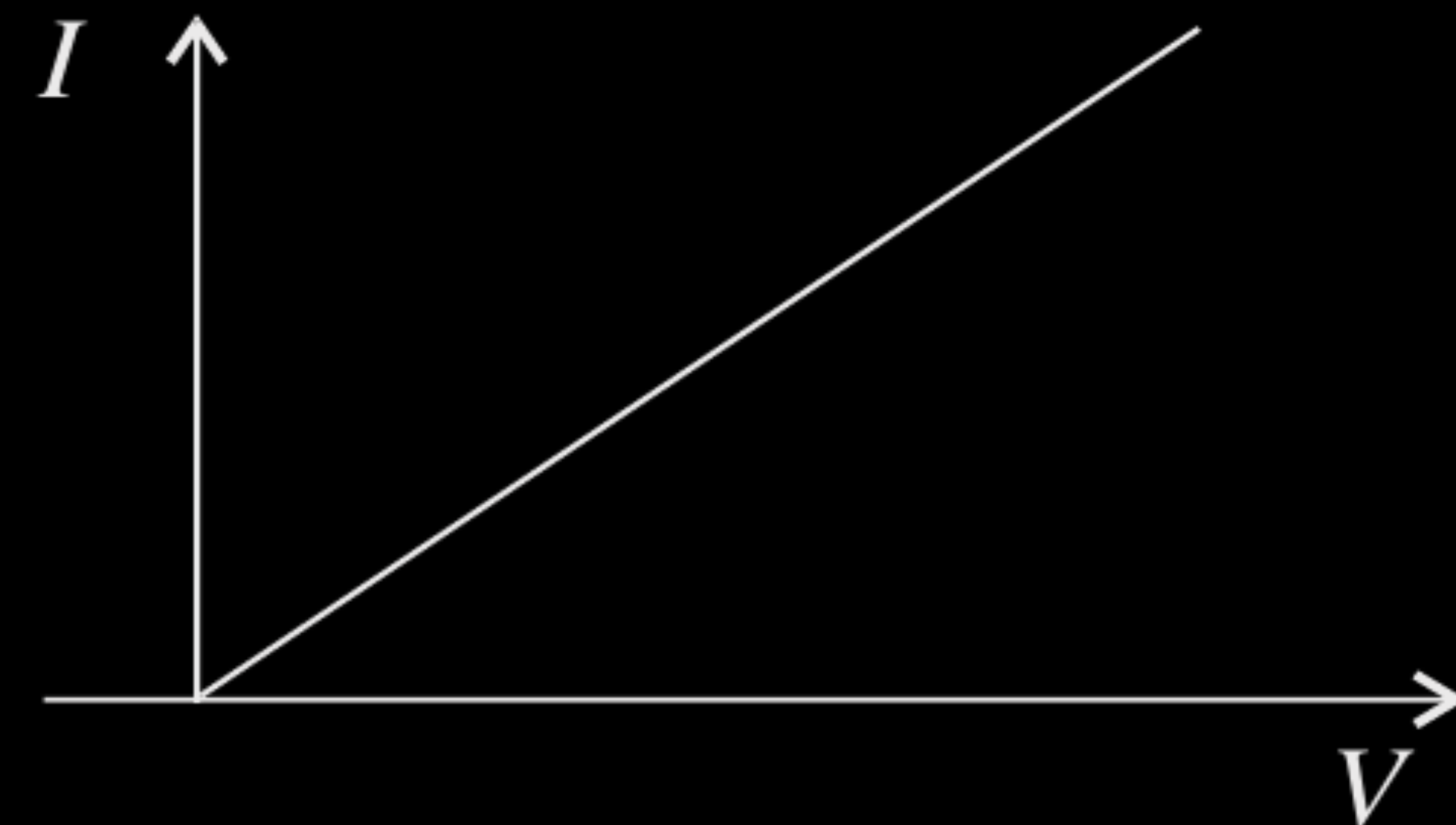


Diagram making

Any two labelling

Potential difference (V) directly proportional to the current (I).



1
 $\frac{1}{2}$, $\frac{1}{2}$
 $\frac{1}{2}$

$\frac{1}{2}$

1. (a) An electric heater rated 1100 W operates at 220 V. Calculate
(i) its resistance, and (ii) the current drawn by it.

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2

1. (a) An electric heater rated 1100 W operates at 220 V. Calculate
(i) its resistance, and (ii) the current drawn by it.

CBSE 2022

2

(a) P (Power) = V (Potential difference) $\times I$ (Current)

Here $P = 1100$ W, $V = 220$ V, $I = ?$, $R = ?$

$$P = \frac{V^2}{R}$$

$\frac{1}{2}$

$$(i) R = \frac{V^2}{P}$$

$$= \frac{220^2 \text{ V} \times 220 \text{ V}}{1100 \text{ W}}$$
$$= 44 \Omega$$

$\frac{1}{2}$

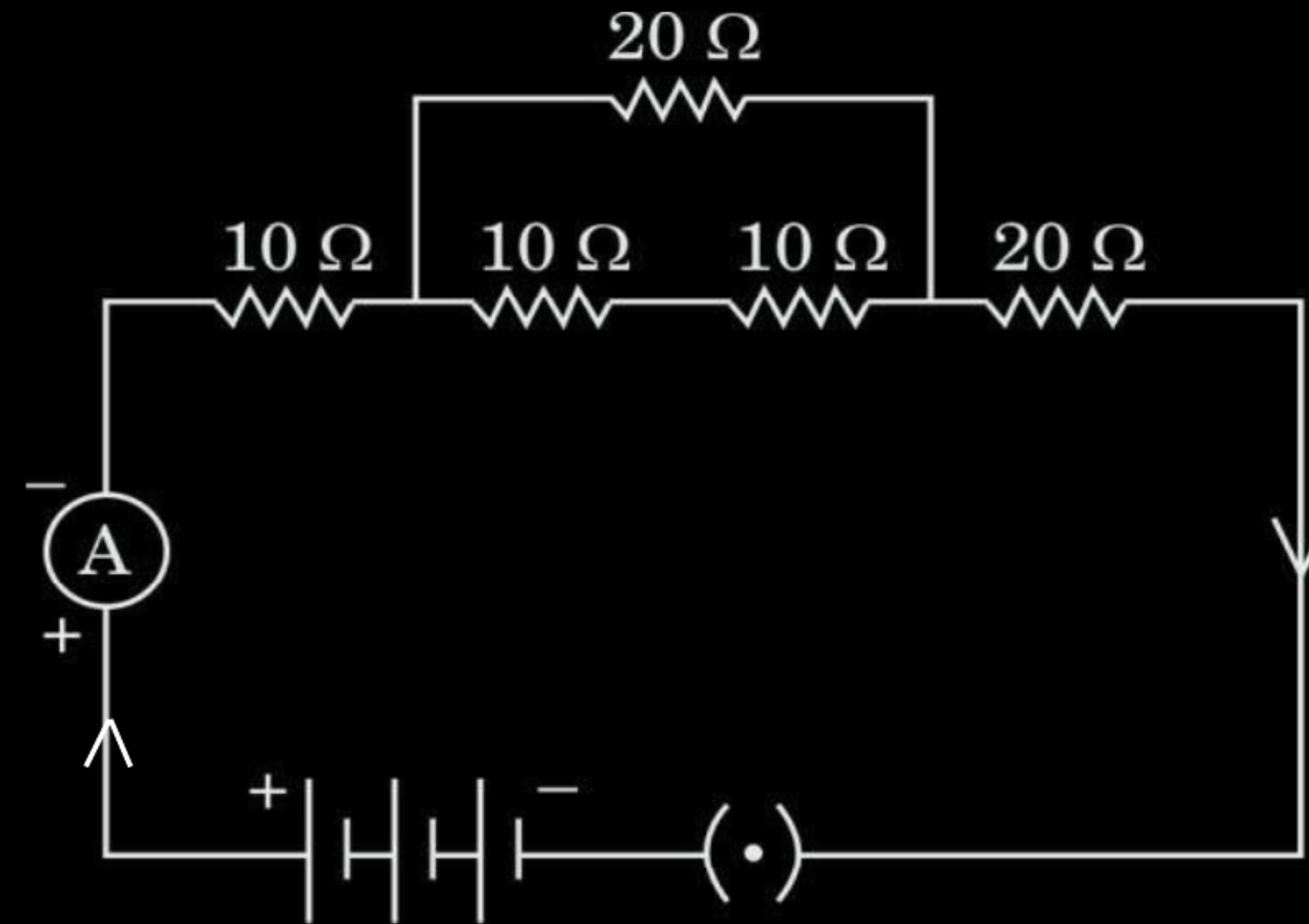
$$(ii) I = \frac{V}{R}$$

$\frac{1}{2}$

$$I(\text{Current}) = \frac{V}{R} = \frac{220 \text{ V}}{44 \Omega} = 5 \text{ A}$$

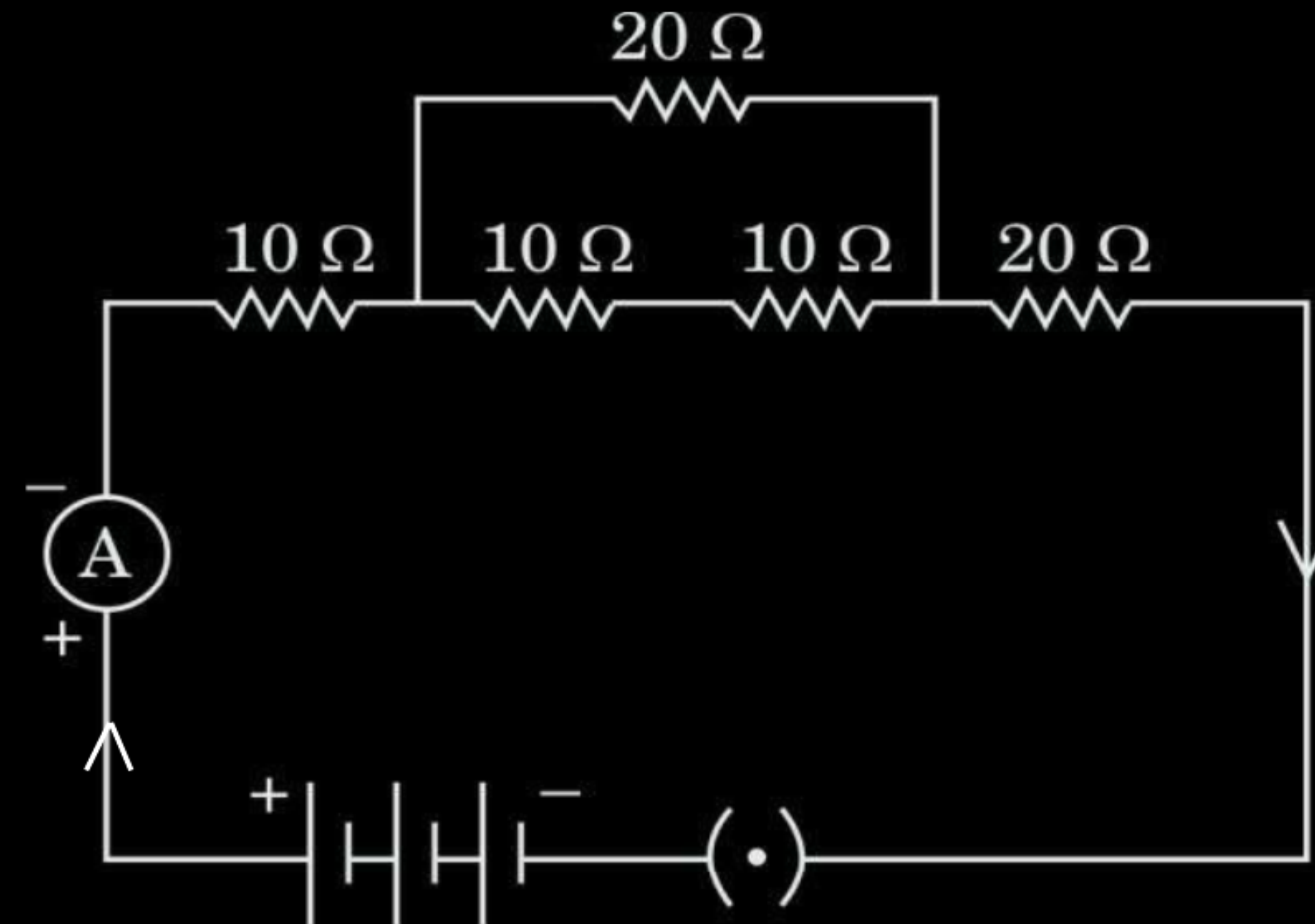
$\frac{1}{2}$

(b) Calculate the equivalent resistance of the following electric circuit : 2



CBSE 2022

(b) Calculate the equivalent resistance of the following electric circuit : 2



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$$(b) R_S = R_3 + R_4 = 10 + 10 = 20 \, \Omega$$

$$\frac{1}{R_P} = \frac{1}{R_2} + \frac{1}{R_S}$$

$$= \frac{1}{20} + \frac{1}{20} = \frac{1}{10} \, \Omega$$

$$R_P = 10 \, \Omega$$

$$\text{Total equivalent resistance} = R = R_1 + R_P + R_5$$

$$= R = 20 + 10 + 10 = 40 \, \Omega$$

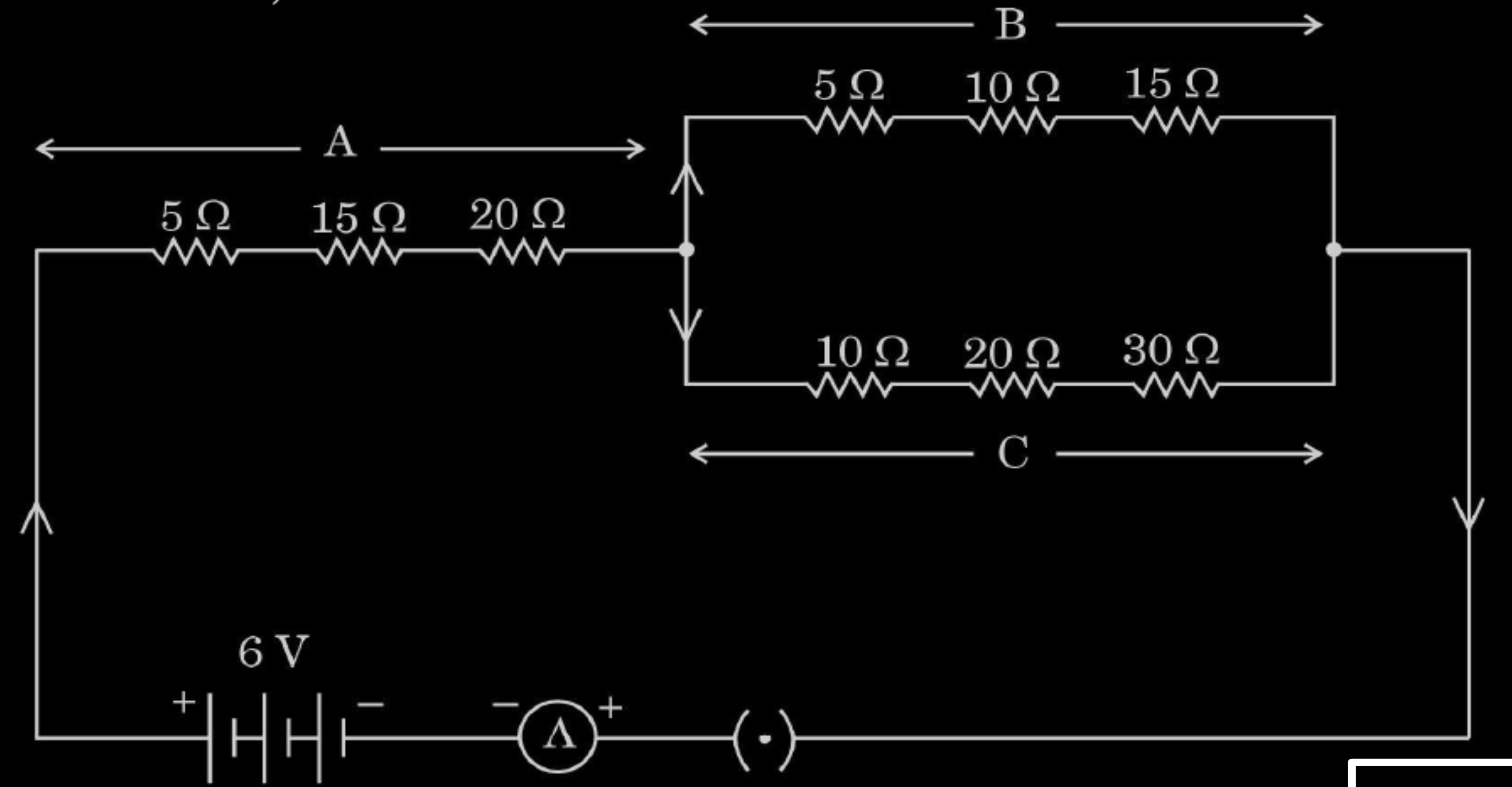
1/2

1/2

1/2

1/2

14. Study the following electric circuit in which the resistors are arranged in three arms A, B and C :



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- (a) Find the equivalent resistance of arm C. 1
- (b) Calculate the equivalent resistance of the parallel combination of the arms B and C. 1
- (c) (i) Determine the current that flows through the ammeter. 2

OR

- (ii) Determine the current that flows in the ammeter when the arm B is withdrawn from the circuit. 2

$$\begin{aligned}
 \text{(a)} \quad R &= R_1 + R_2 + R_3 \\
 &= 10\Omega + 20\Omega + 30\Omega \\
 &= 60\Omega
 \end{aligned}$$

1

$$\text{(b)} \quad \frac{1}{R_p} = \frac{1}{R_B} + \frac{1}{R_C}$$

$\frac{1}{2}$

$$\frac{1}{R_p} = \frac{1}{30\Omega} + \frac{1}{60\Omega}$$

$\frac{1}{2}$

$$R_p = 20\Omega$$

1

$$\begin{aligned}
 \text{(c) (i)} \quad R &= R_s + R_p \\
 &= 40\Omega + 20\Omega = 60\Omega
 \end{aligned}$$

$\frac{1}{2} + \frac{1}{2}$

$$\therefore I = \frac{V}{R} = \frac{6V}{60\Omega} = \frac{1}{10} A = 0.1 A$$

OR

(c) (ii)

1

$$\text{Resistance, } R = 40\Omega + 60\Omega = 100\Omega$$

$\frac{1}{2} + \frac{1}{2}$

$$\therefore I = \frac{V}{R} = \frac{6V}{100\Omega} = 0.06 A$$

13. (a) List the factors on which the resistance of a uniform cylindrical conductor of a given material depends. 2+1
- (b) The resistance of a wire of 0.01 cm radius is $10\ \Omega$. If the resistivity of the wire is $50 \times 10^{-8}\ \Omega\text{ m}$, find the length of this wire. CBSE 2022

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13. (a) List the factors on which the resistance of a uniform cylindrical conductor of a given material depends. 2+1
- (b) The resistance of a wire of 0.01 cm radius is 10 Ω . If the resistivity of the wire is $50 \times 10^{-8} \Omega \text{ m}$, find the length of this wire. CBSE 2022

(a)	(i) Length of the conductor (l)	1
	(ii) Area of cross-section of the conductor (A)	1
(b)	Radius of wire, $r = 0.01 \text{ cm} = 0.01 \times 10^{-2} \text{ m}$ Resistance, $R = 10 \Omega$ Resistivity, $\rho = 50 \times 10^{-8} \Omega \text{ m}$	
	$R = \rho \frac{l}{A} = \rho \frac{l}{\pi r^2} \Rightarrow l = \frac{R\pi r^2}{\rho}$	$\frac{1}{2}$
	$l = \frac{10\Omega \times 22 \times (0.01 \times 10^{-2})^2 \text{ m}}{50 \times 10^{-8} \Omega \text{ m}}$ $= \frac{22}{35} \text{ m} = 0.629 \text{ m} = 0.628 \text{ m} = 0.62 \text{ m}$	$\frac{1}{2}$

- (a) What is the meaning of electric power of an electrical device ? Write its SI unit. **1½**
- (b) An electric kettle of 2kW is used for 2h. Calculate the energy consumed in
- (i) kilowatt hour and
 - (ii) joules. **1½**

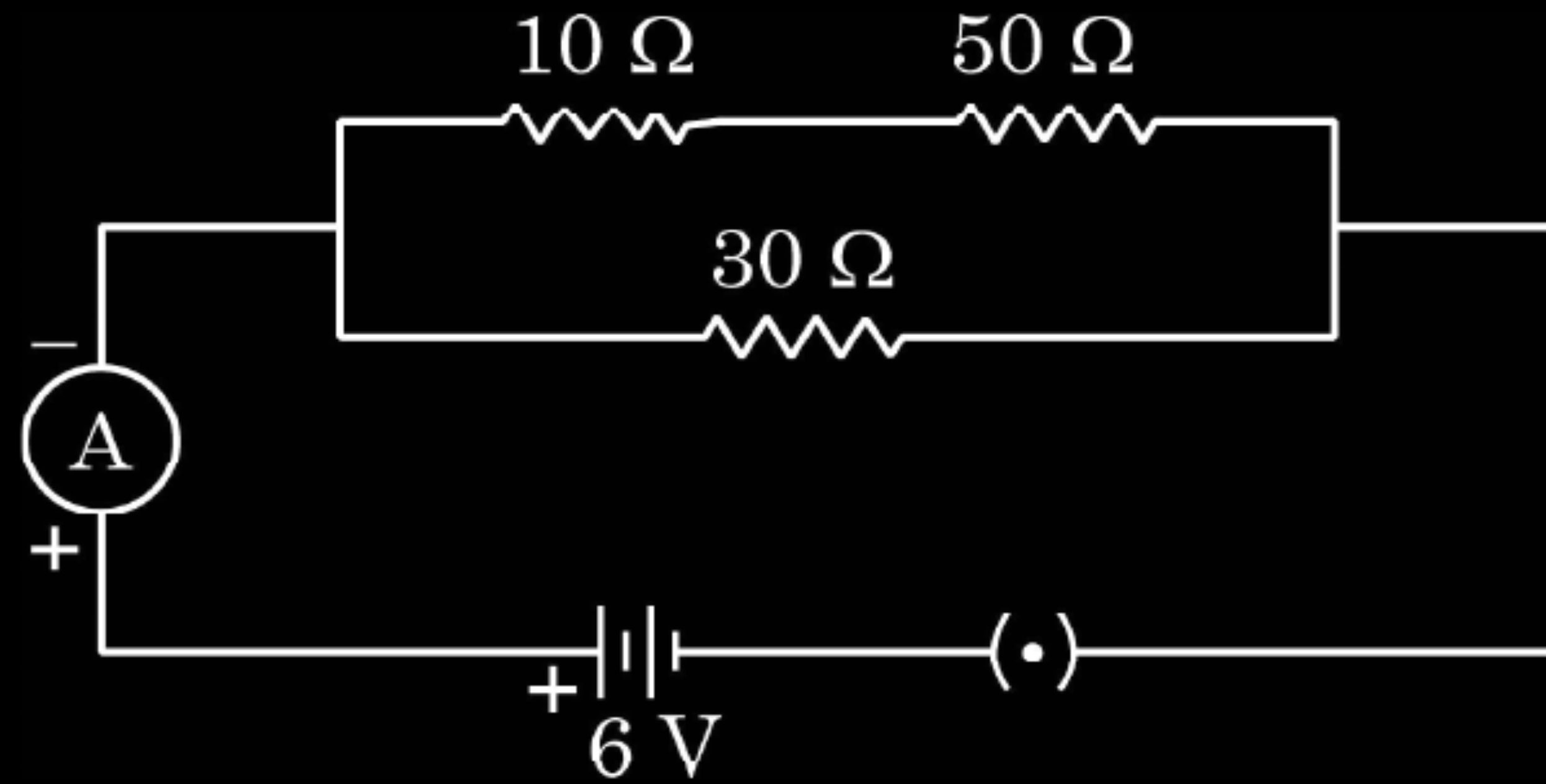
CBSE 2022

- (a) What is the meaning of electric power of an electrical device ? Write its SI unit. 1½
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CBSE 2022

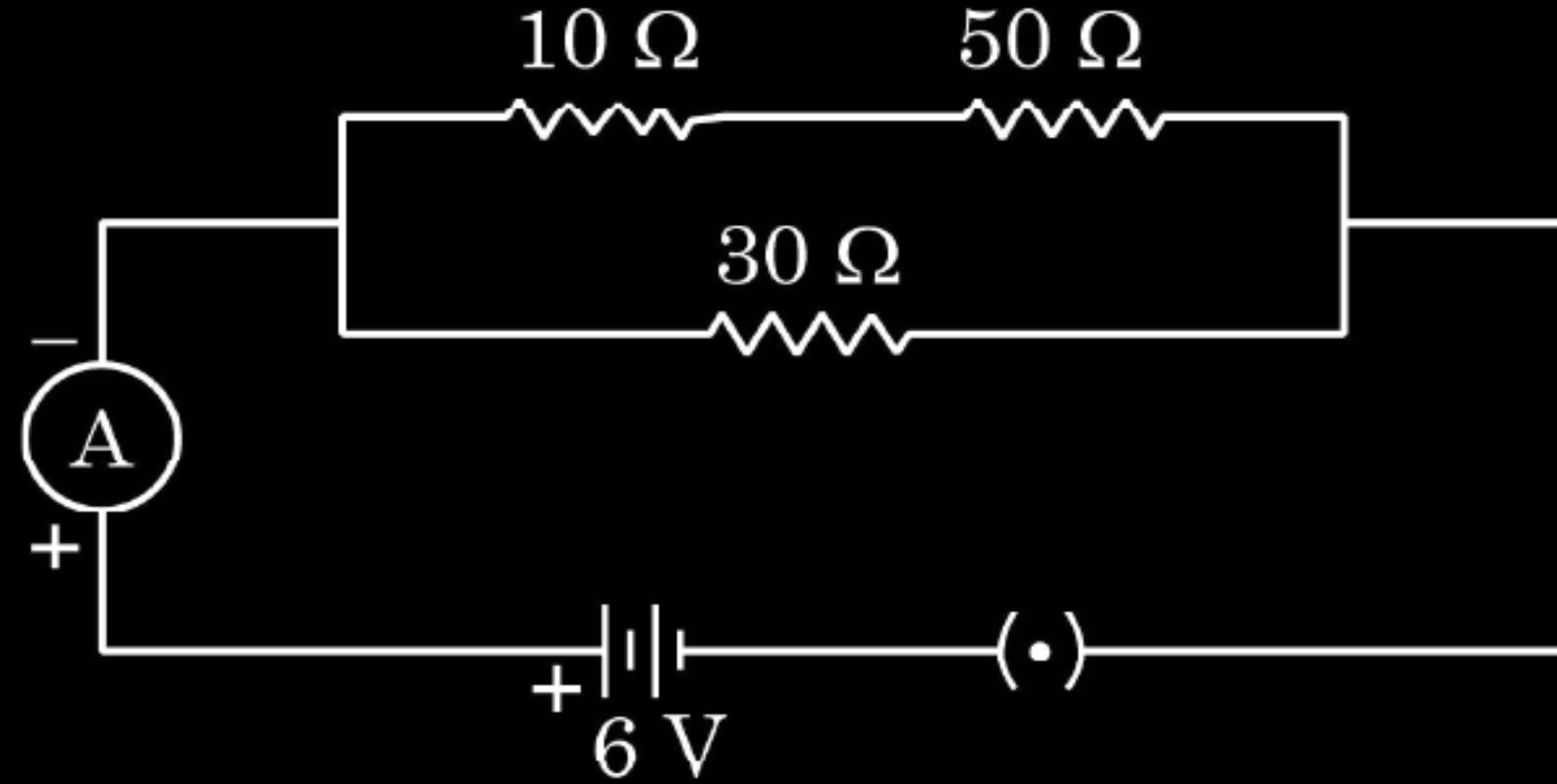
(a) Rate at which electric energy is dissipated / consumed in an electric circuit SI unit : watt / joule per second / volt . ampere	1 ½
(b) $E = P \times t$	½
(i) 2 kW × 2h = 4kWh	½
(ii) $4 \times 3.6 \times 10^6 \text{ joules} = 14.4 \times 10^6 \text{ J} / 1.44 \times 10^7 \text{ J}$	½

10. In the given circuit determine the value of :
- (i) total resistance of the circuit
 - (ii) current flowing through the ammeter.



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10. In the given circuit determine the value of :
- total resistance of the circuit
 - current flowing through the ammeter.



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$$(i) \quad R_S = R_1 + R_2$$

$$\text{Here } R_S = 10 \, \Omega + 50 \, \Omega = 60 \, \Omega$$

$$R_3 = 30 \, \Omega$$

$$\frac{1}{R_P} = \frac{1}{R_S} + \frac{1}{R_3}$$

$$\therefore R = \frac{R_S R_3}{R_S + R_3} = \frac{60 \, \Omega \times 30 \, \Omega}{(60 + 30) \, \Omega} = 20 \, \Omega$$

$$(ii) \quad I = \frac{V}{R}$$

$$= \frac{6 \, V}{20 \, \Omega} = 0.3 \, A$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

11. (a) State Ohm's Law. Represent it mathematically.

3

(b) Define 1 ohm.

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(c) What is the resistance of a conductor through which a current of 0.5 A flows when a potential difference of 2 V is applied across its ends ?

11. (a) State Ohm's Law. Represent it mathematically.

3

(b) Define 1 ohm.

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(c) What is the resistance of a conductor through which a current of 0.5 A flows when a potential difference of 2 V is applied across its ends ?

(a) The potential difference(V) across the ends of a given metallic wire or conductor is directly proportional to the current(I) flowing through it, provided the temperature / physical conditions of the conductor remains the same.

$$V \propto I$$

$$V/I = \text{Constant}$$

$$V = IR$$

$\frac{1}{2}$

$\frac{1}{2}$

(b) If the potential difference across the two ends of a conductor is 1 V and the current through it is 1 A, then the resistance R of the conductor is 1 ohm.

1

$$\text{Alternative answer: } 1 \text{ ohm} = \frac{1 \text{ volt}}{1 \text{ ampere}}$$

(c)

$$\bullet R = \frac{V}{I}$$

$$R = \frac{2V}{0.5 \text{ A}}$$

$\frac{1}{2}$

$\frac{1}{2}$

• 4 ohm

10. For a heater, rated 4 kW and 220 V, calculate the following :

3

(a) The current

(b) Energy consumed in 2 hours

(c) If 1 kWh is priced at ₹ 4.50, then the cost of energy consumed

CBSE 2022

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3

(a) The current

(b) Energy consumed in 2 hours

(c) If 1 kWh is priced at ₹ 4.50, then the cost of energy consumed

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Given : $P = 4 \text{ kW} = 4000 \text{ W}$

$V = 220 \text{ V}$

(a)

$$P = VI$$

$$I = \frac{P}{V} = \frac{4000 \text{ W}}{220 \text{ V}} = 18.18 \text{ A}$$

(b)

$$E = P \times t$$

$$= 4 \text{ kW} \times 2 \text{ h} = 8 \text{ kWh}$$

(c) Cost of 1 kWh = ₹ 4.50

$$\text{Then } 8 \text{ kWh} = 8 \times 4.50 = ₹ 36.0$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

13. (a) (i) Define Electric Power and write its SI unit.

$$\frac{1}{2} + \frac{1}{2}$$

(ii) Two bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

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2

13. (a) (i) Define Electric Power and write its SI unit.

$$\frac{1}{2} + \frac{1}{2}$$

(ii) Two bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

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2

(a) (i)

The rate at which electric energy is dissipated or consumed in an electric circuit.

S.I. unit—watt / V.A / joule per second

$\frac{1}{2}$

$\frac{1}{2}$

(a) (ii) • Current drawn by first bulb

$$I_1 = \frac{100 \text{ W}}{220 \text{ V}} = \frac{100}{220} \text{ ampere}$$

$\frac{1}{2}$

• Current drawn by second bulb

$$I_2 = \frac{60 \text{ W}}{220 \text{ V}} = \frac{60}{220} \text{ ampere}$$

$\frac{1}{2}$

$\frac{1}{2}$

Both the bulbs are in parallel

$$\text{Total current, } I = I_1 + I_2$$

$\frac{1}{2}$

$$= \left(\frac{100}{220} + \frac{60}{220} \right) \text{ ampere} = \frac{160}{220} \text{ A} = 0.73 \text{ A}$$

(Accept any other method)

(b) (i) State Joule's law of heating. Express it mathematically when an appliance of resistance R is connected to a source of voltage V and the current I flows through the appliance for a time t .

(ii) A $5\ \Omega$ resistor is connected across a battery of 6 volts.

Calculate the energy that dissipates as heat in 10 s.

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3

(b) i) This law states that heat produced in a resistor is—

- directly proportional to the square of current for a given resistance / $(H \propto I^2)$
- directly proportional to the resistance for a given current / $(H \propto R)$
- directly proportional to the time for which the current flows through the resistor / $(H \propto t)$
- $H = V I t$

ii)

$$V = 6\ \text{V}; R = 5\ \Omega; t = 10\ \text{s}$$

Energy dissipated as heat in $t = 10\ \text{s}$ is

$$H = \frac{V^2}{R} t$$

$$= \frac{(6\ \text{V})^2}{5\ \Omega} \times 10\ \text{s}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

10. (a) Calculate the resistance of a metal wire of length 2 m and area of cross-section $1.55 \times 10^{-6} \text{ m}^2$.
(Resistivity of the metal is $2.8 \times 10^{-8} \Omega \text{ m}$)

(b) Why are alloys preferred over pure metals to make the heating elements of electrical heating devices ?

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2+1

10. (a) Calculate the resistance of a metal wire of length 2 m and area of cross-section $1.55 \times 10^{-6} \text{ m}^2$.
(Resistivity of the metal is $2.8 \times 10^{-8} \Omega \text{ m}$)
- (b) Why are alloys preferred over pure metals to make the heating elements of electrical heating devices ?

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2+1

(a) Given :

Length of the wire = 2 m

Area of cross section = $1.55 \times 10^{-6} \text{ m}^2$

Resistivity, $\rho = 2.8 \times 10^{-8} \Omega \text{ m}$

$$R = \rho \frac{l}{a}$$

$$= \frac{2.8 \times 10^{-8} \text{ ohm metre} \times 2 \text{ metre}}{1.55 \times 10^{-6} (\text{metre})^2} \Omega$$

$$= 3.6 \times 10^{-2} \Omega$$

(deduct half mark if no or incorrect unit is given)

(b) Alloys used for heating element have generally high melting point / high resistivity / Do not get oxidised at high temperature.

1/2

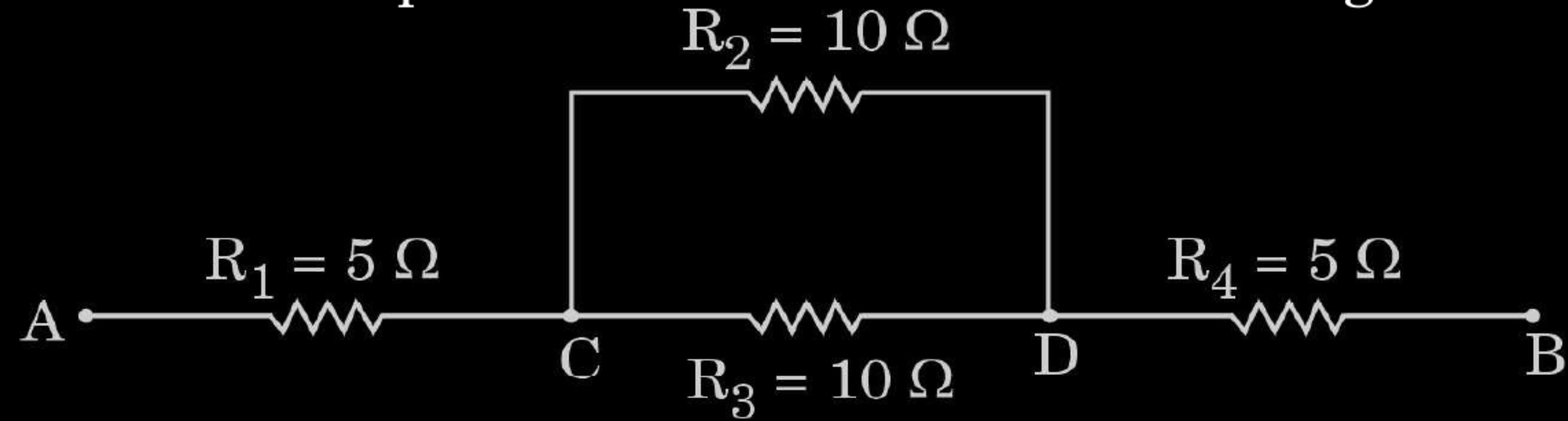
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1

1

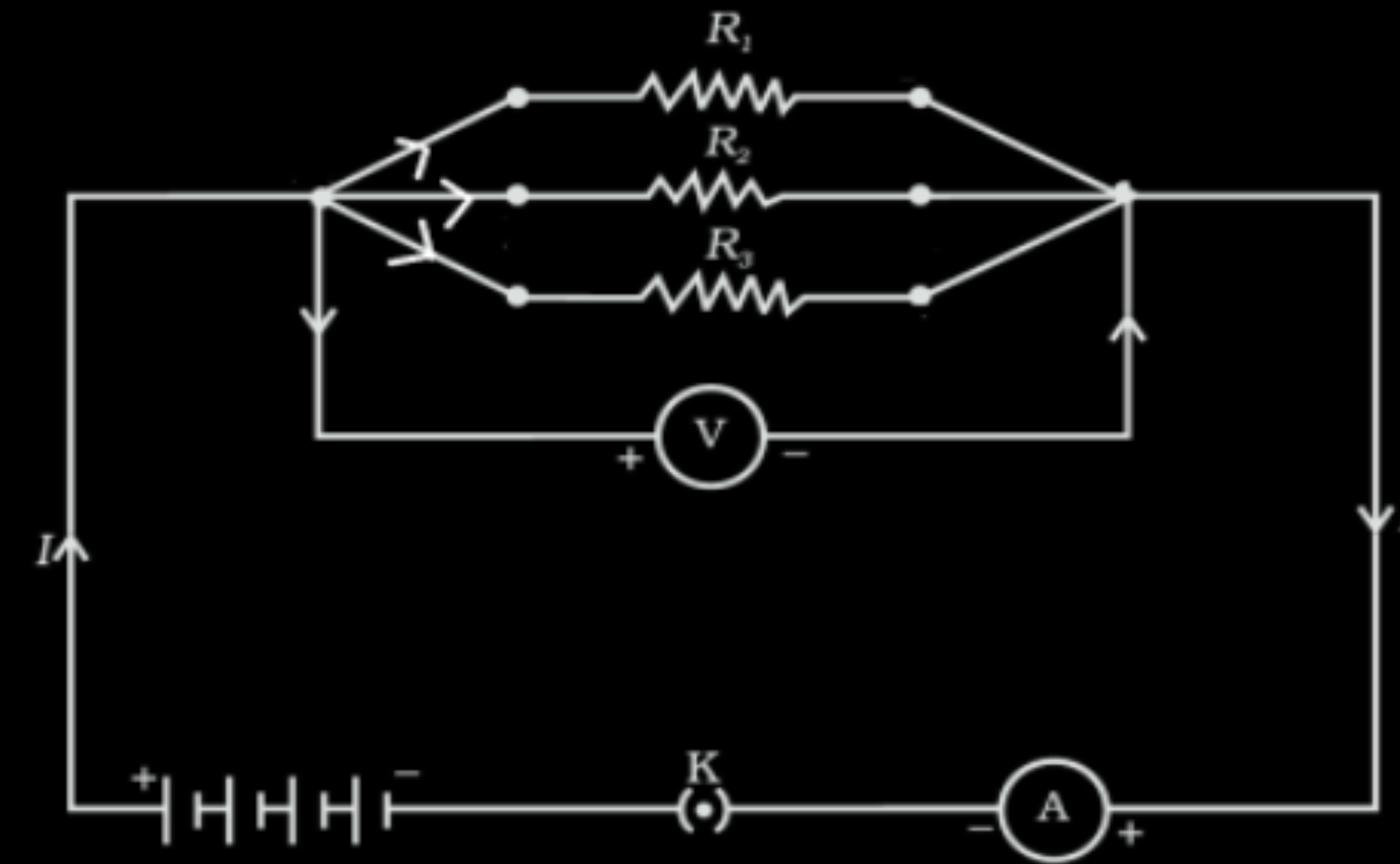
11. (a) Three resistors R_1 , R_2 and R_3 are connected in parallel and the combination is connected to a battery, an ammeter, a voltmeter and a key. Draw suitable circuit diagram to show the arrangement of these circuit components along with the direction of current flowing.
- (b) Calculate the equivalent resistance of the following network :

3



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(a)



Circuit diagram with given components

Direction

(b) Resistance between C and D is given by

$$\frac{1}{R_{CD}} = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \frac{1}{5}$$

$$R_{CD} = 5 \Omega$$

$$D \text{ and } B = R_4 = 5 \Omega$$

$$\therefore \text{Total resistance is } R_S = R_{CD} + R_1 + R_4$$

$$R_{\text{total}} = 5 \Omega + 5 \Omega + 5 \Omega$$

$$= 15 \Omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

6. The values of mA and μA are

1

- (A) 10^{-6} A and 10^{-9} A respectively
- (B) 10^{-3} A and 10^{-6} A respectively
- (C) 10^{-3} A and 10^{-9} A respectively
- (D) 10^{-6} A and 10^{-3} A respectively

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6. The values of mA and μA are

1

- (A) 10^{-6} A and 10^{-9} A respectively
- (B) 10^{-3} A and 10^{-6} A respectively
- (C) 10^{-3} A and 10^{-9} A respectively
- (D) 10^{-6} A and 10^{-3} A respectively

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(B) / 10^{-3} A and 10^{-6} A respectively

7. A cylindrical conductor of length ' l ' and uniform area of cross-section ' A ' has resistance ' R '. Another conductor of length $2.5\ l$ and resistance $0.5\ R$ of the same material has area of cross-section

- (A) $5\ A$
- (B) $2.5\ A$
- (C) $0.5\ A$
- (D) $\frac{1}{5}\ A$

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7. A cylindrical conductor of length ' l ' and uniform area of cross-section ' A ' has resistance ' R '. Another conductor of length $2.5\ l$ and resistance $0.5\ R$ of the same material has area of cross-section

1

- (A) $5\ A$
- (B) $2.5\ A$
- (C) $0.5\ A$
- (D) $\frac{1}{5}\ A$

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(A) / 5A

- 29.** (a) Two lamps rated 100 W, 220 V and 10 W, 220 V are connected in parallel to 220 V supply. Calculate the total current through the circuit.
- (b) Two resistors X and Y of resistances $2\ \Omega$ and $3\ \Omega$ respectively are first joined in parallel and then in series. In each case the voltage supplied is 5 V.
- (i) Draw circuit diagrams to show the combination of resistors in each case.
- (ii) Calculate the voltage across the $3\ \Omega$ resistor in the series combination of resistors.

$$(a) I_1 = \frac{P_1}{V}$$

$$I_1 = \frac{100 \text{ W}}{220 \text{ V}} = \frac{10}{22} \text{ A}$$

$$I_2 = \frac{P_2}{V} = \frac{10}{220} = \frac{1}{22} \text{ A}$$

$$I = I_1 + I_2$$

$$= \left(\frac{10}{22} + \frac{1}{22} \right) \text{ A} = \frac{11}{22} \text{ A} = 0.5 \text{ A}$$

$$(ii) \text{ Net } R = R_1 + R_2 = 2 + 3 = 5 \Omega$$

$$I = \frac{V}{R_{net}} = \frac{5}{5} = 1 \text{ A}$$

∴ Voltage across 3Ω resistor :

$$\therefore V = 1 \times 3 = 3 \text{ V}$$

1/2

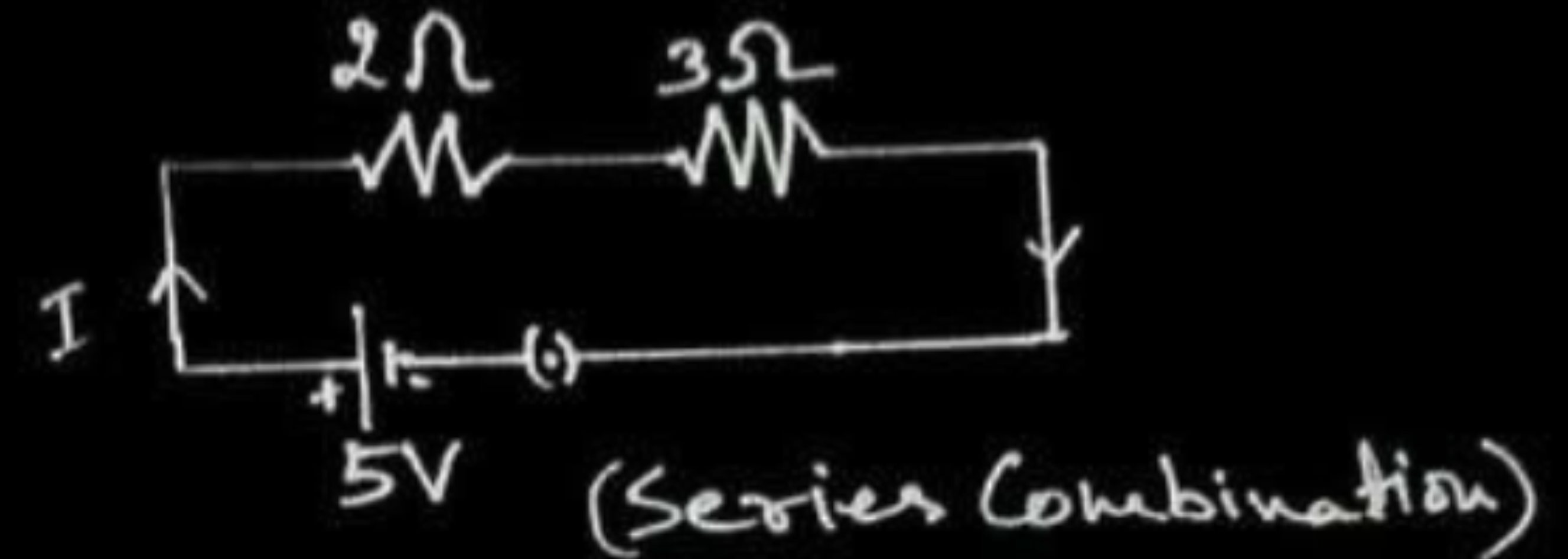
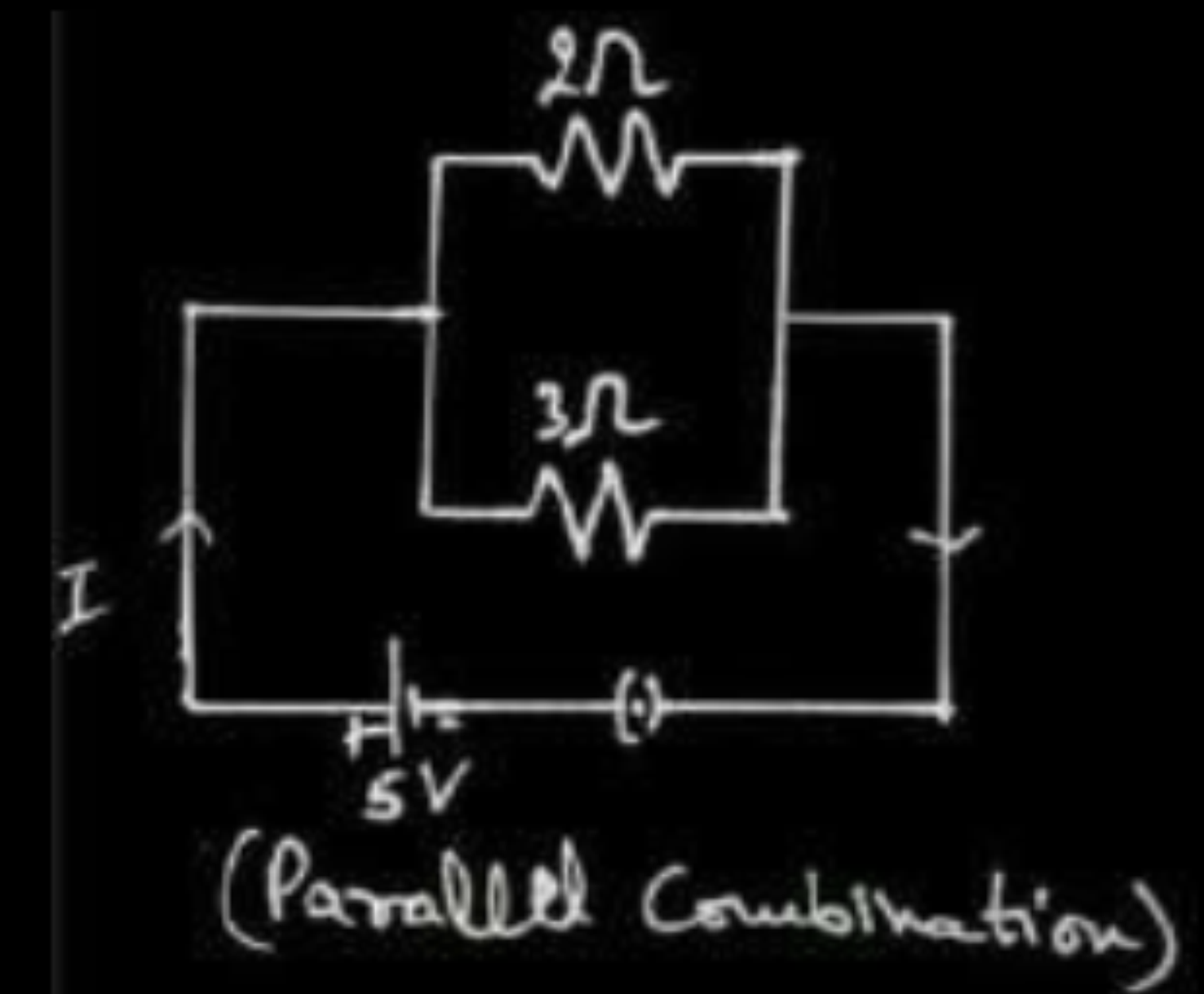
1/2

1/2

1

1

1/2



8. If a person has five resistors each of value $\frac{1}{5} \Omega$, then the maximum resistance he can obtain by connecting them is *1*

(A) 1Ω

(B) 5Ω

(C) 10Ω

(D) 25Ω

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8. If a person has five resistors each of value $\frac{1}{5} \Omega$, then the maximum resistance he can obtain by connecting them is 1

(A) 1Ω

(B) 5Ω

(C) 10Ω

(D) 25Ω

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(A) 1Ω

The resistance of a resistor is reduced to half of its initial value. In doing so, if other parameters of the circuit remain unchanged, the heating effects in the resistor will become

1

- (A) two times.
- (B) half.
- (C) one-fourth.
- (D) four times.

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- (D) four times.

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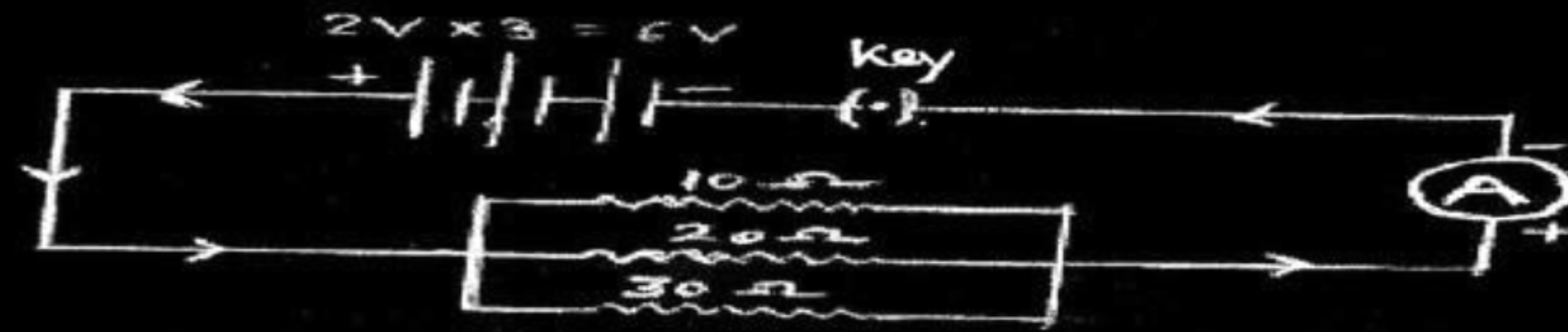
(B) /half

29. Draw a schematic diagram of a circuit consisting of a battery of 3 cells of 2 V each, a combination of three resistors of $10\ \Omega$, $20\ \Omega$ and $30\ \Omega$ connected in parallel, a plug key and an ammeter, all connected in series. Use this circuit to find the value of the following :

- (a) Current through each resistor
- (b) Total current in the circuit
- (c) Total effective resistance of the circuit

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- (a) Current through each resistor
- (b) Total current in the circuit
- (c) Total effective resistance of the circuit



2

(a) Three resistors are connected in parallel hence voltage across each is same i.e. 6V.

$$I_1 = \frac{V}{R_1} = \frac{6}{10} = 0.6 \text{ A}$$

$\frac{1}{2}$

$$I_2 = \frac{V}{R_2} = \frac{6}{20} = 0.3 \text{ A}$$

$\frac{1}{2}$

$$I_3 = \frac{V}{R_3} = \frac{6}{30} = 0.2 \text{ A}$$

$\frac{1}{2}$

$$\text{b) } I = I_1 + I_2 + I_3 = 1.1 \text{ A}$$

$\frac{1}{2}$

$$\text{c) } R_{\text{eff}} = \frac{V}{I}$$

$\frac{1}{2}$

$$= \frac{6}{1.1} = 5.4 \Omega$$

$\frac{1}{2}$

Two identical resistors, each of resistance $15\ \Omega$, are connected in (i) series, and (ii) parallel, in turn to a battery of $6\ \text{V}$. Calculate the ratio of the power consumed in the combination of resistors in each case.

5

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Two identical resistors, each of resistance $15\ \Omega$, are connected in (i) series, and (ii) parallel, in turn to a battery of 6 V . Calculate the ratio of the power consumed in the combination of resistors in each case.

5

$R_1 = R_2 = 15\Omega \quad V = 6V$	
i) In series :	
$R_s = R_1 + R_2 = 15\Omega + 15\Omega = 30\Omega$	$\frac{1}{2}$
$I = \frac{V}{R_s} = \frac{6V}{30\Omega} = 0.2\text{ A}$	$\frac{1}{2}$
$\therefore P_1 = VI = 6V \times 0.2\text{ A} = 1.2\text{ W}$	1
ii) In parallel	
$R_p = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{15 \times 15}{15 + 15} = \frac{225}{30} = 7.5\Omega$	$\frac{1}{2}$
$I = \frac{V}{R_p} = \frac{6V}{7.5\Omega} = 0.8\text{ A}$	$\frac{1}{2}$
$P_2 = VI = 6V \times 0.8\text{ A} = 4.8W$	1
Ratio of power = $\frac{P_1}{P_2} = \frac{1.2W}{4.8W} = \frac{1}{4}$	$\frac{1}{2}$
$\therefore P_1 : P_2 = 1 : 4$	$\frac{1}{2}$

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6. A cylindrical conductor of length ' l ' and uniform area of cross section ' A ' has resistance ' R '. The area of cross section of another conductor of same material and same resistance but of length ' $2l$ ' is

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- (a) $\frac{A}{2}$
- (b) $\frac{3A}{2}$
- (c) $2A$
- (d) $3A$

6. A cylindrical conductor of length ' l ' and uniform area of cross section ' A ' has resistance ' R '. The area of cross section of another conductor of same material and same resistance but of length ' $2l$ ' is

CBSE 2020

- (a) $\frac{A}{2}$
- (b) $\frac{3A}{2}$
- (c) $2A$
- (d) $3A$

1

(c) / 2 A

7. The maximum resistance which can be made using four resistors each of resistance $\frac{1}{2} \Omega$ is

- (a) 2Ω
- (b) 1Ω
- (c) 2.5Ω
- (d) 8Ω

CBSE 2020

7. The maximum resistance which can be made using four resistors each of resistance $\frac{1}{2} \Omega$ is

- (a) 2Ω
- (b) 1Ω
- (c) 2.5Ω
- (d) 8Ω

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1

(a) / 2Ω

14. **Assertion (A)** : Alloys are commonly used in electrical heating devices like electric iron and heater.

Reason (R) : Resistivity of an alloy is generally higher than that of its constituent metals but the alloys have low melting points than their constituent metals.

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14. **Assertion (A)** : Alloys are commonly used in electrical heating devices like electric iron and heater.

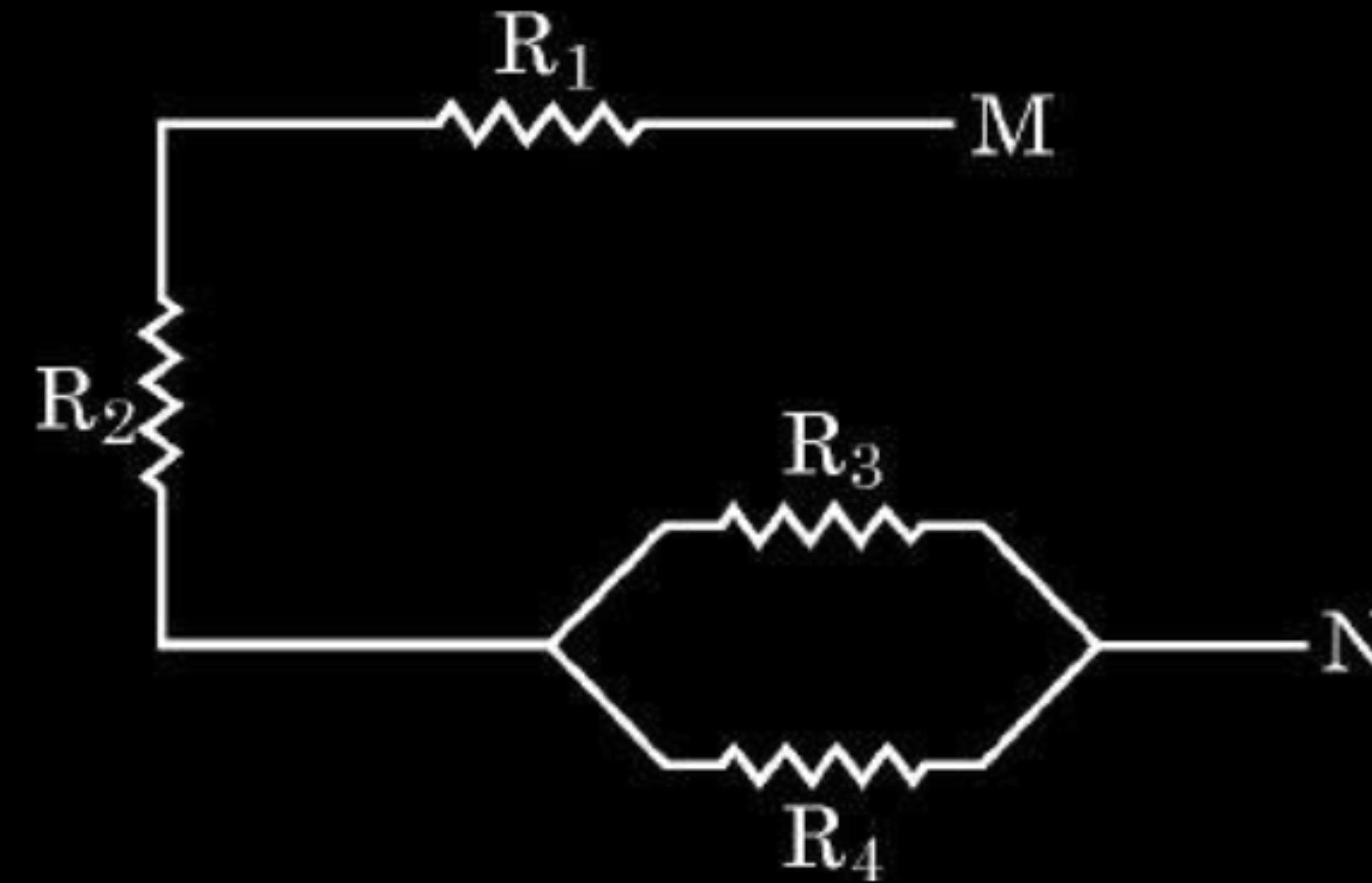
Reason (R) : Resistivity of an alloy is generally higher than that of its constituent metals but the alloys have low melting points than their constituent metals.

1

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(c) / A is true but R is false.

29. (a) For the combination of resistors shown in the following figure, find the equivalent resistance between M & N.



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- (b) State Joule's law of heating.
(c) Why we need a 5 A fuse for an electric iron which consumes 1 kW power at 220 V ?
(d) Why is it impracticable to connect an electric bulb and an electric heater in series ?

(a) R_3 and R_4 are in parallel combination . $\therefore R_{\text{parallel}}$ is given by $\frac{1}{R_p} = \frac{1}{R_3} + \frac{1}{R_4}$ $\frac{1}{R_p} = \frac{R_4 + R_3}{R_3 R_4}$ $R_p = \frac{R_3 R_4}{R_4 + R_3}$ Now, R_1 R_2 and R_p are in series. $\therefore R_{\text{eq}} = R_1 + R_2 + R_p$ $= R_1 + R_2 + \frac{R_3 R_4}{R_4 + R_3}$	1
(b) The heat produced in a resistor is directly proportional to • square of current for a given resistance. • the resistance for a given current and • the time for which the current flows through the resistor. (Note : if a candidate writes $H = I^2 R t$ give full credit).	1
(c) $P = V I$ or $I = \frac{P}{V}$ $I = \frac{1000 \text{ watt}}{220 \text{ volt}} = 4.54 \text{ A}$ Since 4.54 ampere current flows in the circuit , a 5 A fuse must be used.	1
(d) Electric bulb & electric heater will not get currents and voltages as per their requirement.	1

6. The maximum resistance which can be made using four resistors each of $2\ \Omega$ is

(a) $2\ \Omega$

(b) $4\ \Omega$

(c) $8\ \Omega$

(d) $16\ \Omega$

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1

6. The maximum resistance which can be made using four resistors each of $2\ \Omega$ is

(a) $2\ \Omega$

(b) $4\ \Omega$

(c) $8\ \Omega$

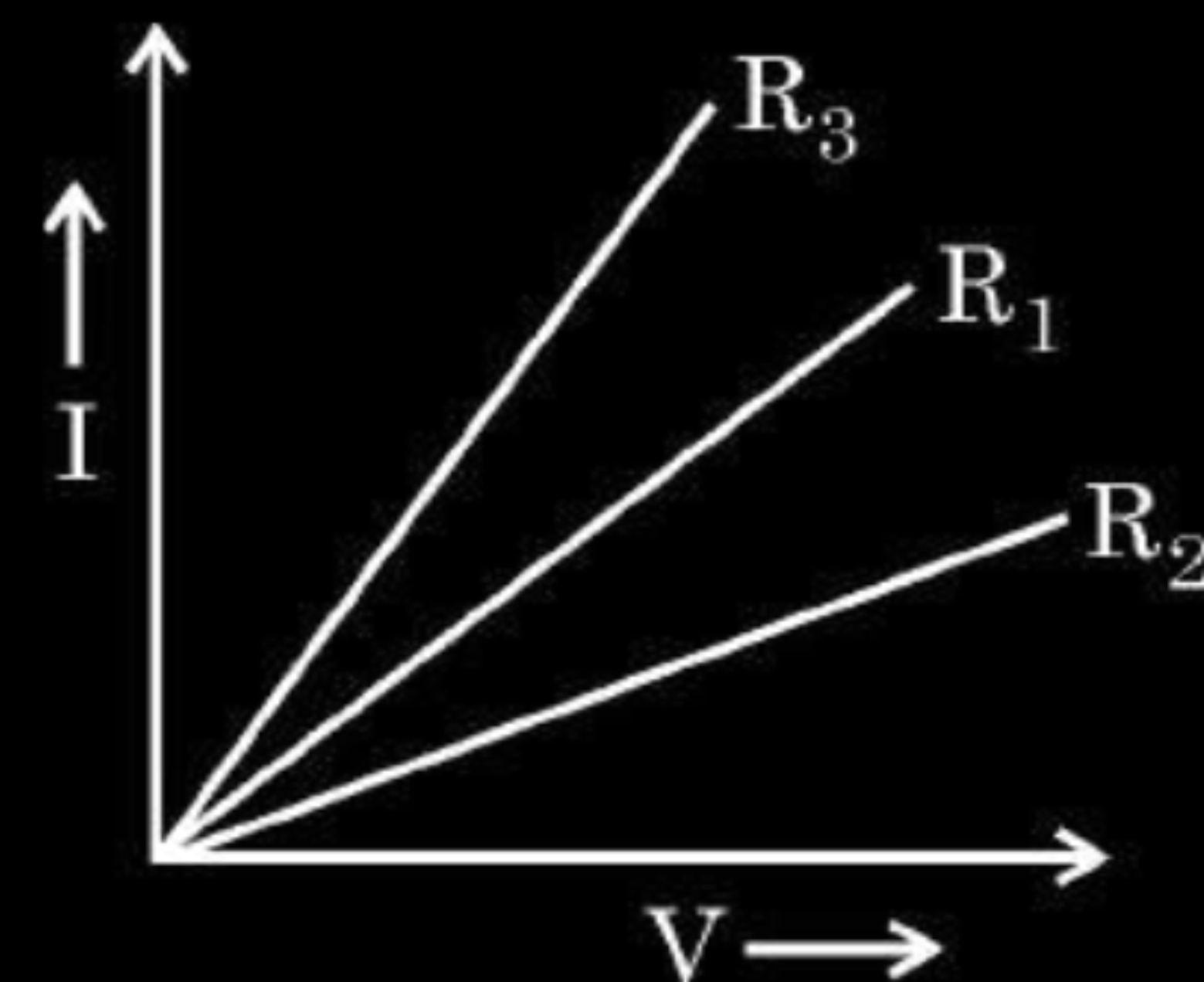
(d) $16\ \Omega$

CBSE 2020

1

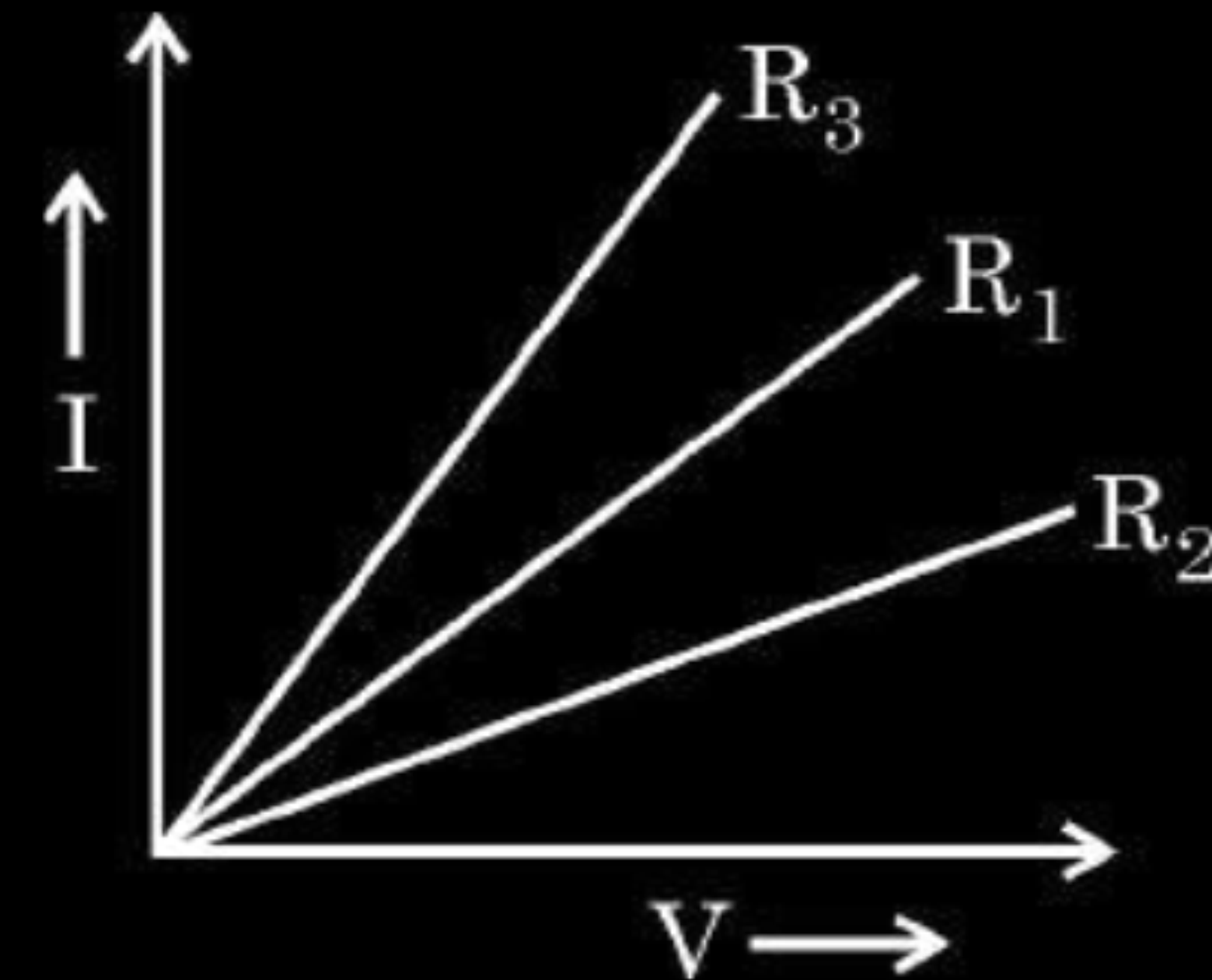
(c) / 8Ω

7. A student plots V-I graphs for three samples of nichrome wire with resistances R_1 , R_2 and R_3 . Choose from the following the statement that holds true for this graph.



- (a) $R_1 = R_2 = R_3$
- (b) $R_1 > R_2 > R_3$
- (c) $R_3 > R_2 > R_1$
- (d) $R_2 > R_1 > R_3$

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- (a) $R_1 = R_2 = R_3$
(b) $R_1 > R_2 > R_3$
(c) $R_3 > R_2 > R_1$
(d) $R_2 > R_1 > R_3$

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1

(d) / $R_2 > R_1 > R_3$

29. (a) Find the ratio of resistances of two copper rods X and Y of lengths 30 cm and 10 cm respectively and having radii 2 cm and 1 cm respectively.
- (b) A current of 500 mA flows in a series circuit containing an electric lamp and a conductor of $10\ \Omega$ when connected to 6 V battery. Find the resistance of the electric lamp.

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(a) $R = \rho \frac{l}{A}$ $R_x = \rho \frac{l_x}{A_x} = \frac{30}{\pi \times 2 \times 2}$ $R_y = \rho \frac{l_y}{A_y} = \rho \frac{10}{\pi \times 1 \times 1}$ $\frac{R_x}{R_y} = \frac{\rho \cdot l_x}{A_x} \times \frac{A_y}{\rho \cdot l_y}$ $= \frac{30}{\pi \times 2 \times 2} \times \frac{\pi \times 1 \times 1}{10}$ $\frac{R_x}{R_y} = \frac{3}{4}$ $R_x : R_y = 3:4$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
(b) Here $I = 500 \text{ mA}$; Resistance of lamp = R_1 Resistance of the conductor (R_2) = 10Ω Pot. Difference (V) = 6 volt Net resistance of the circuit $R = R_1 + R_2 = \frac{V}{I}$ $R_1 + 10 \Omega = \frac{6}{0.5}$ $R_1 + 10 \Omega = 12 \Omega$ $R_1 = (12 - 10) \Omega$	$\frac{1}{2}$ $1 \frac{1}{2}$ 1

Two bulbs of 100 W and 40 W are connected in series. The current through the 100 W bulb is 1 A. The current through the 40 W bulb will be :

(a) 0.4 A

(b) 0.6 A

(c) 0.8 A

(d) 1 A

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Two bulbs of 100 W and 40 W are connected in series. The current through the 100 W bulb is 1 A. The current through the 40 W bulb will be :

(a) 0.4 A

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(c) 0.8 A

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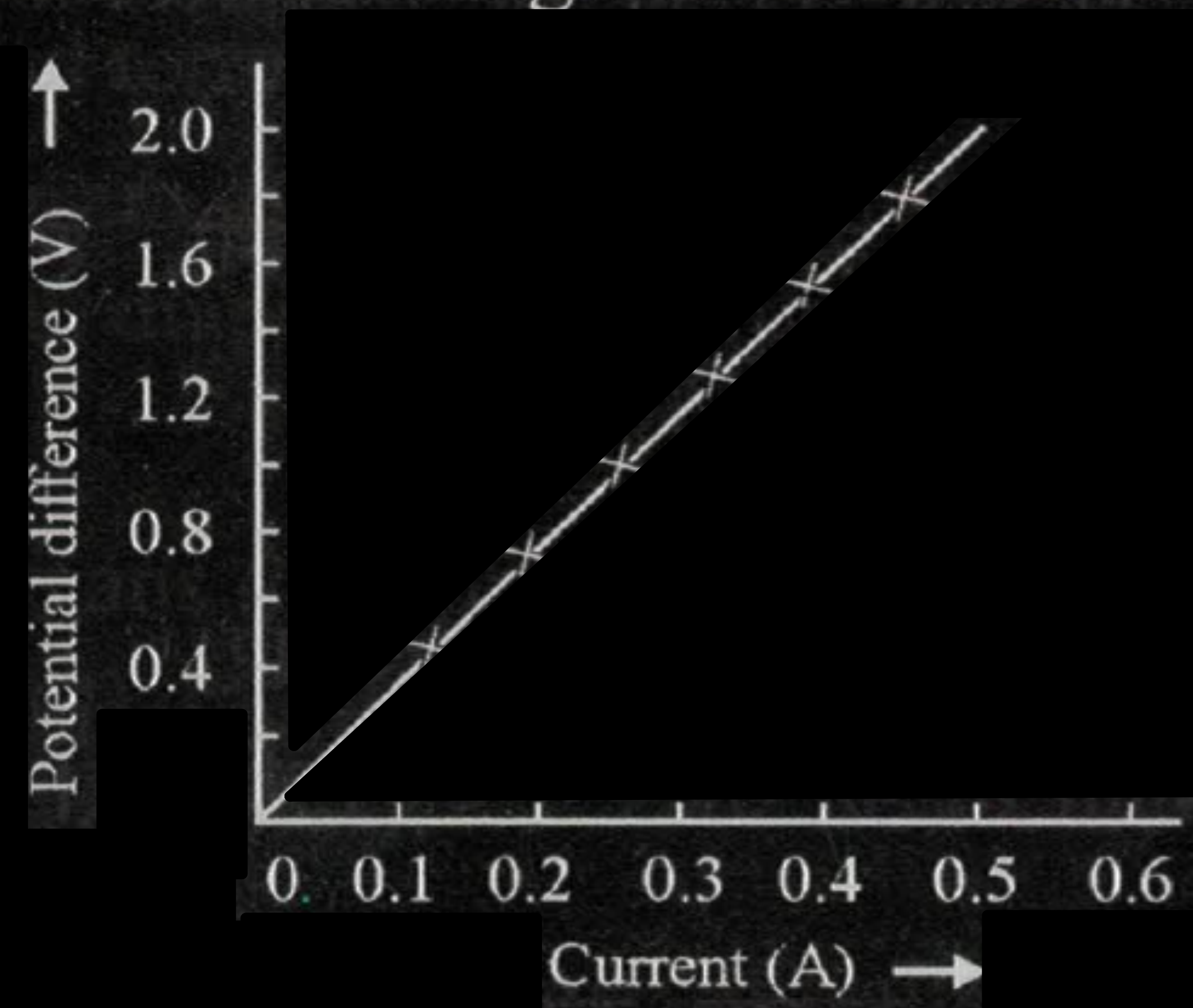
CBSE 2020

1

(d) / 1A

23. A V-I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labelled circuit diagram to obtain such a graph.

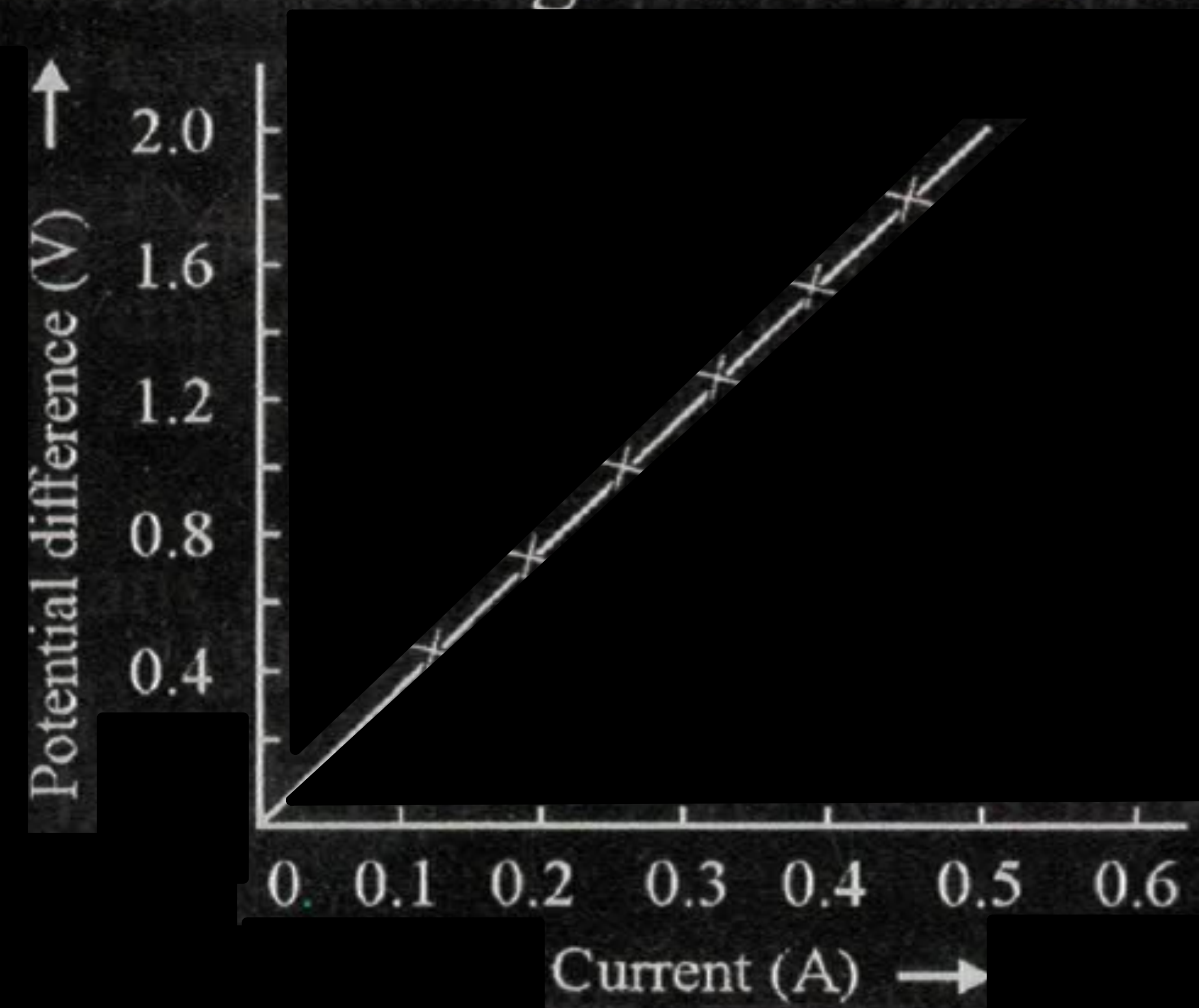
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23. A V-I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labelled circuit diagram to obtain such a graph.

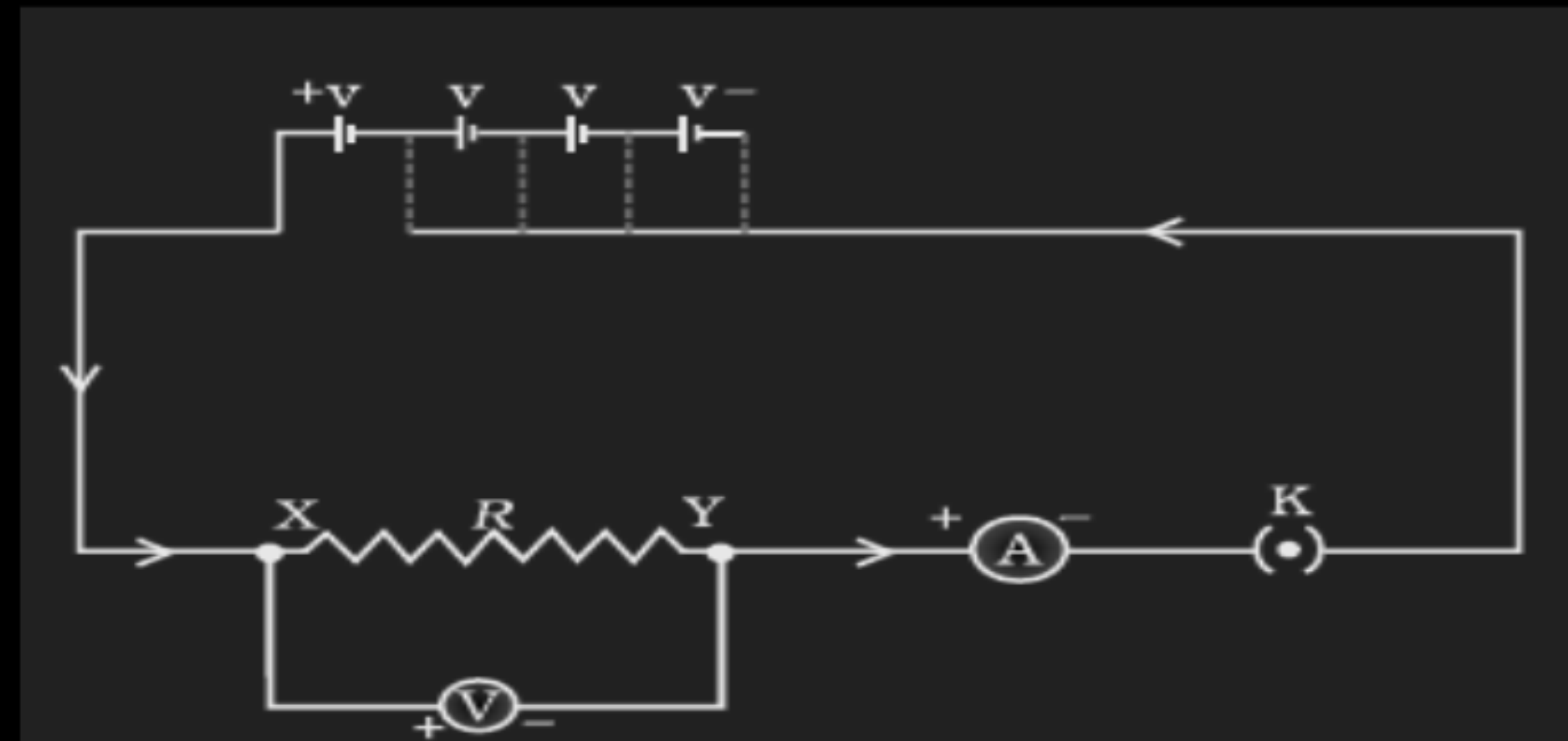
3



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$V \propto I$ or Potential difference is directly proportional to current

1



2

Note: If circuit diagram is correct but labelling of ammeter and voltmeter are incorrect, deduct 1 mark.

24. (a) Write the mathematical expression for Joule's law of heating

(b) Compute the heat generated while transferring 96000 coulomb of charge in two hours through a potential difference of 40 V.

24. (a) Write the mathematical expression for Joule's law of heating.
- (b) Compute the heat generated while transferring 96000 coulomb of charge in two hours through a potential difference of 40 V.

3

i) $H = I^2 R t$

ii) $H = V.I.t$
 $= V.Q$

Given : $V = 40$ volts , $Q = 96000$ C

$H = 40 \text{ V} \times 96000 \text{ C}$

$= 3.84 \times 10^6 \text{ J}$

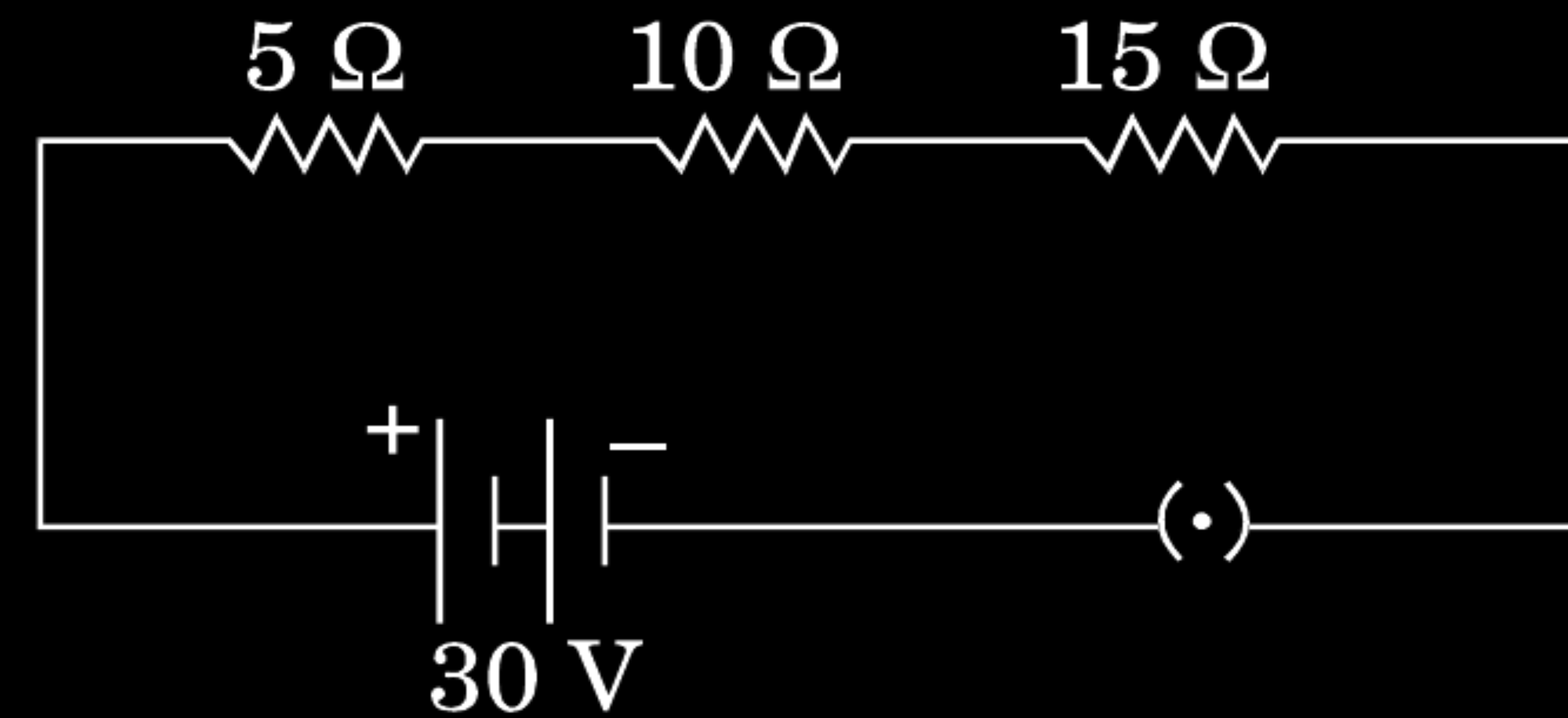
1

 $\frac{1}{2}$

1

 $\frac{1}{2}$

- (a) How will you infer with the help of an experiment that the same current flows through every part of a circuit containing three resistors in series connected to a battery ?
- (b) Consider the given circuit and find the current flowing in the circuit and potential difference across the $15\ \Omega$ resistor when the circuit is closed.



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- (a)
- (i) Join the three resistors of different values in series
 - (ii) Connect them with battery, an ammeter and plug key.
 - (iii) Plug the key and note the ammeter reading
 - (iv) Change the position of ammeter to anywhere in between the resistors and note the ammeter reading each time.
 - (v) The ammeter reading will remain same everytime. Therefore when resistors are connected in series same current flows through all resistors, when it is connected to a battery.

Note: If explained with the help of diagram give full credit

- (b) Total resistance of the circuit =

$$R = R_1 + R_2 + R_3 = 5 + 10 + 15 = 30 \text{ ohm}$$

Potential difference across the circuit / By ohm's law

$$V = IR \text{ or } I = \frac{V}{R} = \frac{30V}{30\text{ohm}} = 1A$$

$$\text{Potential difference across 15 ohm Resistor} = 1A \times 15 \text{ ohm} = 15 \text{ volt}$$

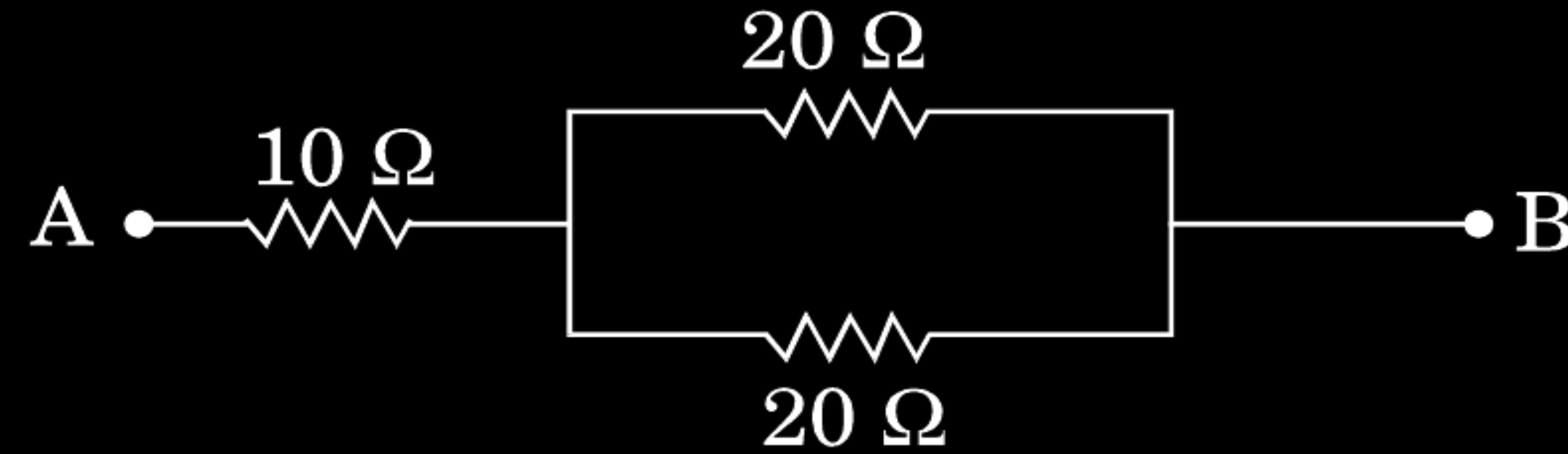
$\frac{1}{2} \times 5$

1

1

$\frac{1}{2}$

- (a) Three resistors R_1 , R_2 and R_3 are connected in parallel and the combination is connected to a battery, ammeter, voltmeter and key. Draw suitable circuit diagram and obtain an expression for the equivalent resistance of the combination of the resistors.
- (b) Calculate the equivalent resistance of the following network :



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(a)

Total current $I = I_1 + I_2 + I_3$

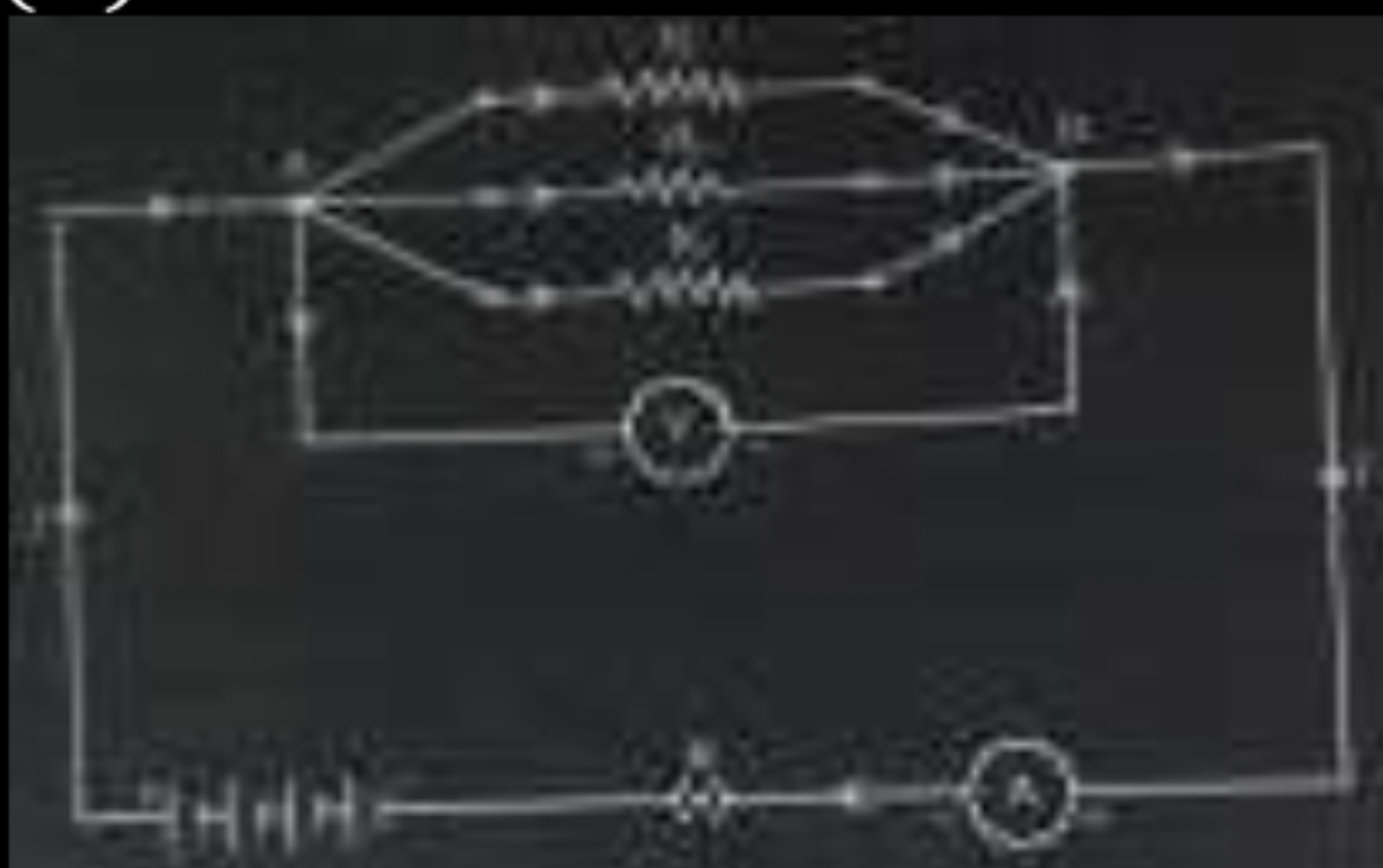
Let R_p be the equivalent resistance of R_1, R_2, R_3 . Then the total current $I = \frac{V}{R_p}$

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}$$

$$\therefore I = V \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) = \frac{V}{R_p}$$

$$\therefore \frac{1}{R_p} + \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(b)



$$\frac{1}{R_p} = \frac{1}{20} + \frac{1}{20} = \frac{2}{20} = \frac{1}{10}$$

$$\Rightarrow R_p = 10 \text{ ohms}$$

$$\text{Equivalent resistance of the network} = R_{eq} = R_1 + R_p = 10 + 10 = 20 \text{ ohm}$$

$$1$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

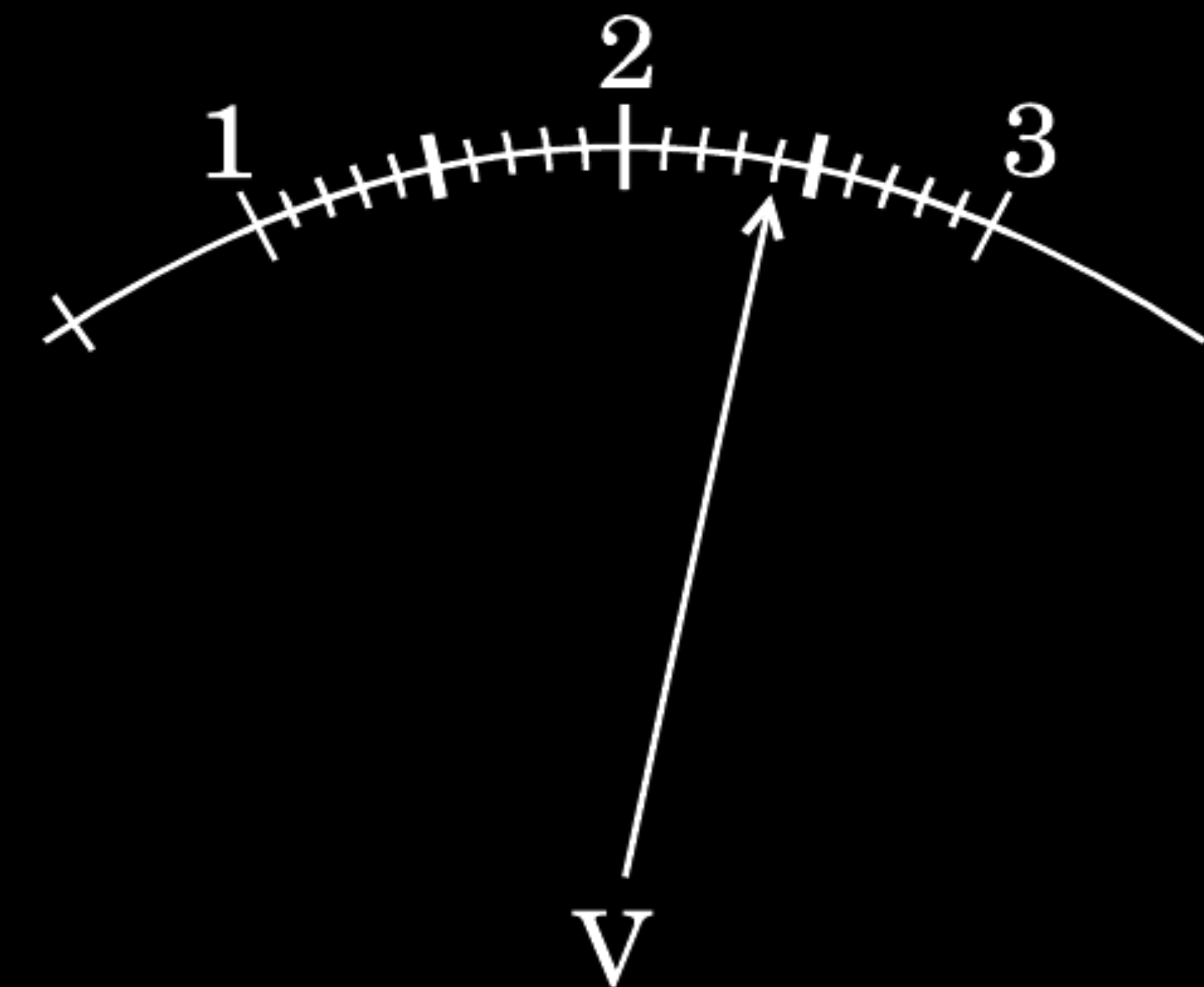
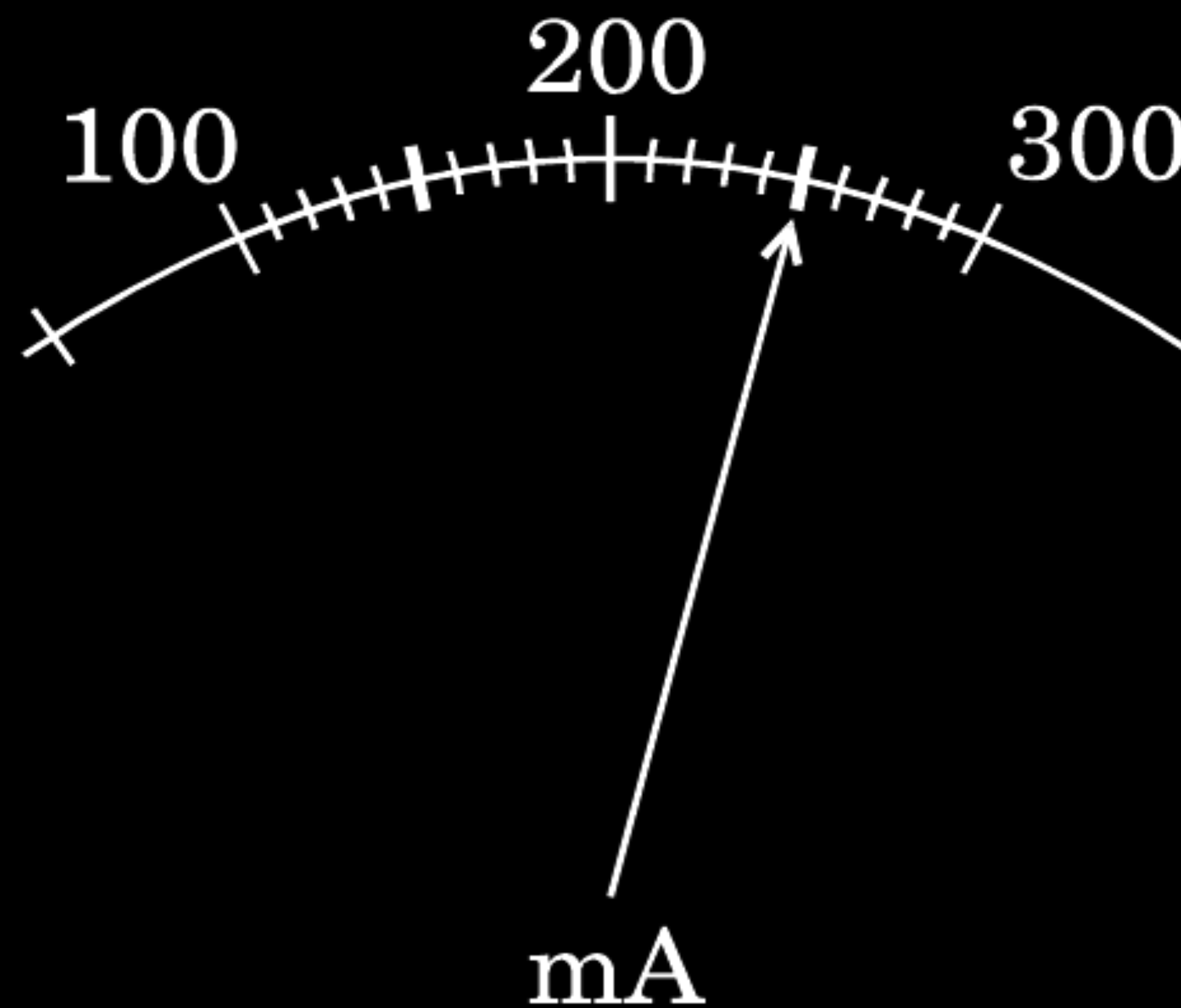
$$1$$

$$1$$

The current flowing through a resistor connected in a circuit and the potential difference developed across its ends are as shown in the diagram by milliammeter and voltmeter readings respectively :

- (a) What are the least counts of these meters ?
- (b) What is the resistance of the resistor ?

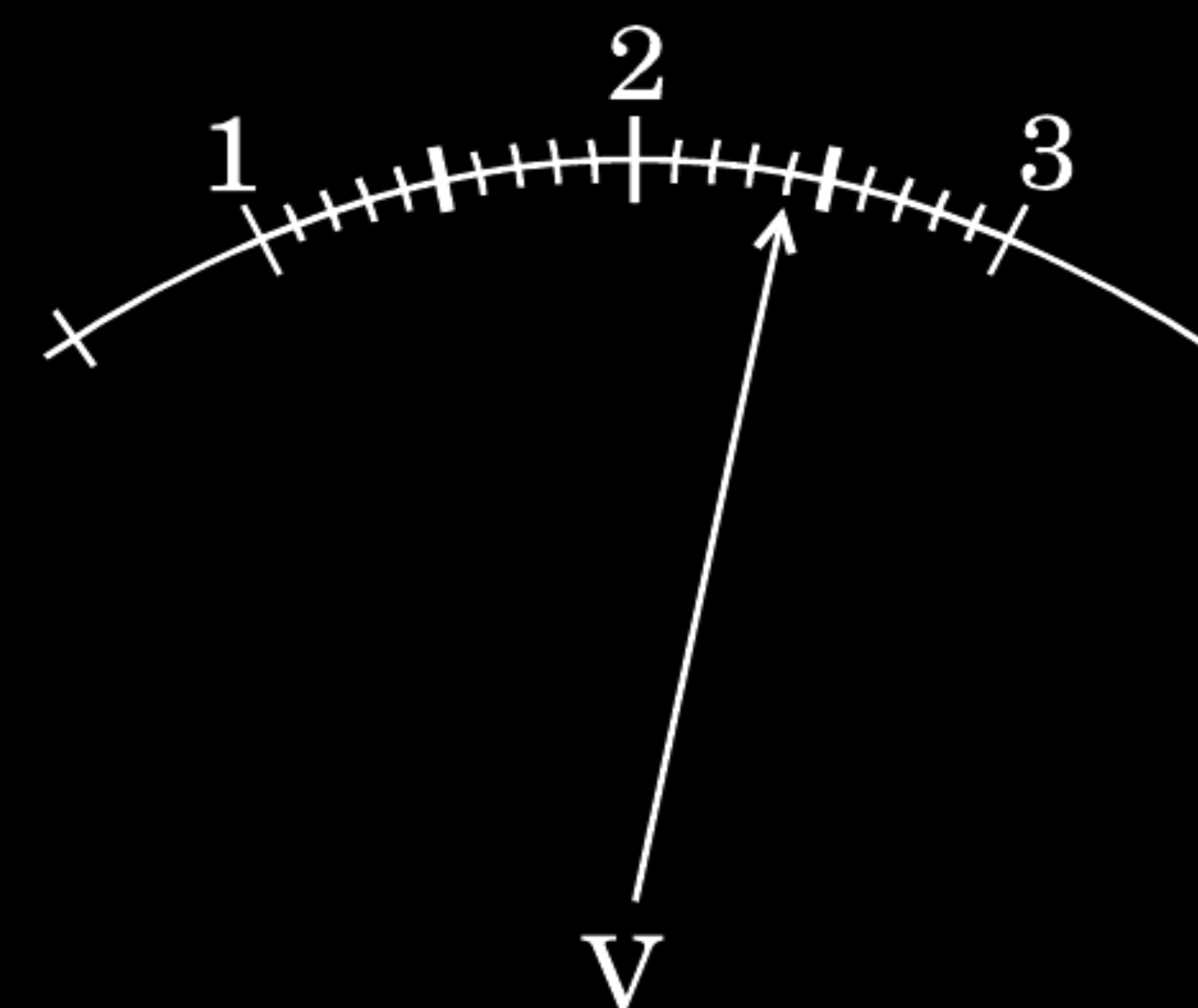
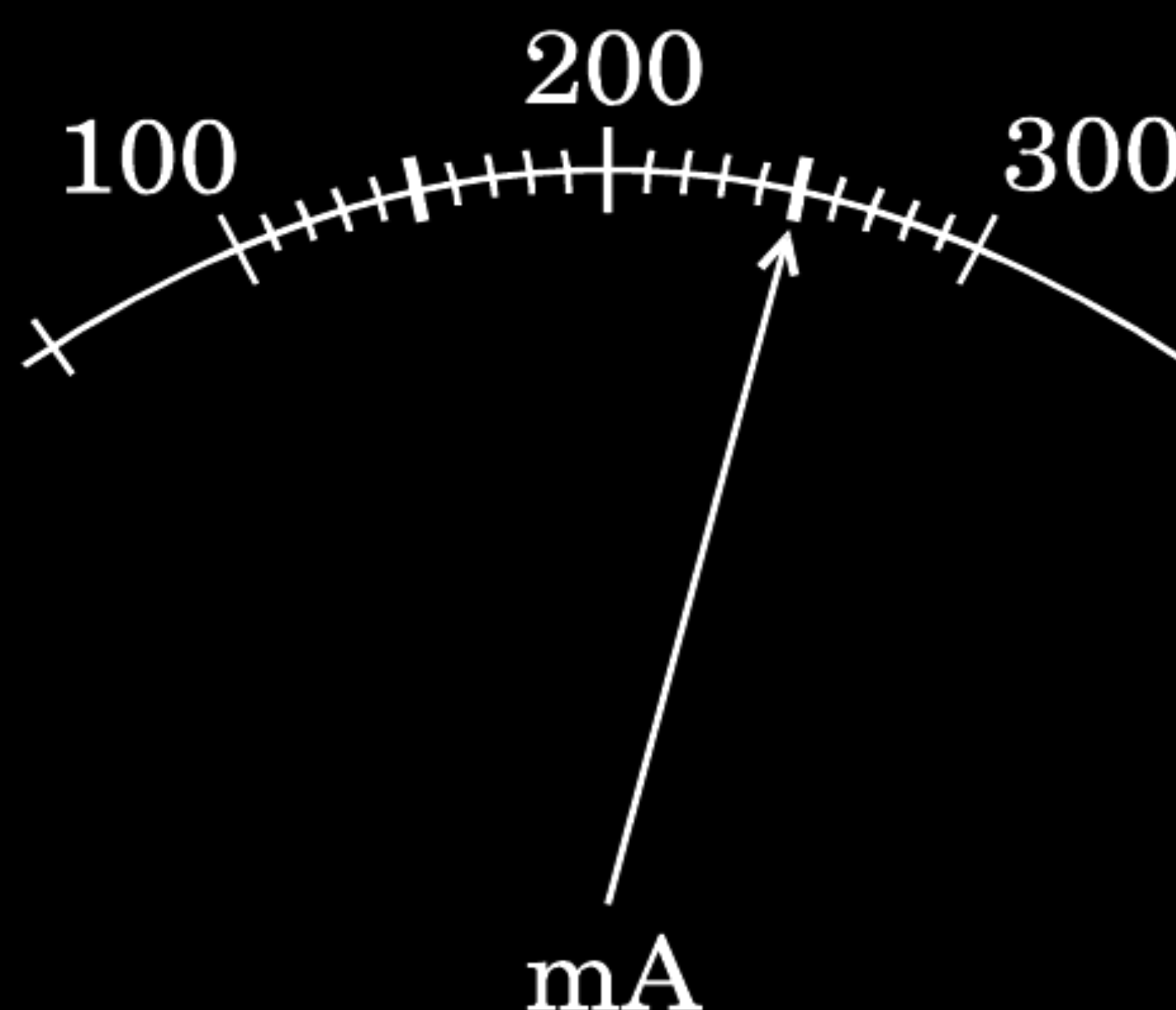
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- (a) What are the least counts of these meters ?
- (b) What is the resistance of the resistor ?

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- (a) $\angle$ least count of ammeter = 10 mA
 $\angle$ least count of Voltmeter = 0.1 V

$\frac{1}{2} + \frac{1}{2}$

- (b) $\frac{2.4}{0.25} = 9.6 \text{ ohm}$ (250mA = 0.25A)

$\frac{1}{2} + \frac{1}{2}$

Should the resistance of a voltmeter be low or high ? Give reason.

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Should the resistance of a voltmeter be low or high ? Give reason.

CBSE 2019

High, In parallel connection, less current passes through high resistance.

$\frac{1}{2} + \frac{1}{2}$

Why does the cord of an electric oven not glow while its heating element does ?

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Why does the cord of an electric oven not glow while its heating element does ?

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Cord is made up of copper wire whereas heating element is made up of alloy.

1

Define the term electrical resistivity of a material.

CBSE 2019

Define the term electrical resistivity of a material.

CBSE 2019

Resistance of material / conductor whose area of cross section is 1m^2 and length 1 meter.

1

- (a) Three resistors of resistances R_1 , R_2 and R_3 are connected (i) in series, and (ii) in parallel. Write expressions for the equivalent resistance of the combination in each case.
- (b) Two identical resistors of $12\ \Omega$ each are connected to a battery of 3 V . Calculate the ratio of the power consumed by the resulting combinations with minimum resistance and maximum resistance.

(a) • In series - $R_S = R_1 + R_2 + R_3$. • In parallel - $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ Resistance is at minimum - $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ $\frac{1}{12} + \frac{1}{12} = \frac{2}{12} = 6\Omega$ Resistance is maximum - $R_S = R_1 + R_2$ $R_S = 12 + 12 = 24\Omega$ $P = \frac{v^2}{R}$ Power ration in parallel and series = 4:1 (b) $\frac{P_{\min}}{P_{\max}} = \frac{V^2 / R_{\min}}{V^2 / R_{\max}} = \frac{R_{\max}}{R_{\min}} = \frac{24}{6} = \frac{4}{1}$	$\frac{1}{2}$ $\frac{1}{2}$ 1 1 2
---	--

- (a) Write the relation between resistance and electrical resistivity of the material of a conductor in the shape of a cylinder of length ' l ' and area of cross-section ' A '. Hence derive the S.I. unit of electrical resistivity.
- (b) Resistance of a metal wire of length 5 m is $100\ \Omega$. If the area of cross-section of the wire is $3 \times 10^{-7}\ \text{m}^2$, calculate the resistivity of the metal.

(a)

$$R \propto l$$

$$R \propto \frac{1}{A}$$

$$R \propto \frac{l}{A}$$

$$R = \rho \frac{l}{A}$$

$$\rho = \frac{RA}{l} = \frac{\text{ohm} \times \text{m}^2}{\text{m}}$$

$$= \text{ohm} \times \text{m}$$

$\frac{1}{2} \times 6$

(b)

$$\rho = \frac{RA}{l}$$

$$= \frac{100 \times 3 \times 10^{-7}}{5}$$

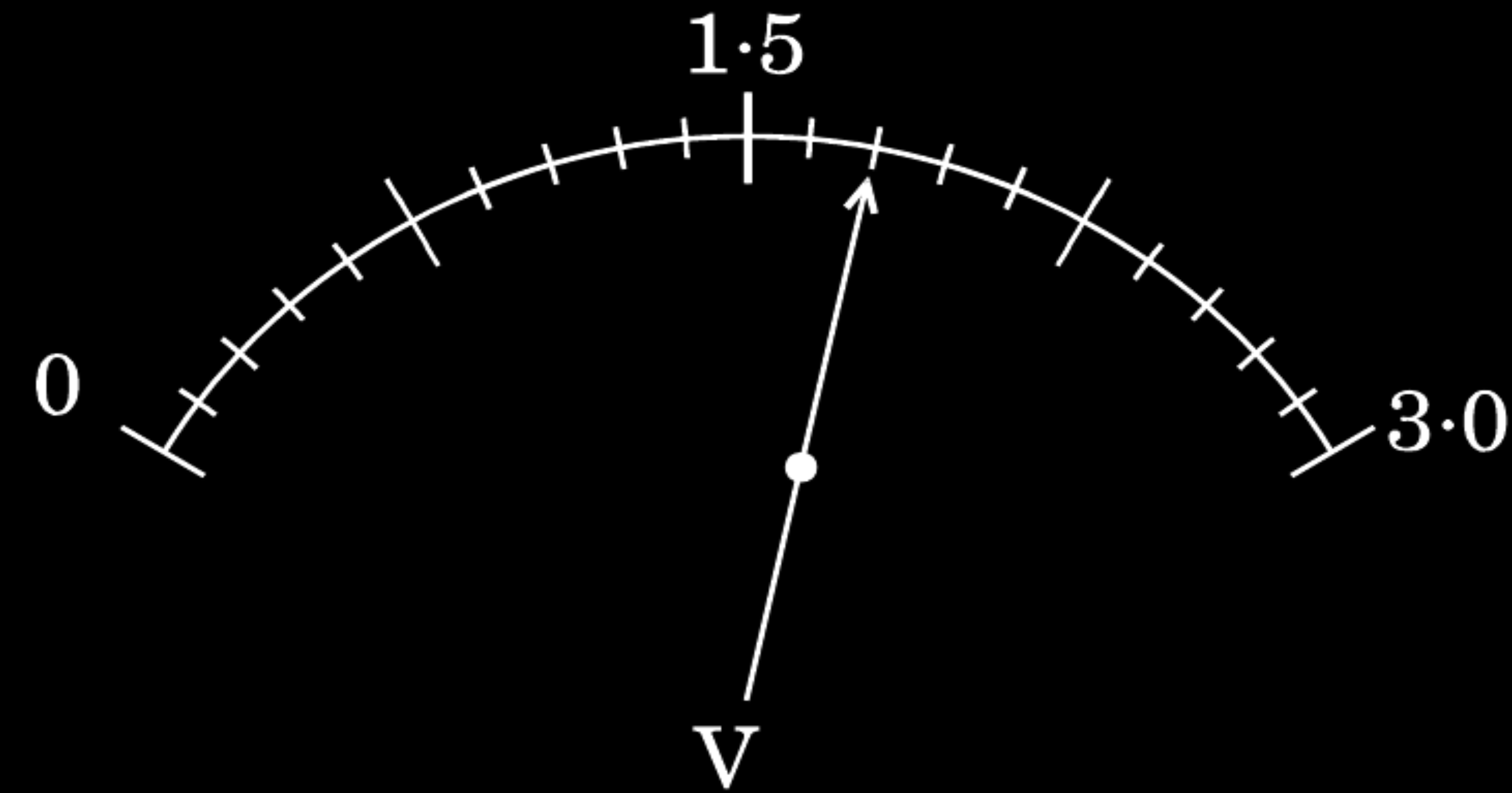
$$= 60 \times 10^{-7} \text{ ohm} \times \text{m}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

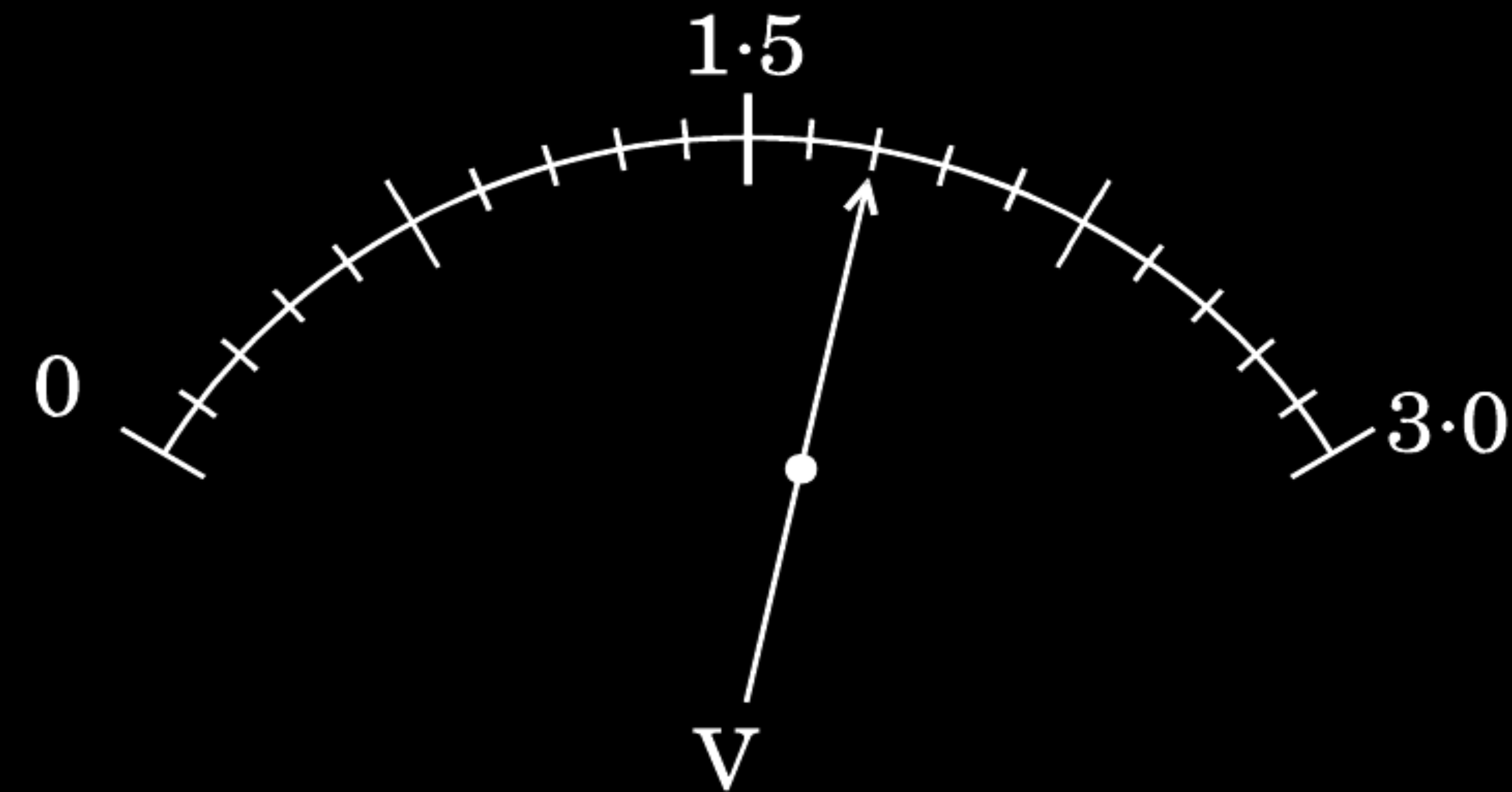
Consider the scale of a voltmeter shown in the diagram and answer the following questions :



- (a) What is the least count of the voltmeter ?
- (b) What is the reading shown by the voltmeter ?
- (c) If this voltmeter is connected across a resistor of $20\ \Omega$, how much current is flowing through the resistor ?

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Consider the scale of a voltmeter shown in the diagram and answer the following questions :



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- (b) What is the reading shown by the voltmeter ?
- (c) If this voltmeter is connected across a resistor of $20\ \Omega$, how much current is flowing through the resistor ?

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a) $0.15V$ is the least count	$\frac{1}{2}$
b) The reading shown is $1.8V$	$\frac{1}{2}$
c) $R = 20\Omega$ $V=1.8V$ $I=\frac{V}{R} = \frac{1.8}{20} = .09amp$	1

When do we say that the potential difference between two points of a circuit is 1 volt ?

CBSE 2019

When do we say that the potential difference between two points of a circuit is 1 volt ?

CBSE 2019

If 1J of work is done to carry 1 coulomb of charge from one point to another.	1
---	---

Define 1 kilowatt hour.

CBSE 2019

Define 1 kilowatt hour.

CBSE 2019

1000 watt of power is used for 1 hour / commercial unit of energy / $3.6 \times 10^6 \text{J}$
--

1

Define resistance. Give its S.I. unit.

CBSE 2019

Define resistance. Give its S.I. unit.

CBSE 2019

• Resistance is the property of a conductor to resist the flow of charges through it • ohm (Ω)	$\frac{1}{2}$ $\frac{1}{2}$
--	--------------------------------

- (a) Define electric power. An electrical device of resistance R is connected across a source of voltage V and draws a current I . Derive an expression for power in terms of current and resistance.
- (b) Two electric bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

- (a) Define electric power. An electrical device of resistance R is connected across a source of voltage V and draws a current I . Derive an expression for power in terms of current and resistance.
- (b) Two electric bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

(a) It is the rate at which electrical energy is dissipated or consumed in an electrical circuit is called electric power .

We know

$$V = W/Q$$

$$W = VQ$$

$$W/t = VQ/t$$

$$P = VI$$

$$P = IR.I$$

$$P = I^2R$$

(b) Bulb I: $I = P/V$, 100W 220V

$$I_1 = 100/220A$$

$$= 5/11A$$

Bulb II : 60W 220V

$$I_2 = 60/220A$$

$$= 3/11 A$$

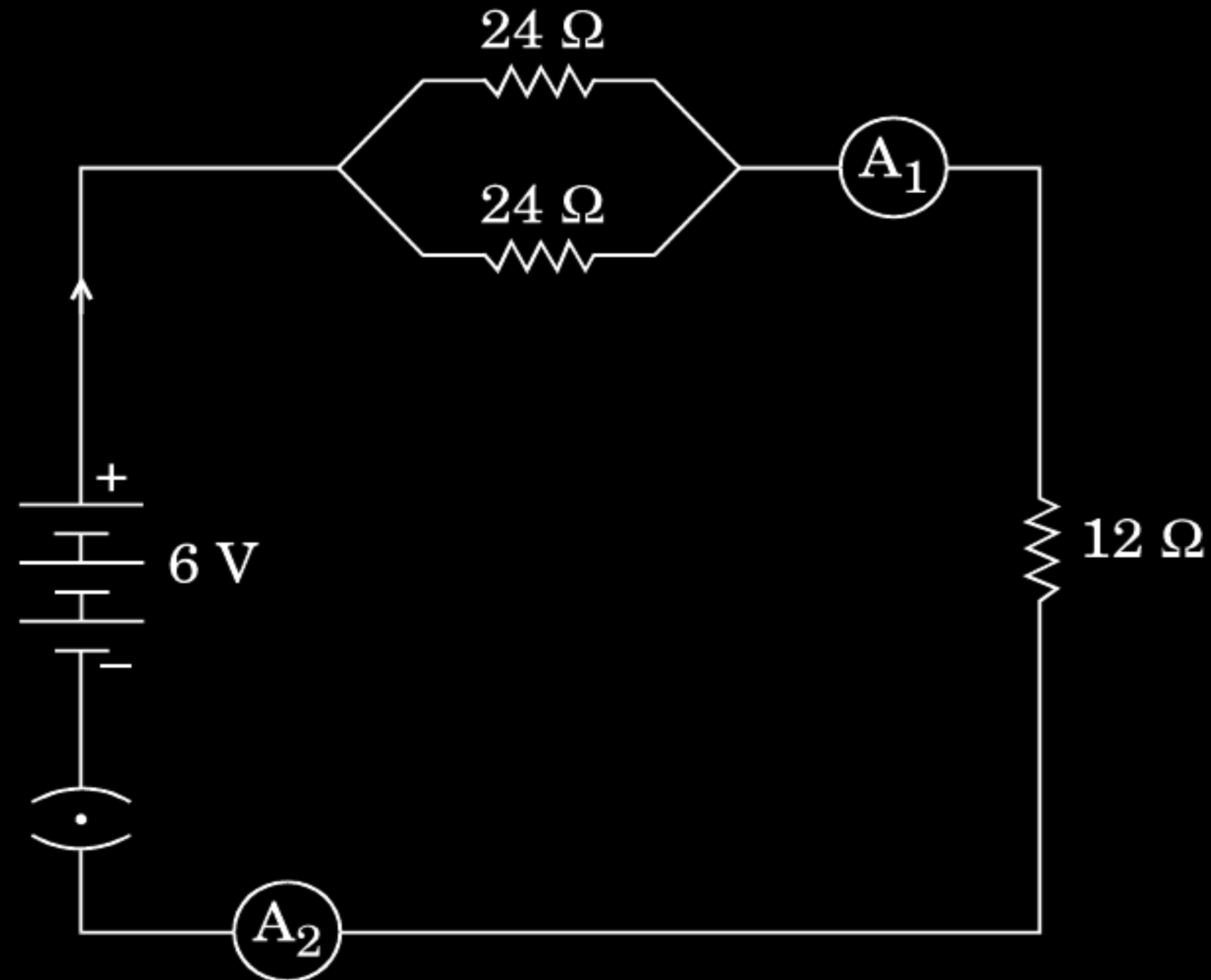
$$\text{Total current } (5/11 + 3/11)A = 8/11A = 0.72A$$

1

2

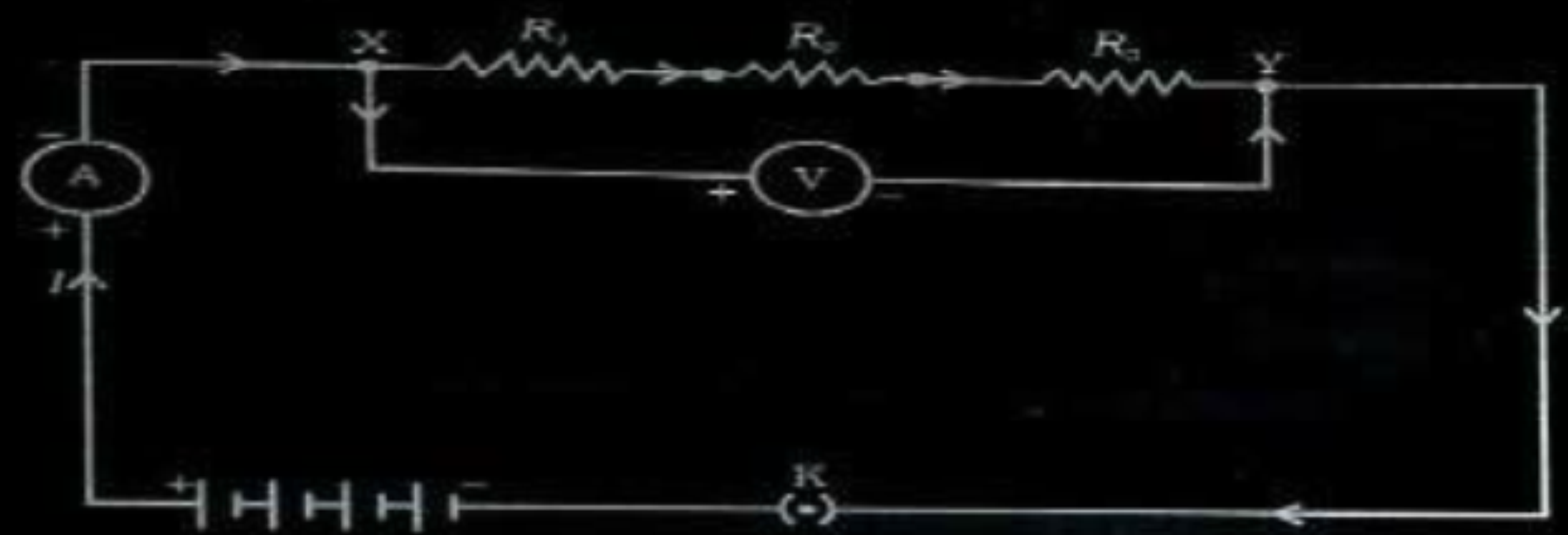
2

- (a) How will you infer with the help of an experiment that the same current flows through every part of the circuit containing three resistors R_1 , R_2 and R_3 in series connected to a battery of V volts ?
- (b) Study the following circuit and find out :
- (i) Current in $12\ \Omega$ resistor.
 - (ii) Difference in the readings of A_1 and A_2 , if any.



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(a)



- Three resistors R_1 , R_2 , R_3 are joined.

- They are connected with

2½

the battery and ammeter and a plug key.

- The ammeter reading is noted.
- Position of ammeter is changed to different position and readings taken each time.
- The reading remain same.

(If it is explained by diagram, give full credit)

(b)

$$(i) \quad \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} \quad \frac{1}{R_1} = \frac{1}{R_1} + \frac{1}{R_1} \quad R_t = R_p + 12 \Omega$$

$$R_t = 24 \Omega$$

1½

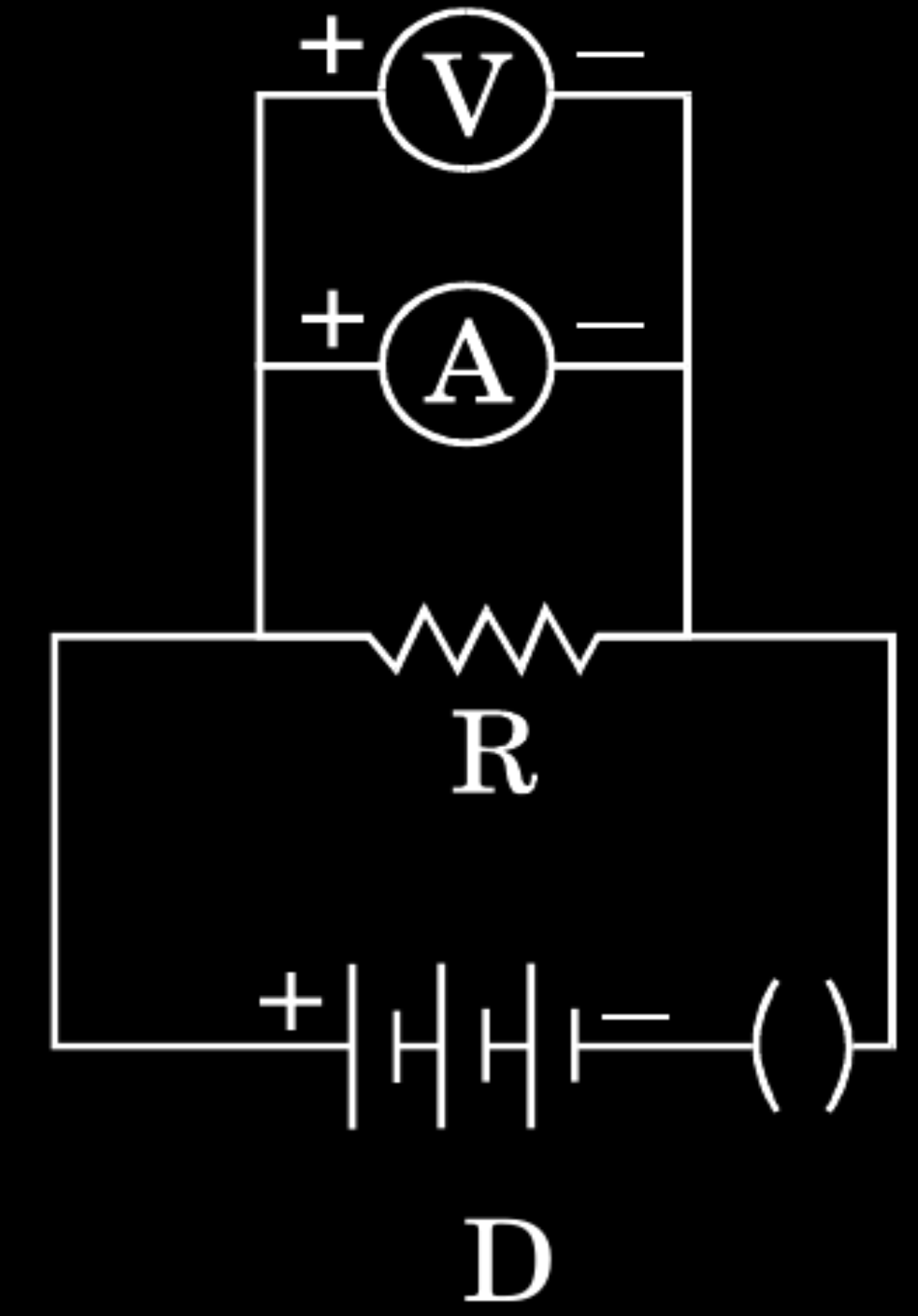
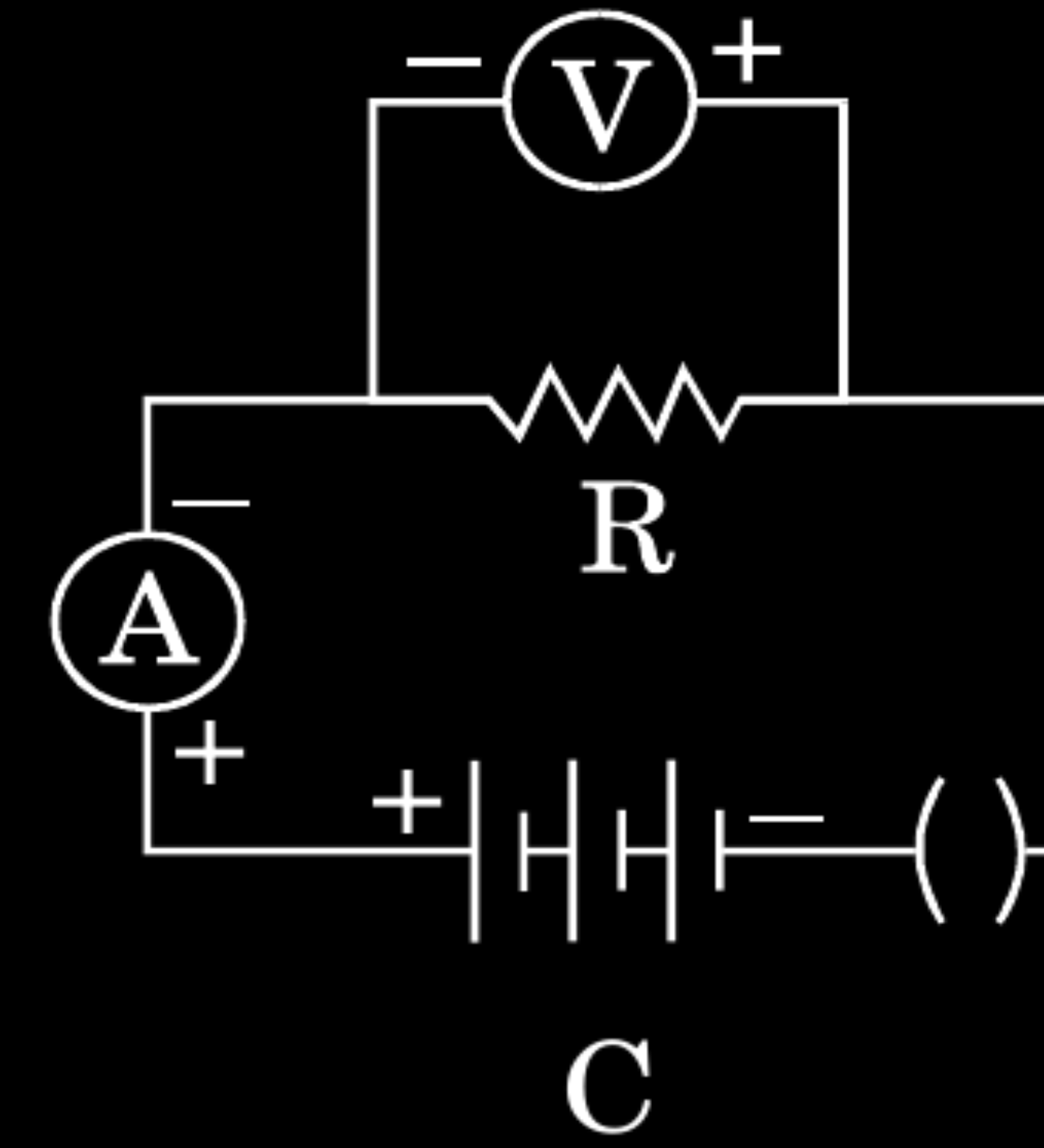
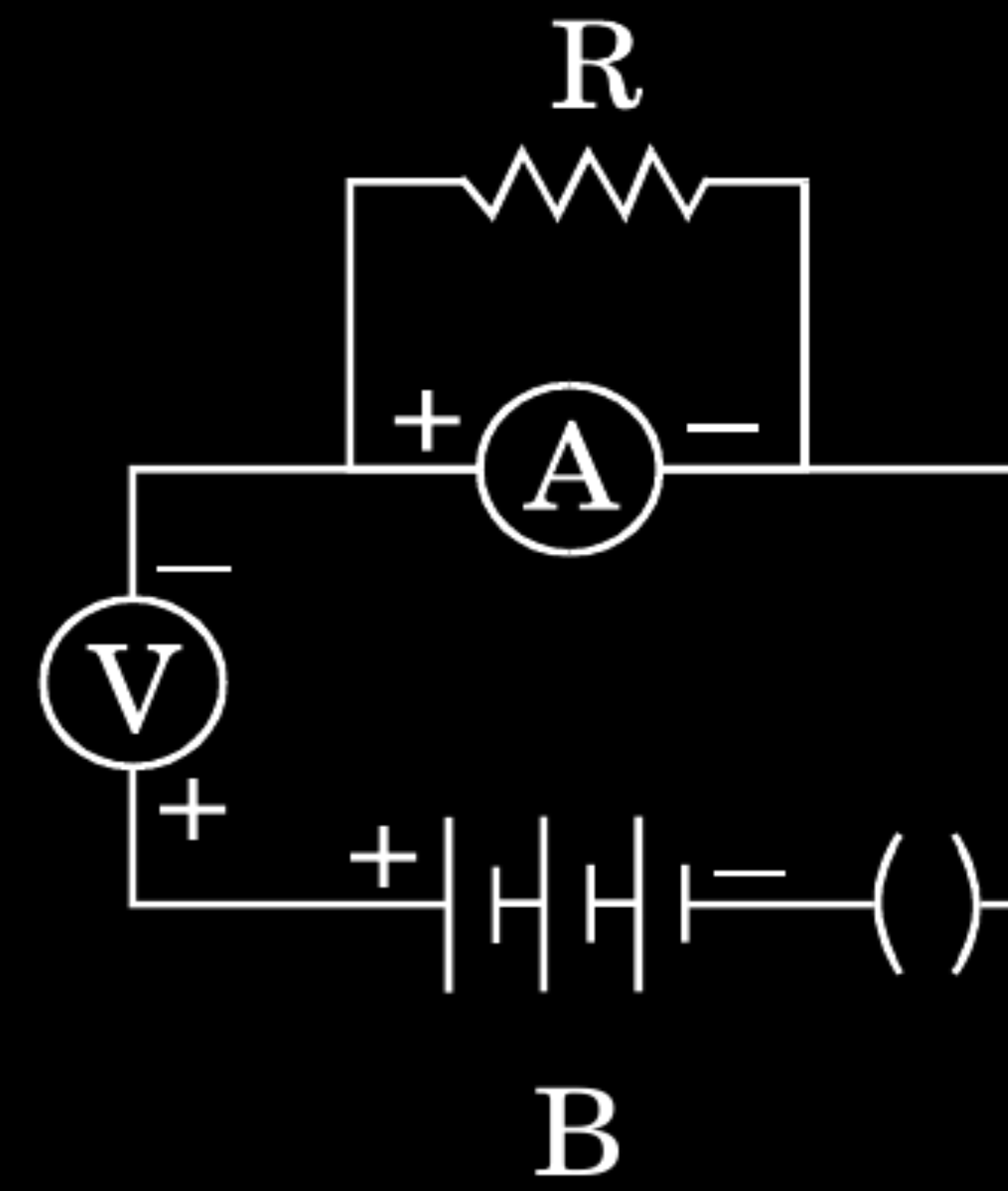
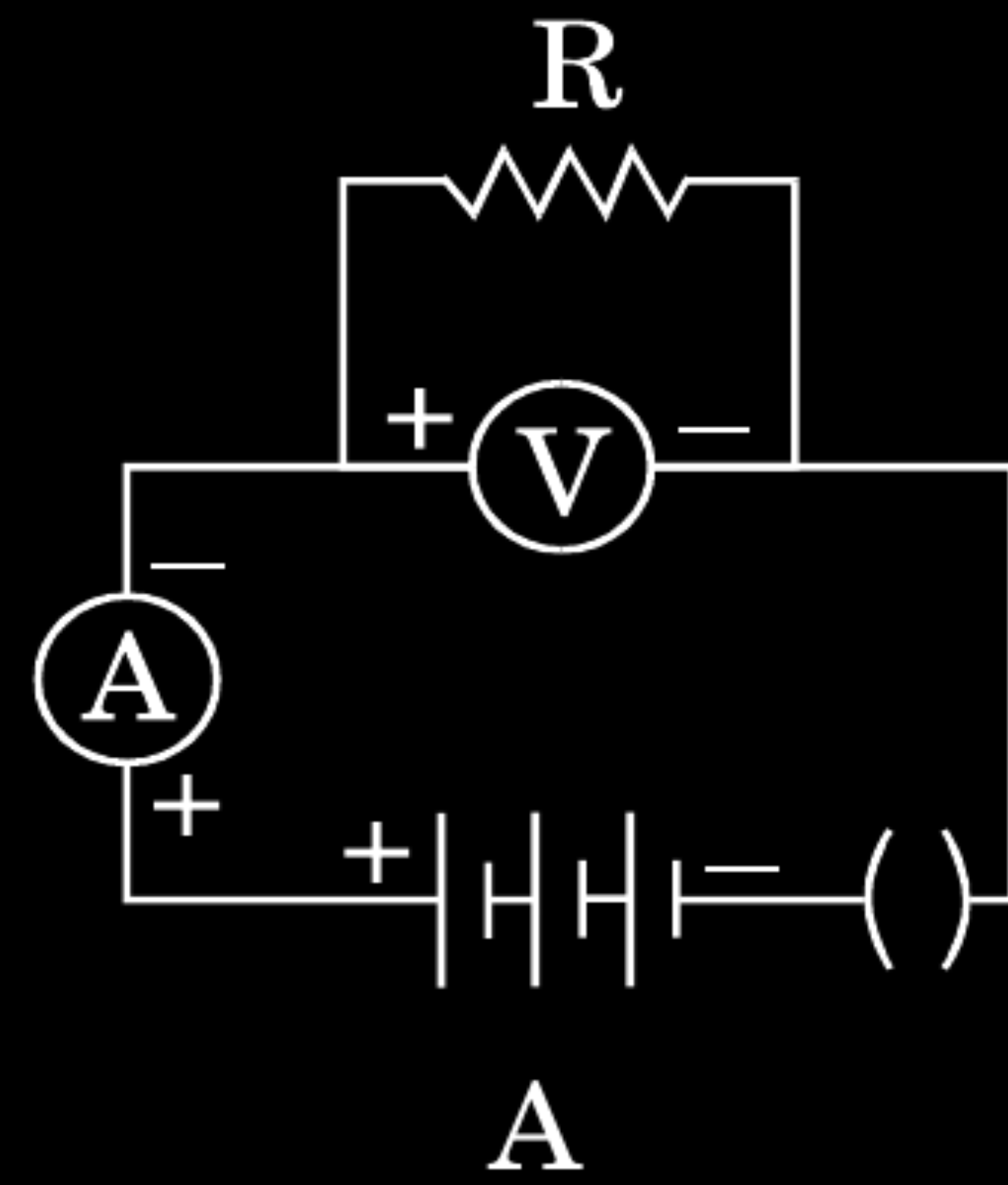
$$V = IR_T$$

$$I = 6/24 = 0.25 \text{ Ampere}$$

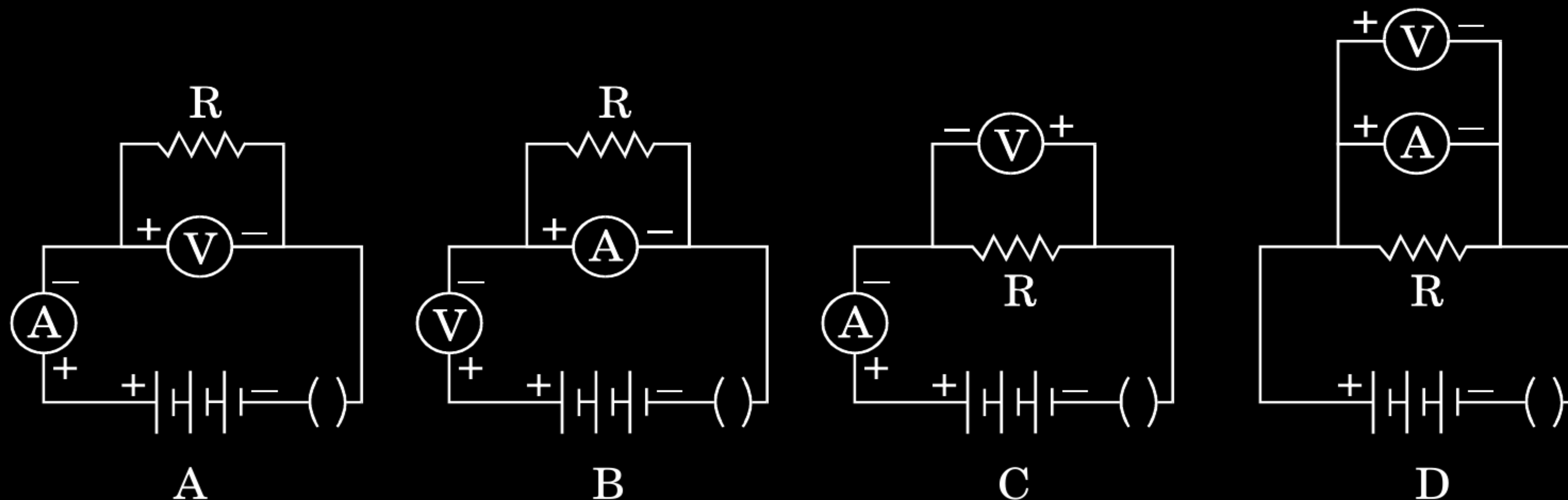
(ii) Same readings of A_1 and A_2

1

Which one of the following is the correct set-up for studying the dependence of the current on the potential difference across a resistor and why ?



Which one of the following is the correct set-up for studying the dependence of the current on the potential difference across a resistor and why ?



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- Set up A is correct.
- Ammeter should be connected in series whereas voltmeter should be connected in parallel to the resistor across which potential difference is to be measured.
- Positive of voltmeter and ammeter should be connected to the positive of supply voltage.

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2}$$

Define current. Give its S.I. unit.

CBSE 2019

Define current. Give its S.I. unit.

CBSE 2019

The rate of flow of charges through a conductor is called electric current. Its S.I. unit is ampere (A)	$\frac{1}{2}$ $\frac{1}{2}$
--	--------------------------------

Mention the condition under which a current can flow in a conductor.

CBSE 2019

Mention the condition under which a current can flow in a conductor.

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Potential difference between ends of conductor	1
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State Ohm's law.

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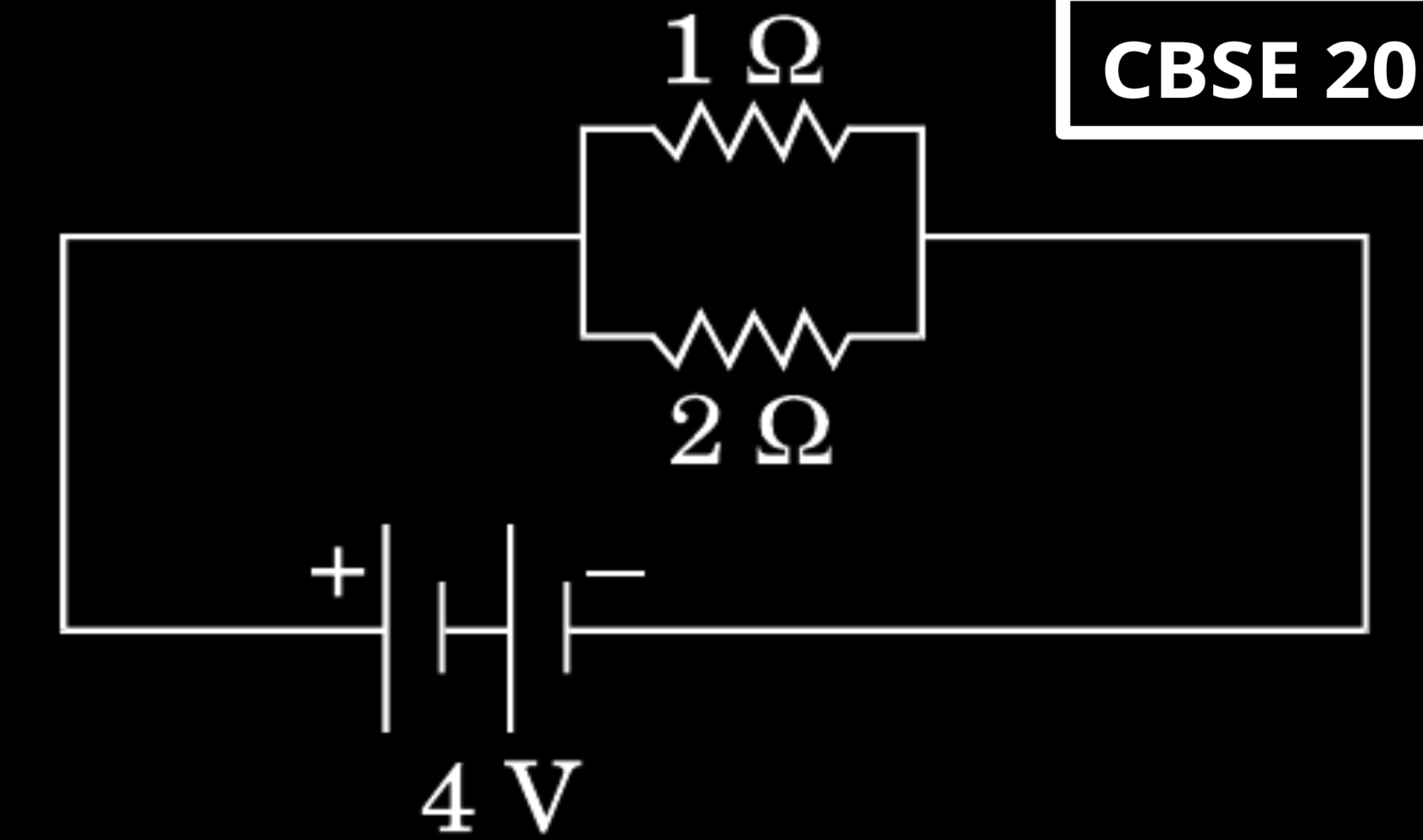
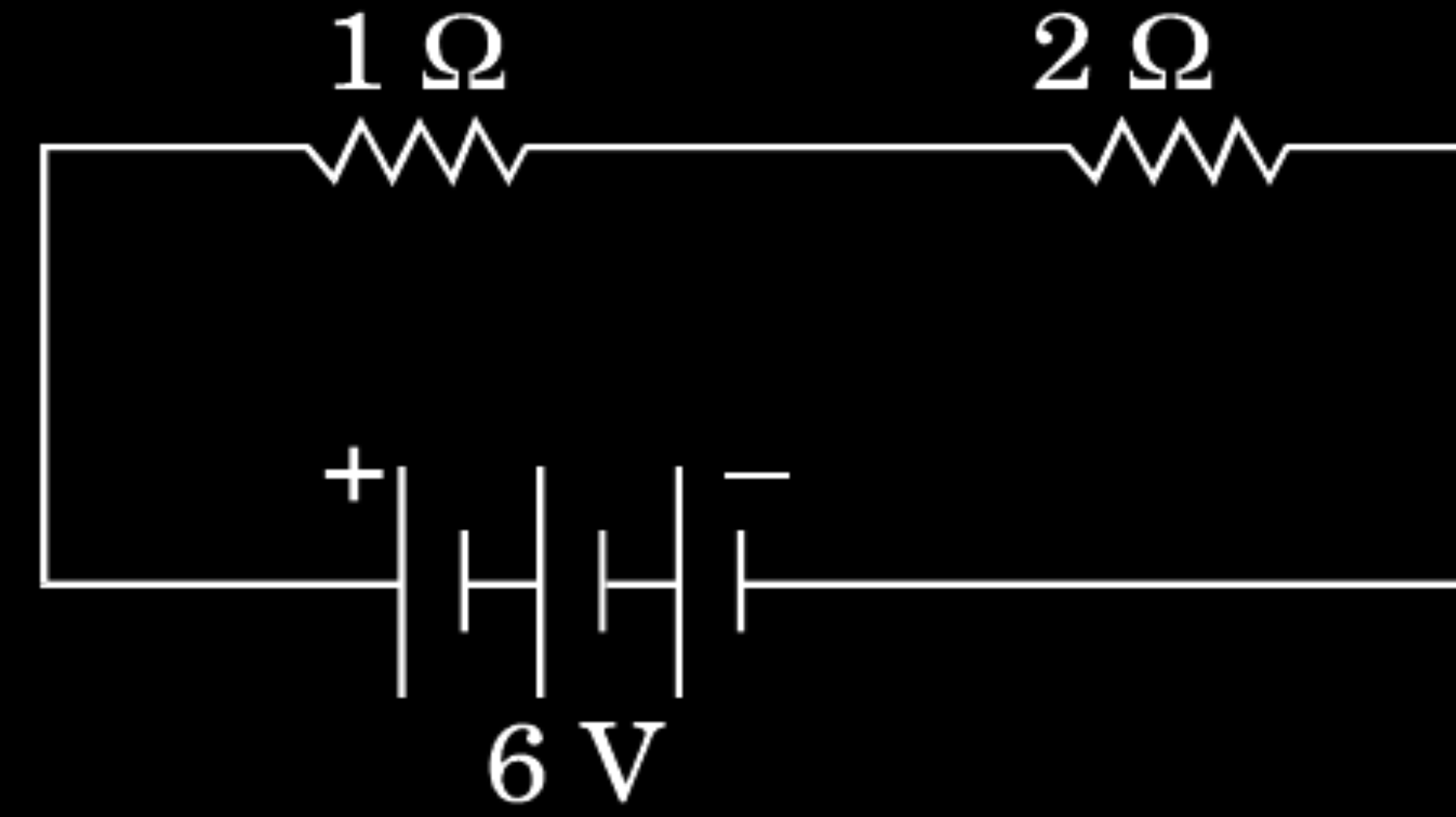
State Ohm's law.

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Ohm's law states that potential difference(V) is directly proportional to the current (I) flowing through the conductor provided the all physical conditions (temperature etc.) remains constant.

1

Compare the power used in $2\ \Omega$ resistor in each of the following circuits :



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(A)

- $R = R_1 + R_2$
 $R = 1\Omega + 2\Omega$
 $R = 3\Omega$
- $V = IR$
 $I = V/R$
 $I = \frac{6V}{3\Omega} = 2 \text{ Ampere or } 2 \text{ A}$
- $P = I^2 R$
 $= 2 \times 2 \times 2$
 $= 8 \text{ W}$

(B) $P = V^2/R$
 $P = \frac{4 \times 4}{2}$
 $P = 8 \text{ W}$

$\frac{1}{2}$

$\frac{1}{2}$

1

1

1

1

A bulb is rated 40 W; 220 V. Find the current drawn by it, when it is connected to a 220 V supply. Also find its resistance. If the given bulb is replaced by a bulb of rating 25 W; 220 V, will there be any change in the value of current and resistance ? Justify your answer and determine the change.

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(i) $P = 40 \text{ W}$ $V = 220 \text{ V}$ $P = VI$ $I = \frac{P}{V} = \frac{40 \text{ W}}{220 \text{ V}}$ $= 0.18 \text{ A}$	1
(ii) $R = \frac{V^2}{P}$ $= \frac{220 \times 220}{40}$ $= 1210 \Omega$	1
(iii) $P = 25 \text{ W}$ $V = 220 \text{ V}$ $P = VI$ $I = \frac{P}{V}$ $= \frac{25}{220} = 0.113 \text{ A}$	1
(iv) $R = \frac{V^2}{P}$ $= \frac{220 \times 220}{25}$ $= 1936 \Omega$	1
(v) Yes there is a change in current and resistance	1

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$$\begin{aligned} \text{(i)} \quad P &= 40 \text{ W} \\ V &= 220 \text{ V} \\ P &= VI \\ I &= \frac{P}{V} = \frac{40 \text{ W}}{220 \text{ V}} \\ &= 0.18 \text{ A} \end{aligned}$$

1

$$\begin{aligned} \text{(ii)} \quad R &= \frac{V^2}{P} \\ &= \frac{220 \times 220}{40} \\ &= 1210 \Omega \end{aligned}$$

1

$$\begin{aligned} \text{(iii)} \quad P &= 25 \text{ W} \\ V &= 220 \text{ V} \\ P &= VI \\ I &= \frac{P}{V} \\ &= \frac{25}{220} = 0.113 \text{ A} \end{aligned}$$

1

$$\begin{aligned} \text{(iv)} \quad R &= \frac{V^2}{P} \\ &= \frac{220 \times 220}{25} \\ &= 1936 \Omega \end{aligned}$$

1

(v) Yes there is a change in current and resistance

1

The rest position of the needles in a milliammeter and voltmeter, not in use, are as shown in Figure A. When a student uses these instruments in his experiment, the readings of the needles are in the positions shown in Figure B. Determine the correct values of current and voltage the student should use in his calculations.

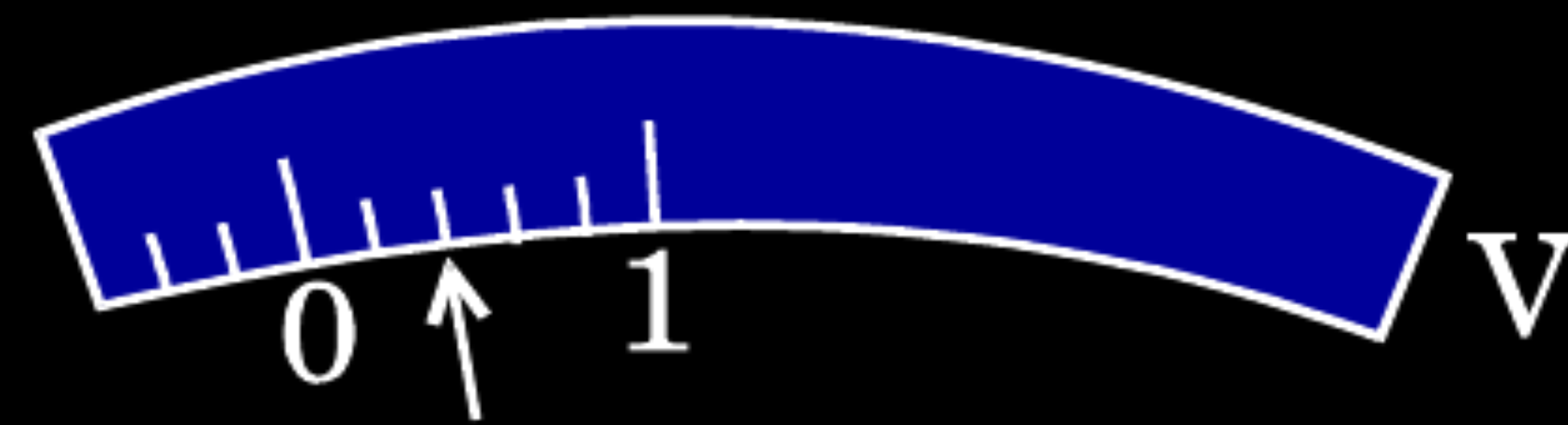
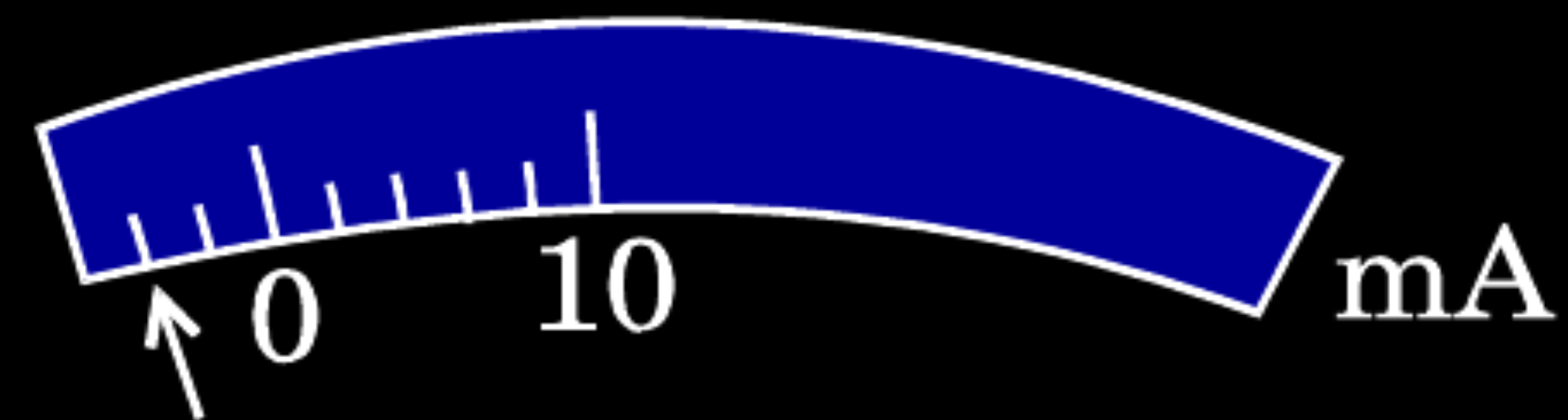


Figure A

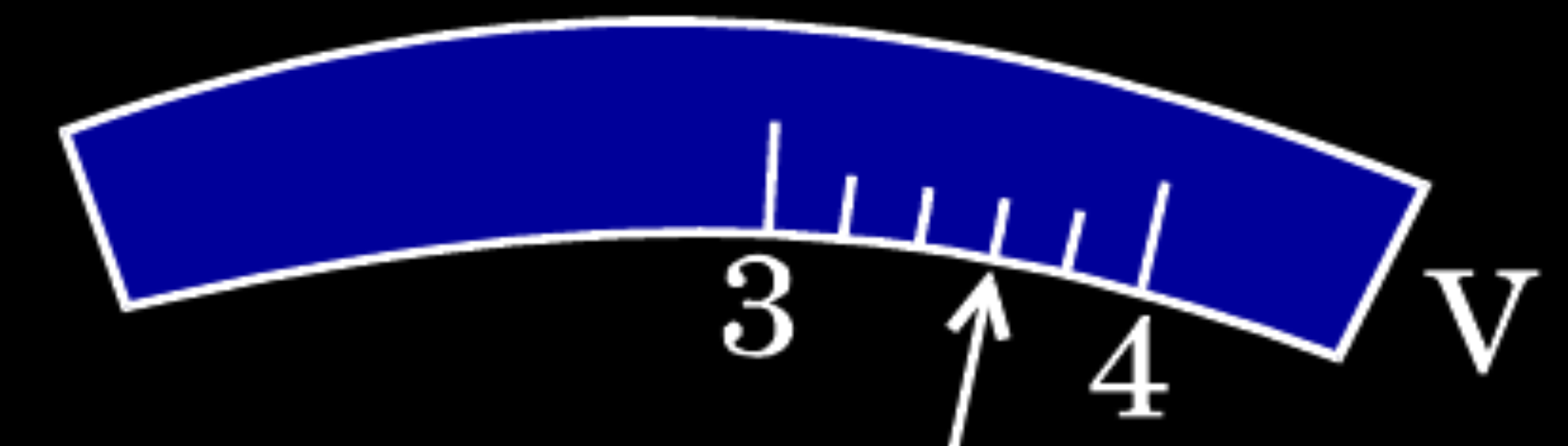
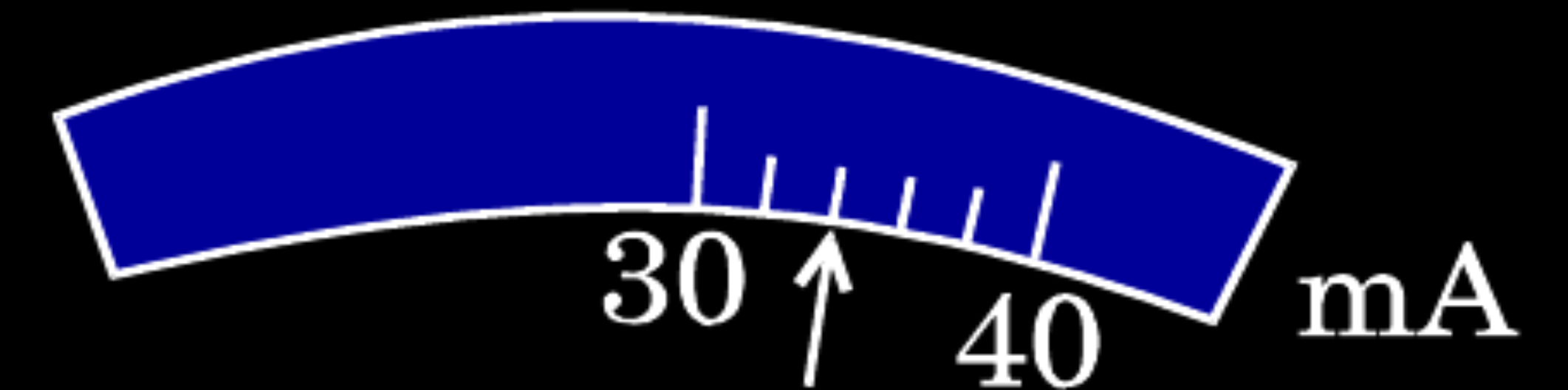


Figure B

The rest position of the needles in a milliammeter and voltmeter, not in use, are as shown in Figure A. When a student uses these instruments in his experiment, the readings of the needles are in the positions shown in Figure B. Determine the correct values of current and voltage the student should use in his calculations.

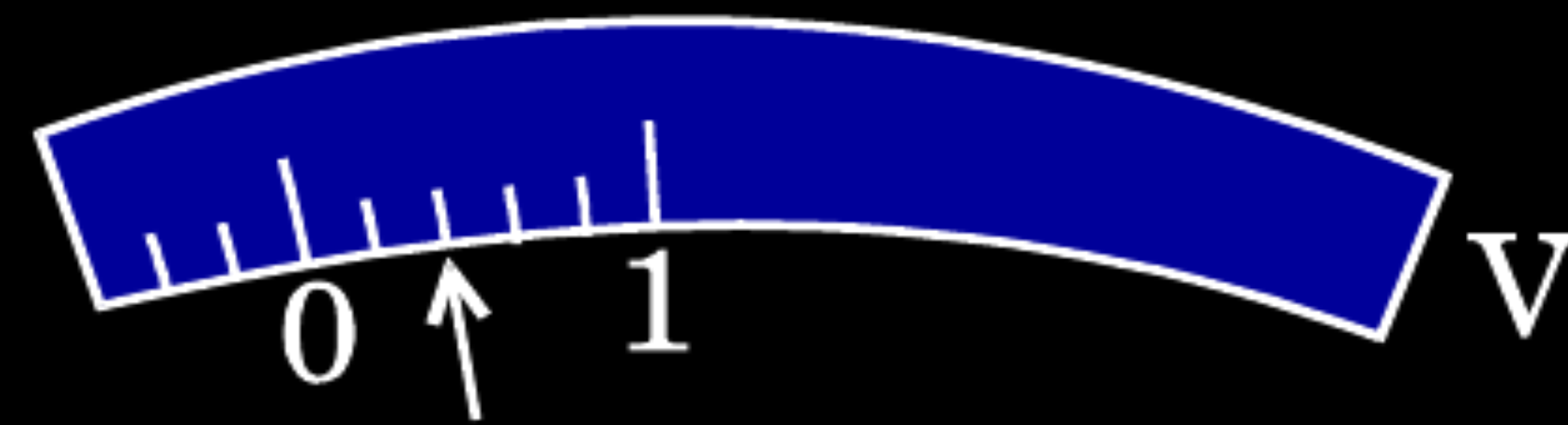
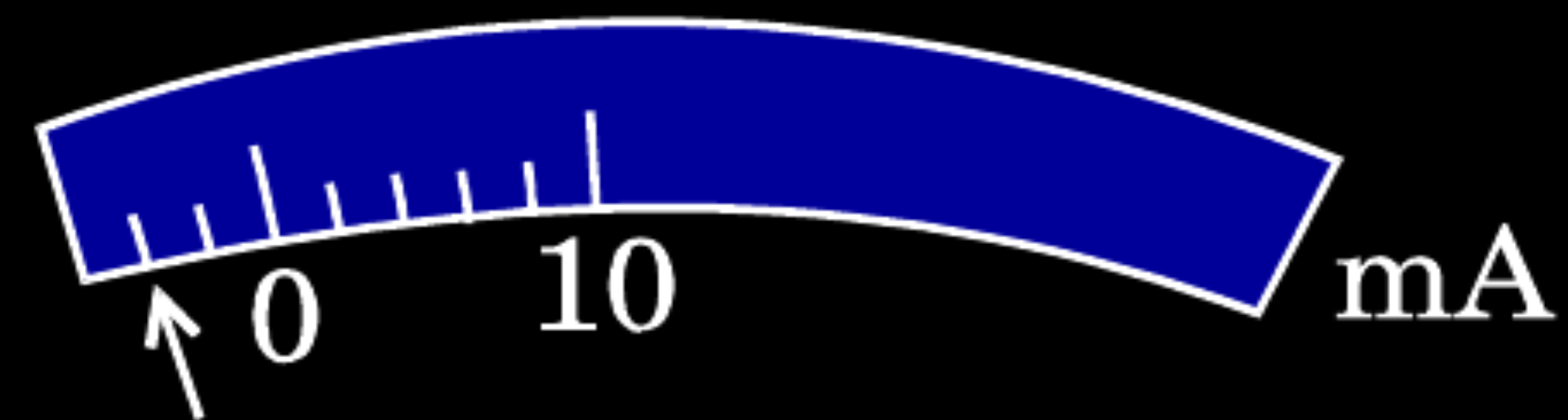


Figure A

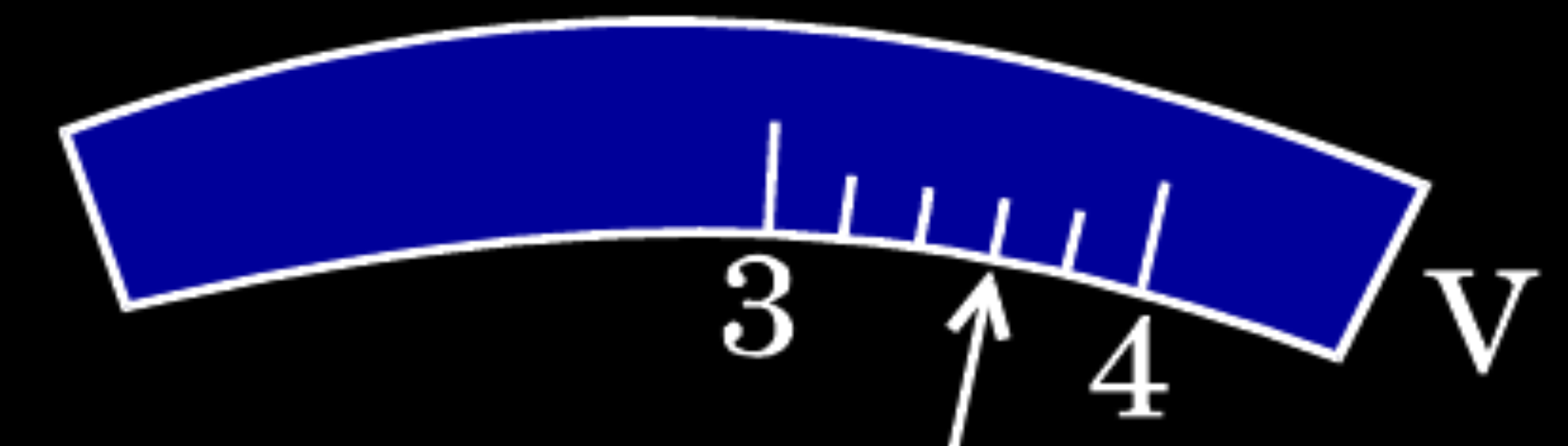
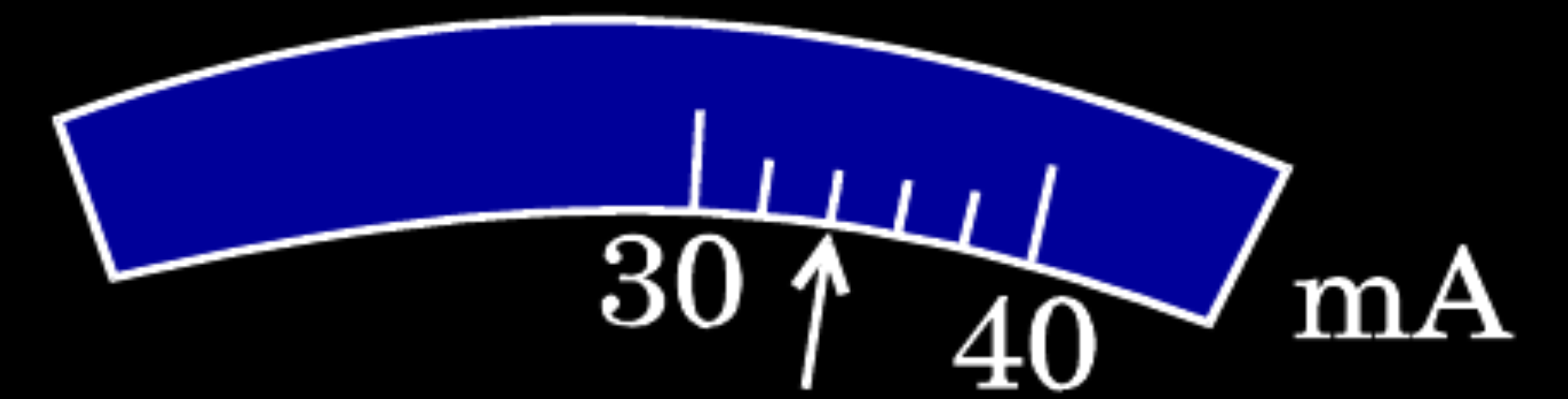


Figure B

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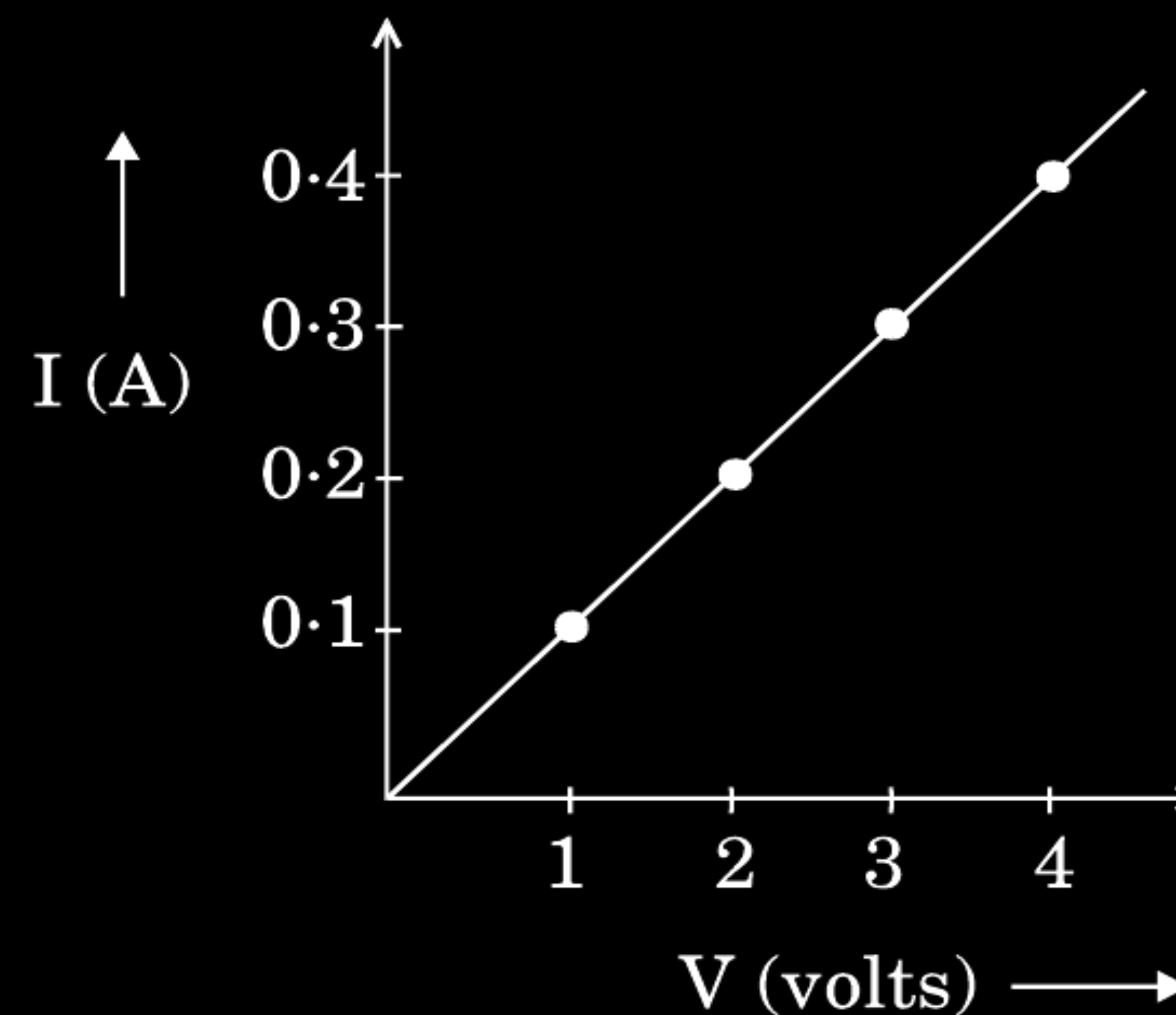
(i) 38mA, 3.2 V

| 1+1

In the experiment to study the dependence of current (I) on the potential difference (V) across a resistor, a student obtained a graph as shown.

- (i) What does the graph depict about the dependence of current on the potential difference ?
- (ii) Find the current that flows through the resistor when the potential difference across it is 2.5 V .

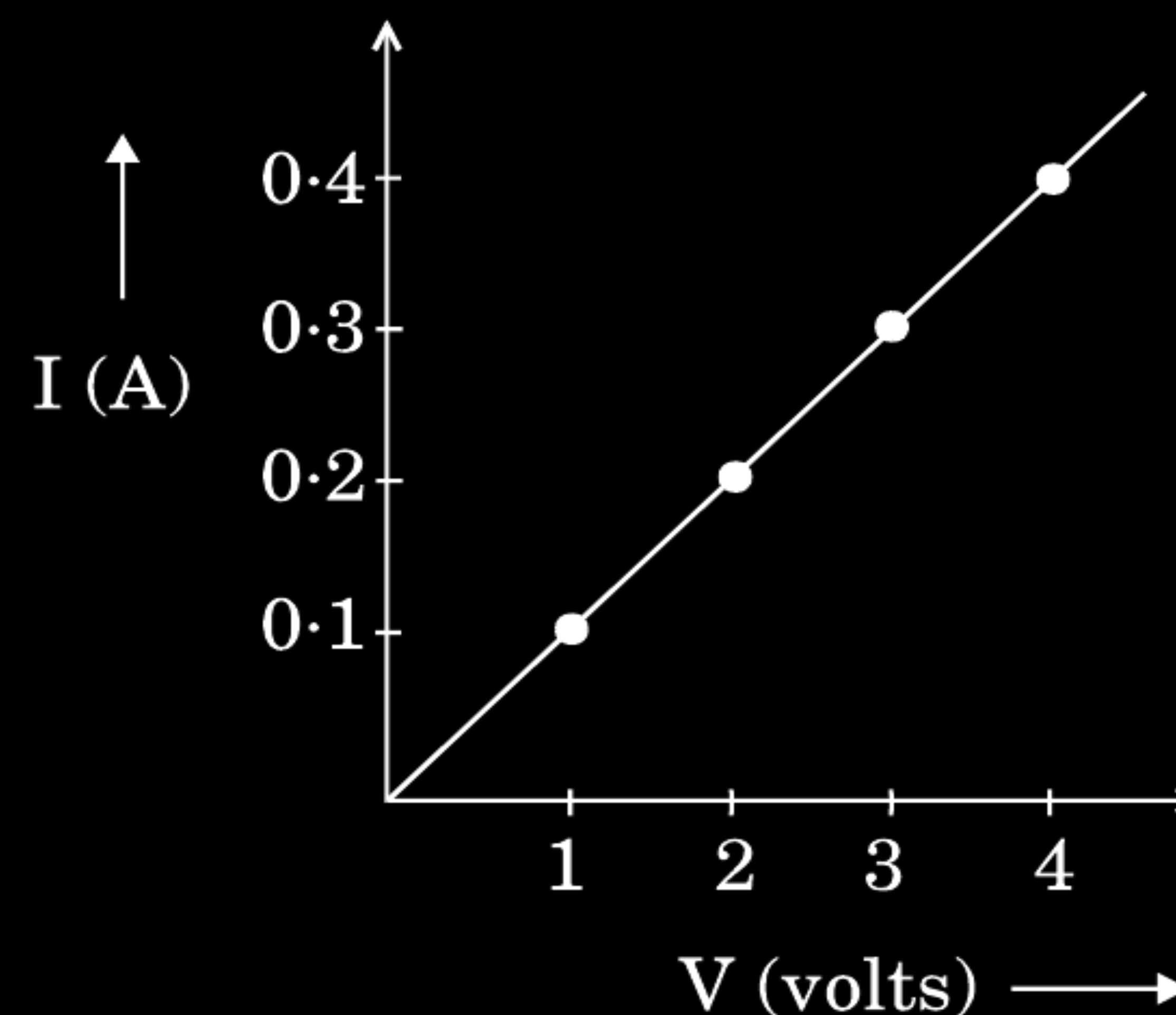
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(i)	$V \propto I$	1
(ii)	at 2.5 V current will be 0.25 A	1

State the S.I. unit of potential difference and name the device used to measure it.

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- Volt
- Voltmeter

$\frac{1}{2}$

$\frac{1}{2}$

- (a) With the help of a suitable circuit diagram prove that the reciprocal of the equivalent resistance of a group of resistances joined in parallel is equal to the sum of the reciprocals of the individual resistances.
- (b) In an electric circuit two resistors of $12\ \Omega$ each are joined in parallel to a $6\ \text{V}$ battery. Find the current drawn from the battery.

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From figure:

$$I = I_1 + I_2 + I_3$$

$$I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}, \quad I_3 = \frac{V}{R_3}$$

$$\therefore \frac{V}{R_P} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\text{b) } R_1 = R_2 = 12 \, \Omega \quad V = 6 \, \text{V}$$

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{12} + \frac{1}{12}$$

$$\therefore R_P = 6 \, \Omega$$

$$I = \frac{V}{R_P} = \frac{6 \, \text{V}}{6 \, \Omega} = 1 \, \text{A}$$

1

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

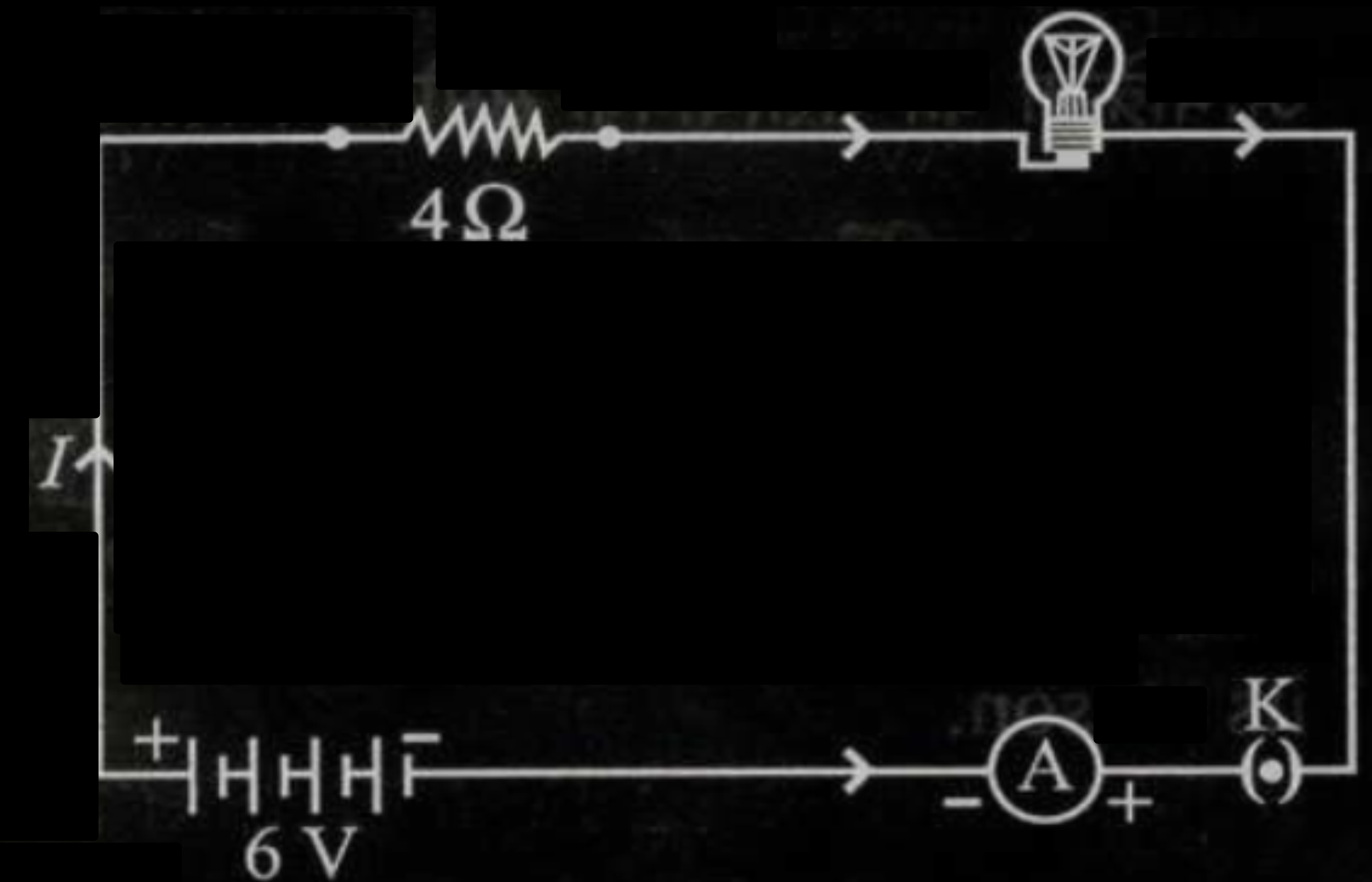
An electric lamp of resistance $20\ \Omega$ and a conductor of resistance $4\ \Omega$ are connected to a $6\ \text{V}$ battery as shown in the circuit. Calculate

(a) the total resistance of the circuit,

(b) the current through the circuit.

(c) the potential difference across the (i) electric lamp and (ii) conductor, and

(d) power of the lamp.



$$\begin{aligned} \text{a) } R &= R_1 + R_2 \\ &= 20 \, \Omega + 4 \, \Omega = 24 \, \Omega \end{aligned}$$

$$\begin{aligned} \text{b) } I &= \frac{V}{R} \\ &= \frac{6V}{24 \, \Omega} = 0.25 \, A \end{aligned}$$

$$\begin{aligned} \text{c) (i) For electric lamp:} \\ V &= IR \\ &= \frac{6}{24} \times 20 = 5 \, V \end{aligned}$$

$$\begin{aligned} \text{(ii) For Conductor:} \\ V &= IR \\ &= \frac{6}{24} \times 4 = 1 \, V \end{aligned}$$

$$\begin{aligned} \text{d) } P &= VI \\ &= 5 \, V \times \frac{6}{24} A = 1.25 \, W \end{aligned}$$

1

1

1

1

While studying the dependence of potential difference (V) across a resistor on the current (I) passing through it, in order to determine the resistance of the resistor, a student took 5 readings for different values of current and plotted a graph between V and I . He got a straight line graph passing through the origin. What does the straight line signify? Write the method of determining resistance of the resistor using this graph

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While studying the dependence of potential difference (V) across a resistor on the current (I) passing through it, in order to determine the resistance of the resistor, a student took 5 readings for different values of current and plotted a graph between V and I. He got a straight line graph passing through the origin. What does the straight line signify? Write the method of determining resistance of the resistor using this graph

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- Potential difference (V) is directly proportional to current (I) or $V \propto I$
- Method: Finding slope of the graph

1

1

What would you suggest to a student if while performing an experiment he finds that the pointer/needle of the ammeter and voltmeter do not coincide with the zero marks on the scales when circuit is open? No extra ammeter/voltmeter is available in the laboratory.

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- | | |
|---|---|
| • Measure the zero error | 1 |
| • Value of zero error should be adjusted to the observed values | 1 |

Write the function of voltmeter in an electric circuit.

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Write the function of voltmeter in an electric circuit.

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To measure potential difference across two points.

Name and define the SI unit of current.

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Name and define the SI unit of current.

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- Ampere
- Flow of 1 coulomb of charge per second / 1 ampere = $\frac{1\text{coulomb}}{1\text{second}}$

If the potential difference across the two ends of a conductor is 5 V and the current through it is 0.2 A, then what is the resistance of the conductor ?

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If the potential difference across the two ends of a conductor is 5 V and the current through it is 0.2 A, then what is the resistance of the conductor ?

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$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{5}{0.2} \\ &= 2.5 \, \Omega \end{aligned}$$

1

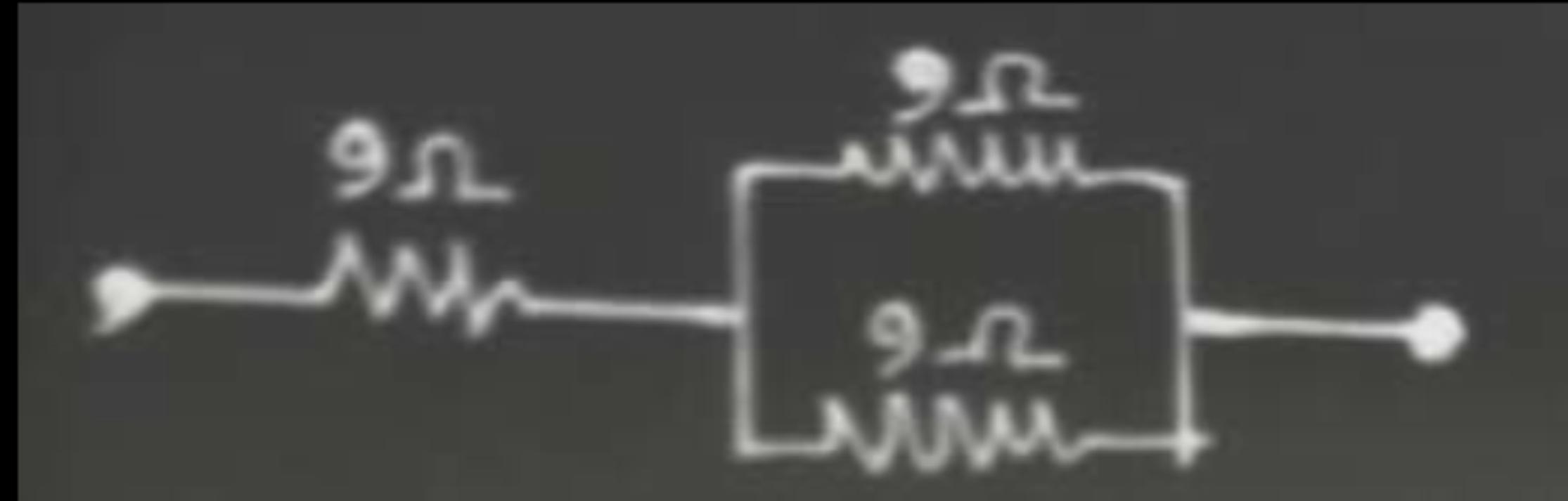
Show how would you join three resistors, each of resistance $9\ \Omega$ so that the equivalent resistance of the combination is (i) $13.5\ \Omega$, (ii) $6\ \Omega$?

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(i)



Two $9\ \text{ohm}$ resistors in parallel connected to one $9\ \text{ohms}$ in series

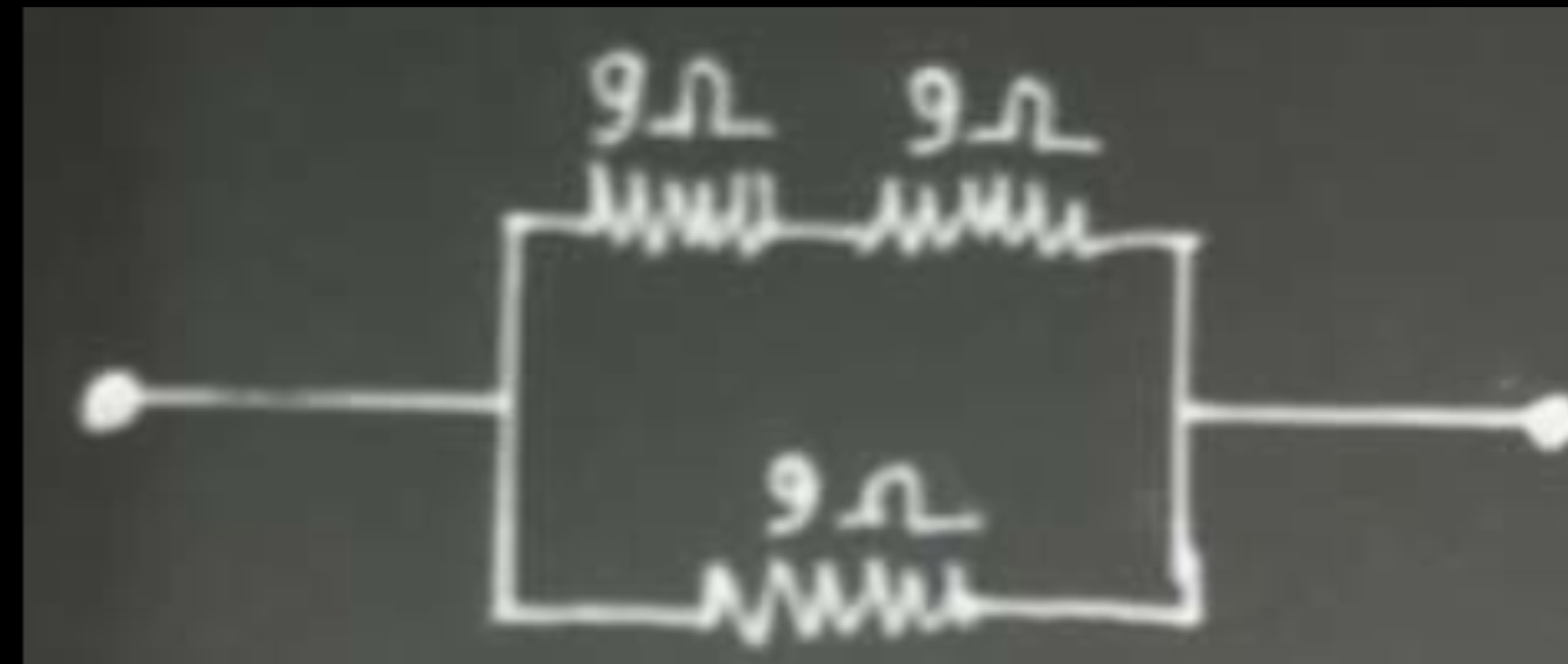
$$\frac{1}{R_p} = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}$$

$$\therefore R_p = \frac{9}{2}\ \Omega$$

$$R = 9\ \Omega + \frac{9}{2}\ \Omega = 13.5\ \Omega$$

$1\ \frac{1}{2}$

(ii)



Two $9\ \text{ohm}$ resistors in series connected to one $9\ \text{ohms}$ in parallel

$$R_s = 9\ \Omega + 9\ \Omega = 18\ \Omega$$

$$\frac{1}{R} = \frac{1}{18} + \frac{1}{9} = \frac{3}{18}$$

$$\therefore R = 6\ \Omega$$

$1\ \frac{1}{2}$

Note: Deduct $\frac{1}{2}$ mark if calculations are not given.

- (a) Write Joule's law of heating.
- (b) Two lamps, one rated 100 W; 220 V, and the other 60 W; 220 V, are connected in parallel to electric mains supply. Find the current drawn by two bulbs from the line, if the supply voltage is 220 V.

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(a) Joule's law of heating – Heat produced in a resistor is (i) directly proportional to the square of current for a given resistance, (ii) directly proportional to the resistance for a given current and (iii) directly proportional to the time for which the current flows through the resistor / $H = I^2Rt$ where, H = Heat produced, I = current, R = Resistance of the conductor and t = Time for which the current flows through the resistor

Note :If the candidate gives only the expression $H = I^2Rt$ award ½ mark only.

(b) Current in 1st bulb, $I_1 = \frac{P_1}{V} = \frac{100}{220} = \frac{5}{11} \text{ A or } 0.45 \text{ A}$
 Current in 2nd bulb, $I_2 = \frac{P_2}{V} = \frac{60}{220} = \frac{3}{11} \text{ A or } 0.27 \text{ A}$

1

1

1

- (a) List the factors on which the resistance of a conductor in the shape of a wire depends.
- (b) Why are metals good conductors of electricity whereas glass is a bad conductor of electricity ? Give reason.
- (c) Why are alloys commonly used in electrical heating devices ? Give reason.

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- (b) Why are metals good conductors of electricity whereas glass is a bad conductor of electricity ? Give reason.
- (c) Why are alloys commonly used in electrical heating devices ? Give reason.

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a. Factors on which resistance of a conductor depends:	
i. Length of conductor [or $R \propto \ell$]	$\frac{1}{2}$
ii. Area of cross-section of the conductor [or $R \propto 1/A$]	$\frac{1}{2}$
b. Metals are good conductor of electricity –as they have low resistivity/ have free electrons	$\frac{1}{2}$
Glass is a bad conductor of electricity – as it has high resistivity/have no free electrons	$\frac{1}{2}$
c. Reason:	
Alloys have high resistivity /high melting point / alloys do not oxidize (Or burn) readily at high temperatures.	
(any one)	1

The values of current (I) flowing through a given resistor of resistance (R), for the corresponding values of potential difference (V) across the resistor are as given below :

V (volts)	0.5	1.0	1.5	2.0	2.5	3.0	4.0	5.0
I (amperes)	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0

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Plot a graph between current (I) and potential difference (V) and determine the resistance (R) of the resistor.

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